CONGRUENCES FOR THE NUMBER OF SUBGROUPS OF ORDER p^k IN A FINITE GROUP

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The method used by Serre [1, p. 147] to prove the Sylow theorem gives results more general than that theorem. We spell them out here.

We use the following notation: p denotes a prime number; for every integer n, $v(n)$ is the exponent of the largest power of p dividing n; finally, G is a finite group of order g.

For completeness, recall the following lemma (Lazard [2, p. 71]).

Lemma 1.

a) *If* $n = \sum c_i p^i$, *with* $0 \leq c_i < p$ *the base-*p *expansion of* n, *then*

$$v(n!) = \frac{1}{p-1}\left(n - \sum c_i\right).$$

b)

$$v\binom{r}{s} \geq v(r) - v(s).$$

For a), one has

$$v(p^k!) = \sum_{i \leq k} i(p^{k-i} - p^{k-i-1}) = \sum_{i=0}^{k-1} p^i = \frac{p^k - 1}{p-1},$$

which proves the assertion in this case. It is clear that, if $0 \leq c < p$, then $v((cp^k)!) = c\,v(p^k!)$, and that, if $m < p^{v(n)}$, then

$$v((n+m)!) = v(n!) + v(m!).$$

These additivity properties are the same as those of the right-hand side, whence the formula.

Part b) follows easily by returning to the definitions and taking account of the algorithm which computes the digits c_i of a sum from the digits of its summands. Equality holds when r is a power of p.

Proposition 1.

$$\binom{p^k n}{p^k m} \equiv \binom{n}{m} \pmod{p^{v(n)+2}}$$

if $p \neq 2$, *or if* n *is odd. The congruence is always true modulo* $p^{v(n)+1}$.

Proof. Let X and Y be two indeterminates. One has

$$(X+Y)^{p^k} = X^{p^k} + Y^{p^k} + pQ,$$

where Q is a polynomial of degree $< p^k$ in each variable. The binomial formula then gives

$$(1) \qquad (X+Y)^{p^k n} = (X^{p^k} + Y^{p^k})^n + np(X^{p^k} + Y^{p^k})^{n-1} Q$$

modulo the largest power of p which divides all the numbers $p^i\binom{n}{i}$, $i \geq 2$. These have valuation at least

$$i + v(n) - v(i) \geq v(n) + 2$$

if $p \neq 2$. If $p = 2$, one always has $i > v(i)$ (Cantor), which gives the exponent $v(n) + 1$.

The binomial coefficient $\binom{p^k n}{p^k m}$ is the coefficient of $X^{p^k m} Y^{p^k(n-m)}$. The second term on the right-hand side of (1) contains no term of this type, since none of its exponents is divisible by p^k. Thus the coefficient of this monomial on the right-hand side of (1) is $\binom{n}{m}$. This proves the assertion when $p \neq 2$. If $p = 2$, the preceding already proves the congruence modulo $2^{v(n)+1}$. In addition, $Q \equiv X^{2^{k-1}} Y^{2^{k-1}} \pmod 2$, so $Q^2 \equiv X^{2^k} Y^{2^k}$ (mod 4), and the next term in the binomial expansion gives

$$\binom{2^k n}{2^k m} \equiv \binom{n}{m} + 2n(n-1)\binom{n-2}{m-1} \pmod{2^{v(n)+2}},$$

which gives the asserted improvement when n is odd. □

Theorem.

Let S be a Sylow p-subgroup of G, let $s = v(g)$, and let d_k be the number of subgroups of G of order p^{s-k}, for $0 \leq k \leq s$. Then:

(i) $d_k \equiv 1 \pmod p$;

(ii) *if S is cyclic or abelian of type $(p, \ldots, p)$, then d_k is congruent modulo p^{k+1} to the number of subgroups of S of index p^k;*

(iii) *if S is not cyclic, then*

$$d_1 \equiv 1 + p \pmod{p^2}.$$

Let G act by left translations on the set of its subsets with p^k elements. If such a subset P of G has stabilizer H, then P is a union of right cosets of H, and hence H is a group of order p^l with $l \leq k$; one has $l = k$ if and only if P is a translate, on the right or on the left, of a subgroup of order p^k. The number of elements in the orbit of P is divisible by p^{s-l}. Therefore $\binom{g}{p^k}$ is congruent modulo p^{s-k+1} to the number of subsets of G which are translates of a subgroup of order p^k. By Proposition 1, it is also congruent to gp^{-k}. There are $d_{s-k}\, g\, p^{-k}$ such subsets. Thus

$$gp^{-k} \equiv d_{s-k}\, g\, p^{-k} \pmod{p^{s-k+1}},$$

and the first assertion follows.

We now consider case (ii), first when S is abelian of type $(p, \ldots, p)$. Let $G_{i,j}$ be the number of subgroups of order p^i in a group of order p^j of this type; this is the number of points of a Grassmannian. For $i > j$, $G_{i,j} = 0$; for $i \leq j$,

$$G_{i,j} = \frac{(p^j - 1)\cdots(p^j - p^{i-1})}{(p^i - 1)\cdots(p^i - p^{i-1})} = G_i G_{j-i} G_j^{-1},$$

where

$$G_i = [(p^i - 1)\cdots(p-1)]^{-1} \quad (i \geq 0), \qquad G_i = 0 \quad (i < 0).$$

We know that p^i divides the number u_i of subsets of G with p^i elements whose stabilizer has order p^{s-i}. Put $g = p^s h$. Counting in two ways the pairs formed by a subgroup of order p^{s-i} and a subset with p^i elements fixed by it gives, for $0 \leq i, j \leq s$,

(2) $$d_i \binom{p^s h}{p^i} = \sum_j G_{s-i,s-j} u_j,$$

(3) $$p^i \mid u_i,$$

(4) $$d_s = 1.$$

Now work in the localization of $\mathbf{Z}$ at p, or in $\mathbf{Z}_p$. Let e_i be the quotient of the left-hand side of (2) by $\binom{p^s h}{p^i} G_i$. By Proposition 1, and since G_i is prime to p for $i \geq 0$, there are numbers v_i, $0 \leq i \leq s$, such that

$$d_i \equiv G_i e_i \pmod{p^{i+1}}, \tag{5}$$

$$e_i = \sum_j G_{i-j} v_j, \tag{2′}$$

$$p^i \mid v_i, \tag{6}$$

$$e_s = G_s^{-1}. \tag{7}$$

Analogous identities hold in the group S. To show that they determine d_i modulo p^{i+1}, it is enough to check that the identities

$$e_i = \sum_j G_{i-j} v_j, \tag{8}$$

$$p^i \mid v_i, \tag{9}$$

$$e_s = 0 \tag{10}$$

for $0 \leq i, j \leq s$ imply $p^{i+1} \mid e_i$.

From $(2'')$ one obtains

$$e_i - e_{i+1} = \sum_j (G_{i-j} - G_{i+1-j}) v_j.$$

Since trivially $G_k^{-1} \equiv G_{k+1}^{-1} \pmod{p^{k+1}}$, one also has

$$p^{i+1-j} \mid G_{i-j} - G_{i+1-j}, \qquad p^{i+1} \mid e_i - e_{i+1}.$$

The conclusion follows from $(4'')$.

The same technique, in a simpler form, applies when S is cyclic.

We turn to case (iii). The subgroups of S of index p are all normal. Their intersection S' is the Frattini subgroup of S. For every subgroup T of S, if $TS' = S$, then $T = S$. In particular, if S is not cyclic, then S/S' is not cyclic either, and the subgroups of S of index p are identified with the hyperplanes of a vector space over $\mathbf{Z}/p$ of dimension > 1; their number is congruent to $1 + p \pmod{p^2}$.

The preceding equations leave

$$\begin{cases} d_0 h = u_0, \\ d_1 \binom{p^s h}{p} = (1+p) u_0 + u_1 \pmod{p^2}, \\ \binom{p^s h}{p^s} = u_0 + u_1 \pmod{p^2}. \end{cases}$$

By Proposition 1, $\binom{p^s h}{p^s} \equiv h \pmod{p^2}$. Hence

$$\begin{aligned} d_1 h &\equiv (1+p) d_0 h + (h - d_0 h) \pmod{p^2}, \quad \text{and} \\ d_1 &\equiv 1 + p d_0 \pmod{p^2}. \end{aligned}$$

The assertion follows because $d_0 \equiv 1 \pmod{p}$.

References

[1] Serre, J.-P., *Corps locaux. Act. sc. et ind.* Hermann, 1962.
[2] Lazard, M., Groupes analytiques p-adiques. *Publ. Math. I.H.E.S.*, no. 26, 1965.

Appendix to *Residues and Duality* (Lect. Notes in Math. 20, Springer-Verlag 1966, pp. 404–421)

APPENDIX: COHOMOLOGY WITH PROPER SUPPORT AND CONSTRUCTION OF THE FUNCTOR $f^!$.

by P. DELIGNE[1]

Verdier has shown that, in the topological case, the formalism of Poincare duality reduces to local problems upstairs (see [1]). To transpose his construction to the schematic setting, one needs a theory of cohomology "with proper support" for coherent sheaves.

Unless explicitly stated otherwise, all preschemes considered are noetherian and all presheaves are quasi-coherent.

No. 1. Formalities of pro-objects.

Proposition 1. *Let C be a U-category*[2] *in which finite inductive limits exist. Let h be a functor $C^\circ \to$ (ens). The following conditions are equivalent:*

(i) *h is an inductive limit, over a small*[3] *filtered ordered set, of representable functors.*
(ii) *h is an inductive limit, over a small filtered category, of representable functors.*
(iii) *h transforms finite projective limits into finite inductive limits, and there is a small set of objects such that every element of an $h(X)$ factors through one of them.*

The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are trivial, and (iii) $\Rightarrow$ (ii) is standard: h is the limit of the h_F over the category of pairs (F, α), where $\alpha \in h(F)$ and F lies in the subcategory of C stable under finite projective limits generated by the given small set. It remains to prove that (ii) $\Rightarrow$ (i). If $\mathcal{L}$ and $\mathcal{M}$ are two filtered categories, a functor $F : \mathcal{L} \to \mathcal{M}$ is called cofinal if $\varinjlim_{\mathcal{L}} h_{F(L)}$ is the final functor. For every $G : \mathcal{M} \to \mathcal{N}$, one then has $\varinjlim_{\mathcal{M}} G = \varinjlim_{\mathcal{L}} GF$. Thus it remains to prove:

Lemma 1. *For every small filtered category $\mathcal{L}$, there exists a cofinal functor from a small filtered ordered set to $\mathcal{L}$.*

[1]This is an expanded version of a letter from P. Deligne to R. Hartshorne (letter of March 3, 1966). The footnotes were added by the copyist.
[2]U denotes a fixed universe throughout.
[3]"small" means "$\in U$".

The first projection $\mathcal{L} \times \mathbb{N} \to \mathcal{L}$ is a cofinal functor, where $\mathbb{N}$ has its natural order. This permits one to reduce to the case where $\operatorname{Ob}\mathcal{L}$, preordered by the relation $\operatorname{Hom}(X, Y) \neq \varnothing$, has no greatest element. Order by inclusion the set E of finite subcategories of $\mathcal{L}$ having a single final object, where finite means having a finite set of arrows. Define a functor from E to $\mathcal{L}$ by sending each of these categories to its final object. Under the hypotheses made, E is filtered and this functor is cofinal.

The functors satisfying the equivalent conditions of Proposition 1 are the Ind-objects of C. They form a full subcategory $\operatorname{Ind} C$ of $\underline{\operatorname{Hom}}(C^\circ, (\mathrm{Ens}))$, stable under inductive limits. The assignment $C \mapsto \operatorname{Ind} C$ is a functor (cat) $\to$ (cat). If filtered inductive limits exist in C, then for every $h \in \operatorname{Ind} C$ the functor $X \mapsto \operatorname{Hom}(h, h_X)$ is corepresentable; hence there is a functor, denoted $\varinjlim$, from $\operatorname{Ind} C$ to C. In particular, one always has a functor $\operatorname{Ind} \operatorname{Ind} C \to \operatorname{Ind} C$, and every functor $C \to \operatorname{Ind} D$ extends to a functor $\operatorname{Ind} C \to \operatorname{Ind} D$. In $\operatorname{Ind} C$, filtered inductive limits are exact and will be denoted "$\varinjlim$"; in particular, identifying C with a full subcategory of $\operatorname{Ind} C$, if $(X_i)_{i \in I}$ is a filtered inductive system in C, then

$$\varinjlim_{i \in I} X_i = \varinjlim_{i \in I} \text{“}\varinjlim\text{”} X_i.$$

A functor on $\operatorname{Ind} C$ is representable if and only if it transforms finite projective limits into finite inductive limits, filtered inductive limits into filtered inductive limits, and its restriction to C satisfies the smallness condition of Proposition 1(iii). Indeed, Proposition 1 then shows that its restriction to C is an Ind-object, to which the functor is everywhere equal because of the condition on limits.

The preceding statements, except Proposition 1(iii) and the preceding assertion, remain true if one uses everywhere limits of sequences, or countable limits, which is the same thing.

By duality one defines the category $\operatorname{pro} C$ of pro-objects, namely the full subcategory of $\underline{\operatorname{Hom}}(C, (\mathrm{Ens}))^\circ$.

Proposition 2. *Let X be a quasi-compact, quasi-separated prescheme, not necessarily noetherian. The category of quasi-coherent sheaves on X is equivalent to the category of Ind-objects of the category of quasi-coherent sheaves of finite presentation on X.*

The map is "$\varinjlim$" $\mathcal{F}_i \mapsto \varinjlim \mathcal{F}_i$. If $\mathcal{F}$ is of finite presentation, then for every filtered inductive system $(\mathcal{G}_i)_{i \in I}$,

$$\operatorname{Hom}_X(\mathcal{F}, \varinjlim \mathcal{G}_i) = \varinjlim \operatorname{Hom}_X(\mathcal{F}, \mathcal{G}_i).$$

By a finite gluing argument, which commutes with $\varinjlim$, one reduces to the affine case. Thus the preceding functor is fully faithful. Its image is stable under filtered inductive limits and finite sums. It remains only to prove that for every $x \in X$, every $\mathcal{F}$ on X, and every $s \in \mathcal{F}_x$, there exists a sheaf $\mathcal{G}$ of finite presentation and a map from $\mathcal{G}$ to $\mathcal{F}$ whose image contains s.

The sheaf $\mathcal{F}$ will be a limit of images of sheaves of finite presentation, and each of these will be a quotient of a sheaf of finite presentation by a limit of sub-sheaves of finite type.

On a quasi-compact neighbourhood U of x, let $f : \mathcal{C} \to \mathcal{F}$ be a map with $\mathcal{C}$ of finite presentation and with s in its image. It is enough to extend $\mathcal{C}$ and f to all of X; proceeding step by step, it is enough to know how to extend them to $U \cup V$ when V is affine. This amounts to carrying out an extension from $U \cap V$ to V.

Lemma 2. *Let $\mathcal{C}$ be a sheaf of finite presentation on a quasi-compact open U of an affine scheme X, let $\mathcal{F}$ be a sheaf on X, and let f be a map from $\mathcal{C}$ to $\mathcal{F}|U$. It is possible to extend $\mathcal{C}$ and f to $\overline{\mathcal{C}}$ and $\overline{f}$, defined on X, with $\overline{\mathcal{C}}$ of finite presentation.*

Let ι be the inclusion of U in X, and let $\mathcal{C}_1$ be the fiber product of $\iota_*\mathcal{C}$ and $\mathcal{F}$ over $\iota_*\iota^*\mathcal{F}$. Then $\mathcal{C}_1$ extends $\mathcal{C}$ and maps to $\mathcal{F}$. If one writes $\mathcal{C}_1$ as a limit of its sub-sheaves of finite type, then, since U is quasi-compact and $\mathcal{C}$ is of finite presentation, one of these already extends $\mathcal{C}$. Writing the latter as a quotient of $\mathcal{O}^n$ by a limit of sub-sheaves of finite type of $\mathcal{O}^n$, one sees in the same way that $\mathcal{C}_1$ may be replaced by a sheaf of finite presentation.

Corollary 1. *A contravariant additive functor $F : (q\ coh) \to Ab$ is representable if and only if it is left exact and transforms filtered inductive limits into filtered inductive limits.*

Corollary 2. *Every sheaf of finite presentation defined on a quasi-compact open U of X extends to a sheaf of finite presentation on X.*

Let $\mathcal{A}$ be an abelian category. We shall have to work essentially in $\operatorname{pro} D^b\mathcal{A}$, the full subcategory of $\operatorname{pro} D(\mathcal{A})$ consisting of the "$\varprojlim$" K_i for which the cohomology of K_i is uniformly bounded as i varies. The functor $\mathbb{H}^p$ extends to a functor $\operatorname{pro} D(\mathcal{A}) \to \operatorname{pro} \mathcal{A}$.

Proposition 3. *The functors $\mathbb{H}^p$ form a conservative[4] family of functors on $\operatorname{pro} D^b\mathcal{A}$.*

[4]That is, if u is an arrow such that all $\mathbb{H}^p(u)$ are invertible, then u is invertible.

Let $f : K \to L$ be an arrow of pro $D^b\mathcal{A}$ such that the maps $\mathbb{H}^p(K) \to \mathbb{H}^p(L)$ are isomorphisms in pro $\mathcal{A}$. We must check that, for every M in $D^b(\mathcal{A})$,

$$\operatorname{Hom}(K, M) \longleftarrow \operatorname{Hom}(L, M)$$

is an isomorphism. Write $K = \text{“}\varprojlim\text{”} K_\alpha$. There is a convergent spectral sequence

$$\operatorname{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \operatorname{Ext}^{p+q}(K_\alpha, M),$$

whence, on passing to the limit, another convergent spectral sequence

$$\varinjlim \operatorname{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \varinjlim \operatorname{Ext}^{p+q}(K_\alpha, M),$$

since q is uniformly bounded. Using the fact that every functor, here Ext, passes to pro-objects, and using the functor $\varinjlim$: Ind Ab $\to$ Ab, we may rewrite this as

$$\varinjlim \operatorname{Ext}^p(\mathbb{H}^q K, M) \Longrightarrow \varinjlim \operatorname{Ext}^{p+q}(K, M) = \operatorname{Hom}(K[-p-q], M).$$

This spectral sequence remains valid for $M \in D(\mathcal{A})$. The proposition follows immediately from the existence of these spectral sequences.

No. 2. Fundamental lemmas.

Recall that all preschemes considered are noetherian.

Proposition 4 (Duality theorem for an open immersion)**.** *Let U be an open subset of X, let j be the inclusion, let $\mathcal{J}$ be an ideal defining the closed complement, let $\mathcal{F}$ be a coherent sheaf on U, and let $\mathcal{G}$ be quasi-coherent on X. Let $\overline{\mathcal{F}}$ be a coherent extension of $\mathcal{F}$. Then*

$$\operatorname{Hom}_U(\mathcal{F}, j^*\mathcal{G}) = \operatorname{Hom}_X(\text{“}\varprojlim\text{”} \mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}). \tag{1}$$

The map $\varinjlim \operatorname{Hom}_X(\mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}) \to \operatorname{Hom}_U(\mathcal{F}, \mathcal{G})$ is injective: if the image under f of $\mathcal{J}^n\overline{\mathcal{F}}$ in $\mathcal{G}$ has its support in the complement of U, then it is annihilated by a power of $\mathcal{J}$, say $\mathcal{J}^k$, and the image of $\mathcal{J}^{n+k}\overline{\mathcal{F}}$ is zero.

Now assume that X is affine, and let $f \in \operatorname{Hom}_U(\mathcal{F}, \mathcal{G})$. For k sufficiently large, every section s of $\mathcal{J}^k\overline{\mathcal{F}}$ over X has an image over U which extends to a section of $\mathcal{G}$ over X. Replacing $\overline{\mathcal{F}}$ by $\mathcal{J}^k\overline{\mathcal{F}}$, we may suppose that we have a diagram

$$\begin{tikzcd}
0 \arrow[r] & R \arrow[r] \arrow[d, dotted] & \mathcal{O}^n \arrow[r] \arrow[d, equal] & \overline{\mathcal{F}} \arrow[r] \arrow[d, dotted] & 0 \\
0 \arrow[r] & R_1 \arrow[r] & \mathcal{O}^n \arrow[r] & \mathcal{G} &
\end{tikzcd}$$

in which the dotted arrows are defined on U.

For ℓ sufficiently large, by Krull's theorem and by the fact that $R/(R \cap R_1)$ is concentrated outside U, one has $R \cap \mathcal{J}^\ell \cdot \mathcal{O}^n \subset R_1$, which permits f to be extended to a map $\overline{f} : \mathcal{J}^\ell \overline{\mathcal{F}} \to \mathcal{G}$.

In the general case, let U_i be a finite affine covering of X, and let U_{ijk} be finite affine coverings of $U_{ij} = U_i \cap U_j$. Since $\varinjlim$ commutes with finite products, one obtains

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hom}_U(\mathcal{F}, j^*\mathcal{G}) \arrow[r] & \prod_i \mathrm{Hom}_{U_i \cap U}(\mathcal{F}, j^*\mathcal{G}) \arrow[r, shift left] \arrow[r, shift right] & \prod_{ijk} \mathrm{Hom}_{U_{ijk} \cap U}(\mathcal{F}, j^*\mathcal{G}) \\
0 \arrow[r] & \varinjlim \mathrm{Hom}_X(\mathcal{J}^n \overline{\mathcal{F}}, j^*\mathcal{G}) \arrow[r] \arrow[u] & \prod_i \varinjlim \mathrm{Hom}_{U_i}(\mathcal{J}^n \overline{\mathcal{F}}, \mathcal{G}) \arrow[r, shift left] \arrow[r, shift right] \arrow[u] & \prod_{ijk} \varinjlim \mathrm{Hom}_{U_{ijk}}(\mathcal{J}^n \overline{\mathcal{F}}, \mathcal{G}) \arrow[u]
\end{tikzcd}$$

which completes the proof.

Proposition 4 gives an explicit formula for the functor (coh on U) $\to$ pro (coh on X) left adjoint to j^*. We shall denote this functor by $j_!$ (extension by zero)[5]; it follows from Krull's theorem that it is exact. This also follows from the fact that, since X is noetherian, an injective object of the category of quasi-coherent sheaves on X, restricted to U, remains injective as a quasi-coherent sheaf on U.

Proposition 5 (Independence of the compactification)**.**

$$\begin{tikzcd}
U \arrow[r, hook] \arrow[d, equal] & X \arrow[d, "f"] \\
U \arrow[r, hook] & Y
\end{tikzcd}$$

Let $f : X \to Y$ be a proper morphism inducing an isomorphism between the open subset U of Y and $f^{-1}(U)$. Let $\mathcal{J}$ be a sheaf of ideals defining $Y \setminus U$, and put $\mathcal{I} = f^*\mathcal{J}$. If $\mathcal{F}$ is coherent on X, then

(i) *for $k > 0$, the projective system $\mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$ is essentially zero, that is, it defines the zero pro-object;*

(ii) *for $k = 0$, for all sufficiently large n, $f_* \mathcal{I}^{n+1} \mathcal{F} = \mathcal{J} \cdot f_* \mathcal{I}^n \mathcal{F}$.*

One has

$$\bigoplus_{n=0}^{\infty} \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \quad \text{is a finite type } \bigoplus_{n=0}^{\infty} \mathcal{J}^n\text{-module (EGA III).}$$

Thus, for n sufficiently large, the map $\mathcal{J} \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \to \mathbb{R}^k f_* \mathcal{I}^{n+1} \mathcal{F}$ is surjective; (ii) follows. If $k > 0$, $\mathbb{R}^k f_*$ is zero on U, so there exists N

[5] This terminology and notation plainly conflict with the generally accepted ones, for example in Godement's book and in SGA 1962. The reader may prefer to read $\mathfrak{J}_!$ instead of $j_!$.

such that $\mathcal{J}^N \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$ is zero for every n. The diagram

$$\begin{tikzcd}
\mathcal{J}^p \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \arrow[r] \arrow[d] & \mathbb{R}^k f_* \mathcal{I}^{n+p} \mathcal{F} \arrow[r] \arrow[d] & 0 \quad (\text{if } n \geq n_0) \\
\mathcal{J}^p \cdot \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \arrow[r] & \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} &
\end{tikzcd}$$

proves that for N sufficiently large and $k > 0$ the map

$$\mathbb{R}^k f_* \mathcal{I}^{n+N} \mathcal{F} \to \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$$

is zero, which is (i).

No. 3. Definition of $\mathbb{R}f_!$.

A morphism f, respectively a composable pair of morphisms, respectively a triple, will be called compactifiable if one can respectively find commutative diagrams

$$\begin{tikzcd}
X \arrow[r, hook] \arrow[d, "f"'] & \overline{X} \arrow[dl] \\
S &
\end{tikzcd}
\qquad
\begin{tikzcd}
X \arrow[r, hook] \arrow[d, "f"'] & \overline{X} \arrow[d, "g"] \\
Y \arrow[r, hook] \arrow[d, "h"'] & \overline{Y} \arrow[dl] \\
S &
\end{tikzcd}
\qquad
\begin{array}{ccc}
X & \hookrightarrow & \overline{X} \\
\downarrow f & & \downarrow g \\
Y & \hookrightarrow & \overline{Y} \\
\downarrow h & & \downarrow \\
Z & \hookrightarrow & \overline{Z} \\
\downarrow & & \swarrow \\
S & &
\end{array}$$

in which the horizontal arrows are open immersions and the oblique arrows are proper. Every compactifiable morphism is separated and of finite type, and Nagata asserts in [2] that his proof gives the converse for integral noetherian schemes, but the hypotheses he imposes on the base are not clear.[6]

We propose, for every compactifiable f, to define

$$\mathbb{R}f_! : \operatorname{pro} D^b_{\mathrm{coh}}(X) \to \operatorname{pro} D^b_{\mathrm{coh}}(S).$$

Recall that the category $D^b_{\mathrm{coh}}(S)$, the bounded derived category of coherent sheaves on S, is a full subcategory of the derived category of all sheaves, not necessarily quasi-coherent, on S; its essential image consists of complexes with bounded coherent cohomology.

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[6]Mumford is said to have checked, via Nagata's proof, that every separated morphism of finite type of noetherian schemes is compactifiable.

If f is proper, one has $\mathbb{R}f_* : D^b_{\mathrm{coh}}(X) \to D^b_{\mathrm{coh}}(S)$ of finite dimension, and it induces $\mathbb{R}f_! : \operatorname{pro} D^b_{\mathrm{coh}}(X) \to \operatorname{pro} D^b_{\mathrm{coh}}(S)$. If f is an open immersion, $f_! : (\text{coh on } X) \to \operatorname{pro}(\text{coh on } S)$ induces

$$D^b(\text{coh on } X) \to D^b \operatorname{pro}(\text{coh on } S),$$

which maps to $\operatorname{pro} D^b_{\mathrm{coh}}(S)$, and this map extends to

$$\mathbb{R}f_! : \operatorname{pro} D^b_{\mathrm{coh}}(X) \to \operatorname{pro} D^b_{\mathrm{coh}}(S).$$

If a coherent complex K on X is extended to $\overline{K}$ on S, its image is "$\varprojlim$" $\mathcal{J}^n\overline{K}$, where $\mathcal{J}$ defines $S \setminus X$.

For a composite $f = gh$, with g proper and h an open immersion, set $\mathbb{R}f_! = \mathbb{R}g_!\,\mathbb{R}h_!$. One must verify:

Independence of the compactification.

Consider a commutative diagram

$$\begin{tikzcd}
& X' \arrow[dd, "g"] \\
X \arrow[ur, "j'"] \arrow[dr, "j''"'] \arrow[ddd, "f"'] & \\
& X'' \arrow[ddl, "\bar{f}"] \\
& \\
S &
\end{tikzcd}$$

where j' and j'' are open immersions, and $\bar{f}$ and g are proper maps. We have $j'X = g^{-1}j''X \cap j'X$. We must prove that

$$\mathbb{R}\bar{f}_*\,\mathbb{R}j''_! = \mathbb{R}(\bar{f}g)_*\,\mathbb{R}j'_!,$$

or equivalently, since $\mathbb{R}(\bar{f}g)_* = \mathbb{R}\bar{f}_*\,\mathbb{R}g_*$, that

$$\mathbb{R}j''_! = \mathbb{R}g_*\,\mathbb{R}j'_!.$$

One reduces to the case where X is dense in X', so as to be in the situation of Proposition 5. Let K be a bounded coherent complex on X, extended to $\overline{K}$ on X'. Then $g_*\overline{K}$ extends K on X'', and one has a map

(2) $$g_*\mathcal{J}'^n\overline{K} \to \mathbb{R}g_*\mathcal{J}'^n\overline{K},$$

where $\mathcal{J}'$ denotes a sheaf of ideals defining $X' \setminus X$. Proposition 5(ii) shows that $j''_!K =$ "$\varprojlim$" $g_*\mathcal{J}'^n\overline{K}$. Passing to the "limit", one obtains

$$\text{“}\varprojlim\text{”}\, \mathbb{R}^p g_*\mathcal{J}'^n\overline{K}^q \Longrightarrow \text{“}\varprojlim\text{”}\, \mathbb{R}^{p+q} g_*\mathcal{J}'^n\overline{K}.$$

Proposition 5(i) shows that this spectral sequence degenerates, and Proposition 3 implies that the arrow deduced from (1) by passing to the "limit" is an isomorphism. This gives the desired formula.

It remains to verify the following.

(a) For every diagram one has a compatibility

$$\begin{array}{ccc} X & \longrightarrow & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle h} \\ S & \longrightarrow & \bar{X} \end{array} \qquad \mathbb{R}\bar{f}_*\,\mathbb{R}j''_! = \mathbb{R}\bar{f}_*\,\mathbb{R}g_*\,\mathbb{R}j''_! = \mathbb{R}\bar{f}_*\,\mathbb{R}g_*\,\mathbb{R}h_*\,\mathbb{R}j'_! = \mathbb{R}\bar{f}_*\,\mathbb{R}(gh)_*\,\mathbb{R}j'_! = \mathbb{R}f_*\,\mathbb{R}j'''_!.$$

Since two compactifications can always be dominated by a third, for example $X' \times_S X''$, this is enough to prove independence of the choices.

(b) For every compactifiable pair, an identification

$$\mathbb{R}(fg)_! = \mathbb{R}f_!\,\mathbb{R}g_!.$$

(c) For every compactifiable triple, a compatibility

$$\begin{array}{ccc} \mathbb{R}(fgh)_! & = & \mathbb{R}(fg)_!\,\mathbb{R}h_! \\ \| & & \| \\ \mathbb{R}f_!\,\mathbb{R}(gh)_! & = & \mathbb{R}f_!\,\mathbb{R}g_!\,\mathbb{R}h_! \end{array}$$

One would then have proved:

Theorem 1. *For $f : X \to Y$ compactifiable, $K \in pro\, D^b_{coh}(X)$ and $L \in pro\, D^b_{coh}(Y)$, put*

$$\mathrm{Hom}_f(K, L) = \mathrm{Hom}_Y(\mathbb{R}f_!K, L).$$

Modulo compactifiability questions, this makes the categories $pro\, D^b_{coh}(X)$ into a cofibrant category over noetherian preschemes, with the arrows compactifiable.

No. 4. Definition of $\mathbb{R}f^!$.

It follows from a general theorem of Verdier [1] that the functor $\mathbb{R}f_* : D(X) \to D(S)$ has a right adjoint $D^+(S) \to D^+(X)$. It deserves a name, say $\mathbb{R}f^!$, only when f is proper, and its definition is not directly suited for computation.

First, one must make explicit a procedure for computing $\mathbb{R}f_*$ at the level of complexes, of the form

$$\mathbb{R}f_*K = f_*C^*(K),$$

where C^* is a finite acyclic resolution depending functorially and exactly on the object to which it is applied, and compatible with filtered inductive limits. Then, for every injective I on S, the functor $\mathrm{Hom}(f_*C^p\mathcal{F}, I)$ is exact in the quasi-coherent sheaf $\mathcal{F}$ on X and transforms $\varinjlim$ into $\varprojlim$; hence it is representable by $f_p^! I$, an injective quasi-coherent sheaf. The $f_p^! I$ form a complex, which defines $\mathbb{R}f^! : D^+S \to D^+X$, right adjoint to $\mathbb{R}f_* : D(X) \to D(S)$.

Here is how to define such a resolution.

1) <u>If S is separated</u>: take a finite covering of X by opens affine over S, for instance affine opens, and put $\mathbb{R}f_* K = f_* \check{C}(\mathcal{U}, K)$, the alternating Cech complex.

2) <u>Otherwise</u>: the canonical flasque resolutions have two defects here.

(a) $f_* C^* \mathcal{F}$ is no longer quasi-coherent. This causes no problem as long as, as here, quasi-coherent injectives are injective as sheaves.

(b) C^* does not commute with filtered inductive limits. This defect is easy to correct: write a quasi-coherent sheaf as $\mathcal{F} = \varinjlim \mathcal{F}_i$, with the $\mathcal{F}_i$ of finite presentation, and put $C'^* \mathcal{F} = \varinjlim C^* \mathcal{F}_i$. This does not depend on the choices.

For an open immersion, take $\mathbb{R}f^! K = f^* K$.

For a composite morphism $f = gh$, with g proper and h an open immersion, take $\mathbb{R}f^! = \mathbb{R}g^! \, \mathbb{R}h^!$. Combining Proposition 4 with the preceding construction gives:

Theorem 2 (Duality)**.** *Let $f : X \to S$ be compactifiable, $f = gh$. The functors*

$$\mathbb{R}f_! : \mathit{pro}\, D^b_{coh}(X) \to \mathit{pro}\, D^b_{coh}(S) \quad \text{and} \quad \mathbb{R}f^! : D^+(S) \to D^+(X)$$

are adjoint to one another.

One should verify the independence of the compactification and a formalism of compatibilities and identifications analogous to the case of $\mathbb{R}f_!$. Particular cases of the formula sheet follow from the adjunction formula, by uniqueness of a genuine adjoint.

In the general case one may use the following proposition.

Proposition 6. *Let $f : X \to S$ be compactifiable and let $L \in D^+_{\mathrm{qc}}(S)$. The fiber at $x \in X$ of $\mathcal{H}^p \mathbb{R}f^! L$ is given by*

$$(\mathcal{H}^p \mathbb{R}f^! L)_x = \varinjlim_{x \in U} \mathrm{Hom}^p_S(\mathbb{R}f_!(U, \mathcal{O}_U), L).$$

One may assume that X is proper and that L is given as a complex of injective quasi-coherent sheaves. Let U be an affine open of X, and let $\mathcal{J}$ be an ideal defining $X \setminus U$. Then

$$\begin{aligned}(\mathcal{H}^p \mathbb{R}f^! L)(U) &= \mathbb{H}^p(\mathbb{R}f^! L(U)) = \mathbb{H}^p \,\mathrm{Hom}_U(\mathcal{O}, \mathbb{R}f^! L) \\ &= \mathbb{H}^p \varinjlim \mathrm{Hom}_X(\mathcal{J}^n, \mathbb{R}f^! L) = \varinjlim \mathrm{Hom}^p_S(\mathbb{R}f_* \mathcal{J}^n, L) \\ &= \mathrm{Hom}^p_S(\mathbb{R}f_!(U, \mathcal{O}_U), L),\end{aligned}$$

and the proposition follows.

We shall assume for $\mathbb{R}f^!$ a variance formalism analogous to that of Theorem 1 for $\mathbb{R}f_!$.

If one wants to make the definition of $\mathbb{R}f^!$ more explicit in the proper case, one can observe that if a quasi-coherent sheaf $\mathcal{F}$ represents a functor F, then for every open U and every $\mathcal{J}$ defining $X \setminus U$, one has

$$\mathcal{F}(U) = \varinjlim \operatorname{Hom}(\mathcal{J}^n, \mathcal{F}) = \varinjlim F(\mathcal{J}^n).$$

No. 5. Computation of $\mathbb{R}f^!$.

For a finite morphism, $\mathbb{R}f^!$ is what one wants; to see this, it is enough to take the identity as resolving functor in the computation of $\mathbb{R}f_*$. For projective space, the uniqueness of an adjoint leaves no choice; one uses the fact that the adjunction formula is available for $\mathbb{R}f_! = \mathbb{R}f_* : D(X) \to D(S)$. We shall return to the arrow. For an open immersion, $\mathbb{R}f^! = \mathbb{R}f^*$.

Thus for every quasi-projective open U of X over S, one can compute the restriction of $\mathbb{R}f^!$ to U. In particular, if one can check locally that $\mathbb{R}^q f^! K = 0$ for $q \neq p$, then one knows $\mathbb{R}f^! K$, since an object of the derived category with only one non-zero cohomology sheaf is determined by that sheaf. Recall an important case where this condition is satisfied, with $K = \mathcal{O}$.

Proposition 7. *Let $f : X \to Y$ be compactifiable. Suppose that locally on X and on Y one can find a diagram*

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "g"'] & & Y \arrow[dl, "h"] \\ & S & \end{tikzcd}$$

where g and h are locally complete intersections of relative dimensions d' and d'', and let $d = d' - d''$.

Then $\mathbb{R}f^!\mathcal{O}$ is reduced to a single cohomology sheaf, situated in degree $-d$, and that sheaf is invertible.

Indeed, $\mathbb{R}g^!\mathcal{O} = \mathbb{R}f^!\,\mathbb{R}h^!\mathcal{O}$. We know, by local calculations in projective space, that $\mathbb{R}g^!\mathcal{O}$ and $\mathbb{R}h^!\mathcal{O}$ are of the indicated type, situated in degrees $-d'$ and $-d''$. Since everything is local, one may suppose that $\mathbb{R}^{-d''}h^!\mathcal{O}$ is isomorphic to $\mathcal{O}$, and the assertion follows.

To pass from this to more general complexes, one would first have to prove formulas of the form

$$\mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L) = \mathbb{R}f^!(K) \overset{\mathbb{L}}{\otimes} \mathbb{L}f^* L.$$

Only the definition of the arrow is problematic; its being an isomorphism is a local question. Of course boundedness restrictions are needed. I have checked none of the following in detail.

First method for defining the arrow.

Assume that K and L lie in $D^b_{\mathrm{coh}}(S)$, not merely in the quasi-coherent setting, and also that

(a) $\mathbb{R}f^!K$ is bounded, it being automatically coherent, which is a local problem;

(b) $K \overset{\mathbb{L}}{\otimes} L$ is bounded;

(c) $\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L$ is bounded.

To define

$$\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L \to \mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L),$$

it is enough by duality to define

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L) \to K \overset{\mathbb{L}}{\otimes} L.$$

By Kunneth,

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L) = \mathbb{R}f_!\,\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} L,$$

and the required arrow is induced by the adjunction arrow $\mathbb{R}f_!\,\mathbb{R}f^!K \to K$.

Second method.

Assume that f is compactifiable, flat, and locally a relative complete intersection. Then $\mathbb{R}f^!\mathcal{O} = \omega[d]$, where ω is invertible. Let $\mathbb{R}f_!$ denote a procedure for computing $\mathbb{R}f_!$ at the level of complexes, namely such a procedure for a compactification $\bar{f} : \bar{X} \to S$ of f, assumed to be of the type described in No. 4 and such that $C^p\mathcal{F}$ is zero for $p > d$, which can be arranged by truncation. Then there is an arrow

$$\mathbb{R}^d\bar{f}_*\mathcal{F}[-d] \to \mathbb{R}\bar{f}_*\mathcal{F} \quad \text{for every } \mathcal{F} \text{ on } \bar{X},$$

since the last cohomology sheaf maps to its complex. From this one obtains morphisms of complexes

$$\mathrm{Hom}_S(\mathbb{R}\bar{f}_*\mathcal{F}, I) \to \mathrm{Hom}_S(\mathbb{R}^d\bar{f}_*\mathcal{F}[-d], I) \quad \text{for every complex } I \text{ on } S.$$

Let I be injective. The functor $\mathbb{R}^d\bar{f}_*[-d]$ is right exact in $\mathcal{F}$, so the functor in $\mathcal{F}$ on the right of the arrow is representable. Define $I_!$ by

$$\mathrm{Hom}_S(\mathbb{R}^d f_*\mathcal{F}, I) = \mathrm{Hom}_{\bar{X}}(\mathcal{F}, I_!) \quad \text{and} \quad \mathbb{R}f^!_!I = I_![d].$$

The preceding construction defines

$$\mathbb{R}f^!I \to \mathbb{R}f^!_!I,$$

and, for a complex of injectives, $\mathbb{R}f^!_!$ is defined term by term.

It remains to evaluate $\mathbb{R}f_!^!I$ on affine opens U of X. It is given by $\varinjlim \operatorname{Hom}(\mathbb{R}^d f_! \mathcal{J}^n, I)$. Since this problem is local upstairs, it is easy to show by projective methods that the dual of the ind-object

$$\text{“}\varprojlim\text{”}\, \mathbb{R}^d f_! \mathcal{J}^n = \mathbb{R}^d f_!(U, \mathcal{O})$$

is $f_*(U, \omega)$, represented as the limit of its submodules of sections having a pole of order at least n outside U, with $n \to \infty$.

This gives

$$\mathbb{R}f_!^!I = \omega \otimes f^* I[d],$$

and the general formula

$$\mathbb{R}f^! K = f^* K \otimes \omega[d].$$

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LEFSCHETZ THEOREM
AND DEGENERATION CRITERIA
FOR SPECTRAL SEQUENCES

by P. DELIGNE[1]

1. General theorems

Let $\mathcal{A}$ be an abelian category and let $D^b(\mathcal{A})$ be the full subcategory of its derived category formed by the bounded complexes. Recall that if T is a cohomological functor (see [8] or [9], p. 10) from $D^b(\mathcal{A})$ to an abelian category $\mathcal{B}$, Verdier [8] has defined a spectral sequence, functorial in $X \in D^b(\mathcal{A})$:

$$E_2^{pq} = T(H^q(X)[p]) \Longrightarrow T(X[p+q]). \tag{1.1}$$

Proposition (1.2). Let $X \in D^b(\mathcal{A})$. The following conditions are equivalent:

(i) for every T as above, the spectral sequence (1.1) degenerates;

(ii) the object X is isomorphic, in $D^b(\mathcal{A})$, to $\sum_i H^i(X)[-i]$.

(i) ⇒ (ii). Denote by T_i the cohomological functor in K defined by

$$T_i(K) = \mathrm{Hom}_{D^b(\mathcal{A})}(H^i(X), K).$$

For each of the T_i ($i \in \mathbb{Z}$), the spectral sequence (1.1) is written

$$E_2^{pq} = \mathrm{Ext}^p(H^i(X), H^q(K)) \Longrightarrow \mathrm{Hom}_{D^b(\mathcal{A})}(H^i(X), K[p+q]), \tag{1.3}$$

and the homomorphism from $\mathrm{Hom}(H^i(X), K[i])$ to

$$E_\infty^{0i} = \mathrm{Hom}(H^i(X), H^i(K))$$

which follows from it is none other than the one coming from the functoriality of H^0.

If the spectral sequences (1.3) degenerate, the maps

$$\mathrm{Hom}(H^i(X)[-i], K) \longrightarrow \mathrm{Hom}(H^i(X), H^i(K))$$

are surjective. By hypothesis, this is so for $K = X$; hence there exist morphisms a_i from $H^i(X)[-i]$ to X, inducing the identity on the i-th cohomology object. Moreover, since $H^i(X) = 0$ except for finitely many values of i, the a_i vanish except for finitely many i, which allows one to define

$$a = \sum_i a_i : \sum_i H^i(X)[-i] \longrightarrow X.$$

By construction, a induces the identity on cohomology, hence is an isomorphism.

(ii) ⇒ (i). The spectral sequence (1.1) is an additive functor in X, and degenerates when X has only one nonzero cohomology object; hence it also degenerates if X is a sum of complexes having only one nonzero cohomology object.

Remark (1.4). It follows from the preceding proof that it is enough to verify condition (i) for cohomological functors of the form $\mathrm{Hom}_{D^b(\mathcal{A})}(A, *)$, with $A \in \mathrm{Ob}(\mathcal{A})$. By duality, it would also be enough to consider only the functors $\mathrm{Hom}_{D^b(\mathcal{A})}(*, A)$, since condition (ii) is self-dual.

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Theorem (1.5). Let $X \in D^b(\mathcal{A})$, let n be an integer, and let u be an endomorphism of degree 2 of X,

$$u : X \longrightarrow X[2],$$

whose iterates induce isomorphisms

$$u^i : H^{n-i}(X) \xrightarrow{\sim} H^{n+i}(X) \qquad (i \geq 0).$$

Under these conditions, X is isomorphic to $\sum_i H^i(X)[-i]$ in the category $D^b(\mathcal{A})$.

We verify condition (i) of (1.2). Let T be a cohomological functor from $D^b(\mathcal{A})$ to $\mathcal{B}$, and put $T^p(K) = T(K[p])$.

The primitive part of $H^{n-i}(X)$, denoted ${}_0H^{n-i}(X)$, is the kernel of the homomorphism

$$u^{i+1} : H^{n-i}(X) \longrightarrow H^{n+i+2}(X).$$

One checks that, for $i \geq 0$, the maps

$$\begin{aligned} \bigoplus_{k\geq 0} u^k : \bigoplus_{k\geq 0} {}_0H^{n-i-2k}(X) \xrightarrow{\sim} H^{n-i}(X), \\ \bigoplus_{k\geq 0} u^{k+i} : \bigoplus_{k\geq 0} {}_0H^{n-i-2k}(X) \xrightarrow{\sim} H^{n+i}(X) \end{aligned} \tag{1.6}$$

are isomorphisms.

Since the spectral sequence (1.1) is functorial in X and compatible with translations of degrees, the endomorphism u of X defines an endomorphism of degree 2 of the spectral sequence,

$$u : E_r^{pq} \longrightarrow E_r^{p,q+2},$$

commuting with the differentials d_r $(r \geq 2)$.

When $r = 2$, the endomorphism u of the spectral sequence

$$u : T^p(H^q(X)) \longrightarrow T^p(H^{q+2}(X))$$

is obtained from the morphism $u : H^q(X) \to H^{q+2}(X)$ by functoriality of T^p.

Let $i \geq 0$. Denote by ${}_0E_r^{p,n-i}$ the kernel of the homomorphism

$$u^{i+1} : E_r^{p,n-i} \longrightarrow E_r^{p,n+i+2}$$

(the primitive part of the spectral sequence). For $r = 2$, the decompositions (1.6) induce, by functoriality of each T^p, decompositions

$$\begin{aligned} \bigoplus_{k\geq 0} u^k : \bigoplus_{k\geq 0} {}_0E_2^{p,n-i-2k} \xrightarrow{\sim} E_2^{p,n-i}, \\ \bigoplus_{k\geq 0} u^{k+i} : \bigoplus_{k\geq 0} {}_0E_2^{p,n-i-2k} \xrightarrow{\sim} E_2^{p,n+i}. \end{aligned} \tag{1.7}$$

We prove by induction on r that the differentials d_r $(r \geq 2)$ are zero. The induction hypothesis implies that $E_2 = E_r$, so that the decomposition (1.7) applies to the term E_r of the spectral sequence. Since u commutes with the d_r, it will be enough to prove that d_r vanishes on the primitive part of E_r.

The following diagram is commutative:

$$\begin{tikzcd} {}_0E_r^{p,n-i} \arrow[r, "d_r"] \arrow[d, "u^{i+1}"'] & E_r^{p+r,n-i-r+1} \arrow[d, "u^{i+1}"] \\ E_r^{p,n+i+2} \arrow[r, "d_r"'] & E_r^{p+r,n+i-r+3}. \end{tikzcd}$$

The left vertical arrow is zero by definition of primitivity. The right vertical arrow is a monomorphism, because the arrow

$$u^{i+r-1} : E_r^{p+r,n-(i+r-1)} \longrightarrow E_r^{p+r,n+(i+r-1)}$$

is an isomorphism and $r - 1 \geq 1$. It follows that d_r is zero.

Remark (1.8). One should take care that in general one cannot choose an isomorphism between X and $\sum_i H^i(X)[-i]$ which transforms u into the sum of the maps $H^i(X) \to H^{i+2}(X)$. One easily obtains a counterexample by taking $H^i(X) = 0$ for $i \neq n$.

One can define an isomorphism, "more beautiful than the others", between X and $\sum_i H^i(X)[-i]$, but it will not be used here.

Remark (1.9). Let $X \in D^b(\mathcal{A})$, let n and s be integers, and let u be an endomorphism of degree s (respectively $2s$) of X. Suppose that $H^i(X) = 0$ when i is not of the form ks $(0 \leq k \leq n)$ (respectively $0 \leq k \leq 2n$), and that u^{n-2k} (respectively u^i) induces an isomorphism between $H^{ks}(X)$ and $H^{(n-k)s}(X)$ (respectively between $H^{(n-i)s}(X)$ and $H^{(n+i)s}(X)$). Under these hypotheses, the proof of (1.5) still shows that X is isomorphic to $\sum_i H^i(X)[-i]$.

Remark (1.10). Let $(X_i)_{i\in\mathbb{Z}}$ be a family of objects of $D^b(\mathcal{A})$ and let u be a family of degree-2 homomorphisms $u_i : X_i \to X_{i+1}[2]$. If, for all i and j, u_i induces an isomorphism between $H^{n-j}(X_i)$ and $H^{n+j}(X_{i+j})$, it is still true that each X_i is isomorphic to the sum of its cohomology objects. To see this, it is enough to apply the preceding arguments formally to

$$u : \sum_i X_i \longrightarrow \sum_i X_i,$$

even if this sum does not exist in $D^b(\mathcal{A})$.

The same remark applies to the variant (1.9).

Theorem (1.11). Let $X \in D^b(\mathcal{A})$ and suppose that X admits degree-0 endomorphisms $\pi_i : X \to X$ such that $H^j(\pi_i) = \delta_{ij}$. Then X is isomorphic to $\sum_i H^i(X)[-i]$ in the category $D^b(\mathcal{A})$.

We again apply criterion (1.2) (i). The diagram

$$\begin{tikzcd} E_r^{p,q} \arrow[r, "d_r"] \arrow[d, "\pi_q"'] & E_r^{p+r,q-r+1} \arrow[d, "\pi_q"] \\ E_r^{p,q} \arrow[r, "d_r"'] & E_r^{p+r,q-r+1} \end{tikzcd}$$

is commutative; the first column is the identity and the second is zero, so that $d_r = 0$.

Corollary (1.12). Let $X \in D^b(\mathcal{A})$ be the sum of a finite family of objects $X_k \in D^b(\mathcal{A})$. If

$$X \simeq \sum_i H^i(X)[-i],$$

then, for every k,

$$X_k \simeq \sum_i H^i(X_k)[-i].$$

Let j_k be the canonical injection of X_k into X, let pr_k be the canonical projection of X onto X_k, and let π_i be an endomorphism of X such that $H^j(\pi_i) = \delta_{ij}$. For every k, the $\mathrm{pr}_k\, \pi_i j_k$ satisfy the hypothesis of (1.11).

Remark (1.13). One checks that, in order that the isomorphism between X and $\sum_i H^i(X)[-i]$ can be chosen so that the π_i are identified with the pr_i, it is necessary and sufficient that the π_i be pairwise orthogonal idempotents. There is then a unique isomorphism with this property and inducing the identity on cohomology.

2. Applications

Let $f : X \to Y$ be a continuous map between topological spaces, or even a morphism of topoi, let A be a sheaf of rings on X, and let $\mathcal{F}$ be a sheaf of A-modules. If $u \in H^2(X, A)$, then u defines a degree-2 endomorphism in $D^b(X)$, again denoted

$$u : \mathcal{F} \longrightarrow \mathcal{F}[2].$$

By functoriality, u defines a degree-2 endomorphism

$$Rf_*(u) : Rf_*\mathcal{F} \longrightarrow Rf_*\mathcal{F}[2].$$

We shall say that $(X, f, u, \mathcal{F}, n)$ satisfies the Lefschetz condition, or that $\mathcal{F}$ satisfies the Lefschetz condition relative to u, if, for every $i \geq 0$, the maps obtained by iterating i times the arrow $Rf_*(u)$,

$$Rf_*(u)^i : R^{n-i}f_*\mathcal{F} \longrightarrow R^{n+i}f_*\mathcal{F} \qquad (i \geq 0),$$

are isomorphisms. Theorem (1.5) gives here:

Proposition (2.1). If $\mathcal{F}$ satisfies the Lefschetz condition relative to u, then, in $D^b(Y)$,

$$Rf_*\mathcal{F} \simeq \sum_i R^i f_*\mathcal{F}[-i]$$

and in particular the Leray spectral sequence

$$H^p(Y, R^q f_*\mathcal{F}) \Longrightarrow H^{p+q}(X, \mathcal{F})$$

degenerates.

If one had wished, one could have supposed that $\mathcal{F}$ was an (A, f^*B)-bimodule, where B is a sheaf of rings on Y, and stated (2.1) in $D^b(Y, B)$.

(2.2) The case of schemes, endowed with the étale topology, does not enter into the framework of (2.1). In what follows we shall suppose that the theory of $\mathbb{Z}_\ell$-sheaves and $\mathbb{Q}_\ell$-sheaves (not necessarily constructible) has been made, and that the derived category of the category of these "sheaves" is reasonable. As far as the degeneration statements for spectral sequences are concerned, the reader who has no confidence in Jouanolou's future work may verify, by returning to the proof of (1.5), that one can dispense with it. Let ℓ be a prime number. Throughout what follows, assume that ℓ is invertible on all schemes under consideration.

Recall that on every scheme X the $\mathbb{Z}_\ell$-sheaf $\mathbb{Z}_\ell(1)$ is defined by the formula

$$\mathbb{Z}_\ell(1) = \varprojlim_n \mu_{\ell^n}.$$

This sheaf is an invertible $\mathbb{Z}_\ell$-module, which permits one to define, for every $\mathbb{Z}_\ell$-sheaf or $\mathbb{Q}_\ell$-sheaf $\mathcal{F}$, the "sheaf" $\mathcal{F}(i)$ by the formula

$$\mathcal{F}(i) = \mathcal{F} \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(1)^{\otimes i} \qquad (i \in \mathbb{Z}).$$

For every morphism $f : X \to Y$ of schemes and every pair of $\mathbb{Z}_\ell$- or $\mathbb{Q}_\ell$-sheaves $\mathcal{F}$ and $\mathcal{G}$ on X and Y respectively, one has

$$Rf_*(\mathcal{F})(i) = Rf_*(\mathcal{F}(i)) \qquad \text{and} \qquad f^*(\mathcal{G}(i)) = f^*(\mathcal{G})(i). \tag{2.3}$$

If $\mathcal{L}$ is an invertible sheaf on X, its first ℓ-adic Chern class lies in $H^2(X, \mathbb{Z}_\ell(1))$.

If $f : X \to Y$ is a morphism of schemes, if $\mathcal{F}$ is a $\mathbb{Z}_\ell$-sheaf, respectively a $\mathbb{Q}_\ell$-sheaf, on X, and if $u \in H^2(X, \mathbb{Z}_\ell(1))$, respectively $u \in H^2(X, \mathbb{Q}_\ell(1))$, then $\mathcal{F}$ will be said to satisfy the Lefschetz condition relative to u if, for every $i \geq 0$, the maps obtained by iterating i times the arrow $Rf_*(u)$,

$$Rf_*(u)^i : R^{n-i}f_*\mathcal{F} \longrightarrow R^{n+i}f_*\mathcal{F}(i) \qquad (i \geq 0),$$

are isomorphisms. By (2.3), this implies that all the arrows

$$Rf_*(u)^i : R^{n-i}f_*\mathcal{F}(k) \longrightarrow R^{n+i}f_*\mathcal{F}(k+i)$$

are isomorphisms. Applying (1.10), one obtains:

Proposition (2.4). With the preceding notation, if $\mathcal{F}$ satisfies the Lefschetz condition relative to u, then, in $D^b(Y, \mathbb{Z}_\ell)$, respectively $D^b(Y, \mathbb{Q}_\ell)$, one has

$$Rf_*\mathcal{F} \simeq \sum_i R^i f_*\mathcal{F}[-i] \tag{2.5}$$

and in particular the Leray spectral sequence

$$H^p(Y, R^q f_*\mathcal{F}) \Longrightarrow H^{p+q}(X, \mathcal{F})$$

degenerates.

If g is a morphism of schemes from Y to S, (2.4) also implies the degeneration of the Leray spectral sequence

$$R^p g_* R^q f_*\mathcal{F} \Longrightarrow R^{p+q}(gf)_*\mathcal{F}.$$

(2.6.1) It remains to review a few interesting cases in which the Lefschetz condition is satisfied; the justifying details are given in (2.7). Let $f : X \to Y$, u, $\mathcal{F}$, and n be given. In the topological case, it is known from Godement [2], II, (4.7.1), that if f is proper and Y locally paracompact, the formation of higher direct images commutes with passage to fibers, so that the Lefschetz condition is checked fiber by fiber. In the case of schemes, the same reduction applies, the reference to Godement being replaced by SGA 4, XII, (5.1). Still in the case of schemes, if f is proper and smooth and $\mathcal{F}$ is twisted constant, that is, smooth in a newer terminology, then the $R^i f_*\mathcal{F}$ are still twisted constant (cf. SGA 4, XVI, (2.2)); hence, if Y is connected, it is even enough to verify the Lefschetz condition on one geometric fiber.

In all the examples given, f will be a proper smooth continuous map, respectively a proper smooth morphism of schemes, of relative topological dimension $2n$, respectively algebraic dimension n. A continuous map is called smooth if locally on the source it is isomorphic to the projection from $\mathrm{R}^n \times Y$ to Y.

(2.6.2) If f is a proper smooth morphism of Kähler varieties, the constant sheaf $\mathbb{C}$ satisfies the Lefschetz condition relative to the fundamental 2-class of X.

(2.6.3) If X is a complex relative scheme (see [3]), projective and smooth over the locally paracompact topological space Y, and endowed with a relatively ample invertible sheaf $\mathcal{O}(1)$, the constant sheaf $\mathbb{C}$ satisfies the Lefschetz condition relative to the first Chern class of $\mathcal{O}(1)$.

(2.6.4) Suppose that f is a projective smooth morphism of schemes, with Y connected, and that a relatively ample invertible sheaf $\mathcal{O}(1)$ on X is given. If one can lift to characteristic 0 a geometric fiber of f, and its polarization, then the constant $\mathbb{Q}_\ell$-adic sheaf $\mathbb{Q}_\ell$ satisfies the Lefschetz condition relative to the ℓ-adic first Chern class of $\mathcal{O}(1)$.

It is known that one can lift to characteristic 0 in the preceding sense in the following cases:

- X is a Severi–Brauer variety over Y;
- X is a nonsingular complete intersection of hypersurfaces in a Severi–Brauer variety over Y, for the polarization induced by the canonical polarization of the Severi–Brauer variety;
- X is a polarized abelian scheme over Y (Mumford, to appear).

It is conjectured that (2.6.4) remains true when no lifting to characteristic 0 is known; this is proved only for surfaces.

(2.6.5) If f is a projective smooth morphism of relative dimension 2 and if X is endowed with a relatively ample invertible sheaf $\mathcal{O}(1)$, then the constant sheaf $\mathbb{Q}_\ell$ satisfies the Lefschetz condition relative to the first Chern class of $\mathcal{O}(1)$.

(2.7) In cases (2.6.2) and (2.6.3), the given cohomology 2-class on X induces on the fibers of f the 2-class defined by the Kähler structure of the fibers. Thus, by (2.6.1), to prove (2.6.2) and (2.6.3) it suffices to prove (2.6.2) in the special case where Y is a point. The assertion then follows from Hodge theory and the Hodge–Lepage decomposition (Weil [10], IV, no. 6, corollary to theorem 5). The proof given in loc. cit. still applies if $\mathcal{F}$ is a locally constant sheaf of finite-dimensional $\mathbb{C}$-vector spaces, endowed with a locally constant positive definite Hermitian structure.

By (2.6.1), to verify (2.6.4) it suffices to treat the case where Y is the spectrum of a field of characteristic 0, and the Lefschetz principle allows one to suppose $Y = \mathrm{Spec}(\mathbb{C})$.

It is then known (cf. SGA 4, XI, (4.4)) that, for every twisted constant $\mathbb{Q}_\ell$-adic sheaf $\mathcal{F}$ on X, if $\mathcal{F}^{an}$ denotes the locally constant sheaf of $\mathbb{Q}_\ell$-vector spaces on the topological space X^{an} deduced from $\mathcal{F}$, then the algebraic and transcendental cohomology $\mathbb{Q}_\ell$-vector spaces of $\mathcal{F}$ and $\mathcal{F}^{an}$ are isomorphic:

$$H^i(X, \mathcal{F}) \xrightarrow{\sim} H^i(X^{an}, \mathcal{F}^{an}). \tag{2.7.2}$$

Choosing an embedding of $\mathbb{Q}_\ell$ into $\mathbb{C}$, one obtains isomorphisms

$$H^i(X, \mathcal{F}) \otimes_{\mathbb{Q}_\ell} \mathbb{C} \xrightarrow{\sim} H^i(X^{an}, \mathcal{F}^{an}) \otimes_{\mathbb{Q}_\ell} \mathbb{C}. \tag{2.7.3}$$

Over $\mathbb{C}$, the exponential map defines the so-called canonical isomorphism between $\mathbb{Z}/\ell^n$ and μ_{ℓ^n}, hence between $\mathbb{Z}_\ell$ and $\mathbb{Z}_\ell(1)$. The image, under the composite isomorphism

$$H^2(X, \mathbb{Z}_\ell(1)) \longrightarrow H^2(X, \mathbb{Z}_\ell) \longrightarrow H^2(X^{an}, \mathbb{Z}) \otimes \mathbb{Z}_\ell,$$

of the ℓ-adic first Chern class of $\mathcal{O}(1)$ coincides with the transcendental first Chern class of $\mathcal{O}(1)$. The isomorphism (2.7.3) therefore reduces the case of (2.6.4) to which we had come to (2.6.3).

For the proof of (2.6.5), see Kleiman [6], pp. 28 ff. The essential point is the following theorem (Weil [11], p. 127).

(2.8) Let S be a smooth projective surface over a field k, and let D be a smooth curve on S such that $\mathcal{O}(D)$ is ample. Then the composite map

$$\mathrm{Pic}^0(S) \longrightarrow \mathrm{Pic}^0(D) = \mathrm{Alb}(D) \longrightarrow \mathrm{Alb}(S)$$

is an isogeny.

Remark (2.9). The degeneration theorem for the spectral sequence (2.2) deduced from (2.6.4) was conjectured by Grothendieck, from considerations of "weights", when Y is projective and smooth over an algebraically closed field.

Remark (2.10). Serre pointed out to me that the degeneration theorems for spectral sequences deduced from (2.6.2) and (2.6.3) can also be proved by an easy extension of the method used by Blanchard ([1], II, 1), at least if Poincaré duality is available on the base. Conversely, the method followed here completes Theorems II (1.1) and II (1.2) of loc. cit.; the proof of the implication (ii) $\Rightarrow$ (i) is essentially Blanchard's, who had to suppose the higher direct-image sheaves constant.

Theorem (2.11). Let $f : X \to Y$ be a proper smooth continuous map (2.6.1) of topological spaces of relative dimension n, with Y locally paracompact, let k be a commutative field of characteristic zero, and let $u \in H^0(Y, R^2 f_* k)$. Suppose that:

a) the sheaf $R^1 f_* k$ is constant, i.e. simple;

b) the iterated cup products

$$u^i \wedge : R^{n-i} f_* k \longrightarrow R^{n+i} f_* k$$

are isomorphisms.

Then the following conditions are equivalent:

(i) the class u is induced by an element of $H^2(X, k)$;

(ii) the transgression

$$d_2 : H^0(Y, R^2 f_* k) \longrightarrow H^2(Y, f_* k)$$

is zero on u;

(iii) in $\mathrm{D}^b(Y,k)$, one has

$$Rf_*k \simeq \bigoplus_i R^i f_*k[-i].$$

The implication (i) ⇒ (iii) is contained in (2.1). Assertion (iii) implies the degeneration of the whole Leray spectral sequence

$$H^p(Y, R^q f_*k) \Longrightarrow H^{p+q}(X,k), \tag{2.12}$$

and a fortiori (ii). It remains to prove that (ii) implies (i).

Let $x \in Y$ and $X_x = f^{-1}(x)$. The induced class $u_x \in H^2(X_x, k)$ is nonzero on every connected component of X_x, which is therefore orientable. Let Tr_u denote the composite map

$$H^{2n}(X_x,k) \xrightarrow{(u_x^n)^{-1}} H^0(X_x,k) = H_0(X_x,k) \xrightarrow{\mathrm{Tr}} k.$$

Poincaré duality implies that the bilinear form

$$H^{n-i}(X_x,k) \times H^{n+i}(X_x,k) \longrightarrow k, \qquad (x,y) \longmapsto \mathrm{Tr}_u(xy)$$

is a perfect duality. This construction globalizes and gives

$$\mathrm{Tr}_u : R^{2n} f_*k \longrightarrow k.$$

First prove that in (2.12), $d_2 u \in H^2(Y, R^1 f_*k)$ is zero. If $x \in H^0(Y, R^1 f_*k)$, then for degree reasons $xu^n = 0$, hence

$$d_2(xu^n) = d_2x\, u^n - nxu^{n-1} d_2 u = 0.$$

By b), the map

$$u^{n-1}\wedge : H^0(Y, R^1 f_*k) \longrightarrow H^0(Y, R^{2n-1} f_*k)$$

is an isomorphism. Thus, if (ii) holds, for every $y \in H^0(Y, R^{2n-1}f_*k)$ one has

$$y\, d_2 u = 0,$$

and in particular

$$\mathrm{Tr}_u(y\, d_2 u) = 0. \tag{2.13}$$

Let V be a k-vector space such that $R^1 f_*k$ is isomorphic to the constant sheaf V. Then $d_2 u \in H^2(Y,V)$ and, by (2.13), for every linear form w on V, the image of $d_2 u$ under w in $H^2(Y,k)$ is zero. Thus $d_2 u = 0$.

If $d_2 u = 0$, the proof of (1.5) shows that all differentials d_2 of (2.12) are zero, so that $E_2 = E_3$. For degree reasons, $u^{n+1} = 0$, hence

$$d_3 u^{n+1} = (n+1) u^n d_3 u = 0.$$

Since $d_3 u \in H^3(Y, R^0 f_*k)$, b) implies that $d_3 u = 0$. For degree reasons $d_r u = 0$ for $r > 3$, and u therefore comes from an element of $H^2(X,k)$.

(2.14) In the scheme-theoretic setting, the preceding result remains valid, but it is usable only in the "geometric" case, more precisely when $\mathbb{Q}_\ell$ is isomorphic to $\mathbb{Q}_\ell(1)$ on Y.

(2.15) Corollary (1.12) allows some of the preceding statements to be extended to certain nonconstant sheaves.

Proposition (2.16). Let g and f be two composable continuous maps, respectively two morphisms of schemes,

$$X \xrightarrow{g} Y \xrightarrow{f} Z,$$

and let $\mathcal{F}$ be a sheaf, respectively a $\mathbb{Z}_\ell$-sheaf or $\mathbb{Q}_\ell$-sheaf, on X. Suppose that

a) $Rg_*\mathcal{F} \in \mathrm{D}^b(Y)$ and $Rg_*\mathcal{F} \simeq \bigoplus_i R^i g_*\mathcal{F}[-i]$;

b) $R(fg)_*\mathcal{F} \in \mathrm{D}^b(Z)$ and $R(fg)_*\mathcal{F} \simeq \bigoplus_i R^i (fg)_*\mathcal{F}[-i]$.

Then, for every k, one has

$$Rf_*(R^k g_*\mathcal{F}) \simeq \bigoplus_i R^i f_*(R^k g_*\mathcal{F})[-i].$$

By b), the complex

$$R(fg)_*\mathcal{F} = Rf_* Rg_*\mathcal{F} \simeq \bigoplus_k Rf_*(R^k g_*\mathcal{F}[-k])$$

is a sum of its cohomology objects; by (1.12), the complexes $Rf_* R^k g_*\mathcal{F}$, which up to a shift are direct factors thanks to a), have the same property.

Proposition (2.17). Let $\mathcal{E}$ be a locally free vector bundle on a scheme S, let $\mathcal{F}$ be a twisted constant ℓ-adic sheaf on $\mathbb{P}(\mathcal{E})$, and let f be the projection from $\mathbb{P}(\mathcal{E})$ to S. In the derived category, one has

$$Rf_*\mathcal{F} \simeq \bigoplus_i R^i f_*\mathcal{F}[-i].$$

Projective space is simply connected and has torsion-free cohomology (SGA 1, XI, 1.1 and SGA 4, XVI, 2.2). Hence

$$\mathcal{F} = f^* f_*\mathcal{F}, \qquad Rf_*\mathcal{F} = Rf_*\mathbb{Z}_\ell \otimes f_*\mathcal{F},$$

which reduces the proof to the case $\mathcal{F} = \mathbb{Z}_\ell$. Let u be the Chern class of the invertible sheaf $\mathcal{O}(1)$ on $\mathbb{P}(\mathcal{E})$. Then $\mathbb{Z}_\ell$ is known to satisfy the Lefschetz condition relative to u, and (2.2) concludes the proof.

(2.18) From (2.17), with the help of (2.16), one obtains the same result for flag bundles of all kinds defined by $\mathcal{E}$. In the topological setting, the preceding applies directly to vector bundles over $\mathbb{C}$, and extends to vector bundles over $\mathbb{H}$ by (1.9). For real vector bundles, one must restrict to the case where $\mathcal{F}$ comes from a sheaf of 2-torsion on the base.

Remark (2.19). Let X be an abelian scheme of relative dimension g over a base Y. For every integer n, multiplication by n, denoted $[n]$, defines an endomorphism

$$[n]^* : Rf_*\mathbb{Q}_\ell \longrightarrow Rf_*\mathbb{Q}_\ell.$$

Taking rational linear combinations, with denominators bounded in terms of g, of these endomorphisms, one constructs endomorphisms

$$\pi_i : Rf_*\mathbb{Q}_\ell \longrightarrow Rf_*\mathbb{Q}_\ell$$

such that $H^j(\pi_i) = \delta_{ij}$. Applying (1.11), one concludes, without any projectivity hypothesis, that

$$Rf_*\mathbb{Q}_\ell \simeq \bigoplus_i R^i f_*\mathbb{Q}_\ell[-i].$$

The idea used here is due to Lieberman.

3. The infinitesimal form of the semicontinuity theorem

I recall in this paragraph some results which one manages to find in EGA III, § 7. The formulation (3.4) was pointed out to me by L. Illusie.

(3.0) Throughout this paragraph, A denotes a fixed noetherian local ring, $\mathfrak{m}$ its maximal ideal, and k its residue field. Let $D_f^-(A)$ denote the subcategory of the derived category of A-modules formed by bounded-above complexes of finite-type A-modules.

Proposition (3.1). Let M be a finite-type module over A.

(i) For every homomorphism, not necessarily local, from A to an Artinian ring B, and every finite-type B-module N, one has

$$\lg_B(M \otimes_A N) \geq \lg_B(N) \cdot \dim_k(M \otimes_A k). \tag{3.1.1}$$

(ii) The following conditions on M are equivalent:

a) M is a free module;

b) for all B and N as in (i), one has

$$\lg_B(M \otimes_A N) = \lg_B(N) \cdot \dim_k(M \otimes_A k); \tag{3.1.2}$$

c) for every $n \in \mathbb{N}$, condition (3.1.2) is satisfied for $B = N = A/\mathfrak{m}^n$;

d) for every $\mathfrak{p} \in \mathrm{Ass}(A)$, $M_\mathfrak{p}$ is flat over $A_\mathfrak{p}$, and condition (3.1.2) is satisfied for $B = N = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$.

If A has no embedded components, these conditions are also equivalent to:

e) for every maximal point $\mathfrak{p}$ of $\mathrm{Spec}(A)$ one has

$$\lg_{A_\mathfrak{p}}(M_\mathfrak{p}) = \lg(A_\mathfrak{p}) \cdot \dim_k(M \otimes_A k). \tag{3.1.2'}$$

The proof, an application of Nakayama's lemma, is left to the reader.

(3.2) For every complex K and every integer n, let $\tau_{\geq n}(K)$ denote the complex defined by

$$\tau_{\geq n}(K)^i = \begin{cases} 0, & i < n, \\ \mathrm{coker}(d^{n-1}), & i = n, \\ K^i, & i > n, \end{cases} \tag{3.2.1}$$

so that

$$H^i(\tau_{\geq n}(K)) = \begin{cases} 0, & i < n, \\ H^i(K), & i \geq n. \end{cases} \tag{3.2.2}$$

Let K be an object of $D^-_{\mathrm{coh}}(A)$ (cf. (3.0)). Replacing K, if necessary, by an isomorphic complex in the derived category, one may suppose the components of K to be free of finite type.

For all B and N as in (3.1) (i), one has at the level of complexes

$$\tau_{\geq n}(K \otimes^L_A N) \simeq \tau_{\geq n}(K) \otimes_A N.$$

Taking the Euler–Poincaré characteristics of the two sides gives

$$\sum_{i \geq 0} (-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) = \lg_B(\mathrm{coker}(d^{n-1}) \otimes_A N) + \sum_{i > 0} (-1)^i \lg_B(K^{n+i} \otimes_A N).$$

Subtract from this identity the analogous identity for $B = N = k$, multiplied by $\lg_B(N)$. By (3.1) (ii), $a \Leftrightarrow b$, and since the K^i are free, one obtains

$$\begin{aligned} &\sum_{i \geq 0} (-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) - \lg_B(N) \sum_{i \geq 0} (-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big) \\ &\qquad = \lg_B(\mathrm{coker}(d^{n-1}) \otimes_A N) - \lg_B(N) \dim_k(\mathrm{coker}(d^{n-1}) \otimes_A k), \end{aligned} \tag{3.2.3}$$

which allows one to apply (3.1) to the first member of (3.2.3).

Theorem (3.3) (exchange and semicontinuity theorems). Let $K \in \operatorname{Ob} D^-_{\mathrm{coh}}(A)$ (cf. (3.0)) and $n \in \mathbb{Z}$.

(i) For every homomorphism from A to an Artinian ring B and every finite-type B-module N, one has

$$\sum_{i\geq 0}(-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) \leq \lg_B(N) \sum_{i\geq 0}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.3.1}$$

(ii) The following conditions on K and n are equivalent:

a) The cohomological functor in the A-module N, $H^*(K \otimes^L_A N)$, satisfies the exchange property in degree $* = n$, i.e. the following equivalent conditions on K and n are satisfied:

a.1) the functor $H^n(K \otimes^L_A N)$ is left exact;
a.2) there is a module Q such that this functor is isomorphic to $\operatorname{Hom}_A(Q, N)$;
a.3) the functor $H^{n-1}(K \otimes^L_A N)$ is right exact;
a.4) there is a module P such that this functor is isomorphic to $P \otimes_A N$;
a.5) the map $H^{n-1}(K) \to H^{n-1}(K \otimes^L_A k)$ is surjective;
a.6) there exists a bounded-above complex K', quasi-isomorphic to K and with flat components, respectively free components of finite type, such that $d'^{\,n-1} = 0$.

b) For all B and N as in (i), one has

$$\sum_{i\geq 0}(-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) = \lg_B(N) \sum_{i\geq 0}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.3.2}$$

If A has no embedded components, these conditions are also equivalent to:

c) for every maximal point $\mathfrak{p}$ of $\operatorname{Spec}(A)$ one has

$$\sum_{i\geq 0}(-1)^i \lg_{A_{\mathfrak{p}}} H^{n+i}(K_{\mathfrak{p}}) = \lg(A_{\mathfrak{p}}) \sum_{i\geq 0}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.3.2'}$$

Remark (3.3.3). One could have lengthened the list by adding the conditions analogous to (3.1) (ii) c) and d).

Assertion (i) follows from (3.2.3) and (3.1) (i). By EGA III, (7.4.2), and its proof, conditions a.1), a.3), and a.6) are equivalent and, if K is represented by a bounded-above complex of finite-type free modules, they also mean that $\operatorname{coker}(d^{n-1})$ is free; taking (3.1) (ii) $a \Leftrightarrow b \Leftrightarrow e$ and (3.2.3) into account proves the equivalence of a.1), a.3), a.6), b), and c). Trivially, a.6) $\Rightarrow$ a.2) $\Rightarrow$ a.1) and a.6) $\Rightarrow$ a.4) $\Rightarrow$ a.3). The equivalence of condition a.5) with the others is contained in EGA III, (7.5.2).

Corollary (3.4). Let A, K, and n be as in (3.3), and let $k \in \mathbb{N}$.

(i) For all B and N as in (3.3) (i), one has

$$\sum_{0\leq i\leq 2k}(-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) \leq \lg_B(N) \sum_{0\leq i\leq 2k}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.4.1}$$

(ii) The following conditions on K, n, and k are equivalent:

a) The cohomological functor in N, $H^*(K \otimes^L_A N)$, satisfies the exchange property in degrees $* = n$ and $* = n + 2k + 1$.

a' The complex K is quasi-isomorphic to a bounded-above complex K' with flat components, respectively free components of finite type, such that $d'^{\,n-1} = d'^{\,n+2k} = 0$.

b) For all B and N as in (i), equality holds in (3.4.1).

If A has no embedded components, these conditions are also equivalent to:

c) for every maximal point $\mathfrak{p}$ of $\operatorname{Spec}(A)$, equality holds in (3.4.1) for $B = N = A_{\mathfrak{p}}$.

(3.4.2) For $k = 0$, these are properties only of the single hypertor $H^n(K \otimes^L_A N)$, considered as a functor in the A-module N. They can also be expressed by saying that $H^n(K)$ is free and that, for all N, the canonical map from $H^n(K) \otimes_A N$ to $H^n(K \otimes^L_A N)$ is an isomorphism.

(3.4.3) On the other hand, if K is a perfect complex, one deduces from (3.4), for $n \ll 0$, a result analogous to (3.3), with the summation over $i \geq 0$ in (3.3.1, 2, 2') replaced by a summation over $i \leq 0$.

Assertion (i) and the equivalence of a), b), and c) follow from (3.3) and from the identity

$$\sum_{0 \leq i \leq 2k} (-1)^i x_i = \sum_{0 \leq i} (-1)^i x_i + \sum_{0 \leq i} (-1)^i x_{i+2k+1}.$$

To prove a'), one successively applies (3.3) (ii) $a \Leftrightarrow a.6$) to (K, n) and to $(\tau_{\geq n}(K), n+2k+1)$.

(3.5) When A is Artinian, (3.4) gives, for $k = 0$, the inequality

$$\lg_A H^n(K) \leq \lg(A) \dim_k\big(H^n(K \otimes_A^L k)\big), \tag{3.5.1}$$

and, if equality holds in (3.5.1), then the exchange condition is satisfied in degrees n and $n+1$, and $H^n(K)$ is a free A-module.

4. The Gysin morphism in Hodge cohomology

This paragraph is written from notes of A. Grothendieck.

(4.0) Let $f : X \to Y$ be a proper morphism between schemes X and Y which are smooth and purely of relative dimensions n and m over a base S. We suppose that S is noetherian and has a dualizing complex, in order to refer to [4] for the definition of $\mathrm{R}f^!$; this restriction is probably unnecessary. Finally, put $d = n - m$.

(4.1) Every relative differential form on Y defines, by inverse image, a relative differential form on X, hence a morphism

$$f^*\Omega^q_{Y/S} = Lf^*\Omega^q_{Y/S} \longrightarrow \Omega^q_{X/S}. \tag{4.1.1}$$

By the duality theory for smooth morphisms and the transitivity of $\mathrm{R}f^!$, one has canonically

$$\mathrm{R}f^!\Omega^m_{Y/S} = \Omega^n_{X/S}[d]. \tag{4.1.2}$$

Apply to $\Omega^q_{Y/S}$ and $\Omega^m_{Y/S}$ the induction formula

$$\mathrm{R}f_*\,\mathrm{RHom}(K, L) = \mathrm{RHom}(Lf^*K, \mathrm{R}f^!L),$$

which is obtained readily from the definition of $\mathrm{R}f^!$ by biduality (Hartshorne [4], chap. VII, § 3). Since

$$\mathrm{RHom}(\Omega^q_{Y/S}, \Omega^m_{Y/S}) \simeq \Omega^{m-q}_{Y/S},$$

one obtains

$$\mathrm{R}f^!\Omega^{m-q}_{Y/S} = \mathrm{RHom}(Lf^*\Omega^q_{Y/S}, \Omega^n_{X/S})[d].$$

The arrow (4.1.1) therefore defines, by transposition, an arrow

$$\Omega^{n-q}_{X/S} = \mathrm{RHom}(\Omega^q_{X/S}, \Omega^n_{X/S}) \longrightarrow \mathrm{R}f^!\Omega^{m-q}_{Y/S}[d]. \tag{4.1.3}$$

By duality theory, giving an arrow (4.1.3) amounts to giving an arrow

$$\mathrm{R}f_*\Omega^{n-q}_{X/S} \longrightarrow \Omega^{m-q}_{Y/S}[-d]. \tag{4.1.4}$$

Definition (4.2). Let X, Y, f, S, n, m, and d be as in (4.0). The arrow (4.1.4)

$$\mathrm{Tr}_f : \mathrm{R}f_*\Omega^q_{X/S} \longrightarrow \Omega^{q-d}_{Y/S}[-d]$$

is called the Gysin morphism, as are the induced maps

$$\mathrm{Tr}_f : H^p(X, \Omega^q_{X/S}) \longrightarrow H^{p-d}(Y, \Omega^{q-d}_{Y/S}).$$

Proposition (4.3). Let $f : X \to Y$ be a proper birational morphism of smooth schemes over a field k. The maps

$$f^* : H^p(Y, \Omega^q_Y) \longrightarrow H^p(X, \Omega^q_X)$$

are injective.

One reduces to the case where X and Y are purely of dimension n. The composite morphism

$$\mathrm{Tr}_f \circ f^* : \Omega^q_Y \longrightarrow \mathrm{R}f_* f^* \Omega^q_Y \longrightarrow \mathrm{R}f_* \Omega^q_X \longrightarrow \Omega^q_Y$$

is the identity, because it coincides with the identity on a dense open subset. Applying the functor $H^p(Y,\ )$, one finds that the composite morphism

$$\mathrm{Tr}_f \circ f^* : H^p(Y, \Omega^q_Y) \longrightarrow H^p(X, \Omega^q_X) \longrightarrow H^p(Y, \Omega^q_Y)$$

is the identity, which proves (4.3).

5. Hodge and De Rham cohomology

(5.1) Let $f : X \to S$ be a smooth morphism of schemes. By definition, the relative De Rham cohomology sheaves of X over S are given by

$$H^*_{\mathrm{DR}}(X/S) = \mathrm{R}^* f_*(\Omega^\bullet_{X/S}),$$

where $\Omega^\bullet_{X/S}$, the relative De Rham complex, is an S-linear differential complex. There is a spectral sequence

$$E_1^{pq} = \mathrm{R}^q f_* \Omega^p_{X/S} \Longrightarrow H^{p+q}_{\mathrm{DR}}(X/S). \tag{5.1.1}$$

(5.2) When $S = \mathrm{Spec}(\mathbb{C})$ and X is proper and smooth, GAGA ([7]) shows that the cohomology of each coherent sheaf Ω^p_X, and hence the hypercohomology of $\Omega^\bullet_X$, are the same in the algebraic sense (Zariski topology) and in the transcendental sense (usual topology of X^{an}); in particular, since $\Omega^\bullet_{X^{\mathrm{an}}}$ is a resolution of the constant sheaf $\mathbb{C}$, one has

$$H^n_{\mathrm{DR}}(X) = H^n(X^{\mathrm{an}}, \mathbb{C}),$$

and complex conjugation on the constant sheaf $\mathbb{C}$ induces a complex conjugation, of transcendental nature, on $H^n_{\mathrm{DR}}(X)$. The spectral sequence (5.1.1) is here

$$E_1^{pq} = H^q(X, \Omega^p_X) \Longrightarrow H^{p+q}_{\mathrm{DR}}(X); \tag{5.2.1}$$

let $F^p(H^n(X))$ denote the decreasing filtration which it defines on its abutment. By Weil [10], chap. IV, no. 4, th. 2, the spectral sequence (5.2.1) can be computed as the spectral sequence of the double complex $H^0(X^{\mathrm{an}}, \Omega^{p,q})$, filtered by the first degree, where $\Omega^{p,q}$ denotes the sheaf of C^∞ differential forms bihomogeneous of bidegree (p, q) (Weil [10], chap. II, no. 1).

Suppose now that X is projective, so that X^{an} is a Kähler variety. The harmonic forms then form a double subcomplex (Weil [10], chap. II, no. 6, cor. 1 to th. 2) of the preceding double complex, on which d' and d'' vanish. The inclusion of this subcomplex induces an isomorphism on the E_1 terms of the spectral sequences defined by these double complexes, filtered by the first degree, as one sees by considering the operator H (harmonic component), a retraction of the double complex $H^0(X^{\mathrm{an}}, \Omega^{p,q})$ onto the subcomplex of harmonic forms and satisfying

$$1 - H = d''(2\delta' G) + (2\delta' G)d''$$

(Weil [10], chap. IV, no. 1: (I) and lemma 3; no. 3: cor. 2 and chap. II, no. 5, th. 2 (IX)).

If $H^{p,q}(X)$ denotes the image in $H^{p+q}(X^{\mathrm{an}}, \mathbb{C})$ of the space of harmonic forms of type (p, q), one therefore has

$$H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X), \tag{5.2.2}$$

$$F^p(H^n(X)) = \bigoplus_{i \geq p} H^{i,n-i}(X), \tag{5.2.3}$$

$$\overline{H^{p,q}(X)} = H^{q,p}(X) \quad \text{(complex conjugation)}, \tag{5.2.4}$$

and the spectral sequence (5.2.1) degenerates.

Returning to the general case where X is only assumed proper and smooth over $\mathbb{C}$, put

$$H^{p,q}(X) = F^p(H^{p+q}(X)) \cap \overline{F^q(H^{p+q}(X))}, \tag{5.2.5}$$

so that

$$\overline{H^{p,q}(X)} = H^{q,p}(X). \tag{5.2.6}$$

By (5.2.3), this notation is compatible with the one used when X is projective. Put also

$$h^{p,q} = \dim H^q(X, \Omega_X^p). \tag{5.2.7}$$

Proposition (5.3). Let X be a proper smooth scheme over $\mathbb{C}$:

(i) the spectral sequence (5.2.1) degenerates;

(ii) one has

$$F^p(H^n(X)) = \bigoplus_{i \geq p} H^{i,n-i}(X), \tag{5.3.1}$$

and in particular

$$H^n(X) = \bigoplus_{p+q=n} H^{p,q}(X) \quad \text{(direct sum)}, \tag{5.3.2}$$

$$h^{p,q} = h^{q,p} = \dim H^{p,q}(X). \tag{5.3.3}$$

One reduces to the case where X is connected. Let N be its dimension. By Chow's lemma and resolution of singularities (Hironaka [5]), there exists a projective birational morphism $g : X' \to X$, with X' projective and smooth over $\mathbb{C}$. The spectral sequence

$$E_1^{pq} = H^q(X, \Omega_X^p) \Longrightarrow H_{\mathrm{DR}}^{p+q}(X) \tag{5.3.4}$$

maps, by inverse image, into the analogous spectral sequence

$$E_1'^{\,pq} = H^q(X', \Omega_{X'}^p) \Longrightarrow H_{\mathrm{DR}}^{p+q}(X'), \tag{5.3.5}$$

which degenerates by (5.2). By (4.3), the morphism $g^* : E_1 \to E_1'$ is injective. One concludes, by induction on $r \geq 1$, that $d_r = 0$, that $E_r = E_{r+1}$, and that g^* injects E_{r+1} into E'_{r+1}:

$$\begin{array}{ccc} E_r^{pq} & \xrightarrow{d_r} & E_r^{p+r,q-r+1} \\ {\scriptstyle g^*}\downarrow & & \downarrow{\scriptstyle g^*} \\ E_r'^{\,pq} & \xrightarrow[0]{} & E_r'^{\,p+r,q-r+1}. \end{array}$$

Since the spectral sequences (5.3.4) and (5.3.5) degenerate, and since g^* is injective on their initial term, it is still injective on their abutment. From formulas (5.2.3), (5.2.4), applied to X', it follows that

$$F^p(H^n(X')) \cap \overline{F^{n-p+1}(H^n(X'))} = \{0\};$$

hence one obtains the analogous formula

$$F^p(H^n(X)) \cap \overline{F^{n-p+1}(H^n(X))} = \{0\}, \tag{5.3.6}$$

because g^* sends $F^p(H^n(X))$ to $F^p(H^n(X'))$ and commutes with complex conjugation. In particular,

$$\sum_{i \geq p} h^{i,n-i} + \sum_{i \geq n-p+1} h^{i,n-i} \leq \dim H^n(X) \leq \sum_i h^{i,n-i},$$

which may be written

$$\sum_{i \geq p} h^{i,n-i} \leq \sum_{i \leq n-p} h^{i,n-i}. \tag{5.3.7}$$

By Serre duality, $h^{i,j} = h^{N-i,N-j}$, inequality (5.3.7), for $N - n$, implies the opposite inequality, so that

$$\dim F^p(H^n(X)) + \dim \overline{F^{n-p+1}(H^n(X))} = \dim H^n(X).$$

By (5.3.6), therefore,

$$H^n(X) = F^p(H^n(X)) \oplus \overline{F^{n-p+1}(H^n(X))}. \tag{5.3.8}$$

The decomposition (5.3.8) induces on $F^{p-1}(H^n(X))$, which contains one of the factors, a decomposition

$$F^{p-1}(H^n(X)) = F^p(H^n(X)) \oplus H^{p-1,n-p+1}(X),$$

and formula (5.3.1) follows by descending induction on p; (5.3.2) and (5.3.3) follow at once from (5.3.1) and (5.2.6), completing the proof of (5.3).

Corollary (5.4). Under the hypotheses of (5.3), one has:

(i) A complex cohomology class a on X lies in $H^{p,q}(X)$ if and only if it can be represented by a closed form α of type (p, q).

(ii) Every cohomology class can be represented by a form α satisfying $d'\alpha = d''\alpha = 0$.

(iii) If the form α satisfies $d'\alpha = d''\alpha = 0$, the following conditions are equivalent:

a) $(\exists\beta)\ \alpha = d\beta$;

b) $(\exists\beta)\ \alpha = d'\beta$;

c) $(\exists\beta)\ \alpha = d''\beta$.

By definition, $a \in H^{p,q}(X)$ if and only if in the class of a there exist closed forms α_1 and α_2 whose components of type (p', q') are zero for $p' < p$ (respectively $q' < q$). Then there exists β such that $\alpha_1 - \alpha_2 = d\beta$. Let β_1 (respectively β_2) be the sum of the components of β of type (p', q') with $p' \geq p$ (respectively $q' \geq q$); one has $\beta = \beta_1 + \beta_2$ and

$$\alpha = \alpha_1 - d\beta_1 = \alpha_2 + d\beta_2$$

is a closed form of type (p, q) in the class of a. This proves (i). It suffices to prove (ii) for a bihomogeneous, hence represented by a closed bihomogeneous form α by (i). Such a form automatically satisfies $d'\alpha = d''\alpha = 0$.

If α satisfies $d'\alpha = d''\alpha = 0$, its bihomogeneous components are closed, and each of the conditions a), b), c) is equivalent to the conjunction of the analogous conditions for the various components of α, trivially for b) and c), and by (5.3) (ii) and (5.4) (i) for a). To prove (iii), suppose therefore that α is closed of type (p, q). The cohomology class a defined by α lies in $H^{p,q}(X)$, hence in $F^p(H^{p+q}(X))$, and is zero if and only if it already lies in $F^{p+1}(H^{p+q}(X))$, i.e. if its image in $H^q(X, \Omega^p_X)$ is zero. This proves a) $\Leftrightarrow$ c), and the equivalence a) $\Leftrightarrow$ b) follows by conjugation.

(Added in proofs.) One can show that these conditions are also equivalent to the existence of a form β such that $\alpha = d'd''\beta$.

Theorem (5.5). Let S be a scheme of characteristic 0 and let $f : X \to S$ be a proper smooth morphism. Then:

(i) the sheaves $\mathrm{R}^q f_* \Omega^p_{X/S}$ are locally free of finite type and their formation is compatible with arbitrary base change;

(ii) the spectral sequence (5.1.1)

$$E_1^{pq} = \mathrm{R}^q f_* \Omega^p_{X/S} \Longrightarrow H^{p+q}_{\mathrm{DR}}(X/S)$$

degenerates;

(iii) at every point of S, the sheaves $\mathrm{R}^q f_* \Omega^p_{X/S}$ and $\mathrm{R}^p f_* \Omega^q_{X/S}$ have the same rank (Hodge symmetry).

The second assertion of (i) follows from the first and from EGA III, (7.8.5), applied to each of the $\Omega^p_{X/S}$; it implies that it suffices to verify (iii) for S the spectrum of a field.

Standard arguments allow one to reduce successively to the case where S is affine, since the question is local on S; noetherian, by passage to an inductive limit of rings; the spectrum of a noetherian local ring, since the question is local; the spectrum of a complete noetherian local ring, by faithful flatness of completion; and finally the spectrum of an Artinian local ring. To make this last reduction, write the complete noetherian local ring R

as the projective limit of its Artinian quotients $R/\mathfrak{m}^n$. By the comparison theorem EGA III, (4.1.5), the E_1 term of the spectral sequence (5.1.1) is the projective limit of the E_1 terms of the analogous spectral sequences over the $R/\mathfrak{m}^n$, so that if these degenerate, then (5.1.1) also degenerates. If assertion (i) is true over the $R/\mathfrak{m}^n$, these E_1 terms form a projective system of free modules, obtained from one another by reduction modulo $\mathfrak{m}^n$, so that their projective limit is a free module.

Suppose therefore that $S = \mathrm{Spec}(A)$ with A an Artinian local ring with residue field k of characteristic 0. This ring admits a field of representatives (EGA, 0_{IV}, (19.8.6), (i), for $W = k$, $u =$ identity) and can therefore be regarded as a local k-algebra of finite dimension. The Lefschetz principle permits one to suppose $k = \mathbb{C}$, in which case (iii) follows from (i) and (5.3.3).

To complete the proof of (5.5), it remains to prove the first assertion of (i), and (ii), under the additional hypothesis:

$$S = \mathrm{Spec}(A) \quad \text{for a finite-dimensional local } \mathbb{C}\text{-algebra } A \text{ with maximal ideal } \mathfrak{m}. \tag{5.5.1}$$

The spectral sequence (5.1.1) becomes

$$E_1^{pq} = H^q(X, \Omega^p_{X/S}) \Longrightarrow \mathrm{R}^{p+q}\Gamma(\Omega^\bullet_{X/S}). \tag{5.5.2}$$

Lemma (5.5.3). Under the hypotheses (5.5) and (5.5.1), the complex $\Omega^{\mathrm{an},\bullet}_{X/S}$ is a resolution of the constant sheaf A on the topological space X^{an}.

Since the complex $\Omega^{\mathrm{an},\bullet}_{X/S}$ is A-linear, its $\mathfrak{m}$-adic filtration is a filtration by subcomplexes. For this filtration,

$$\mathrm{Gr}\,\Omega^{\mathrm{an},\bullet}_{X/S} = \mathrm{Gr}\,A \otimes_{\mathbb{C}} \mathrm{Gr}^0\,\Omega^{\mathrm{an},\bullet}_{X/S}.$$

The complex $\mathrm{Gr}^0\,\Omega^{\mathrm{an},\bullet}_{X/S}$ is a resolution of the constant sheaf $\mathbb{C}$, since it is just the De Rham complex of X_{red}. Thus the complex $\mathrm{Gr}\,\Omega^{\mathrm{an},\bullet}_{X/S}$ is a resolution of $\mathrm{Gr}\,A$, and the assertion follows.

The space X is compact, so by GAGA (cf. (5.2)) the algebraic and transcendental hypercohomologies of $\Omega^\bullet_{X/S}$ coincide, and by (5.5.3) they also coincide with the cohomology of X^{an} with values in A. Hence

$$\mathrm{lg}_A\,\mathrm{R}^n\Gamma(\Omega^\bullet_{X/S}) = \mathrm{lg}(A)\dim_{\mathbb{C}}\mathrm{R}^n\Gamma(\Omega^\bullet_{X_{\mathrm{red}}}). \tag{5.5.4}$$

By (3.5.1) and EGA III, (6.10.5), one knows that

$$\mathrm{lg}_A\,H^q(X, \Omega^p_{X/S}) \leq \mathrm{lg}(A)\dim_{\mathbb{C}} H^q(X_{\mathrm{red}}, \Omega^p_{X_{\mathrm{red}}}), \tag{5.5.5}$$

with equality possible only if $H^q(X, \Omega^p_{X/S})$ is A-free.

From (5.5.2) one obtains

$$\sum_{p+q=n} \mathrm{lg}_A\,H^q(X, \Omega^p_{X/S}) \geq \mathrm{lg}_A\,\mathrm{R}^n\Gamma(\Omega^\bullet_{X/S}), \tag{5.5.6}$$

equality being possible for all n only if the spectral sequence (5.5.2) degenerates. Chaining these inequalities, one finds

$$\mathrm{lg}(A)\sum_{p+q=n}\dim_{\mathbb{C}} H^q(X_{\mathrm{red}}, \Omega^p_{X_{\mathrm{red}}}) \geq \mathrm{lg}(A)\mathrm{R}^n\Gamma(\Omega^\bullet_{X_{\mathrm{red}}}), \tag{5.5.7}$$

equality being possible only if the conclusions of (5.4) are satisfied. This equality follows from (5.3) (i) applied to X_{red}.

6. Application to coherent cohomology

Theorem (6.1). Let S be a scheme of characteristic 0, and let $f : X \to S$ be a projective smooth morphism of relative dimension n. For every p, for example $p = 0$, one has in $D^b(S, \mathcal{O}_S)$

$$\mathrm{R}f_*\Omega^p_{X/S} \simeq \bigoplus_q \mathrm{R}^q f_*\Omega^p_{X/S}[-q].$$

Let u denote the first Chern class of a relatively ample invertible sheaf on X, with values in $H^1(X, \Omega^1_{X/S})$. By cup product, the class u defines homomorphisms in $D^b(S)$,

$$u\wedge : \mathrm{R}f_*\Omega^p_{X/S} \longrightarrow \mathrm{R}f_*\Omega^{p+1}_{X/S}[1],$$

and a degree-two endomorphism

$$u\wedge : \bigoplus_p \mathrm{R}f_*\Omega^p_{X/S}[-p] \longrightarrow \left(\bigoplus_p \mathrm{R}f_*\Omega^p_{X/S}[-p]\right)[2]. \tag{6.1.1}$$

Lemma (6.2). The endomorphism (6.1.1) satisfies the hypotheses of (1.5).

By (5.4) (i) and the Lefschetz principle, it suffices to prove (6.2) when $S = \mathrm{Spec}(\mathbb{C})$. The assertion then follows from the Lefschetz theorem (see (2.6.2) or (2.6.3)) and from the Hodge decomposition.

By (1.5), the complex $\bigoplus_p \mathrm{R}f_*\Omega^p_{X/S}[-p]$ is isomorphic, in the derived category, to the sum of its cohomology objects, and the same is true for each of the $\mathrm{R}f_*\Omega^p_{X/S}$, which, up to a shift, are direct factors of it (1.12).

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Ordinary abelian varieties over a finite field

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We give a down-to-earth description of the category of ordinary abelian varieties over a finite field $\mathbb{F}_q$. The result obtained was inspired by Ihara [2], ch. V (see also [3]).

1. Let p be a prime number, let $\mathbb{F}_p$ be the field $\mathbb{Z}/(p)$, let $\overline{\mathbb{F}_p}$ be an algebraic closure of $\mathbb{F}_p$, and, for each power q of p, let $\mathbb{F}_q$ be the subfield with q elements of $\overline{\mathbb{F}_p}$. For every algebraic extension k of $\mathbb{F}_p$, let $W_0(k)$ denote the henselian discrete valuation ring, essentially of finite type over $\mathbb{Z}$, absolutely unramified, with residue field k; let $W(k)$ be the ring of Witt vectors over k, i.e. the completion of $W_0(k)$, let $W = W(\overline{\mathbb{F}_p})$, and let φ be an embedding of W in the field $\mathbb{C}$ of complex numbers. We denote by $\mathbb{Z}(1)$ the subgroup $2\pi i\mathbb{Z}$ of $\mathbb{C}$. The exponential map defines an isomorphism between $\mathbb{Z}(1) \otimes \mathbb{Z}_\ell$ and

$$\mathbb{Z}_\ell(1)(\mathbb{C}) = \varprojlim_m \mu_{\ell^m}(\mathbb{C}).$$

We denote by A^* the abelian variety dual to an abelian variety A. For every field k, let $\bar{k}$ denote an algebraic closure of k.

2. Let A be an abelian variety of dimension g, defined over a field k of characteristic p. Recall that A is called ordinary if the following equivalent conditions are satisfied:

(I) A has p^g points of order dividing p with values in $\bar{k}$.

(II) The "Hasse–Witt matrix"

$$F^* : H^1(A^{(p)}, \mathcal{O}_{A^{(p)}}) \longrightarrow H^1(A, \mathcal{O}_A)$$

is invertible.

(III) The neutral component of the group scheme A_p, the kernel of multiplication by p, is of multiplicative type, hence geometrically isomorphic to a power of μ_p.

If $k = \mathbb{F}_q$, if F is the Frobenius endomorphism of A, and if $P_{cA}(F;x)$ is its characteristic polynomial, these conditions are further equivalent to:

(IV) At least half of the roots of $P_{cA}(F;x)$ in $\overline{\mathbb{Q}}_p$ are p-adic units. In other words, if $n = \dim A$, the reduction modulo p of the polynomial $P_{cA}(F;x)$ is not divisible by x^{n+1}.

3. Let A be an ordinary abelian variety over $\overline{\mathbb{F}_p}$. Let $\widetilde{A}$ denote the canonical Serre–Tate lift [4] of A over W. Recall that $\widetilde{A}$ depends functorially on A and is characterized by the fact that the p-divisible group $T_p(\widetilde{A})$ over W attached to $\widetilde{A}$ [5] is the product of the p-divisible groups, uniquely determined by 2. (III), lifting respectively the neutral component and the largest étale quotient of $T_p(A)$. The canonical lift $\widetilde{A}$ is also the unique lift of A such that every endomorphism of A lifts to $\widetilde{A}$. Let $T(A)$ denote the integral homology of the complex abelian variety $A_{\mathbb{C}}$ deduced from $\widetilde{A}$ and φ by extension of scalars from W to $\mathbb{C}$:

$$T(A) = H_1(\widetilde{A} \otimes_\varphi \mathbb{C}).$$

It is known that $\widetilde{A}$ descends uniquely to $W_0(\overline{\mathbb{F}_p})$, so that $A_{\mathbb{C}}$ depends only on A and on the restriction of φ to $W_0(\overline{\mathbb{F}_p})$. The free $\mathbb{Z}$-module $T(A)$ has rank $2\dim(A)$; it is functorial in A. Moreover, if ℓ is a prime number distinct from p, one has functorially

$$T(A) \otimes \mathbb{Z}_\ell = T_\ell(A). \tag{3.1}$$

The canonical lift of the abelian variety A^* dual to A is the dual of $\widetilde{A}$, so that $(A_{\mathbb{C}})^* = A^*_{\mathbb{C}}$ and $T(A)$ and $T(A^*)$ are in perfect duality with values in $\mathbb{Z}(1)$:

$$T(A) \otimes T(A^*) \longrightarrow \mathbb{Z}(1). \tag{3.2}$$

It is indispensable to use $\mathbb{Z}(1)$ rather than $\mathbb{Z}$ in order to obtain a theory invariant under complex conjugation. The pairings (3.2) are compatible, via (3.1), with the pairings

$$T_\ell(A) \otimes T_\ell(A^*) \longrightarrow \mathbb{Z}_\ell(1);$$

a morphism $\xi : A \to A^*$ defines a polarization of A if and only if $\xi_{\mathbb{C}} : A_{\mathbb{C}} \to A^*_{\mathbb{C}}$ defines a polarization of $A_{\mathbb{C}}$. Put

$$T'_p(A) = \mathrm{Hom}(\mathbb{Q}_p/\mathbb{Z}_p, A(\overline{\mathbb{F}_p}))$$

and

$$T''_p(A) = \mathrm{Hom}_{\mathbb{Z}_p}(T'_p(A^*), \mathbb{Z}(1) \otimes \mathbb{Z}_p).$$

These $\mathbb{Z}_p$-modules are covariant functors of A.

By definition of the canonical lift, the p-divisible group $T_p(\widetilde{A})$ is the sum of the constant pro-étale group $T'_p(A)$ and of the Cartier dual of $T'_p(A^*)$. For every morphism $u : A \to B$, the induced morphism

$$u : T_p(\widetilde{A}) \to T_p(\widetilde{B})$$

is identified with the sum of $u|T'_p(A) : T'_p(A) \to T'_p(B)$ and the Cartier transpose of ${}^t u|T'_p(B^*) : T'_p(B^*) \to T'_p(A^*)$. Over $\mathbb{C}$, one has canonically $\mathbb{Z}(1)/(p^n) \sim \mu_{p^n}$, whence an isomorphism of functors:

$$T_{(p)}(A) = T(A) \otimes \mathbb{Z}_p = T'_p(A) \oplus T''_p(A). \tag{3.2}$$

4. Recall that, if $\varphi : X \to Y$ is an isogeny between complex abelian varieties, the exact homotopy sequence reduces to a short exact sequence

$$0 \longrightarrow H_1(X) \longrightarrow H_1(Y) \longrightarrow \mathrm{Ker}(\varphi) \longrightarrow 0.$$

The abelian varieties which are quotients of X by a finite subgroup, and these finite subgroups of X, correspond bijectively to the sublattices of $H_1(X) \otimes \mathbb{Q}$ containing $H_1(X)$.

Let A be an ordinary abelian variety over $\overline{\mathbb{F}_p}$. If n is an integer prime to p, the finite group subschemes of order n of A, of $\widetilde{A}$, and of $A_{\mathbb{C}}$ correspond bijectively, and still correspond to the overlattices R of $T(A)$ such that $[R : T(A)] = n$.

Put

$$V'_p = T'_p(A) \otimes \mathbb{Q}_p, \qquad V''_p(A) = T''_p(A) \otimes \mathbb{Q}_p.$$

The finite group subschemes of order p^k of A are products of an étale subgroup and an infinitesimal subgroup. The étale subgroups of order p^k of A correspond to those subgroups of order p^k of $A_{\mathbb{C}}$ for which the corresponding overlattice R of $T(A)$ is contained in $T_{(p)}(A) + V'_p(A)$. By duality, the infinitesimal subgroups of A correspond to the overlattices R of $T(A)$, p-isogenous to $T(A)$, i.e. such that $[R : T(A)]$ is a power of p, and contained in $T_{(p)}(A) + V''_p(A)$.

In all, the finite subgroups of A, or the abelian varieties quotient of A, correspond bijectively to the overlattices R of $T(A)$ such that

$$R \otimes \mathbb{Z}_p = (R \otimes \mathbb{Z}_p \cap V'_p) + (R \otimes \mathbb{Z}_p \cap V''_p). \tag{4.1}$$

5. In particular, $A^{(p)}$, the quotient of A by the largest infinitesimal subgroup of A killed by p for A ordinary, is defined by the overlattice $T(A)^{(p)}$ of $T(A)$, p-isogenous to $T(A)$, such that

$$T(A)^{(p)} \otimes \mathbb{Z}_p = T'_p(A) + \frac{1}{p} T''_p(A).$$

6. Let A be an abelian variety over $\mathbb{F}_q$ and let $F : x \mapsto x^q$ be its Frobenius endomorphism. Recall that A is uniquely determined by the pair $(\overline{A}, F)$ deduced from (A, F) by extension of scalars from $\mathbb{F}_q$ to $\overline{\mathbb{F}_q}$; the endomorphism F of $\overline{A}$ factors through the relative Frobenius

$$\mathrm{Fr}^{(q)} : \overline{A} \longrightarrow \overline{A}^{(q)}$$

and an isomorphism

$$F' : \overline{A}^{(q)} \longrightarrow \overline{A}.$$

If A is ordinary, $T(A)$ will denote the $\mathbb{Z}$-module $T(\overline{A})$ endowed with the endomorphism F induced by the Frobenius endomorphism of A. By 5, by what precedes, and by (3.2), the lattices $T(A)$ and $F(T(A))$ are p-isogenous and one has

$$F(T'_p(A)) = T'_p(A), \tag{6.1}$$

$$F(T''_p(A)) = qT''_p(A). \tag{6.2}$$

7. Theorem. The functor

$$A \longmapsto (T(A), F)$$

is an equivalence of categories between the category of ordinary abelian varieties over $\mathbb{F}_q$ and the category of finite-type free $\mathbb{Z}$-modules T endowed with an endomorphism F satisfying the following conditions:

(a) F is semisimple, and its eigenvalues have complex absolute value $q^{1/2}$.

(b) At least half of the roots in $\overline{\mathbb{Q}}_p$ of the characteristic polynomial of F are p-adic units; in other words, if T has rank d, the reduction modulo p of the polynomial $P_{cT}(F; x)$ is not divisible by $x^{\lfloor d/2 \rfloor + 1}$.

(c) There exists an endomorphism V of T such that $FV = q$.

If condition (a) is satisfied, conditions (b) and (c) are equivalent to:

(d) The module $T \otimes \mathbb{Z}_p$ admits a decomposition, stable under F, into two $\mathbb{Z}_p$-submodules T'_p and T''_p of the same dimension, such that $F|T'_p$ is invertible and $F|T''_p$ is divisible by q.

Proof. (A) We prove that (a)+(b)+(c) imply (d). If α is a complex eigenvalue of F, then $\bar{\alpha}$ is another, with the same multiplicity, and $\alpha\bar{\alpha} = q$. Apart from those equal to $\pm q^{1/2}$, the eigenvalues of F in $\mathbb{C}$, hence in $\overline{\mathbb{Q}}_p$, fall into pairs of roots α and q/α. The roots α and q/α cannot both be p-adic units; it follows from (b) that $\pm q^{1/2}$ is not an eigenvalue of F, that half of the eigenvalues of F in $\overline{\mathbb{Q}}_p$ are p-adic units, say $\alpha_1, \ldots, \alpha_{d/2}$, and that the other half consists of $\beta_1 = q/\alpha_1, \ldots, \beta_{d/2} = q/\alpha_{d/2}$. Let

$$T_{(p)} = T \otimes \mathbb{Z}_p, \qquad V_p = T \otimes \mathbb{Q}_p,$$

let V'_p be the subspace of V_p which is the kernel of $\prod_i (F - \alpha_i)$, and let V''_p be the kernel of the endomorphism $\varphi = \prod_i (F - \beta_i)$. One has

$$V_p = V'_p \oplus V''_p.$$

Let T'_p be the projection of $T_{(p)}$ onto V'_p and let $T''_p = T_{(p)} \cap V''_p$. Since φ kills V''_p and preserves T, it sends T'_p into $T_{(p)} \cap V'_p \subset T'_p$. On the other hand,

$$\det(\varphi | V'_p) = \prod_{i,j} (\alpha_i - \beta_j)$$

is a p-adic unit, hence $\varphi(T'_p) = T'_p$ and $T_{(p)} \cap V'_p = T'_p$, so that

$$T_{(p)} = T'_p \oplus T''_p.$$

(B) *Full faithfulness.* Let A and B be two abelian varieties over $\mathbb{F}_q$, and let ψ be the map

$$\psi : \mathrm{Hom}(A, B) \longrightarrow \mathrm{Hom}_F(T(A), T(B)).$$

By Tate's theorem [7] and (3.1), the map

$$\psi_\ell : \mathrm{Hom}(A, B) \otimes \mathbb{Z}_\ell \longrightarrow \mathrm{Hom}_F(T(A), T(B)) \otimes \mathbb{Z}_\ell$$

is an isomorphism for $(\ell, p) = 1$, so that $\psi \otimes \mathbb{Q}$ is an isomorphism. It is known that $\mathrm{Hom}(A, B)$ is torsion-free, hence ψ is injective. Let $u : A \to B$ be a morphism such that $T(u)$ is divisible by n. The induced morphism

$$u_{\mathbb{C}} : A_{\mathbb{C}} \longrightarrow B_{\mathbb{C}}$$

is then divisible by n, hence so is $\widetilde{u} : \widetilde{A} \to \widetilde{B}$ at the generic point of W. The kernel of multiplication by n is flat over W; thus u vanishes on this kernel, $\widetilde{u}$ and u are divisible by n, and ψ is bijective.

(C) *Necessity.* That $(T(A), F)$ satisfies (a) follows from Weil; condition (d), which implies (b) and (c), follows from 6.

(D) *Isogenies.* Let (T_0, F) satisfy (a) (d) and let T be a lattice in $T_0 \otimes \mathbb{Q}$, stable under F, still satisfying (d). Suppose that (T_0, F) is the image of an abelian variety A over $\mathbb{F}_q$ and let us prove that (T, F) comes from an isogenous abelian variety. Replacing T by $\frac{1}{k}T$, which is isomorphic to it, one may suppose that $T \supset T_0$. Condition (d) implies that T satisfies (4.1), and T defines a subgroup H of $\overline{A}$, which is defined over $\mathbb{F}_q$, and

$$(T, F) = T(A/H).$$

(E) *Surjectivity.* The functor T induces a functor $T_{\mathbb{Q}}$ from the category of ordinary abelian varieties up to isogeny over $\mathbb{F}_q$ to the category of finite-rank $\mathbb{Q}$-vector spaces endowed with an automorphism F satisfying (a) (b). By (D), it is enough to prove that this functor $T_{\mathbb{Q}}$ is essentially surjective. It is even enough to show that every simple object (V, F) of the image category is attained. By Honda [1] (see also [6]), there exists an abelian variety A over $\mathbb{F}_q$ such that the characteristic polynomial of the Frobenius F_A of A is a power of that of F. The third characterization in 1. of ordinary abelian varieties shows that A is ordinary. Moreover, $(T(A) \otimes \mathbb{Q}, F)$ is a sum of copies of (V, F), hence, by (B), the abelian variety up to isogeny $A \otimes \mathbb{Q}$ is a sum of copies of an abelian variety B satisfying

$$T(B) \otimes \mathbb{Q} = (V, F).$$

8. Let (T, F) be a pair satisfying the hypotheses of the theorem, let $2g$ be the rank of T, let A be the corresponding abelian variety over $\mathbb{F}_q$, and let $A_{\mathbb{C}}$ be the complex abelian variety deduced from it (3.). One has

$$T = H_1(A_{\mathbb{C}}),$$

so that $T \otimes \mathbb{R}$ is identified with the Lie algebra of $A_{\mathbb{C}}$ and, as such, is endowed with a complex structure. Here is, according to J.-P. Serre, how to reconstruct this complex structure in terms of T, F, and the restriction of φ to $W_0(\overline{\mathbb{F}_p})$.

Proposition. The complex structure defined above on $T \otimes \mathbb{R}$ is characterized by the following properties:

(I) The endomorphism F is $\mathbb{C}$-linear.

(II) If v is the valuation of the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ in $\mathbb{C}$ extending the valuation of $W_0(\overline{\mathbb{F}_p})$, the valuations of the g eigenvalues of this endomorphism are strictly positive.

Condition (I) is evident, and condition (II) follows from the fact that the action of F on the Lie algebra of A is congruent to zero modulo p. The uniqueness of the complex structure satisfying (I) and (II) follows easily from condition (b) satisfied by (T, F).

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THE IRREDUCIBILITY OF THE SPACE OF CURVES OF GIVEN GENUS

by P. DELIGNE and D. MUMFORD[1]

Fix an algebraically closed field k. Let M_g be the moduli space of curves of genus g over k. The main result of this note is that M_g is irreducible for every k. Of course, whether or not M_g is irreducible depends only on the characteristic of k. When the characteristic is 0, we can assume that $k = \mathbb{C}$, and then the result is classical. A simple proof appears in Enriques–Chisini [E, vol. 3, chap. 3], based on analyzing the totality of coverings of $\mathbf{P}^1$ of degree n, with a fixed number d of ordinary branch points. This method has been extended to char. p by William Fulton [F], using specializations from char. 0 to char. p provided that $p > 2g + 1$. Unfortunately, attempts to extend this method to all p seem to get stuck on difficult questions of wild ramification. Nowadays, the Teichmüller theory gives a thoroughly analytic but very profound insight into this irreducibility when $k = \mathbb{C}$. Our approach however is closest to Severi's incomplete proof ([Se], Anhang F; the error is on pp. 344–345 and seems to be quite basic) and follows a suggestion of Grothendieck for using the result in char. 0 to deduce the result in char. p. The basis of both Severi's and Grothendieck's ideas is to construct families of curves X, some singular, with $p_a(X) = g$, over non-singular parameter spaces, which in some sense contain enough singular curves to link together any two components that M_g might have.

The essential thing that makes this method work now is a recent "stable reduction theorem" for abelian varieties. This result was first proved independently in char. 0 by Grothendieck, using methods of etale cohomology (private correspondence with J. Tate), and by Mumford, applying the easy half of Theorem (2.5), to go from curves to abelian varieties (cf. [M_2]). Grothendieck has recently strengthened his method so that it applies in all characteristics (SGA 7, 1968). Mumford has also given a proof using theta functions in char. $\neq 2$. The result is this:

Stable Reduction Theorem. — Let R be a discrete valuation ring with quotient field K. Let A be an abelian variety over K. Then there exists a finite algebraic extension L of K such that, if $R_L =$ integral closure of R in L, and if $\mathcal{A}_L$ is the Néron model of $A \times_K L$ over R_L, then the closed fibre $\mathcal{A}_{L,s}$ of $\mathcal{A}_L$ has no unipotent radical.

We shall give two related proofs of our main result. One of these is quite elementary, and follows by quite standard techniques once the Stable Reduction Theorem for abelian varieties is applied, in § 2, to deduce an analogous stable reduction theorem for curves. The other proof is more powerful, and is based on the use of a larger category than the category of schemes, and on proving for the objects of this category many of the standard theorems for schemes, especially the Enriques–Zariski connectedness theorem (EGA 3, (4.3)). Unfortunately, this larger category is not quite a category — it is a simple type of 2-category; in fact, if X, Y are objects, then $\mathrm{Hom}(X, Y)$ is itself a category, but one in which all morphisms are isomorphisms. The objects of this 2-category we call algebraic stacks[2]. The moduli space M_g is just the "underlying coarse variety" of a more fundamental object, the moduli stack $\mathcal{M}_g$, studied in [M_3]. Full details on the basic properties and theorems for algebraic stacks will be given elsewhere. In this paper, we will only give definitions and state without proof the general theorems which we apply. Using the method of algebraic stacks, we can prove not only the irreducibility of M_g itself, but of all higher level moduli spaces of curves too (cf. § 5 below).

§ 1. Stable curves and their moduli.

The key definition of the whole paper is this:

Definition (1.1). — Let S be any scheme. Let $g \geq 2$. A stable curve of genus g over S is a proper flat morphism $\pi : C \to S$ whose geometric fibres are reduced, connected, 1-dimensional schemes C_s such that:

(i) *C_s has only ordinary double points;*

(ii) *if E is a non-singular rational component of C_s, then E meets the other components of C_s in more than 2 points;*

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[2] A slightly less general category of objects, called algebraic spaces, has been introduced and studied very deeply by M. Artin [A_1] and D. Knutson [K]. The idea of enlarging the category of varieties for the study of moduli spaces is due originally, we believe, to A. Weil.

(iii) $\dim H^1(\mathcal{O}_{C_s}) = g$.

We will study in this section three aspects of the theory of stable curves: their pluri-canonical linear systems, their deformations, and their automorphisms.

Suppose $\pi : C \to S$ is a stable curve. Since π is flat and its geometric fibres are local complete intersections, the morphism π is locally a complete intersection (i.e., locally, C is isomorphic as S-scheme to $V(f_1, \ldots, f_{n-1}) \subset \mathbf{A}^n \times U$, where $U \subset S$ is open, and $f_1, \ldots, f_{n-1} \in \Gamma(\mathcal{O}_{\mathbf{A}^n \times U})$ are a regular sequence). Therefore, by the theory of duality of coherent sheaves [H], there is a canonical invertible sheaf $\omega_{C/S}$ on C — the unique non-zero cohomology group of the complex of sheaves $f^!(\mathcal{O}_S)$. We need to know the following facts about $\omega_{C/S}$:

a) for all morphisms $f : T \to S$, $\omega_{C \times_S T/T}$ is canonically isomorphic to $f^*(\omega_{C/S})$;

b) if $S = \operatorname{Spec}(k)$, k algebraically closed, let $f : C' \to C$ be the normalization of C, $x_1, \ldots, x_n, y_1, \ldots, y_n$ the points of C' such that the $z_i = f(x_i) = f(y_i)$, $1 \le i \le n$, are the double points of C. Then $\omega_{C/S}$ is the sheaf of 1-forms η on C' regular except for simple poles at the x's and y's and with $\operatorname{Res}_{x_i}(\eta) + \operatorname{Res}_{y_i}(\eta) = 0$;

c) if $S = \operatorname{Spec}(k)$, and $\mathcal{F}$ is a coherent sheaf on C, then

$$\operatorname{Hom}(H^1(C, \mathcal{F}), k) \cong \operatorname{Hom}_{\mathcal{O}_C}(\mathcal{F}, \omega_{C/S}).$$

Theorem (1.2). — If $g \ge 2$ and C is a stable curve of genus g over an algebraically closed field k, then $H^1(C, \omega_{C/k}^{\otimes n}) = (0)$ if $n \ge 2$, and $\omega_{C/k}^{\otimes n}$ is very ample if $n \ge 3$.

Proof. — Since C is stable, of genus $g \ge 2$, every irreducible component E of C either 1) has (arithmetic) genus ≥ 2 itself, 2) has genus 1, but meets other components of C in at least one point, or 3) is non-singular, rational and meets other components of C in at least three points. But by b) above, $\omega_{C/k} \otimes \mathcal{O}_E$ is isomorphic to $\omega_{E/k}(\sum Q_i)$, where $\{Q_i\}$ are the points where E meets the rest of C. Since the degree of $\omega_{E/k}$ is $2g_E - 2$, it follows that in any of the cases 1, 2 or 3, $\omega_{C/k} \otimes \mathcal{O}_E$ has positive degree. This shows immediately that $\omega_{C/k}$ is ample on each component E of C, hence is ample.

Next, by c) above, $H^1(\omega_{C/k}^{\otimes n})$ is dual to $H^0(\omega_{C/k}^{\otimes(1-n)})$. Since $\omega_{C/k} \otimes \mathcal{O}_E$ has positive degree, $\omega_{C/k}^{\otimes(1-n)} \otimes \mathcal{O}_E$ has no sections for any E, any $n \ge 2$; therefore $H^0(\omega_{C/k}^{\otimes(1-n)}) = (0)$ if $n \ge 2$, and so $H^1(\omega_{C/k}^{\otimes n}) = (0)$ if $n \ge 2$.

To prove that an invertible sheaf $\mathcal{L}$ on any scheme C, proper over k, is very ample, it suffices to show

a) for all closed points $x \ne y$
$$H^0(C, \mathcal{L}) \to H^0(C, \mathcal{L} \otimes k(x)) \oplus H^0(C, \mathcal{L} \otimes k(y))$$
is surjective,

b) for all closed points x,
$$H^0(C, \mathcal{L}) \to H^0(C, \mathcal{L} \otimes \mathcal{O}_x/\mathfrak{m}_x^2)$$
is surjective.

Using the exact sequence of cohomology, these both follow if $H^1(C, \mathfrak{m}_x \cdot \mathfrak{m}_y \cdot \mathcal{L}) = (0)$ for all closed points $x, y \in C$. In our case, $\mathcal{L} = \omega_C^{\otimes n}$, $n \ge 3$, so if we use duality, we must show:

$$(*) \qquad \operatorname{Hom}(\mathfrak{m}_x \cdot \mathfrak{m}_y, \omega_C^{\otimes -n}) = (0), \qquad \text{if } n \ge 2.$$

If x is a non-singular point, $\mathfrak{m}_x$ is an invertible sheaf. If x is a double point, let $\pi : C' \to C$ be the result of blowing up x, and let $x_1, x_2 \in C'$ be the two points in $\pi^{-1}(x)$. Then it is easy to check that for any invertible sheaf $\mathcal{L}$ on C:

$$\operatorname{Hom}(\mathfrak{m}_x, \mathcal{L}) \cong H^0(C', \pi^*\mathcal{L}),$$
$$\operatorname{Hom}(\mathfrak{m}_x^2, \mathcal{L}) \cong H^0(C', \pi^*\mathcal{L}(x_1 + x_2)).$$

Therefore, we have 3 cases of (*) to check:

Case 1. — x, y non-singular points of C, then

$$H^0(\omega_C^{-n}(x + y)) = (0), \qquad \text{if } n \ge 2.$$

Case 2. — x double point of C, $\pi : C' \to C$ blowing up x, $\{x_1, x_2\} = \pi^{-1}(x)$, and y a non-singular point of C. Then

$$H^0(\pi^*(\omega_C)^{-n}(y)) = (0), \qquad H^0(\pi^*(\omega_C)^{-1}(x_1 + x_2)) = (0), \qquad \text{if } n \ge 2.$$

Case 3. — x, y double points of C, $\pi : C' \to C$ blowing up x and y. Then

$$H^0(\pi^*(\omega_C)^{-n}) = (0) \qquad \text{if } n \geq 2.$$

Now since the degree of ω_C^{-n} $(n \geq 2)$ on all components E of C is less than or equal to -2, all of this is clear, except in those cases where $n = 2$, the degree of ω_C on some E is 1, and in which two poles are allowed on E. This occurs if:

(i) case 1, $p_a(E) = 1$, E meets $C - E$ at only one point, $x, y \in E$.

(ii) case 1, $p_a(E) = 0$, E meets $C - E$ at only three points, $x, y \in E$.

(iii) case 2, E a rational curve with one double point meeting $C - E$ at one point, $x =$ double point of E.

But in all these cases, C has components besides E and a section in the H^0 in question must definitely vanish on all these other components. So at the points where E meets $C - E$, the section has extra zeroes. Since the sheaf in question has degree 0 on E, the section is zero on E too. Q.E.D.

Corollary. — Let $\pi : C \to S$ be any stable curve of genus $g \geq 2$. Then $\omega_{C/S}^{\otimes n}$ is relatively very ample if $n \geq 3$ and $\pi_*(\omega_{C/S}^{\otimes n})$ is a locally free sheaf on S of rank $(2n-1)(g-1)$.

Proof. — In fact, since for all $s \in S$, $H^1(\omega_{C/S}^{\otimes n} \otimes \mathcal{O}_{C_s}) = (0)$, it follows from [EGA, chap. 3, § 7], that $\pi_*(\omega_{C/S}^{\otimes n})$ is locally free and that $\pi_*(\omega_{C/S}^{\otimes n}) \otimes k(s) \cong H^0(\omega_{C/S}^{\otimes n} \otimes \mathcal{O}_{C_s})$. Therefore the corollary follows. Q.E.D.

Taking $n = 3$, it follows that every stable curve C/S can be realized as a family of curves in $\mathbf{P}^{5g-6}$ with Hilbert polynomial:

$$P_g(n) = (6n-1)(g-1).$$

Following standard arguments ([M$_1$], p. 99), it is easy to prove that there is a subscheme

$$H_g \subset \mathrm{Hilb}^{P_g}_{\mathbf{P}^{5g-6}}$$

of "all" tri-canonically embedded stable curves. (Hilb is the Hilbert scheme over $\mathbb{Z}$.) To be precise, there is an isomorphism of functors:

$$\mathrm{Hom}(S, H_g) \cong \left\{ \begin{array}{l} \text{set of stable curves } \pi : C \to S, \text{ plus isomorphisms:} \\ \mathbf{P}(\pi_*(\omega_{C/S}^{\otimes 3})) \cong \mathbf{P}^{5g-6} \times S\,, \quad \text{(modulo isomorphism).} \end{array} \right.$$

We will denote by $Z_g \subset H_g \times \mathbf{P}^{5g-6}$ the universal tri-canonically embedded stable curve. The functor of stable curves itself is the sheafification of the quotient of functors:

$$H_g / \mathrm{PGL}(5g-6).$$

We now consider the deformation theory of stable curves. Let k be any ground field. The deformation theory of X's smooth over k can be found in [SGA, 60–61]; for singular S's, the theory has been worked out in [Sc]. We shall indicate here the results of this theory for a scheme X which is

(i) one-dimensional;

(ii) generically smooth over k;

(iii) locally a complete intersection.

The advantages of this case are two-fold: first, the "cotangent complex" of Grothendieck, Lichtenbaum and Schlessinger reduces, in view of (ii) and (iii), to the single coherent sheaf $\Omega_{X/k}$, the Kähler differentials. Secondly, we have:

Lemma (1.3). — $\mathrm{Ext}^2(\Omega_{X/k}, \mathcal{O}_X) = (0)$.

Proof. — Use the spectral sequence:

$$H^p(X, \mathcal{E}xt^q(\Omega_{X/k}, \mathcal{O}_X)) \Rightarrow \mathrm{Ext}^{p+q}(\Omega_{X/k}, \mathcal{O}_X).$$

Then (i) $H^2(X, \mathcal{E}xt^0) = (0)$ since $\dim X = 1$. (ii) Since $\Omega_{X/k}$ is locally free except at a finite number of points, $\mathcal{E}xt^1(\Omega_{X/k}, \mathcal{O}_X)$ has 0-dimensional support, hence $H^1(X, \mathcal{E}xt^1) = (0)$. (iii) Locally, if we embed $X \subset \mathbf{A}^n$, then $\Omega_{X/k}$ has a free resolution of length 2:

$$0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_{\mathbf{A}^n} \otimes \mathcal{O}_X \to \Omega_{X/k} \to 0$$

where $\mathcal{I}$ = sheaf of ideals defining X. Therefore $\mathcal{E}xt^2 = (0)$. Q.E.D.

In Schlessinger's theory, the significance of Lemma (1.3) is that all obstructions vanish, i.e., deformations of X over base schemes $\mathrm{Spec}(A/\mathcal{J})$ (A = local Artin ring with residue field k) can always be embedded in deformations over $\mathrm{Spec}(A)$. Moreover, the theory says that there is a canonical one-one correspondence between $\mathrm{Ext}^1(\Omega_{X/k}, \mathcal{O}_X)$ and the first order deformations of X, i.e., proper, flat morphisms p, and isomorphisms α as follows:

$$\begin{array}{ccccc}
X_1 & \supset & X_1 \times_{\mathrm{Spec}\, k[\varepsilon]} \mathrm{Spec}\, k & \xrightarrow{\sim_\alpha} & X \\
\downarrow p & & \downarrow & & \downarrow \\
\mathrm{Spec}\, k[\varepsilon]/\varepsilon^2 & \supset & \mathrm{Spec}\, k & = & \mathrm{Spec}\, k.
\end{array}$$

Since the obstructions vanish, there is a versal formal deformation $\mathfrak{X}$ of X over the base scheme

$$\mathfrak{M} = \mathrm{Spec}\, o_k[[t_1, \ldots, t_N]],$$

where $o_k = k$ if the char. is 0, or o_k is the complete regular local ring, with maximal ideal $p \cdot o_k$ and residue field k, if $\mathrm{char}(k) = p$, unique (by Cohen's structure theorem), and

$$N = \dim_k \mathrm{Ext}^1(\Omega_X, \mathcal{O}_C).$$

This means that $\mathfrak{X}$ is a formal scheme, proper and flat over $\mathcal{M}$, with fibre X over $\mathrm{Spec}(k)$ and the two properties:

a) Every deformation of X is induced from $\mathfrak{X}$, i.e., if A is a local Artin o_k-algebra with residue field k, and $p : Y \to \mathrm{Spec}(A)$ is proper and flat with fibre X over $\mathrm{Spec}(k)$, then there is a commutative diagram:

$$\begin{tikzcd}
Y \simeq \mathfrak{X} \times_{\mathcal{M}} \mathrm{Spec}(A) \arrow[rr] \arrow[dd] & & \mathfrak{X} \arrow[dd] \\
& X \arrow[ul] \arrow[ur] \arrow[dd] & \\
\mathrm{Spec}(A) \arrow[rr] & & \mathcal{M} \\
& \mathrm{Spec}(k) \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

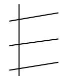
b) If $A = k[\varepsilon]/(\varepsilon^2)$, the above morphism f is uniquely determined by the diagram.

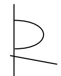
This implies that the tangent space to $\mathcal{M} \times_{o_k} k$ at its closed point is canonically isomorphic to $\mathrm{Ext}^1(\Omega_{X/k}, \mathcal{O}_X)$.

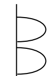
In case $\mathrm{Ext}^0(\Omega_{X/k}, \mathcal{O}_X) = (0)$, $\mathfrak{X}/\mathcal{M}$ is, in fact, *universal*: i.e., in property a), f is always unique, which means that the functor represented by $\mathcal{M}$ in the category of artin, local o_k-algebras is isomorphic to the functor of deformations Y/A of X. This fortunately holds for stable curves:

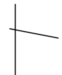
Lemma (1.4). — If X is a stable curve, $\mathrm{Ext}^0(\Omega_{X/k}, \mathcal{O}_X) = (0)$.

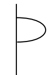
Proof. — We may assume that k is algebraically closed. Now a homomorphism from Ω_X to $\mathcal{O}_X$ is given by an everywhere regular vector field D on X. Such a vector field is given, in turn, by a regular vector field D' on the normalization X' of X which vanishes at all points of X' lying over the double points of X. In particular, D' and hence D vanishes identically on all components E of X whose normalization E' has genus ≥ 2. There remain the following possibilities for E:


E non-singular rational


E' rational, E one double pt.


E' rational, $E \geq 2$ double pts.


E non-singular elliptic


E' elliptic, $E \geq 1$ double points

In all cases where E' is rational, note that D' has to have at least 3 zeroes; and where E' is elliptic, D' has to have at least one zero. So D' vanishes on all components E'. This proves that $\mathrm{Ext}^0(\Omega_{X/k}, \mathcal{O}_X) = (0)$. Q.E.D.

Schlessinger's theory also allows us to trace what happens to the singularities of X in this deformation $\mathcal{M}$. For each closed point $x \in X$, he studies deformations of the complete local ring $\widehat{\mathcal{O}}_{x,X}$ alone, i.e., flat A-algebras $\mathcal{O}$ plus isomorphisms:

$$\mathcal{O} \otimes_A k \simeq \widehat{\mathcal{O}}_{x,X}.$$

This is a functor of A exactly as before. Since $\mathrm{Ext}^2_{\widehat{\mathcal{O}}_x}(\widehat{\Omega}_{x/k}, \widehat{\mathcal{O}}_x) = (0)$, there are no obstructions to extending deformations. First order deformations with $A = k[\varepsilon]/(\varepsilon^2)$ are classified by $\mathrm{Ext}^1_{\widehat{\mathcal{O}}_x}(\widehat{\Omega}_{x/k}, \widehat{\mathcal{O}}_x)$. And there is a versal formal deformation $\widehat{\mathcal{O}}$, which is a complete local ring and a flat $o_k[[t_1, \ldots, t_N]]$-algebra, $N = \dim_k \mathrm{Ext}^1_{\widehat{\mathcal{O}}_x}(\widehat{\Omega}_{x/k}, \widehat{\mathcal{O}}_x)$, such that

$$\widehat{\mathcal{O}}/(p, t_1, \ldots, t_N) = \widehat{\mathcal{O}}_{x,X}.$$

Whenever X is smooth over k at x, $N = 0$, and the theory is uninteresting. The first non-trivial example is a k-rational ordinary double point with k-rational tangent lines:

$$\widehat{\mathcal{O}}_{x,X} \simeq k[[u, v]]/(u \cdot v).$$

Then $N = 1$, and

$$\widehat{\mathcal{O}} \simeq o_k[[u, v, t_1]]/(uv - t_1).$$

In other words, if $\mathcal{O}$ is any deformation of $\widehat{\mathcal{O}}_{x,X}$ and u, v are lifted suitably into $\mathcal{O}$, then $u \cdot v = w \in A$ and $\mathcal{O}$ is induced from $\widehat{\mathcal{O}}$ via the homomorphism $o_k[[t_1]] \to A$ taking t_1 to w. This is easy to prove.

Finally, Schlessinger's theory connects global deformations to local ones. Let $\mathfrak{X}/\mathcal{M}_{gl}$ with $\mathcal{M}_{gl} = \mathrm{Spec}(A)$ be the versal global deformation, let $x_1, \ldots, x_k \in X$ be the points where X is not smooth over k, and let $\widehat{\mathcal{O}}_i$ as A_i-algebra be the versal deformation of the local ring $\widehat{\mathcal{O}}_{x_i,X}$. Let $\mathcal{M}_{lo} = \mathrm{Spec}(A_1 \widehat{\otimes}_{o_k} \cdots \widehat{\otimes}_{o_k} A_k)$. Then we may consider the local rings

$$\widehat{\mathcal{O}}_{x_i,\mathfrak{X}}$$

of $\mathfrak{X}$ at x_i. There are o_k-homomorphisms $\varphi_i : A_i \to A$ such that $\widehat{\mathcal{O}}_{x_i,\mathfrak{X}} \simeq \widehat{\mathcal{O}}_i \otimes_{A_i} A$. Dualizing, we obtain a morphism

$$\Phi = \prod_i \mathrm{Spec}(\varphi_i) : \mathcal{M}_{gl} \to \mathcal{M}_{lo}$$

which describes exactly how the various singularities of X behave in the versal deformation $\mathfrak{X}$. The final fact that we need is:

Proposition (1.5). — $\Phi : \mathcal{M}_{gl} \to \mathcal{M}_{lo}$ is formally smooth, i.e., there are isomorphisms

$$\mathcal{M}_{gl} \simeq \mathrm{Spec}\, o_k[[t_1, \ldots, t_{N+M}]], \qquad \mathcal{M}_{lo} \simeq \mathrm{Spec}\, o_k[[t_1, \ldots, t_N]]$$

such that $\Phi^*(t_i) = t_i$, $1 \leq i \leq N$.

Proof. — In view of the functorial significance of $\mathcal{M}_{gl}$ and $\mathcal{M}_{lo}$, this follows if we prove that the natural map:

$$(*) \qquad \mathrm{Ext}^1_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X) \longrightarrow \prod_{i=1}^{k} \mathrm{Ext}^1_{\widehat{\mathcal{O}}_{x_i}}(\widehat{\Omega}_{x_i}, \widehat{\mathcal{O}}_{x_i})$$

is surjective; $(*)$ is the induced map $d\Phi$ on the tangent spaces to $\mathcal{M}_{gl}$ and $\mathcal{M}_{lo}$. Since Ω_X is an invertible $\mathcal{O}_X$-Module outside the x_i's, it follows that:

$$\begin{aligned}\prod_{i=1}^{k} \mathrm{Ext}^1_{\widehat{\mathcal{O}}_{x_i}}(\widehat{\Omega}_{x_i}, \widehat{\mathcal{O}}_{x_i}) &\simeq \prod_{i=1}^{k} \mathrm{Ext}^1_{\mathcal{O}_{x_i}}(\Omega_{x_i}, \mathcal{O}_{x_i}) \\ &\simeq H^0\big(X, \mathcal{E}xt^1_{\mathcal{O}_X}(\Omega_X, \mathcal{O}_X)\big).\end{aligned}$$

Therefore, $(*)$ is surjective by the spectral sequence used in Lemma (1.3) and the fact that $H^2(X, \mathcal{E}xt^0(\Omega_X, \mathcal{O}_X)) = (0)$. Q.E.D.

In particular, suppose C is a stable curve over an algebraically closed ground field k, and let $x_1, \ldots, x_k$ be the double points of C. Let $N = \dim \mathrm{Ext}^1(\Omega_X, \mathcal{O}_X)$. Then C has a universal formal deformation $\mathcal{C}/\mathcal{M}$ where $\mathcal{M} = \mathrm{Spec}\, o_k[[t_1, \ldots, t_N]]$. Note that since in this case, the invertible sheaf $\omega_{\mathcal{C}/\mathcal{M}}$ is relatively ample, $\mathcal{C}$ is not only a formal scheme over $\mathcal{M}$, but also the formal completion of a unique scheme proper and flat over $\mathcal{M}$, which we will also denote by $\mathcal{C}$. $\mathcal{C}$ is clearly a stable curve over $\mathcal{M}$. Now each double point x_i has one modulus (cf. our example above) so the versal deformation space of the rings $\mathcal{O}_{x_i,C}$ is $o_k[[t'_1, \ldots, t'_k]]$. By Proposition (1.5), we may identify t_i with t'_i, and we conclude that for suitable u_i, v_i:

$$\widehat{\mathcal{O}}_{x_i,\mathcal{C}} \simeq o_k[[u_i, v_i, t_1, \ldots, t_N]]/(u_i v_i - t_i).$$

In particular, $t_i = 0$ is the locus in $\mathcal{M}$ where "x_i remains a double point".

The relation between the formal moduli space $\mathcal{M}$ of C and the local structure of H_g at a point x with $\kappa(x) = k$ corresponding to some tri-canonical model of C is exactly the same as in the case of non-singular curves ([M$_1$], chap. 5, § 2). Let $\widehat{\mathcal{O}}_x$ be the completion of the local ring $\mathcal{O}_{x,H_g}$, and let $T = \mathrm{Spec}(\widehat{\mathcal{O}}_x)$. Let x denote the closed point of T too. The universal family of stable curves $Z_g \subset H_g \times \mathbf{P}^{5g-6}$ induces a family $Z' \subset T \times \mathbf{P}^{5g-6}$, whose fibre Z'_x over x is C. Then there is a unique morphism $f : T \to \mathcal{M}$ such that $Z' \simeq \mathcal{C} \times_{\mathcal{M}} T$, with this isomorphism restricting to the identity on the fibres over x, both of which are C. I claim that via f, T is formally smooth over $\mathcal{M}$, i.e., $\widehat{\mathcal{O}}_x \simeq o_k[[t_1, \ldots, t_N, t_{N+1}, \ldots, t_m]]$. In fact, by choosing an isomorphism $\mathbf{P}(\pi_*(\omega_{\mathcal{C}/\mathcal{M}}^{\otimes 3})) \simeq \mathbf{P}^{5g-6} \times \mathcal{M}$, we obtain a tri-canonical embedding $\mathcal{C} \subset \mathbf{P}^{5g-6} \times \mathcal{M}$ of $\mathcal{C}$, hence a morphism $s : \mathcal{M} \to H_g$ such that $\mathcal{C}$, with this embedding, is the pull-back of Z_g. Then s factors through T and $s : \mathcal{M} \to T$ is a section of f. On the other hand, consider the action of $\mathrm{PGL}(5g-6)$ on H_g. Let S_x be the stabilizer of the k-valued point x. Then S_x is finite and reduced. Because if it were not, S_x would have a non-trivial tangent space at the origin, i.e., there would be a $k[\varepsilon]/(\varepsilon^2)$-valued point of $\mathrm{PGL}(5g-6)$ centered at the identity, which maps the embedded stable curve $C \subset \mathbf{P}^{5g-6}$ corresponding to x into itself. But this action is given by an everywhere regular derivation on C, and we have

seen that all such vanish. This means that this $k[\varepsilon]/(\varepsilon^2)$-valued automorphism is the identity at all points of C and, since C is connected and spans $\mathbf{P}^{5g-6}$, the automorphism is the identity everywhere. Thus S_x is finite and reduced. It follows that the action of $\mathrm{PGL}(5g-6)$ on T is formally free, and hence that T is formally a principal fibre bundle over $\mathcal{M}$ with group $\mathrm{PGL}(5g-6)$. Therefore T is formally smooth over $\mathcal{M}$ as required.

Putting this together with what we know about $\mathcal{M}$, we conclude the following:

Let k be any algebraically closed field,
let $H'_g = H_g \times \mathrm{Spec}(o_k)$, $Z'_g = Z_g \times \mathrm{Spec}(o_k)$,
let $x \in H'_g$ be a closed point,
let $C \subset Z'_g$ be the stable curve over x,
let $x_1, \ldots, x_k \in C$ be its double points.

Then

Theorem (1.6). — There are isomorphisms

$$\widehat{\mathcal{O}}_{x,H'_g} \simeq o_k[[t_1, \ldots, t_N]],$$

$$\widehat{\mathcal{O}}_{x_i,Z'_g} \simeq o_k[[u_i, v_i, t_1, \ldots, t_N]]/(u_i v_i - t_i).$$

Corollary (1.7). — H_g is smooth over $\mathbb{Z}$. In particular, for all algebraically closed fields k, $H_g \times \mathrm{Spec}(k)$ is a disjoint union of a finite number of non-singular algebraic varieties over k.

Let

$$H_g^0 = \{x \in H_g \mid \text{the corresponding stable curve } (Z_g)_x \text{ is non-singular}\},$$

$$S = \{x \in Z_g \mid \text{the projection } \pi : Z_g \to H_g \text{ is not smooth at } x\}.$$

Definition (1.8). — Let $p : X \to Y$ be a smooth morphism of finite type, with Y a noetherian scheme, and let $D \subset X$ be a relative Cartier divisor. Then D has normal crossings relative to Y if for all $x \in D$, the local equation $d = 0$ of D decomposes in the strict completion[3] $\widehat{\mathcal{O}}_{x,X}$ of $\mathcal{O}_{x,X}$ as $d = d_1 \cdots d_k$, where $d_1, \ldots, d_k$ are linearly independent in

$$\widehat{\mathfrak{m}}_{x,X}/\big(\widehat{\mathfrak{m}}_{x,X}^2 + \mathfrak{m}_{y,Y}\widehat{\mathcal{O}}_{x,X}\big),$$

with $y = p(x)$.

[3]The complete local ring, formally etale over $\mathcal{O}_{x,X}$ with residue field the separable closure of $\mathcal{O}_{x,X}/\mathfrak{m}_{x,X}$.

Corollary (1.9). — $H_g^0 = H_g - S^*$*, where* S^* *is a divisor with normal crossings relative to* $\mathbb{Z}$*.* Z_g *and* S *are smooth over* $\mathbb{Z}$*, and the projection* $p : S \to S^*$ *is finite and an isomorphism at all points where* S^* *is smooth over* $\mathbb{Z}$*, i.e.,* S *is the normalization of* S^**.*

Proof. — In the notation of Theorem (1.6), S^* is defined in $\widehat{\mathcal{O}}_{x,H'_g}$ by the local equation $t_1 \cdots t_k = 0$. And

$$\widehat{\mathcal{O}}_{x_i,Z'_g} \simeq o_k[[u_i, v_i, t_1, \ldots, t_{i-1}, t_{i+1}, \ldots, t_N]],$$

$$\widehat{\mathcal{O}}_{x_i,S} \simeq \widehat{\mathcal{O}}_{x_i,Z'_g}/(u_i, v_i) \simeq o_k[[t_1, \ldots, t_{i-1}, t_{i+1}, \ldots, t_N]].$$

Q.E.D.

Next we take up the isomorphisms and automorphisms of stable curves. Suppose $p : X \to S$, $q : Y \to S$ are two stable curves:

Definition (1.10). — $\underline{\mathrm{Isom}}_S(X, Y)$ *is the functor on* (Sch/S) *associating to each* S*-scheme* S' *the set of* S'*-isomorphisms between* $X \times_S S'$ *and* $Y \times_S S'$*. If* $X = Y$*, we denote* $\underline{\mathrm{Isom}}_S(X, X)$ *by* $\underline{\mathrm{Aut}}_S(X)$*.*

Since both X and Y have the canonical polarizations $\omega_{X/S}$, $\omega_{Y/S}$ respectively, any isomorphism $f : X \to Y$ must satisfy $f^*(\omega_{Y/S}) \simeq \omega_{X/S}$. Therefore, by Grothendieck's results on the representability of the Hilbert scheme and related functors [Gr$_1$], we conclude that $\underline{\mathrm{Isom}}_S(X, Y)$ is represented by a scheme $\mathrm{Isom}_S(X, Y)$, quasi-projective over S. Concerning this scheme, we have:

Theorem (1.11). — $\mathrm{Isom}_S(X, Y)$ *is finite and unramified over* S*.*

Proof. — To check that $\mathrm{Isom}_S(X, Y)$ is unramified, we may take S to be the spectrum of an algebraically closed field k, in which case $\mathrm{Isom}_S(X, Y)$ is either empty or isomorphic to $\mathrm{Aut}_k(X)$. A point of $\mathrm{Aut}_k(X)$ with values in $k[\varepsilon]/(\varepsilon^2)$ with image the identity may be identified with a vector field on X. By Lemma (1.4), stable curves have no non-zero vector fields. This proves that $\mathrm{Isom}_S(X, Y)$ is unramified over S, and since it is also of finite type over S, it is quasi-finite over S. It remains to check that $\mathrm{Isom}_S(X, Y)$ is proper over S.

Locally over S, X and Y are the pull-backs of the universal tri-canonically embedded stable curve by some morphisms from S to H_g, so that it suffices to prove the properness of $\mathrm{Isom}_S(X, Y)$ in the "universal" case where $S = H_g \times H_g$, X and Y being the two inverse images of the universal curve on H_g. In that case, the open subset of $\mathrm{Isom}_S(X, Y)$ corresponding to smooth curves is dense, so that the Theorem follows from the valuative criterion of properness which holds by:

Lemma (1.12). — *Let* X *and* Y *be two stable curves over a discrete valuation ring* R*, with algebraically closed residue field. Denote by* η *and* s *the generic and closed points of* $\mathrm{Spec}(R)$*, and assume that the generic fibres* X_η *and* Y_η *of* X *and* Y *are smooth. Then any isomorphism* φ_η *between* X_η *and* Y_η *extends to an isomorphism* φ *between* X *and* Y*.*

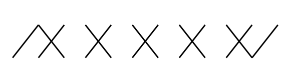
(A posteriori, it follows from Theorem (1.11) that the lemma holds for any valuation ring R and without assuming X_η or Y_η smooth.)

Proof. — Another way to put the lemma is that if we start with a smooth curve X_η of genus $g \geq 2$ over the quotient field K of R, there is, up to canonical isomorphism, at most one stable curve X over R with X_η as its generic fibre. We shall deduce this from the analogous uniqueness assertion for minimal models ([L] and [S]): given a smooth curve X_η of genus $g \geq 1$ over K, there is, up to canonical isomorphism, at most one regular 2-dimensional scheme $\widetilde{X}$, proper and flat over R, with X_η as its generic fibre, without exceptional curves of the first kind in $\widetilde{X}_s$.

Let z denote a generator of the maximal ideal of R and consider the affine plane curve C_n over R given by:

$$xy = z^n.$$

Let $\widetilde{C}_n$ denote the scheme obtained by: 1) blowing up the maximal ideal at the unique singularity of C_n; 2) blowing up the maximal ideal at the unique singularity of this scheme..., and so on $\left[\frac{n}{2}\right]$ times. It is easy to check that $\widetilde{C}_n$ is a regular scheme whose special fibre is the same as that of C_n except that the singular point is replaced by a sequence of $n-1$ projective lines as follows:



Now suppose x is a singular point of the stable curve X over R. At x, X is formally isomorphic as scheme over R to one of the schemes C_n, so we may blow up X the same way we blew up C_n. If we do this for all singular points of X, we get a regular scheme $\widetilde{X}$ with generic fibre X_η. In addition, any non-singular rational component of $\widetilde{X}_s$ is linked to the other irreducible components by at least two points, hence it is not exceptional of first kind. Therefore $\widetilde{X}$ is the minimal model of X_η. Note finally that C_n is a normal scheme, hence so is X; therefore X is the unique normal scheme obtained from $\widetilde{X}$ by contracting all non-singular rational components of $\widetilde{X}_s$ linked to the other irreducible components by exactly two points. This proves that X is essentially unique. Q.E.D.

Another important fact about the automorphisms of stable curves is:

Theorem (1.13). — Let k be an algebraically closed field and X a stable curve over k. Let $\mathrm{Pic}^0(X)$ denote the group of invertible sheaves on X of degree 0 on each component. Then the map (of ordinary groups):

$$\mathrm{Aut}_k(X) \longrightarrow \mathrm{Aut}_k(\mathrm{Pic}^0(X))$$

is injective.

Proof. — Let φ be an automorphism of X inducing the identity on $\mathrm{Pic}^0 X$.

Lemma (1.14). — If X is smooth, then φ is the identity.

Proof. — If not, by the Lefschetz-Weil fixed point formula, the number n of fixed points of φ, counted with their multiplicities, is

$$n = 1 - \mathrm{Tr}(\varphi, T_l(\mathrm{Pic}^0(X))) + 1 = 2 - 2g < 0,$$

which is absurd. Q.E.D.

Lemma (1.15). — If X is irreducible, then φ is the identity.

Proof. — Let φ' be the action of φ on the normalization X' of X. Each singular point of X, together with an ordering of its two inverse images in X', defines a distinct morphism from $\mathbb{G}_m$ to $\mathrm{Pic}^0(X)$, so that the inverse image S of the singular locus of X is pointwise fixed by φ'. One has either

a) genus$(X') \geq 2$: then conclude by Lemma (1.14);

b) genus$(X') = 1$, $|S| \geq 2$, then φ' is a translation on X' leaving a point fixed, and so φ' is the identity;

c) X' is the projective line, $|S| \geq 4$ and φ' is a projectivity leaving more than three points fixed, so is the identity. Q.E.D.

Let Γ be the following (unoriented) graph:

(i) The set of vertices of Γ is the set Γ^0 of irreducible components of X,

(ii) the set of edges of Γ is the set Γ^1 of the singular points of X which lie on two distinct irreducible components,

(iii) an edge $x \in \Gamma^1$ has for extremities the irreducible components on which x lies.

Lemma (1.16). — If φ induces the identity on Γ, then φ is the identity.

Proof. — If X_1 is an irreducible component of X, then $\varphi(X_1) = X_1$ and φ leaves fixed the points of intersection of X_1 with the other components. In addition, $\mathrm{Pic}^0(X)$ maps onto $\mathrm{Pic}^0(X_1)$ so that φ acts trivially on $\mathrm{Pic}^0(X_1)$. Either:

a) genus$(X_1) \geq 2$ and $\varphi | X_1$ is the identity by Lemma (1.15);

b) genus$(X_1) = 1$, φ acts by a translation and leaves a point fixed, so is the identity on X_1;

c) X_1 is the projective line and φ leaves fixed at least three points, so is the identity on X_1. Q.E.D.

Lemma (1.17). — (i) Any edge in Γ has distinct extremities.

(ii) Any vertex which is the extremity of 0, 1 or 2 edges is fixed by φ.

(iii) φ acts trivially on $H^1(\Gamma, \mathbb{Z})$.

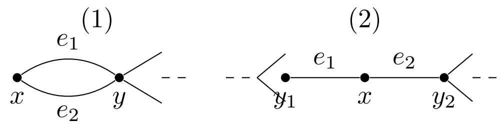
Proof. — It is easy to check that the subgroup of $\mathrm{Pic}^0(X)$ corresponding to invertible sheaves whose restriction to each irreducible component of X is trivial is canonically isomorphic to

$$H^1(\Gamma, \mathbb{Z}) \otimes \mathbb{G}_m.$$

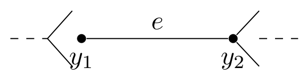
This implies (iii), and (i) is trivial.

The morphism from $\mathrm{Pic}^0(X)$ to the product $\prod_i \mathrm{Pic}^0(X_i)$, extended over the irreducible components of X, is surjective, so that if $\mathrm{Pic}^0(X_i) \neq \{e\}$, then $\varphi(X_i) = X_i$. This is the case, unless X_i is a projective line, linked to the other components in at least three points. Q.E.D.

We prove now that if an automorphism φ of any finite graph Γ has the properties stated in Lemma (1.17), it is the identity. Make induction on the sum of the number of vertices and edges of Γ. If Γ has an isolated point x, then $\varphi(x) = x$ so let $\Gamma^* = \Gamma - \{x\}$. Then $\varphi =$ identity on Γ^* by induction, so $\varphi =$ identity on Γ too. If Γ has an extremity x, then $\varphi(x) = x$, and again let Γ^* be Γ minus x and the edge abutting at x. Then Γ^* has all the properties Γ has, so $\varphi =$ identity on Γ^*, hence $\varphi =$ identity on Γ. If Γ has a vertex x on which only 2 edges abut, we have one of the two cases:



In the first case, $\varphi(y) = y$, and let Γ^* be Γ minus x, e_1 and e_2. Then $\varphi =$ identity on Γ^* and $\varphi(e_i) = e_i$ too, since if φ reverses the e_i's, this contradicts (iii). In the second case, let Γ^* be Γ minus x, and with e_1 and e_2 identified:



Then $\varphi =$ identity on Γ^*, so $\varphi =$ identity on Γ. Next, say Γ has an edge e with extremities x and y such that $\varphi(x) = x$, $\varphi(y) = y$, $\varphi(e) = e$. Let Γ^* be Γ minus e. Then $\varphi =$ identity on Γ^*, so $\varphi =$ identity on Γ. If none of these reductions are possible, we must be in a situation where a) every vertex is the abutment of at least three edges and b) no edge is left fixed. It is easily seen that the first Betti number b_1 of any of the connected components of Γ is at least 2. Let

$$n_0 = \text{number of fixed vertices,}$$
$$n_1 = \text{number of edges reversed by } \varphi.$$

Then, unless $\Gamma = \varnothing$, the Lefschetz fixed point formula reads:

$$n_0 + n_1 = b_0 - b_1 < 0,$$

which is impossible. Q.E.D.

§2. Degenerations of curves and their jacobians.

We consider the situation:

$K =$ discretely-valued field;
$R =$ integers in K, $\quad k = R/\mathfrak{M} =$ residue field (assumed algebraically closed);
$S = \mathrm{Spec}(R)$, $\quad \eta$ and s its generic and closed points respectively;
$C =$ a curve, smooth, geometrically irreducible and proper over K, of genus $g \geq 2$;
$J =$ the jacobian variety of C;
$\mathscr{J} =$ the Néron model of J over R (cf. [N]);
$\mathscr{J}^0 \subset \mathscr{J}$ the open subgroup scheme with $\mathscr{J}_s^0 =$ identity component of $\mathscr{J}_s$;
$\mathscr{C} =$ the minimal model of C over R.

A word about the existence and uniqueness of $\mathscr{C}$ is needed. We recall that $\mathscr{C}$ is to be a regular scheme, flat and proper over R, with generic fibre $\mathscr{C}_\eta = C$ such that for any other regular scheme $\mathscr{C}'$, flat over R, with generic fibre $\mathscr{C}'_\eta = C$, the birational map $\mathscr{C}' \to \mathscr{C}$ is a morphism. Safarevich in [S] and Lichtenbaum [L] have proven that such a $\mathscr{C}$ (which is obviously unique) exists, provided that there is some regular $\mathscr{C}'$, proper and flat over R, with generic fibre C. And, in fact, that $\mathscr{C}$ is projective over R. To construct such a $\mathscr{C}$, proceed as follows: first let $\mathscr{C}''$ be any scheme, projective and flat over R with generic fibre C. Let $\widehat{R}$ be the completion of R, and let

$$\widehat{\mathscr{C}}'' = \mathscr{C}'' \times_{\mathrm{Spec}\, R} \mathrm{Spec}\, \widehat{R}.$$

Then $\widehat{\mathscr{C}}''$ is an excellent surface, so by [Ab] and by unpublished results of Hironaka, there is a sheaf of ideals $\widehat{\mathcal{J}}$ with support in the singular locus of $\widehat{\mathscr{C}}''$ such that blowing up $\widehat{\mathcal{J}}$ leads to a regular surface $\widehat{\mathscr{C}}'$. But then

$$\widehat{\mathcal{J}} \supset \mathfrak{M}^n \cdot \mathcal{O}_{\widehat{\mathscr{C}}''}$$

for some n, so $\widehat{\mathcal{J}}$ is induced by a unique sheaf of ideals $\mathcal{J} \subset \mathcal{O}_{\mathscr{C}''}$. Let $\mathscr{C}'$ be obtained by blowing up $\mathcal{J}$. Then

$$\widehat{\mathscr{C}}' \simeq \mathscr{C}' \times_{\operatorname{Spec} R} \operatorname{Spec} \widehat{R}$$

so $\mathscr{C}'$ is regular. $\mathscr{C}'$ is also projective over R, hence a projective $\mathscr{C}$ exists.

Definition (2.1). — J has stable reduction if $\mathscr{J}_s$ has no unipotent radical.

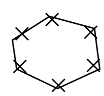
Definition (2.2). — C has stable reduction in sense 1 if $\mathscr{C}_s$ is reduced and has only ordinary double points. C has stable reduction in sense 2 if there is a stable curve $\mathscr{C}'$ over R with generic fibre $\mathscr{C}'_\eta = C$.

Note that if a stable $\mathscr{C}'$ exists, then by Theorem (1.11) it is unique.

Proposition (2.3). — The two senses of stable reduction for C are equivalent.

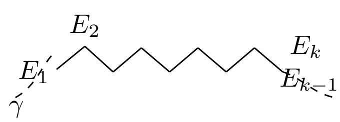
Proof. — Say a stable $\mathscr{C}'$ exists. Blowing up the singularities of $\mathscr{C}'$ as in Lemma (1.12), we obtain the minimal model $\mathscr{C}$ of C and it is seen that $\mathscr{C}_s$ is reduced with only ordinary double points. Conversely, suppose the minimal model $\mathscr{C}$ has this property. Let $E_1, \dots, E_n$ be the non-singular rational components of $\mathscr{C}_s$ which meet the other components in only two points. Then the E_i's divide into several chains of the type:

unless the entire fibre consisted of E_i's and has the type:

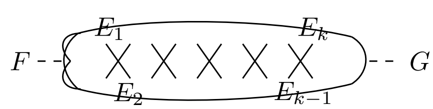


$n \geq 2$
$\mathscr{C}_s =$ loop of E_i's
$(E_i^2)_{\text{on } \mathscr{C}} = -2.$

But in this case genus(C) = genus$(\mathscr{C}_s) = 1$, which contradicts our assumption. Now, according to Theorem (27.1) of Lipman [Li] (generalizing a result of Artin [A_2], which works for surfaces of finite type over a field) any set of k non-singular rational curves connected in a chain as above on a regular surface X, with self-intersection -2 on X, can be blown down to a rational double point P of type A_k on a normal surface X_0. Reversing the process and blowing up a rational double point of type A_k, it is easy to see that non-singular branches γ on X, crossing transversally only the first or the last rational curve in the chain:



are still non-singular branches when mapped to X_0; and if γ_1, γ_2 intersect E_1 or E_k in distinct points, then they cross transversally on X_0. Therefore, suppose we blow down all the chains of E_i's on $\mathscr{C}$. Let $\mathscr{C}'$ be the normal surface so obtained. Then if one of these chains fits into $\mathscr{C}_s$ like this:



Contract to P

then on $\mathscr{C}'_s$, the images of F and G still have only ordinary double points, each has one non-singular branch through the singular point P, and these branches cross transversally. Therefore $\mathscr{C}'$ is a stable curve over R. Q.E.D.

We are now ready to prove the key result on which our proof of irreducibility depends:

Theorem (2.4). — J has stable reduction if and only if C has stable reduction.

Proof. — The connection between $\mathscr{C}$ and $\mathscr{J}$ is based on the following result of Raynaud [R]:

Theorem (2.5). — If $\mathscr{C}$ and $\mathscr{J}$ are as above, and the greatest common denominator d of the multiplicities of the components of $\mathscr{C}_s$ is 1, then $\mathscr{J}^0$ represents the functor $\operatorname{Pic}^0(\mathscr{C}/S)$.

(This result is not stated as such in [R]. It comes out like this, in the terminology of his paper:

a) Condition (N) is verified and $p_*(\mathcal{O}_{\mathscr{C}}) = \mathcal{O}_S$, so

b) $\mathscr{C}$ is cohomologically flat over S in dim. 0 by Theorem 4;

c) therefore Pic^0 is representable and separated over S by Theorem 3;

d) since $E = \{0\}$ and $Q = P/E$, we find $P^0 = Q^0$, and since $\mathscr{C}$ is 1-dimensional over S, P^0 and Q^0 are smooth over S; therefore $R^0 = Q^0$ and $\widetilde{R} = R$.

e) Then by Theorem 5, Pic^0 is the identity component of the Néron model of $\mathrm{Pic}^0(\mathscr{C}_\eta) = J$.)

Now assume that C has stable reduction. Then $\mathscr{C}_s$ is reduced so $d = 1$. By Theorem (2.5), $\mathscr{J}^0 = \mathrm{Pic}^0(\mathscr{C}/S)$. Therefore $\mathscr{J}^0_s = \mathrm{Pic}^0(\mathscr{C}_s/k)$. Since $\mathscr{C}_s$ is reduced with only ordinary double points, its generalized jacobian $\mathrm{Pic}^0(\mathscr{C}_s/k)$ is an extension of an abelian variety by a torus, i.e., has no unipotent radical. Therefore J has stable reduction.

The converse is more difficult. Assume J has stable reduction. We first prove that C has stable reduction under the additional hypothesis that C has a K-rational point.[4] In this case, $\mathscr{C}$ has an R-rational point, and since $\mathscr{C}$ is regular, sections of $\mathscr{C}$ over R pass through components of $\mathscr{C}_s$ of multiplicity one. Therefore $d = 1$, and

Theorem (2.5) applies. In particular, $\mathrm{Pic}^0(\mathscr{C}_s/k) = \mathscr{J}^0_s$, so $\mathrm{Pic}^0(\mathscr{C}_s/k)$ has no unipotent radical. We apply:

Lemma (2.6). — Let D be a complete 1-dimensional scheme over k such that $H^0(\mathcal{O}_D) = k$, and such that the generalized jacobian of D has no unipotent radical. Then

(i) $\chi(\mathcal{O}_D) = \chi(\mathcal{O}_{D_{\mathrm{red}}})$;

(ii) *the singularities of D_{red} are all transversal crossings of a set of non-singular branches (i.e., analytically isomorphic to the union of the coordinate axes in $\mathbf{A}^n$).*

Proof. — Let $\mathcal{J} \subset \mathcal{O}_D$ be the ideal of nilpotent elements. Filtering $\mathcal{J}$ by a chain of ideals $\mathcal{J}_k$ such that $\mathcal{J}\mathcal{J}_k \subset \mathcal{J}_{k+1}$, and using the exact sequence

$$0 \longrightarrow \mathcal{J}_k/\mathcal{J}_{k+1} \xrightarrow{a \mapsto 1+a} (\mathcal{O}_D/\mathcal{J}_{k+1})^* \longrightarrow (\mathcal{O}_D/\mathcal{J}_k)^* \longrightarrow 0,$$

it is easy to deduce that $\mathrm{Pic}^0(D/k)$ is an extension of $\mathrm{Pic}^0(D_{\mathrm{red}}/k)$ by a unipotent group. Therefore, since by assumption $\mathrm{Pic}^0(D/k)$ has no unipotent subgroups,

$$\mathrm{Pic}^0(D/k) \simeq \mathrm{Pic}^0(D_{\mathrm{red}}/k).$$

Since $H^1(\mathcal{O}_D)$, resp. $H^1(\mathcal{O}_{D_{\mathrm{red}}})$, is naturally isomorphic to the Zariski tangent space to $\mathrm{Pic}^0(D/k)$, resp. $\mathrm{Pic}^0(D_{\mathrm{red}}/k)$, it follows that

$$H^1(\mathcal{O}_D) \simeq H^1(\mathcal{O}_{D_{\mathrm{red}}}),$$

hence (i) is proven. Let $\pi : C \to D_{\mathrm{red}}$ be the normalization of D_{red} and let D^* be the local ringed space which, as topological space is D, and whose structure sheaf is given by

$$\Gamma(U, \mathcal{O}_{D^*}) = \{f \in \Gamma(U, \pi_*(\mathcal{O}_C)) \mid x_1, x_2 \in \pi^{-1}(U),\ f(x_1) = f(x_2) \text{ if } \pi(x_1) = \pi(x_2)\}.$$

It is easy to check that D^* is a 1-dimensional scheme whose singularities are all transversal crossings of a set of non-singular branches and that π factors:

$$C \xrightarrow{\pi'} D^* \xrightarrow{\pi''} D_{\mathrm{red}}.$$

I claim that $\pi'' : D^* \to D_{\mathrm{red}}$ is an isomorphism. Filter $\mathcal{O}_{D^*}/\mathcal{O}_{D_{\mathrm{red}}}$ so as to obtain a chain of coherent $\mathcal{O}_{D_{\mathrm{red}}}$-algebras

$$\mathcal{O}_{D^*} = \mathcal{O}^{(0)} \supset \mathcal{O}^{(1)} \supset \cdots \supset \mathcal{O}^{(k)} = \mathcal{O}_{D_{\mathrm{red}}}$$

such that $\ell(\mathcal{O}^{(n)}/\mathcal{O}^{(n+1)}) = 1$. Equivalently, this factors π'':

$$D^* = D_0 \longrightarrow D_1 \longrightarrow \cdots \longrightarrow D_k = D_{\mathrm{red}},$$

where $\mathcal{O}^{(n)} \simeq \mathcal{O}_{D_n}$, and all arrows are homeomorphisms. If $\{x_n\} = \mathrm{Supp}(\mathcal{O}^{(n)}/\mathcal{O}^{(n+1)})$, then we get an exact sequence:

$$0 \longrightarrow \mathcal{O}^*_{D_{n+1}} \longrightarrow \mathcal{O}^*_{D_n} \longrightarrow k_{x_n} \longrightarrow 0$$

[4] This is in fact the only case which will be needed in our application to questions of irreducibility.

(k_y denotes the residue field at y, as sheaf on D), and hence:

$$0 \longrightarrow k \longrightarrow H^1(\mathcal{O}^*_{D_{n+1}}) \longrightarrow H^1(\mathcal{O}^*_{D_n}) \longrightarrow 0.$$

It follows easily that $\mathrm{Pic}^0(D_{n+1}/k)$ is an extension of $\mathrm{Pic}^0(D_n/k)$ by G_a. But $\mathrm{Pic}^0(D_{\mathrm{red}}/k)$ has no unipotent subgroups, and this can only happen if $D_{\mathrm{red}} = D^*$. Q.E.D. for lemma.

We apply the lemma to $\mathscr{C}_s$. According to Lichtenbaum [L] and Safarevich [S], there is a divisor K on $\mathscr{C}$ such that for all positive divisors D lying over the closed point of $\mathrm{Spec}(R)$, we have:

$$\chi(\mathcal{O}_D) = -\frac{(D \cdot (D+K))}{2}.$$

Let $E_1, \ldots, E_n$ be the components of $\mathscr{C}_s$, $d_1, \ldots, d_n$ their multiplicities. Then conclusion (i) of the lemma implies that:

$$\Big(\sum_i d_i E_i \cdot \big(\sum_i d_i E_i + K\big)\Big) = \Big(\sum_i E_i \cdot \big(\sum_i E_i + K\big)\Big).$$

But $((\sum_i d_i E_i) \cdot E_k) = 0$, all k, since $\sum_i d_i E_i$ is the divisor of a function $\pi \in R$ if $(\pi) =$ maximal ideal of R. Therefore:

$$(*) \qquad \Big(\big(\sum_i (d_i - 1)E_i\big) \cdot K\Big) = \Big(\sum_i E_i \cdot \sum_i E_i\Big).$$

Note that at least one d_i equals 1 since $\mathscr{C}$ has a section over $\mathrm{Spec}(R)$ and every section must pass through a component of $\mathscr{C}_s$ of multiplicity 1. Moreover the intersection matrix $(E_i \cdot E_j)$ is negative indefinite, with one-dimensional degenerate subspace generated by $\sum_i d_i E_i$, hence if some $d_i > 1$, it follows that $(\sum_i E_i \cdot \sum_i E_i) < 0$. Therefore, by $(*)$, $(E_{i_0} \cdot K) < 0$ for some i_0. Then we have:

a) $(E_{i_0} \cdot K) < 0$;

b) $(E_{i_0} \cdot E_{i_0}) < 0$;

c) $(E_{i_0} \cdot (E_{i_0} + K)) = -2\chi(\mathcal{O}_{E_{i_0}}) \geq -2$;

hence in fact $(E_{i_0} \cdot E_{i_0}) = (E_{i_0} \cdot K) = -1$, so E_{i_0} is an exceptional curve of the first kind. This contradicts our assumption that on $\mathscr{C}$ all possible curves have been blown down. Therefore $d_i = 1$, all i.

This proves that $\mathscr{C}_s$ is reduced. By conclusion (ii) of the lemma, plus the fact that the dimension of its Zariski tangent-space is everywhere one or two (since $\mathscr{C}_s$ lies on a regular surface $\mathscr{C}$), we deduce that $\mathscr{C}_s$ has only double points. This proves that C has stable reduction in the case that $C(K) \neq \varnothing$.

In the general case, C will acquire a rational point in a finite extension K' of K. Let S' be the spectrum of the localisation at some maximal ideal of the integral closure of R in K'; S' is the spectrum of a valuation ring and is faithfully flat over S. We put $S'' = S' \times_S S'$ and denote by C' and C'' the inverse images of C on $S'_\eta = \mathrm{Spec}(K')$ and S''_η respectively.

$$\begin{array}{ccccc} C'' & \longrightarrow & C' & \longrightarrow & C \\ \downarrow & & \downarrow & & \downarrow \\ S'' & \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\longrightarrow}} & S' & \overset{p}{\longrightarrow} & S. \end{array}$$

Let $\overline{C}'$ be the stable curve on S' having C' as generic fibre. The restriction, C', of $\overline{C}'$ to S'_η carries a descent datum with respect to p, i.e. an isomorphism, φ_η, between the restrictions of $\mathrm{pr}_1^*(\overline{C}')$ and $\mathrm{pr}_2^*(\overline{C}')$ to S''_η. It remains only to extend the isomorphism φ_η to an isomorphism, φ, between the $\mathrm{pr}_i^*(\overline{C}')$. Then φ will be a descent datum for $\overline{C}'$ with respect to p. As $\overline{C}'$ is canonically polarized (1.2), this descent datum will be effective, and so define a stable curve, $\overline{C}$, over S with generic fibre C.

Because $\mathscr{J}_s^0$ has no unipotent radical, the inverse image, $p^*\mathscr{J}^0$, of $\mathscr{J}^0$ on S' is the identity component of the Néron Model of the jacobian of C', so that, defining $q = p \circ \mathrm{pr}_1 = p \circ \mathrm{pr}_2$, one has

$$\mathrm{Pic}^0(\overline{C}'/S') = p^*\mathscr{J}^0,$$

$$\mathrm{Pic}^0(\mathrm{pr}_1^*\overline{C}') = \mathrm{Pic}^0(\mathrm{pr}_2^*\overline{C}') = q^*\mathscr{J}^0.$$

We denote by T the closed subscheme of $\mathrm{Isom}(S'', \mathrm{pr}_1^*(\overline{C}'), \mathrm{pr}_2^*(\overline{C}'))$ corresponding to those isomorphisms which induce (via the preceding identifications) the identity on the inverse image of $\mathscr{J}^0$.

By (1.11), T is finite and unramified over S'', and by (1.13) T is radicial over S''. We conclude that the morphism from T to S'' identifies T with a closed subscheme X of S''. As X contains S''_η, which is schematically dense in S'', we have that X is S'' and T "is" the desired section, φ, of $\mathrm{Isom}(S'', \mathrm{pr}_1^*(\overline{C}'), \mathrm{pr}_2^*(\overline{C}'))$ over S'' which extends φ_η. Q.E.D.

Combining Proposition (2.3), Theorem (2.4), and the stable reduction theorem for abelian varieties quoted in the introduction, we obtain the most important consequence:

Corollary (2.7). — Let R be a discrete valuation ring with quotient field K. Let C be a smooth geometrically irreducible curve over K of genus $g \geq 2$. Then there exists a finite algebraic extension L of K and a stable curve $\mathscr{C}_L$ over R_L, the integral closure of R in L, with generic fibre $\mathscr{C}_{L,\eta} = C \times_K L$.

§3. Elementary derivation of the theorem.

Let k be an algebraically closed field of char. $p \neq 0$. We use the notation of § 1, except that we will now denote by H_g the product $H_g \times \mathrm{Spec}(k)$ of the previous H_g with $\mathrm{Spec}(k)$: it is a disjoint union of non-singular varieties $H_{g,1}, \ldots, H_{g,n}$ over k and is the subscheme of $\mathrm{Hilb}^{P_g}_{\mathbf{P}^{5g-6}/k}$ of tri-canonical stable curves. Similarly, H_g^0 is the open dense subset of H_g of tri-canonical non-singular curves. By the results of [M$_1$], we know that a coarse geometric quotient

$$M_g^0 = H_g^0 / \mathrm{PGL}(5g-6)$$

exists, that it is a disjoint union of normal varieties over k and is the coarse moduli space for non-singular curves of genus g. Let $S^* = H_g - H_g^0$. Then everything decomposes into the same set of components.

Let

$$\begin{array}{rrcl} & H_{g,i} & = & \text{components of } H_g, \\ \text{then} & H_{g,i}^0 & = & H_{g,i} \cap H_g^0 = \text{components of } H_g^0, \\ \text{and} & M_{g,i}^0 & = & H_{g,i}^0 / \mathrm{PGL}(5g-6) = \text{components of } M_g^0, \\ \text{and} & S^* & = & \text{disjoint union of } S_1^*, \ldots, S_n^*, \quad S_i^* = H_{g,i} \cap S^*. \end{array}$$

We want to prove that M_g^0, or equivalently H_g^0, or equivalently H_g is irreducible. We shall use: (i) the fact that these statements are true in char. 0; (ii) the inductive assumption that these statements are true for smaller genus.

Step I. — No component of M_g^0 is complete (i.e., proper over k).
Proof. — Here we use the char. 0 result. By [M$_1$], there is a scheme X, quasi-projective over $\mathrm{Spec}(W(k))$, $W(k)$ the Witt vectors, whose closed fibre is M_g^0, and whose generic fibre X_η is the char. 0 coarse moduli space over the quotient field of $W(k)$. In particular, X_η is known to be connected. Since X is quasi-projective over $W(k)$, we can embed X as an open dense subset of a scheme $\overline{X}$ projective over $W(k)$. $\overline{X}_\eta$ is still connected, hence by the connectedness theorem of Enriques-Zariski [EGA 3], the closed fibre $\overline{X}_0$ of $\overline{X}$ is connected. But if Y were a complete variety which is a component of M_g^0, then: a) Y is an open subset of M_g^0, which is an open subset of $\overline{X}_0$, and: b) since Y is proper/k, Y would be a closed subset of $\overline{X}_0$ too. Therefore, $Y = \overline{X}_0$, hence M_g^0 is itself irreducible and complete. On the other hand, if A_g is the coarse moduli space of principally polarized g-dimensional abelian varieties, then the map associating to each curve its jacobian defines a morphism:

$$\theta : M_g^0 \to A_g.$$

If M_g^0 were complete, the image of θ would be closed. But it is well known that the closure of the image of θ contains all products of lower dimensional jacobians too, so it is not closed. Q.E.D.

Step II. — No component of H_g consists entirely of non-singular curves, i.e., $S_i^* \neq \varnothing$ for all i.
Proof. — Here we combine Step I with the result of § 2. Take any i. Let $T = \mathrm{Spec}\, k[[t]]$. Since $M_{g,i}^0$ is not complete, there is a morphism φ of the generic point T_η of T into $M_{g,i}^0$ which does not extend to a morphism of T into $M_{g,i}^0$. Now we replace T by its normalization T' in a finite algebraic extension and let $\varphi' : T'_\eta \to M_{g,i}^0$ be the induced morphism, which still does not extend to a morphism from T' to $M_{g,i}^0$. By the results of § 2, if T' is chosen suitably there exists a stable curve $\pi : C' \to T'$ over T' whose generic fibre C'_η is a non-singular curve corresponding to the morphism $\varphi' : T'_\eta \to M_g^0$ via the functorial properties of the moduli space. Since T' is the spectrum of a local ring, we can choose an isomorphism

$$\mathbf{P}\big(\pi_*(\omega_{C'/T'}^{\otimes 3})\big) \simeq \mathbf{P}^{5g-6} \times T'$$

and get a tri-canonical embedding $C' \subset \mathbf{P}^{5g-6} \times T'$. C', with this embedding, is then induced from the universal tri-canonically embedded stable curve by a morphism $\psi : T' \to H_g$. Since the generic fibre of C' is C'_η, we get a commutative diagram:

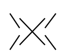
$$\begin{array}{ccc} T' & \xrightarrow{\psi} & H_g \\ \cup & & \cup \\ T'_\eta & \xrightarrow{\mathrm{res}(\psi)} & H_g^0 \\ & \varphi' \searrow & \downarrow \\ & M_{g,i}^0 & \subset M_g^0. \end{array}$$

If $\mathrm{Im}(\psi) \subset H_g^0$, then φ' would extend to a morphism from T' to M_g^0, hence from T' to $M_{g,i}^0$, and this is a contradiction. Moreover, $\psi(T_\eta)$ must be a point of $H_{g,i}^0$, since its image in M_g^0 is in $M_{g,i}^0$. Therefore the image x of the closed point of T' by ψ is in the closure of $H_{g,i}^0$, i.e., in $H_{g,i}$, but not in $H_{g,i}^0$ itself. Q.E.D.

Step III. — S^* is connected.

Proof. — This will follow using only the induction assumption of irreducibility for lower genera. Let $Z \subset H_g \times \mathbf{P}^{5g-6}$ be the universal tri-canonically embedded stable curve. Let $S \subset Z$ be the set of points where Z is not smooth over H_g. As we proved in § 1, S is non-singular and is the normalization of S^*. In particular, this shows that if $x \in S^*$, then the corresponding curve Z_x has exactly one double point if and only if x is a non-singular point of S^*. Stable curves C of genus g with exactly one double point belong to one of the following types:

type 0 :  C irreducible
normalization C' of C has genus $g-1$;

type k :  C has two non-singular components C_1, C_2
genus$(C_1) = k$
genus$(C_2) = g-k$

where $1 \leq k \leq \left[\frac{g}{2}\right]$. If $0 \leq k \leq \left[\frac{g}{2}\right]$, let

$$S^*(k) = \{x \in S^* \mid Z_x \text{ has one double point and is of type } k\}.$$

Then the open dense subset of S^* of non-singular points is the disjoint union of open subsets $S^*(0), \ldots, S^*(\left[\frac{g}{2}\right])$. We first check:

$(*)$ Each set $S^*(k)$ is irreducible.

Proof of $(*)$. — Take the case $k=0$: the cases $1 \leq k \leq \left[\frac{g}{2}\right]$ are similar. Let

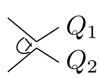
$$T = \{\{x_1, x_2\} \mid x_1 \neq x_2 \text{ and } \pi(x_1) = \pi(x_2) \in H_{g-1}^0\} \subset Z_{g-1} \times_{H_{g-1}} Z_{g-1}.$$

T is smooth with irreducible fibres over H_{g-1}^0, hence T is irreducible. Consider the correspondence relating $S^*(0)$ and T:

$$W = \left\{ \begin{array}{l} \text{set of pairs } x \in S^*(0),\ \{x_1, x_2\} \in T \text{ such that if } y = \pi(x_1) = \pi(x_2), \\ \text{then there exists a birational morphism } f : (Z_{g-1})_y \to (Z_g)_x \\ \text{such that } f(x_1) = f(x_2). \end{array} \right.$$

It is easy to check that W is Zariski-closed. Moreover, for any $\{x_1, x_2\} \in T$, $W \cap (S^*(0) \times \{x_1, x_2\})$ is an orbit in $S^*(0)$ under $\mathrm{PGL}(5g-6)$: so these are non-empty irreducible subsets all of the same dimension. It follows that W itself is irreducible. But the projection from W to $S^*(0)$ is surjective, so $S^*(0)$ is irreducible. Q.E.D. for $(*)$.

Now for any k, $1 \leq k \leq \left[\frac{g}{2}\right]$, choose any stable curve $C(k)$ of the type:



one non-singular component $C(k)'$ of genus k,
one component $C(k)''$ with one double point, normalization of genus $g-k-1$.

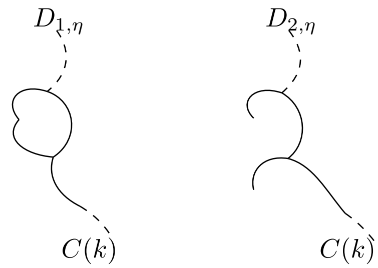
Let $P(k)$ be a point of H_g such that $Z_{P(k)} \simeq C(k)$. Step III will be completed if we prove:

$(**)$ $P(k)$ is in the closure of $S^*(0)$ and of $S^*(k)$, hence $S^*(0)$, $S^*(k)$ both lie in the same topological component of S^*.

Proof of $(**)$. — Let $T = \operatorname{Spec} k[[t]]$. Using the fact that S^* has two branches through $P(k)$, one for each of the double points of $C(k)$, we see that there exist two morphisms

$$f_1, f_2 : T \to S^*,$$

$$f_1(T_s) = f_2(T_s) = P(k), \qquad T_s = \text{closed point of } T,$$



such that if $\pi_1 : D_1 \to T$, $\pi_2 : D_2 \to T$ are the two stable curves over T induced by f_1 and f_2, then: a) the closed fibres $D_{1,s}$, $D_{2,s}$ are $C(k)$; b) there are sections $s_1 : T \to D_1$, $s_2 : T \to D_2$ whose images are non-smooth points of π_1, π_2 and such that $s_1(T_s)$ and $s_2(T_s)$ are the two double points Q_1 and Q_2 of $C(k)$ respectively; and c) the generic fibres $D_{1,\eta}$, $D_{2,\eta}$ have only one double point (cf. figure). I claim:

A) $D_{1,\eta}$ is of type (k);

B) $D_{2,\eta}$ is of type (0).

To prove A), let D_1' be the result of blowing up the subscheme $s_1(T)$ of D_1. Then D_1' is still flat and proper over T and its special fibre is the special fibre of D_1 with Q_1 blown up (use the fact that formally at Q_1, D_1 is isomorphic to $k[[t,x,y]]/(xy)$ with the section given by $x = y = 0$). Therefore the special fibre of D_1' is the disjoint union of $C(k)'$, $C(k)''$, so the general fibre of D_1' is the disjoint union of two irreducible curves which specialize to $C(k)'$, $C(k)''$ respectively. Since $D_{1,\eta}$ has only one double point, $D_{1,\eta}'$ is non-singular, so $D_{1,\eta}'$ is the disjoint union of two non-singular irreducible curves which must then have the same genera as $C(k)'$ and $C(k)''$, i.e., k, $g-k$. Thus $D_{1,\eta}$ has type (k).

To prove B), it suffices to check that $D_{2,\eta}$ is geometrically irreducible. If not, $D_{2,\eta}$ would have two components meeting at the single point $s_2(T_\eta)$. Since D_2 is smooth over T at each generic point of its special fibre, distinct geometric components of $D_{2,\eta}$ have to have specializations which are distinct components of $(D_2)_s$. Then $(D_2)_s$ would have two components meeting at the point $Q_2 = s_2(T_s)$. This is false, so B) is proven.

Now because of A), $f_1(T_\eta) \in S^*(k)$, hence $P(k) = f_1(T_s)$ is in the closure of $S^*(k)$. And because of B), $f_2(T_\eta) \in S^*(0)$, hence $P(k) = f_2(T_s)$ is in the closure of $S^*(0)$ too. Q.E.D. for $(**)$.

This completes the proof of Step III since we now see that all irreducible components of S^* are part of the same topological component. Finally, from Steps II and III, we see that

a) S^*, being connected, is part of a single component of H_g, while

b) each component of H_g contains part of S^*.

Thus H_g is irreducible, as was to be proven.

§4. Some results on algebraic stacks.

The proofs of the results stated in this section will be given elsewhere.

Let C be a category and let $p : \mathscr{S} \to C$ be a category over C. For each $U \in \operatorname{Ob} C$, we denote by $\mathscr{S}_U$ the fibre $p^{-1}(U)$. The category $\mathscr{S}$ is *fibered in groupoids* over C if the following two conditions are verified:

a) For all $\varphi : U \to V$ in C and $y \in \operatorname{Ob} \mathscr{S}_V$ there is a map $f : x \to y$ in $\mathscr{S}$ with $p(f) = \varphi$.

b) Given a diagram

$$\begin{tikzcd} x \arrow[dr, "f"] & \\ y \arrow[r, "g"'] & z \end{tikzcd}$$

in $\mathscr{S}$, let

$$\begin{tikzcd} U \arrow[dr, "\varphi"] & \\ V \arrow[r, "\psi"'] & W \end{tikzcd}$$

be its image in C. Then for all $\chi : U \to V$ such that $\varphi = \psi\chi$, there is a unique $h : x \to y$ such that $f = g \cdot h$ and $p(h) = \chi$.

Condition b) implies that the $f : x \to y$ whose existence is asserted in a) is unique up to canonical isomorphism.

Assume that for each $\varphi : U \to V$ in C and each $Y \in \operatorname{Ob} \mathscr{S}_V$, such an $f : x \to y$ has been chosen. This x will be written as $\varphi^* y$. Then, φ^* "is" a functor from $\mathscr{S}_V$ to $\mathscr{S}_U$ and if $\varphi\psi$ is a composite morphism in C, the functors $(\varphi\psi)^*$ and $\psi^*\varphi^*$ are canonically isomorphic.

We propose the terminology "stack" for the French word "champ" of non-abelian cohomology (Giraud [G]).

Definition (4.1). — Let C be a category with a Grothendieck topology. We assume products and fibre products exist in C. A stack in groupoids over C is a category over C, $p : \mathscr{S} \to C$ such that:

(i) *$\mathscr{S}$ is fibered in groupoids over C.*

(ii) *For any $U \in \operatorname{Ob} C$ and any objects x, y in $\mathscr{S}_U$ the functor from C/U to (sets) which to any $\varphi : V \to U$ associates $\operatorname{Hom}_{\mathscr{S}_V}(\varphi^* x, \varphi^* y)$ is a sheaf.*

(iii) *If $\varphi_i : V_i \to U$ is a covering family in C, any descent datum relative to the φ_i, for objects in $\mathscr{S}$, is effective.*

For each $x \in \operatorname{Ob} \mathscr{S}_U$, there are given isomorphisms between the inverse images of $x_i = \varphi_i^* x$ and $x_j = \varphi_j^* x$ over $V_{ij} = V_i \times_U V_j$, and the pull-backs of these isomorphisms on $V_{ijk} = V_i \times_U V_j \times_U V_k$ satisfy a "cocycle" condition. In (iii) it is required that reciprocally, any such "descent datum" be defined by some $x \in \mathscr{S}_U$.

In what follows, for the sake of brevity, we will use "stack" to mean "stack in groupoids".

If $U \in \operatorname{Ob} C$ and if $\mathscr{S}$ is a stack over C, the fibre $\mathscr{S}_U$ will be called the category of sections of $\mathscr{S}$ over U.

Let C be as in (4.1). The stacks over C are the objects of a 2-category [B] (stacks/C): 1-morphisms are functors from one stack to another, compatible with the projection into C; and 2-morphisms are morphisms of functors. In this 2-category every 2-morphism is an isomorphism. Products and 2-fibre products exist in this 2-category.

To each $X \in \operatorname{Ob} C$ is associated the "representable" stack over C whose category of sections over U is the discrete category whose objects are the morphisms from U to X. This stack will be denoted simply X. For any stack $\mathscr{S}$, the category

$$\operatorname{Hom}(X, \mathscr{S})$$

is canonically equivalent to the category of sections of $\mathscr{S}$ over X. Because of this, $\mathscr{S}$ is sometimes said to "classify" its sections over variable $X \in \operatorname{Ob} C$. In the category $\operatorname{Hom}(\mathscr{S}, X)$, all morphisms are identities, i.e., $\operatorname{Hom}(\mathscr{S}, X)$ is just a set.

Let us denote also by C the 2-category having the same objects and morphisms as C, and in which the identities are the only 2-morphisms. The above construction then identifies C with a full sub-2-category of (stacks/C).

For each $S \in \operatorname{Ob} C$, the category C/S satisfies the assumptions of (4.1). Any stack $\mathscr{S}_0$ over C/S (an "S-stack") gives rise to a stack $\mathscr{S}$ over C; a section (φ, η) of $\mathscr{S}$ over $U \in \operatorname{Ob} C$ consists of

(i) a morphism $\varphi : U \to S$;

(ii) a section η of $\mathscr{S}_0$ over (U, φ).

Definition (4.2). — A 1-morphism of stacks over C, $F : \mathscr{S}_1 \to \mathscr{S}_2$, will be called representable if for any X in C and any 1-morphism $x : X \to \mathscr{S}_2$, the fibre product $X \times_{\mathscr{S}_2} \mathscr{S}_1$ is a representable stack.

In down to earth terms, this means the following:

(i) for any $f : Y \to X$ in C, the category whose objects are pairs

$$\{\text{a section, } y, \text{ of } \mathscr{S}_1 \text{ over } Y; \text{ an isomorphism } F(y) \xrightarrow{\sim} f^*(x)\}$$

is equivalent to a category $S(f)$ in which all morphisms are identities;

(ii) the functor $f \mapsto \operatorname{Ob} S(f)$ is representable by some $g : Z \to X$. Such a Z represents the fibre product $X \times_{\mathscr{S}_2} \mathscr{S}_1$.

Let P be a property of morphisms in C, stable by change of base and of a local nature on the target.

Definition (4.3). — A representable morphism $F : \mathscr{S}_1 \to \mathscr{S}_2$ of stacks over C has property P if for any 1-morphism $x : X \to \mathscr{S}_2$ the morphism in C deduced by base change: $F' : X \times_{\mathscr{S}_2} \mathscr{S}_1 \to X$ has that property.

Proposition (4.4). — Let $\mathscr{S}$ be a stack. The diagonal map

$$\mathscr{S} \to \mathscr{S} \times \mathscr{S}$$

is representable if and only if for all $X, Y \in \operatorname{Ob} C$ and 1-morphisms $x : X \to \mathscr{S}$, $y : Y \to \mathscr{S}$, the fibre product $X \times_{\mathscr{S}} Y$ is representable.

If $X \in \mathrm{Ob}\, C$ and x, y are sections of $\mathscr{S}$ over X, we denote by $\underline{\mathrm{Isom}}(X, x, y)$ the sheaf on C/X which to every Z over X associates the set of isomorphisms between the inverse images of x and y over Z. Then the object representing $\mathscr{S} \times_{(\mathscr{S}\times\mathscr{S})} X$ (the product taken with the map $(x, y) : X \to \mathscr{S} \times \mathscr{S}$) is just $\mathrm{Isom}(X, x, y)$. If $X \to \mathscr{S}$, $Y \to \mathscr{S}$ are 1-morphisms, then the object representing $X \times_{\mathscr{S}} Y$ is just $\mathrm{Isom}(X \times Y, p_1^* x, p_2^* y)$.

Henceforth, C will be the category of schemes with the etale topology (SGAD, IV, (6.3)).

Definition (4.5). — A stack $\mathscr{S}$ is quasi-separated if the diagonal morphism from $\mathscr{S}$ to $\mathscr{S} \times \mathscr{S}$ is representable, quasi-compact and separated.

Definition (4.6). — A stack $\mathscr{S}$ is an algebraic stack[5] *if*

(i) *$\mathscr{S} \to \mathscr{S} \times \mathscr{S}$ is representable;*

(ii) *there exists a 1-morphism $x : X \to \mathscr{S}$ such that for all $y : Y \to \mathscr{S}$, the projection morphism $X \times_{\mathscr{S}} Y \to Y$ is surjective and etale (i.e., x is etale and surjective).*

The 2-category of algebraic stacks contains the representable stacks and is stable under products and fibre products. If $\mathscr{S}$ is a quasi-separated algebraic stack, the diagonal map is unramified and quasi-affine.

Definition (4.7). — An algebraic stack $\mathscr{S}$ is separated if $\mathscr{S} \to \mathscr{S} \times \mathscr{S}$ is proper (or, equivalently, finite). A 1-morphism $f : \mathscr{S}_1 \to \mathscr{S}_2$ is separated (resp. quasi-separated) if for any morphism $x : X \to \mathscr{S}_2$ from a separated scheme X to $\mathscr{S}_2$, the fibre product $\mathscr{S}_1 \times_{\mathscr{S}_2} X$ is separated (resp. quasi-separated).

Example (4.8). — Let X be a scheme over S. Let G be a group scheme over S, etale, separated and of finite type over S, which operates on X. We will denote by $[X/G]$ the S-stack whose category of sections over an S-scheme T is the category of principal homogeneous spaces (p.h.s.) E over T, with structural group G (i.e., a p.h.s. under G_T), provided with a G-morphism $\varphi : E \to X$. The principal homogeneous space $G \times X$ over X (G acting only on the first factor) plus the G-morphism $G \times X \to X$ (given by the action of G on X) is a section of $[X/G]$ over X. The corresponding morphism $q : X \to [X/G]$ is etale and surjective, so that $[X/G]$ is an algebraic stack. In addition, X is a principal homogeneous space over $[X/G]$; the stack $[X/G]$ is representable if and only if X is a principal homogeneous space over a scheme Y, in which case

$$[X/G] \sim Y.$$

If $X = S$, then $[X/G] = [S/G]$ might be called the "classifying stack" of G over S.

Example (4.9). — Suppose a stack $\mathscr{S}$ has the property that in each category $\mathscr{S}_X$ the only morphisms are the identity morphisms. Then $\mathscr{S}_X$ is just a set $\mathscr{F}(X) = \mathrm{Ob}\, \mathscr{S}_X$, and this set, under pull-back, is a contravariant functor $\mathscr{F}$ in X. Conditions (ii) and (iii) of (4.1) assert that the functor $\mathscr{F}$ is a sheaf on C. Artin and Knutson [K] have defined an *algebraic space* to be a sheaf $\mathscr{F}$ such that:

(i) for any morphisms $X \to \mathscr{F}$, $Y \to \mathscr{F}$ of representable functors to $\mathscr{F}$, the fibre product $X \times_{\mathscr{F}} Y$ is representable;

(ii) there exists a morphism $X \to \mathscr{F}$, represented by surjective, etale morphisms of schemes.

This is exactly what we have called an algebraic stack in this case.

Definition (4.10). — Let $\mathscr{S}$ be an algebraic stack. The etale site $\mathscr{S}_{et}$ of $\mathscr{S}$ is the category with objects the etale morphisms

$$x : X \to \mathscr{S}$$

and where a morphism from (X, x) to (Y, y) is a morphism of schemes $f : X \to Y$ plus a 2-morphism between the 1-morphism $x : X \to \mathscr{S}$ and $y \cdot f : X \to \mathscr{S}$. A collection of morphisms $f_i : (X_i, x_i) \to (X, x)$ is a covering family if the underlying family of morphisms of schemes is surjective.

The site $\mathscr{S}_{et}$ is in a natural way ringed. When we speak of sheaves on $\mathscr{S}$ we mean sheaves on $\mathscr{S}_{et}$.

We now explain how many concepts from the theory of schemes may be applied to algebraic stacks.

Let $\mathcal{P}$ be a property of morphisms of schemes, stable by etale change of base, and of a local nature (for the etale topology) on the target.

For instance: being an open immersion with dense image, being dominant, birational...

A representable morphism of algebraic stacks $f : T_1 \to T_2$ is said to have property $\mathcal{P}$ if for one (and hence for every) surjective etale morphism $x : X \to T_2$, the morphism of schemes deduced by base change

$$f' : X \times_{T_2} T_1 \longrightarrow X$$

[5] This definition is the "right" one only for quasi-separated stacks. It will however be sufficient for our purposes.

has that property.

Let $\mathcal{P}$ be a property of morphisms of schemes, which, at source and target, is of a local nature for the etale topology. This means that, for any family of commutative squares

$$\begin{tikzcd} X_i \arrow[r, "g_i"] \arrow[d, "f_i"'] & X \arrow[d, "f"] \\ Y_i \arrow[r, "h_i"'] & Y \end{tikzcd}$$

where the g_i (resp. h_i) are etale and cover X (resp. Y):

$$\mathcal{P}(f) \Longleftrightarrow \forall i\, \mathcal{P}(f_i).$$

For instance: f flat, smooth, etale, unramified, normal, locally of finite type, locally of finite presentation.

If $f : T_1 \to T_2$ is a morphism of algebraic stacks, we say f has property $\mathcal{P}$ if for one, then necessarily for every, commutative diagram

$$\begin{tikzcd} X \arrow[r, "x"] \arrow[d, "f'"'] & T_1 \arrow[d, "f"] \\ Y \arrow[r, "y"'] & T_2 \end{tikzcd}$$

where X and Y are schemes and x, y are etale and surjective, f' has property $\mathcal{P}$.

Similarly, if $\mathcal{P}$ is a property of schemes, of a local nature for the etale topology, an algebraic stack T will be said to have property $\mathcal{P}$ if for one (and hence for every) surjective etale morphism $x : X \to T$, X has property $\mathcal{P}$. This applies to, for instance, the properties of being regular, normal, locally noetherian, of characteristic p, reduced, Cohen–Macaulay...

An algebraic stack T will be called *quasi-compact* if there exists a surjective etale morphism $x : X \to T$ with X quasi-compact. A morphism $f : T_1 \to T_2$ of algebraic stacks will be called *quasi-compact* if for any quasi-compact scheme X over T_2, the fibre product $T_1 \times_{T_2} X$ is quasi-compact. It is enough to test the condition for a surjective family $f_i : X_i \to T_2$. We define a morphism $f : T_1 \to T_2$ to be of *finite type*, if it is quasi-compact, and locally of finite type; of *finite presentation*, if it is quasi-compact, quasi-separated, and locally of finite presentation. An algebraic stack is *noetherian*, if it is quasi-compact, quasi-separated, and locally noetherian.

The key point in what follows will be the definition of a "proper morphism" and the analogue of Chow's lemma.

A morphism $f : T_1 \to T_2$ is said to have a property $\mathcal{P}$, locally on T_2, if there exists a surjective etale morphism $x : X \to T_2$ such that the morphism f' deduced from f by the change of base by x has property $\mathcal{P}$.

Definition (4.11). — A morphism $f : T_1 \to T_2$ is proper if it is separated, of finite type and if, locally over T_2, there exists a commutative diagram

$$\begin{tikzcd} T_3 \arrow[rr, "g"] \arrow[dr, "h"'] & & T_1 \arrow[dl, "f"] \\ & T_2 & \end{tikzcd}$$

with g surjective and h representable and proper.

The following form of Chow's lemma will be sufficient for our purposes.

Theorem (4.12). — Let S be a noetherian scheme and f be a morphism from an etale site T to S. We assume f to be separated and of finite type. Then, there exists a commutative diagram

$$\begin{tikzcd} T \arrow[dr] & T' \arrow[l, "g"'] \arrow[r, "j"] \arrow[d, "f'"] & T'' \arrow[dl, "f''"] \\ & S & \end{tikzcd}$$

in which T' and T'' are schemes and such that

(i) *g is proper, surjective and generically finite;*

(ii) *j is an open immersion;*

(iii) *f'' is projective.*

Using (4.12), it is easy to extend the cohomological theory of coherent sheaves to the present situation. In fact if $f : T_1 \to T_2$ is a proper morphism of noetherian algebraic stacks and if $\mathscr{F}$ is a coherent sheaf on T_1, then the $R^i f_*(\mathscr{F})$ are coherent sheaves on T_2.

However, the $R^i f_*(\mathscr{F})$ don't need to be zero for i large enough. Let S be a scheme and G a finite group. We denote by p the projection $p : [S/G] \to S$. Quasi-coherent (resp. coherent) sheaves of modules on $[S/G]$ may (and will) be identified with quasi-coherent (resp. coherent) sheaves of modules on S, on which G acts. One has

$$R^i p_*(\mathscr{F}) \simeq H^i(G, \mathscr{F}).$$

In general, the (quasi-coherent sheaf) cohomology of algebraic stacks appears as a mixture of finite group cohomology and of scheme cohomology.

The *disjoint sum* T of a family $(T_i)_{i \in I}$ of stacks is the stack a section of which over a scheme X consists of

(i) a decomposition $X = \coprod_i X_i$ of X;

(ii) a section of T_i over X_i for each i.

The *void stack* $\varnothing$ is the one represented by the void scheme.

A stack is *connected* if it is non-void and is not the disjoint sum of two non-void stacks.

Proposition (4.13). — A locally noetherian algebraic stack is in one and only one way the disjoint sum of a family of connected algebraic stacks (called its connected components).

We denote by $\pi_0(T)$ the set of connected components of a locally noetherian algebraic stack T. If $x : X \to T$ is surjective and etale, $\pi_0(T)$ is the cokernel of the two maps

$$\pi_0(X \times_T X) \rightrightarrows \pi_0(X) \longrightarrow \pi_0(T).$$

Proposition (4.14). — Let T be an algebraic stack of finite type over a field k. Then, T is connected if and only if there exists a connected scheme X, of finite type over k, and a surjective morphism from X to T.

An open subset U of an algebraic stack T is a full subcategory $U \subset T$ which is an algebraic stack, which contains together with any $t \in \mathrm{Ob}(T)$ all isomorphic t' and such that the inclusion $j : U \to T$ is representable by open immersions. The open subsets of T correspond bijectively to the open subsets of its etale site.

For each open subset U of T, there exists one and only one full subcategory $T - U$ of T, which is an algebraic stack, which contains together with any $t \in \mathrm{Ob}(T)$ all isomorphic t' and such that

(i) $T - U$ is reduced;

(ii) the inclusion map $i : T - U \to T$ is representable by closed immersions;

(iii) for any etale surjective morphism $x : X \to T$, the inverse image of $T - U$ on X is the complement of the inverse image of U.

An algebraic stack F in T satisfying (i) and (ii) is a *closed subset* of T, and the functor $U \mapsto T - U$ is an isomorphism of the set of open and the set of closed subsets of T. If F satisfies only (ii), F_{red} satisfies (i) and (ii) so that F defines a closed subset of T.

An algebraic stack T is *irreducible* if it is not the union of two closed subsets, non-void and distinct from T.

Proposition (4.15). — A noetherian algebraic stack T is in one and only one way the union of irreducible closed subsets, none of which contains any other. They are called the irreducible components of T. If U is an open dense subset of T, the irreducible components of U are the non-void intersections of U with the irreducible components of T.

Each irreducible component of T is contained in a connected component of T. Conversely:

Proposition (4.16). — The connected components of a normal noetherian algebraic stack are irreducible.

Theorem (4.17). — Let f be a morphism of finite type from an algebraic stack T to a noetherian scheme S. For $s \in S$, let $n(s)$ be the number of connected components of the geometric fibre of T at s. Then

(i) *$n(s)$ is a constructible function of s;*

(ii) *if f is proper and flat, then $n(s)$ is lower-semi-continuous;*

(iii) *if f is proper flat, and has geometrically normal fibres, then $n(s)$ is constant.*

Let $f : T \to S$ be a morphism of finite type from an algebraic stack T to a noetherian scheme S. Assume that the diagonal map $T \to T \times_S T$ is separated and quasi-compact.

Theorem (4.18) (Valuative criterion for separation). — The morphism f is separated if and only if, for any complete discrete valuation ring with algebraically closed residue field and any commutative diagram

$$\begin{tikzcd} & T \arrow[dr, "f"] & \\ \mathrm{Spec}(V) \arrow[ur, shift left, "g_1"] \arrow[ur, shift right, "g_2"'] \arrow[rr] & & S \end{tikzcd}$$

any isomorphism between the restrictions of g_1 and g_2 to the generic point of $\mathrm{Spec}(V)$ *can be extended to an isomorphism between g_1 and g_2.*

This criterion is nothing other but the valuative criterion of properness (EGA, II, 7.3.8) applied to the (representable) diagonal morphism.

Theorem (4.19) (Valuative criterion for properness). — If f is separated, then f is proper if and only if, for any discrete valuation ring V with field of fractions K and any commutative diagram

$$\begin{tikzcd} & & T \arrow[d, "f"] \\ \mathrm{Spec}(K) \arrow[urr, "g"] \arrow[r] & \mathrm{Spec}(V) \arrow[r] & S \end{tikzcd}$$

there exists a finite extension K' of K such that g extends to $\mathrm{Spec}(V')$*, where V' is the integral closure of V in K'*

$$\begin{tikzcd} & & T \arrow[dd, "f"] \\ \mathrm{Spec}(K') \arrow[r] \arrow[d] & \mathrm{Spec}(V') \arrow[ur, dashed] \arrow[d] & \\ \mathrm{Spec}(K) \arrow[uurr] \arrow[r] & \mathrm{Spec}(V) \arrow[r] & S \end{tikzcd}$$

To prove a given f is proper, it suffices to verify the above criterion under the additional hypothesis that V is complete and has an algebraically closed residue field. Further, given a dense open subset U of T, it is enough to test only g's which factor through U.

Proposition (4.20). — Let $\mathscr{S}$ be an algebraic stack. The functor which, to any algebraic stack over $\mathscr{S}$, $f : \mathscr{T} \to \mathscr{S}$, associates the $\mathcal{O}_{\mathscr{S}}$-sheaf of algebras $f_*\mathcal{O}_{\mathscr{T}}$ induces an equivalence of categories between:

(i) *the category of algebraic stacks representable and affine over $\mathscr{S}$;*

(ii) *the dual of the category of quasi-coherent $\mathcal{O}_{\mathscr{S}}$-algebras.*

Let $\mathscr{A}$ be a quasi-coherent sheaf of $\mathcal{O}_{\mathscr{S}}$-algebras on an algebraic stack $\mathscr{S}$. For each etale morphism $x : U \to \mathscr{S}$, with U affine, let $\mathscr{A}'(U)$ be the integral closure of $\Gamma(U, \mathcal{O}_{\mathscr{S}})$ in $\mathscr{A}(U)$. By (EGA, II, 6.3.4), the $\mathscr{A}'(U)$ for variable U are the sections over U of a quasi-coherent sheaf $\mathscr{A}'$ on $\mathscr{S}$, which will be called the integral closure of $\mathcal{O}_{\mathscr{S}}$ in $\mathscr{A}$.

Let $f : \mathscr{T} \to \mathscr{S}$ be representable and affine. The algebraic stack which is associated by (4.20) to the integral closure of $\mathcal{O}_{\mathscr{S}}$ in $f_*\mathcal{O}_{\mathscr{T}}$ will be called the normalisation of $\mathscr{S}$ with respect to $\mathscr{T}$. Its formation is compatible with any etale change of basis.

Theorem (4.21). — Let $\mathscr{S}$ be a quasi-separated stack over a noetherian scheme S. Assume that

(i) *the diagonal map $\mathscr{S} \to \mathscr{S} \times_S \mathscr{S}$ is representable and unramified;*

(ii) *there exists a scheme X of finite type over S and a smooth and surjective S-morphism from X to $\mathscr{S}$.*

Then, $\mathscr{S}$ is an algebraic stack of finite type over S.

M. Artin has developed powerful methods to relate pro-representability of a stack to the existence of etale surjective maps $x : X \to \mathscr{S}$.

§ 5. Second proof of the irreducibility theorem.

Let $\mathcal{M}_g$ ($g \geq 2$) be the stack whose category of sections over a scheme S is the category of stable curves of genus g over S, the morphisms being the isomorphisms of schemes over S. By (1.11), the diagonal morphism

$$\Delta : \mathcal{M}_g \longrightarrow \mathcal{M}_g \times \mathcal{M}_g$$

is representable, finite and unramified.

We saw in § 1 that the stack classifying the tricanonically embedded stable curves of genus g is represented by a scheme H_g, smooth and of finite type over $\operatorname{Spec}(\mathbb{Z})$. The "forgetful" morphism

$$H_g \longrightarrow \mathcal{M}_g$$

is representable, smooth and surjective. Indeed, if $p : C \to S$ is a stable curve over S, defining a morphism

$$\gamma : S \longrightarrow \mathcal{M}_g,$$

then the fibre product $H_g \times_{\mathcal{M}_g} S$ is the scheme, smooth over S, of isomorphisms between the standard projective space of dimension $5g - 6$ over S and $\mathbf{P}(p_*(\omega_{C/S}^{\otimes 3}))$. We deduce from this and (4.21) that:

Proposition (5.1). — $\mathcal{M}_g$ is a separated algebraic stack of finite type over $\operatorname{Spec}(\mathbb{Z})$.

Let us denote by $\mathcal{M}_g^0$ the open subset of $\mathcal{M}_g$ which "consists of" smooth curves, and by $\mathcal{Z}_g$, the "universal curve" over $\mathcal{M}_g$, the algebraic stack classifying pointed stable curves.

Theorem (5.2). — The algebraic stacks $\mathcal{M}_g$ and $\mathcal{Z}_g$ are proper and smooth over $\operatorname{Spec}(\mathbb{Z})$ *and the complement of $\mathcal{M}_g^0$ in $\mathcal{M}_g$ is a divisor with normal crossings relative to* $\operatorname{Spec}(\mathbb{Z})$.

Proof. — Let $x : X \to \mathcal{M}_g$ be etale and consider the following commutative diagram of algebraic stacks of finite type over $\operatorname{Spec}(\mathbb{Z})$:

$$\begin{tikzcd}
X \times_{\mathcal{M}_g} \mathcal{Z}_g \arrow[rr] \arrow[dr] \arrow[dd] & & \mathcal{Z}_g \arrow[d] \\
& X \times_{\mathcal{M}_g} H_g \arrow[r, "x'"] \arrow[dd] & H_g \arrow[dd, bend left, "q"] \\
X \times_{\mathcal{M}_g} \mathcal{Z}_g \arrow[rr] \arrow[dr] & & \mathcal{Z}_g \arrow[d, "p"] \\
& X \arrow[r, "x"] & \mathcal{M}_g
\end{tikzcd}$$

In this diagram, the four horizontal arrows are etale and the four vertical arrows are smooth and surjective. As H_g and Z_g (p. 78) are smooth over $\operatorname{Spec}(\mathbb{Z})$, so are X and $X \times_{\mathcal{M}_g} \mathcal{Z}_g$. In addition,

$$q'^{-1} x^{-1}(\mathcal{M}_g^0) = x'^{-1}(H_g^0)$$

is the complement of a divisor with normal crossings relative to $\operatorname{Spec}(\mathbb{Z})$, so that the inclusion of $x^{-1}(\mathcal{M}_g^0)$ in X has the same property. This being true for any x, $\mathcal{M}_g$ and $\mathcal{Z}_g$ are smooth over $\operatorname{Spec}(\mathbb{Z})$ and $\mathcal{M}_g^0$ is the complement of a divisor with normal crossings relative to $\operatorname{Spec}(\mathbb{Z})$. In particular $\mathcal{M}_g^0$ is dense in $\mathcal{M}_g$.

We may now use the valuative criterion for properness in its modified form (4.19) to deduce the properness of $\mathcal{M}_g$ from the stable reduction theorem. The properness of $\mathcal{Z}_g$ then results from that of p.

(5.3) Let $p : X \to S$ be a stable smooth curve of genus $g \geq 2$ over S. If $k \in \mathbb{N}$ is invertible in $\mathcal{O}_S$, the sheaf $R^1 p_*(\mathbb{Z}/k\mathbb{Z})$ on S_{et} is locally free over $\mathbb{Z}/k\mathbb{Z}$, of rank $2g$, and the cup-product is a non degenerate alternating form

$$R^1 p_*(\mathbb{Z}/k\mathbb{Z}) \otimes R^1 p_*(\mathbb{Z}/k\mathbb{Z}) \longrightarrow R^2 p_*(\mathbb{Z}/k\mathbb{Z}) \simeq \mu_k^{\otimes -1}.$$

Locally on S_{et}, μ_k is isomorphic to $\mathbb{Z}/k\mathbb{Z}$, and thus, locally, $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ is provided with a non degenerate symplectic form with values in $\mathbb{Z}/k\mathbb{Z}$, which is determined up to a unit in $\mathbb{Z}/k\mathbb{Z}$, or, as we shall say, $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ is provided with an homogeneous symplectic structure. The constant sheaf $(\mathbb{Z}/k\mathbb{Z})^{2g}$ will be provided with the homogeneous symplectic structure induced by the standard symplectic structure of $(\mathbb{Z}/k\mathbb{Z})^{2g}$.

Definition (5.4). — A Jacobi structure of level k on X is an isomorphism (respecting their homogeneous symplectic structures) between $R^1p_*(\mathbb{Z}/k\mathbb{Z})$ and $(\mathbb{Z}/k\mathbb{Z})^{2g}$.

(5.5) Given a section s of p, and a set of prime numbers $\mathbf{P}$ including all residue characteristics of S, the specialisation theorem for the fundamental group enables one to construct a pro-object $\pi_1(X/S,s)^{(\mathbf{P})}$ of the category (l.c. $\mathrm{gr}^{(\mathbf{P})}$) of locally constant sheaves of finite groups of order prime to $\mathbf{P}$, with the properties

(i) $\mathrm{Hom}_S(\pi_1(X/S,s)^{(\mathbf{P})},G) = R^1p_*(X \bmod s, p^*G) = R^1p_*(\mathrm{Ker}(p^*G \to_s G))$ functorially in $G \in$ (l.c. $\mathrm{gr}^{(\mathbf{P})}$);

(ii) the formation of $\pi_1(X/S,s)^{(\mathbf{P})}$ is compatible with any change of base.

If G and H are two sheaves of groups on S_{et}, we define the sheaf of exterior morphisms from H to G, $\mathrm{Hom}^{\mathrm{ext}}(H,G)$, as the quotient of $\mathrm{Hom}(H,G)$ by the action of H induced by its action on itself by inner automorphism.

The sheaf

$$\mathrm{Hom}^{\mathrm{ext}}_S(\pi_1(X/S,s)^{(\mathbf{P})},G) \simeq R^1p_*(p^*G) \qquad (\text{for } G \in (\text{l.c. gr}^{(\mathbf{P})}))$$

is "independent" of the choice of s.

We shall denote it as

$$\mathrm{Hom}^{\mathrm{ext}}_S(\pi_1(X/S)^{(\mathbf{P})},G).$$

As $p : X \to S$ admits sections locally for the etale topology, this sheaf makes sense without assuming p to have a global section. It is independent of the choice of $\mathbf{P}$, so long as $\mathbf{P}$ is prime to the order of G and includes all residue characteristics of S.

Definition (5.6). — Let G be a finite group of order n prime to $\mathbf{P}$. A Teichmüller structure of level G on X is a surjective exterior homomorphism from $\pi_1(X/S)$ to G.

The finite generation of $\pi_1(X/S)$ (SGA, 60/61, exp. 10) implies:

Lemma (5.7). — The sheaf on (Sch/S) *of the Teichmüller structures of level G on X is represented by an etale covering of S.*

We denote by ${}_G\mathcal{M}_g^0$ the stack classifying the stable smooth curves of genus g and characteristic prime to n, with a Teichmüller structure of level G.

For any algebraic stack $\mathscr{M}$, we denote by $\mathscr{M}[1/n]$ its open subset $\mathscr{M} \times \mathrm{Spec}(\mathbb{Z}[1/n])$.

Lemma (5.7) may now be rephrased:

Proposition (5.8). — The "forgetful" morphism

$${}_G\mathcal{M}_g^0 \longrightarrow \mathcal{M}_g^0[1/n]$$

is representable, finite and etale.

The stack ${}_G\mathcal{M}_g^0$ is thus an algebraic stack. Let ${}_G\mathcal{M}_g$ be the normalisation of $\mathcal{M}_g[1/n]$ with respect to ${}_G\mathcal{M}_g^0$. The stack ${}_G\mathcal{M}_g$, being representable and finite over $\mathcal{M}_g[1/n]$, is proper over $\mathrm{Spec}(\mathbb{Z}[1/n])$.

Theorem (5.9). — The geometric fibres of the projection of ${}_G\mathcal{M}_g$ onto $\mathrm{Spec}(\mathbb{Z}[1/n])$ *are normal, and, fibre by fibre, ${}_G\mathcal{M}_g^0$ is dense in ${}_G\mathcal{M}_g$.*

Proof. — We will use Abhyankar–Artin's lemma, in its "absolute" form:

Lemma (5.10). — Let D be a divisor with normal crossings on an excellent regular scheme X, Y an etale covering of $X-D$ and $\overline{Y}$ the normalisation of X with respect to Y. Assume that the generic points of the irreducible components of D are all of characteristic 0. *Then every geometric point of X has an etale neighbourhood $x : X' \to X$ such that, on X':*

(i) *D becomes a union of regular divisors $(D_i)_{i \in I}$, D_i of equation $t_i = 0$.*

(ii) *There exists an integer k prime to residue characteristics of X' such that $\overline{Y}$ becomes isomorphic to a disjoint union of quotients (by subgroups of μ_k^I) of the covering of X' obtained by extracting the k^{th}-roots of the t_i's.*

If $x : X \to \mathcal{M}_g[1/n]$ is an etale morphism, then $X^0 = x^{-1}(\mathcal{M}_g^0)$ is the complement in X of a divisor with normal crossing relative to $\mathrm{Spec}(\mathbb{Z})$, $X_1^0 = X \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g^0)$ is an etale covering of X^0 and $X_1 = X \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g)$ is the normalisation of X with respect to this covering of X^0.

By the explicit local description (5.10), for any prime number l prime to n, $X_1 \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ is the normalisation of $X \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ with respect to $X_1^0 \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$. As this is true for any modular family, we get (5.9).

Corollary (5.11). — The geometric fibres of the projection

$$
{}_G\mathcal{M}_g \longrightarrow \mathrm{Spec}(\mathbb{Z}[1/n])
$$

all have the same number of connected components.

Proof. — By (4.17), all geometric fibres $\mathcal{M}_g \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ of $\mathcal{M}_g$ over $\mathrm{Spec}(\mathbb{Z}[1/n])$ have the same number of connected components. These connected components are irreducible (4.16). Furthermore ${}_G\mathcal{M}_g^0 \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ is dense in ${}_G\mathcal{M}_g \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ and thus has the same number of connected (or irreducible) components as ${}_G\mathcal{M}_g \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$, and this number is independent of l.

(5.12) Let us denote by Π the fundamental group of an oriented closed differentiable surface S_0 of genus g. The group Π may be defined by generators and relations as follows:

(i) generators: x_i for $1 \le i \le 2g$;

(ii) relation: $(x_1, x_{g+1}) \cdots (x_i, x_{g+i}) \cdots (x_g, x_{2g}) = e$, where $(a, b) = aba^{-1}b^{-1}$.

Let (e_i) be the standard basis of $\mathbb{Z}^{2g}$. The morphism φ from Π to $\mathbb{Z}^{2g}$ with $\varphi(x_i) = e_i$ identifies $\Pi/(\Pi,\Pi) = H_1(\Pi, \mathbb{Z})$ with $\mathbb{Z}^{2g}$.

The surface S_0 is a $K(\Pi, 1)$ and thus

$$
H^2(\Pi, \mathbb{Z}) = H^2(S_0, \mathbb{Z}) = \mathbb{Z},
$$

and the cup product defines a symplectic structure on $\Pi/(\Pi,\Pi)$. This structure is identified by φ with the standard symplectic structure of $\mathbb{Z}^{2g}$.

We denote by $\mathrm{Aut}^0(\Pi)$ the subgroup of $\mathrm{Aut}(\Pi)$ which acts trivially on $H^2(\Pi, \mathbb{Z})$. Dehn has proved that each exterior automorphism of Π is induced by a diffeomorphism of S_0 onto itself and that the map induced by φ:

$$
\mathrm{Aut}^0(\Pi) \longrightarrow \mathrm{Sp}_{2g}(\mathbb{Z}),
$$

is surjective (see [Ma]).

Theorem (5.13). — The number of connected components of any geometric fibre of the projection of ${}_G\mathcal{M}_g^0$ *onto* $\mathrm{Spec}(\mathbb{Z}[1/n])$ *is equal to the number of orbits of* $\mathrm{Aut}^0(\Pi)$ *in the set of exterior epimorphisms from* Π *to* G.

Proof. — By (5.13), it suffices to prove that ${}_G\mathcal{M}_g^0 \times \mathrm{Spec}(\mathbb{C})$ has the said number of connected components. As results from (4.14),

$$
\pi_0({}_G\mathcal{M}_g^0 \times \mathrm{Spec}(\mathbb{C})) = \pi_0({}_GM_g)
$$

where ${}_GM_g$ is the coarse moduli scheme classifying stable smooth curves of genus g over $\mathbb{C}$ with a Teichmüller structure of level G.

Recall that a Teichmüller curve of genus g is a stable smooth curve C of genus g over $\mathbb{C}$ together with an exterior isomorphism φ of the transcendental fundamental group $\pi_1(C)$ with Π, which induces a symplectic isomorphism[6] between

$$
H_1(C, \mathbb{Z}) \simeq \pi_1(C)/(\pi_1(C), \pi_1(C))
$$

and $\Pi/(\Pi,\Pi)$. By Teichmüller's theory [W], the analytic space T_g classifying Teichmüller curves of a given genus $g \ge 2$ is homeomorphic to a ball, and hence connected.

If ψ is a surjective homomorphism from Π to G, the map

$$
(C, \varphi) \longmapsto (C, \psi\varphi)
$$

defines a morphism $t_\psi : T_g \to {}_GM_g$. Two such maps t_ψ and $t_{\psi'}$ have the same image if and only if $\psi = \psi'\sigma$ for $\sigma \in \mathrm{Aut}^0(\Pi)$, and

$$
{}_GM_g = \coprod_{\psi \bmod \mathrm{Aut}^0(\Pi)} t_\psi(T_g)
$$

[6] An arbitrary isomorphism φ induces an isomorphism between $H_1(C, \mathbb{Z})$ and $\Pi/(\Pi,\Pi)$ which is always symplectic up to sign.

which implies (5.13).

(5.14) Let us denote by ${}_n\mathcal{M}_g^0$ the algebraic stack classifying stable smooth curves with a Jacobi structure of level n. This algebraic stack "is" a true scheme for $n \geq 3$ (by [S]).

If φ is a Jacobi structure of level n on a stable smooth curve $p : X \to S$, we define the "multiplicator" $\mu(\varphi)$ of φ by the commutative diagram

$$\begin{tikzcd}
\Lambda^2(\mathbb{Z}/n\mathbb{Z})^{2g} \arrow[rr, "A"] & & \mathbb{Z}/n\mathbb{Z} \\
\Lambda^2 R^1 p_*(\mathbb{Z}/n\mathbb{Z}) \arrow[u, "\Lambda^2\varphi"] \arrow[r, "\Lambda"'] & \mu_n^{\otimes -1} \simeq R^2 p_*(\mathbb{Z}/n\mathbb{Z}) \arrow[ur, "\mu(\varphi)"'] &
\end{tikzcd}$$

The scheme of isomorphisms between $\mathbb{Z}/n\mathbb{Z}$ and $\mu_n^{\otimes -1}$ is $\operatorname{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$; thus $\varphi \mapsto \mu(\varphi)$ defines a morphism μ from S to $\operatorname{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$. This being true for any X and S, μ is induced by

$$\mu : {}_n\mathcal{M}_g^0 \longrightarrow \operatorname{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n]).$$

Theorem (5.15). — The geometric fibres of the morphism μ are connected.

Proof. — By definition, ${}_n\mathcal{M}_g^0$ is open and closed in ${}_G\mathcal{M}_g^0$ for $G = (\mathbb{Z}/n\mathbb{Z})^{2g}$. The group $\mathrm{GL}_{2g}(\mathbb{Z}/n\mathbb{Z})$ acts on G, and thus on ${}_G\mathcal{M}_g^0$. One has:

(i) the open subset ${}_n\mathcal{M}_g^0$ of ${}_G\mathcal{M}_g^0$ is stable under the subgroup $H = \mathrm{CSp}_{2g}(\mathbb{Z}/n\mathbb{Z})$ of symplectic similitudes;

(ii) ${}_G\mathcal{M}_g^0 = \coprod_{\sigma/H} \sigma({}_n\mathcal{M}_g^0)$.

It now results from (5.11) that all geometric fibres of μ have the same number of connected components.

Consider the geometric fibre of μ at the standard complex place of $\operatorname{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$. This fibre is the algebraic stack classifying complex stable smooth curves C provided with a symplectic isomorphism

$$H^1(C, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} (\mathbb{Z}/n\mathbb{Z})^{2g}.$$

Reasoning as in (5.13), we are reduced to proving

Lemma (5.16). — The homomorphism

$$\operatorname{Aut}^0(\Pi) \longrightarrow \mathrm{Sp}_{2g}(\mathbb{Z}/n\mathbb{Z})$$

is surjective.

Proof. — This results from Dehn's theory (5.12) and from the fact that the groups Sp_{2g}, being split semi-simple simply connected groups, are generated by their unipotent elements, so that the reduction map

$$\mathrm{Sp}_{2g}(\mathbb{Z}) \longrightarrow \mathrm{Sp}_{2g}(\mathbb{Z}/n\mathbb{Z})$$

is surjective.

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MODULAR FORMS AND ℓ-ADIC REPRESENTATIONS

Séminaire Bourbaki, 21st year, 1968/69, no. 355

Pierre Deligne

1 Introduction

Let

$$D(q) = q\prod_{n=1}^{\infty}(1-q^n)^{24} = \sum_{n=1}^{\infty}\tau(n)q^n \qquad (|q|<1)$$

and

$$\Delta(z) = D(e^{2\pi i z}) \qquad (\operatorname{Im} z > 0).$$

It is known that the function Δ is, up to a constant factor, the unique cusp form of weight 12 for the group $\mathrm{SL}_2(\mathbb{Z})$. Put, for p prime,

$$H_p(X) = 1 - \tau(p)X + p^{11}X^2.$$

Following Hecke's theory, the Dirichlet series

$$L_\tau(s) = \sum \tau(n)n^{-s} = \prod_p \frac{1}{H_p(p^{-s})}$$

extends to an entire function of s, and the function

$$(2\pi)^{-s}\Gamma(s)L_\tau(s)$$

is invariant under $s \mapsto 12-s$. Ramanujan's conjecture asserts that the roots of the polynomial H_p have absolute value $p^{-11/2}$; equivalently, $|\tau(p)| < 2p^{11/2}$.

These properties, proven or conjectural, are similar to conjectural properties of zeta functions of algebraic varieties over $\mathbb{Q}$. Suggested by this, as a first approximation, one is tempted to interpret L_τ as the zeta function of such a variety.

For each prime number ℓ, let K_ℓ be the maximal extension of $\mathbb{Q}$ unramified away from ℓ and, for $p \neq \ell$, let F_p be the inverse in $\mathrm{Gal}(K_\ell/\mathbb{Q})$ of the Frobenius element φ_p relative to p. This object is well defined up to conjugation.

Translating the preceding in terms of ℓ-adic cohomology, Serre conjectured the existence, for each ℓ, of a representation of $\mathrm{Gal}(K_\ell/\mathbb{Q})$ in a $\mathbb{Q}_\ell$-vector space W_ℓ of rank 2 such that, for each $p \neq \ell$,

$$H_p(X) = \det(1 - F_pX; W_\ell).$$

Moreover, the representation W_ℓ should lie in the range of application of the Weil conjectures, and Ramanujan's conjecture should be a particular case of them.

This program has been carried out by Kuga–Shimura [4] in the analogous case of modular forms relative to certain subgroups of $\mathrm{SL}_2(\mathbb{R})$ with compact quotient. Reduced to the present case, the fundamental idea of Sato–Kuga–Shimura is the following: if E is the universal elliptic curve over the moduli scheme S of elliptic curves (forgetting that it does not exist) and if E^k is the fiber product of E with itself over S, repeated k times, then $L_\tau(s)$ is essentially the zeta function of E^k for $k = 10 = 12-2$.

We show below how to resolve the difficulties created by the points at infinity, and how to construct the representations W_ℓ having the properties indicated above. For more historical details and for applications, see Serre [6].

Notation.

- We denote by $\mathbb{A}$ the ring of adeles of $\mathbb{Q}$, by $\mathbb{A}_f$ the ring of finite adeles, the restricted product over all prime numbers of the fields $\mathbb{Q}_p$, and, for S a finite set of prime numbers, put

$$\mathbb{A}_f^S = \prod_{p\in S}\mathbb{Q}_p \times \prod_{p\notin S}\mathbb{Z}_p \subset \mathbb{A}_f.$$

  For $S = \varnothing$, one writes $\widehat{\mathbb{Z}} = \mathbb{A}_f^{\varnothing}$.

- If X is a topological space, or the étale site of a scheme, and G is a set, G also denotes the constant sheaf on X defined by G.
- We denote by $\mathbb{G}_a$ and $\mathbb{G}_m$ the additive and multiplicative groups.
- An elliptic curve is an abelian variety of dimension one, and in particular is equipped with an origin.
- If L is an invertible sheaf and $n \in \mathbb{Z}$, L^n denotes the nth tensor power $L^{\otimes n}$.
- We denote by $\overline{\mathbb{Q}}$ the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$.
- The symbol $\square$ marks the end of a proof, or its absence.

2 The Shimura Isomorphism

(2.1) An elliptic curve over a complex analytic space S is a proper and flat morphism of analytic spaces $f : E \to S$, equipped with a section e, whose fibers are elliptic curves. An elliptic curve over S admits one and only one S-group law

$$\mu : E \times_S E \to E$$

whose unit section is e. Associated with an elliptic curve are the following.

(a) The invertible sheaf

$$\omega_E = e^* \Omega^1_{E/S}.$$

The relative Lie algebra $\mathrm{Lie}_S(E)$ is the invertible sheaf ω^{-1} dual to ω. One has

$$f_* \Omega^1_{E/S} \xrightarrow{\sim} \omega.$$

(b) The local system of free rank-2 $\mathbb{Z}$-modules $R^1 f_* \mathbb{Z}$. One puts

$$T_{\mathbb{Z}}(E) = (R^1 f_* \mathbb{Z})^\vee, \qquad T_{\mathbb{Q}}(E) = T_{\mathbb{Z}}(E) \otimes \mathbb{Q},$$

the local system of homology of E over S.

The exponential mapping defines an exact sequence of sheaves of sections

$$0 \longrightarrow T_{\mathbb{Z}}(E) \xrightarrow{\alpha} \omega^{-1} \longrightarrow E \longrightarrow 0,$$

so that the elliptic curve E can be reconstructed from the map α.

The local system $\bigwedge^2 R^1 f_* \mathbb{Z} \simeq R^2 f_* \mathbb{Z}$ is canonically isomorphic to $\mathbb{Z}$. An isomorphism between $\mathbb{Z}^2$ and $R^1 f_* \mathbb{Z}$ is said to be admissible if it induces 1 on second exterior powers. Denote by $\mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C})$ the set of $\mathbb{R}$-vector-space isomorphisms between $\mathbb{R}^2$ and $\mathbb{C}$ which respect the natural orientations of $\mathbb{R}^2$ and $\mathbb{C}$, defined by $e_1 \wedge e_2 > 0$ and $1 \wedge i > 0$. Such a homomorphism is determined by its restriction to $\mathbb{Z}^2$, and one puts

$$\mathrm{Hom}^+(\mathbb{Z}^2, \mathbb{C}) = \mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C}).$$

This space is equipped with the complex structure obtained from its inclusion in the complex vector space $\mathrm{Hom}(\mathbb{Z}^2, \mathbb{C})$. Over this space one arranges a "universal" exact sequence

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{\alpha} \mathbb{G}_a \longrightarrow E_0 \longrightarrow 0.$$

Proposition (2.2)**.**

(i) The functor which associates with each analytic space S the set of isomorphism classes of elliptic curves E over S, equipped with isomorphisms

$$\omega_E \simeq \mathbb{G}_a, \qquad R^1 f_* \mathbb{Z} \simeq \mathbb{Z}^2$$

(the last being admissible), is represented by the analytic space $\mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C})$*, equipped with a universal elliptic curve E_0.*

(ii) The functor which associates with each analytic space S the set of isomorphism classes of elliptic curves over S, equipped with an admissible isomorphism $R^1 f_* \mathbb{Z} \simeq \mathbb{Z}^2$, is represented by the analytic space

$$X = \mathbb{C}^\times \backslash \mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C}),$$

the Poincaré half-plane.

(iii) The space $\mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C})$ *is a principal homogeneous space of the group* $\mathbb{G}_m$ *over* X*.*

One can again regard X as the set of complex structures on $\mathbb{R}^2$. This space is, by (ii), equipped with a universal elliptic curve E_X, whose local system of real cohomology is canonically isomorphic to $\mathbb{R}^2$. Let ω be the invertible sheaf associated with E_X.

The coherent analytic sheaf

$$R^1 f_* \mathbb{R} \otimes_{\mathbb{R}} \mathcal{O}_X$$

is the relative de Rham cohomology sheaf of E_X over X, and as such it is inserted into an exact sequence, the Hodge filtration,

$$0 \longrightarrow \omega \longrightarrow R^1 f_* \mathbb{R} \otimes_{\mathbb{R}} \mathcal{O}_X \xrightarrow{q} \omega^{-1} \longrightarrow 0$$

since, by Serre duality, $\omega^{-1} \simeq R^1 f_* \mathcal{O}_{E_X}$.

The functorial description in 2.2(ii) evidently gives a right action of the group $\mathrm{SL}_2(\mathbb{Z})$ on (X, E_X): to $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and to the elliptic curve E equipped with $\alpha : \mathbb{Z}^2 \xrightarrow{\sim} R^1 f_* \mathbb{Z}$, one associates $(E, \alpha \circ \gamma)$. If one regards X, equipped with

$$q : \mathbb{R}^2 \otimes_{\mathbb{R}} \mathcal{O}_X \simeq R^1 f_* \mathbb{R} \otimes_{\mathbb{R}} \mathcal{O}_X \longrightarrow \omega^{-1},$$

as classifying the complex structures on $\mathbb{R}^2$, one has the same evident action of the group $\mathrm{GL}_2^+(\mathbb{R})$ on $(X, \mathbb{R}^2, \omega, q)$.

(2.3) Choose a basis (x_1, x_2) of $\mathbb{R}^2$ such that $x_1 \wedge x_2 > 0$. A point

$$(f : \mathbb{R}^2 \to \mathbb{C}) \quad \mod \mathbb{C}^\times$$

of X is located by

$$z = \frac{f(x_1)}{f(x_2)} \qquad (\mathrm{Im}\, z > 0),$$

and the map q is identified with

$$q : \mathbb{R}^2 \to \mathbb{G}_a, \qquad ax_1 + bx_2 \longmapsto az + b.$$

This gives an evident trivialization of ω^{-1} over X, which is not equivariant. Relative to this trivialization, a section $f(z)$ of ω^k over X is transformed by an element

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \in \mathrm{GL}^+(\mathbb{R}^2)$$

(with respect to the basis (x_1, x_2)) by

$$f \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} (z) = (cz + d)^{-k} f\left(\frac{az+b}{cz+d}\right).$$

From the identity

$$dz = (cz+d)^2\, d\left(\frac{az+b}{cz+d}\right) \begin{vmatrix} a & b \\ c & d \end{vmatrix}^{-1},$$

one deduces that dz is a section of $\omega^{-2}\Omega^1_X$ invariant under $\mathrm{SL}_2(\mathbb{R})$. This section is everywhere nonzero and defines an equivariant isomorphism of $\mathrm{SL}_2(\mathbb{R})$-sheaves between ω^2 and Ω^1_X.

(2.4) Let Γ be a discrete subgroup of $\mathrm{SL}_2(\mathbb{R})$ which has no elements of finite order and has quotient of finite volume. One then knows that the quotient space X/Γ is identified with a smooth projective curve $\overline{X/\Gamma}$ minus a finite number of points. The group Γ acts on X without fixed points. The equivariant local system $\mathbb{R}^2$ on X, as well as the exact sequence

$$0 \to \omega \to \mathbb{R}^2 \otimes_{\mathbb{R}} \mathcal{O} \to \omega^{-1} \to 0,$$

therefore define a local system U over X/Γ and an exact sequence

$$0 \to \omega \to U \otimes_{\mathbb{R}} \mathcal{O} \to \omega^{-1} \to 0.$$

(2.5) In the particular case where $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$, these structures are obtained from an elliptic curve E over X/Γ whose inverse image is the equivariant elliptic curve E_X over X.

(2.6) The points at infinity of X/Γ are described as follows (see [9]).

(a) They correspond to conjugacy classes in Γ of nontrivial subgroups of Γ which are maximal among the subgroups of Γ consisting of unipotent elements.

(b) Let $\Gamma_0 \subset \Gamma$ be one such subgroup and choose a basis (x_1, x_2) of $\mathbb{R}^2$ such that, in this basis, Γ_0 is represented as the set of matrices

$$\left\{ \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix} \,\middle|\, n \in \mathbb{Z} \right\}.$$

Let z be the coordinate (2.3) on X defined by (x_1, x_2). There exists N such that the subset

$$X_N = \{z \mid \operatorname{Im} z > N\}$$

of X is disjoint from all of its conjugates γX_N for $\gamma \in \Gamma/\Gamma_0$, so that $X_N/\Gamma_0 \hookrightarrow X/\Gamma$. The function $q = e^{2\pi i z}$ establishes an isomorphism between X_N/Γ_0 and the punctured disk

$$0 < |q| < e^{-2\pi N}.$$

If P_{Γ_0} is the point of $\overline{X/\Gamma} - X/\Gamma$ associated with Γ_0, this isomorphism extends to an isomorphism of a neighborhood of P_{Γ_0} with the disk $0 \leq |q| < e^{-2\pi N}$.

By virtue of (2.3), the sections of ω over X_N which are invariant under Γ_0 are identified with periodic holomorphic functions of period one on X_N. One again denotes by ω the invertible sheaf over $\overline{X/\Gamma}$ which extends ω and such that, in a neighborhood of a point P_{Γ_0}, the section of ω over X_N/Γ_0 defined by the function 1 extends to an invertible section on X_N/Γ_0.

(2.7) Over X/Γ one has the two invertible sheaves Ω^1 and ω^2, and an isomorphism φ (2.3) between their restrictions to X/Γ. From the formula

$$dq = d(e^{2\pi i z}) = 2\pi i e^{2\pi i z}\, dz = 2\pi i q\, dz$$

it follows that the map

$$\varphi : \Omega^1 \to \omega^2$$

extends to $\overline{X/\Gamma}$, and has a simple zero at each of the points at infinity.

Definition (2.8)**.** *The space of cusp forms of weight $k+2$, relative to the group Γ, is the space of global sections*

$$H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k).$$

By virtue of (2.7), this space is again identified with the space of global sections of ω^{k+2} vanishing at infinity, that is, at each "point."

(2.9) Denote by U^k the kth symmetric power of the local system U over X/Γ. The map (2.5) induces a map

$$\iota_k : \omega^k \to U^k \otimes_{\mathbb{R}} \mathcal{O},$$

hence a map, again denoted by ι_k,

$$\iota_k : \Omega^1 \otimes \omega^k \to \Omega^1(U^k),$$

where the target is the sheaf of holomorphic differential forms over X/Γ with coefficients in the local system U^k.

The de Rham resolution of $U^k \otimes_{\mathbb{R}} \mathbb{C}$,

$$0 \to U^k \otimes_{\mathbb{R}} \mathbb{C} \to U^k \otimes_{\mathbb{R}} \mathcal{O} \xrightarrow{d} U^k \otimes_{\mathbb{R}} \Omega^1 \to 0,$$

induces a map

$$\delta : H^0(X/\Gamma, \Omega^1(U^k)) \to H^1(X/\Gamma, U^k \otimes \mathbb{C}).$$

Moreover, the cohomology space $H^1(X/\Gamma, U^k \otimes \mathbb{C})$ is equipped with a natural complex conjugation, such that δ defines a conjugate-linear mapping $\overline{\delta}$ from the complex conjugate of $H^0(X/\Gamma, \Omega^1(U^k))$ into $H^1(X/\Gamma, U^k \otimes \mathbb{C})$. One thus obtains a map

$$\mathrm{sh}_0 = \delta \circ H^0(\iota_k) \oplus \overline{\delta} \circ H^0(\overline{\iota_k})$$

$$\mathrm{sh}_0 : H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k)} \longrightarrow H^1(X/\Gamma, U^k \otimes \mathbb{C}).$$

For an arbitrary sheaf F on a space Y, one denotes by $\widetilde{H}^i(Y, F)$ the image of cohomology with compact supports $H^i_c(Y, F)$ in cohomology without supports $H^i(Y, F)$.

Theorem 4.2.6 of [9] is essentially equivalent to the following theorem; in loc. cit. k is supposed even, but the same proof works in general.

Theorem (2.10, Shimura [7])**.** *There exists an isomorphism* sh *making the following diagram commutative:*

$$\begin{array}{ccc} H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k)} & \xrightarrow{\mathrm{sh}} & \widetilde{H}^1(X/\Gamma, U^k \otimes \mathbb{C}) \\ \downarrow & & \downarrow \\ H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k)} & \xrightarrow{\mathrm{sh}_0} & H^1(X/\Gamma, U^k \otimes \mathbb{C}). \end{array}$$

One calls sh *the Shimura isomorphism.*

(2.11) In the particular case where Γ is a subgroup of finite index of $\mathrm{SL}_2(\mathbb{Z})$, the elliptic curve E over X/Γ furnishes an elliptic curve scheme over the algebraic curve $\overline{X/\Gamma}$; equivalently, its modular invariant is meromorphic at infinity. It therefore admits a Néron model E over $\overline{X/\Gamma}$. One can show that the fibers of E at the points at infinity are of multiplicative type, and that over all of $\overline{X/\Gamma}$ one has

$$\omega = e^*\Omega^1_{E/\overline{X/\Gamma}}.$$

In this particular case, one has $U = R^1 f_* \mathbb{Z} \otimes \mathbb{R}$, so that the target of the Shimura isomorphism can be rewritten as

$$\widetilde{H}^1(X/\Gamma, U^k \otimes \mathbb{C}) \simeq \widetilde{H}^1(X/\Gamma, \mathrm{Sym}^k(R^1 f_* \mathbb{Z})) \otimes_{\mathbb{Z}} \mathbb{C}.$$

3 Hecke Operators and the Fundamental ℓ-adic Representation

(3.1) Recall (cf. [3]) that the category of locally constant constructible $\mathbb{Z}_\ell$-sheaves, abbreviated lcc, over a scheme S is the category of projective systems $(F_n)_{n \in \mathbb{N}}$ on the étale site $S_{\text{ét}}$ of S satisfying:

(i) F_n is a locally constant sheaf of $\mathbb{Z}/(\ell^n)$-modules of finite type;

(ii) if $n \le m$, then
$$F_m \otimes_{\mathbb{Z}/(\ell^m)} \mathbb{Z}/(\ell^n) \xrightarrow{\sim} F_n.$$

The lcc $\mathbb{Z}_\ell$-sheaves form a stack of abelian categories over S; the stack of lcc $\mathbb{Q}_\ell$-sheaves is the quotient of this stack by the thick substack of lcc $\mathbb{Z}_\ell$-sheaves killed by a power of ℓ. One denotes by $\otimes \mathbb{Q}_\ell$ the canonical functor from the category of lcc $\mathbb{Z}_\ell$-sheaves into that of lcc $\mathbb{Q}_\ell$-sheaves.

If S is connected and equipped with a geometric point s, the category of lcc $\mathbb{Z}_\ell$-sheaves, respectively $\mathbb{Q}_\ell$-sheaves, over S is equivalent, by the functor "fiber over s," to the category of continuous representations of the fundamental group $\pi_1(S, s)$ on $\mathbb{Z}_\ell$-modules of finite type, respectively on finite-dimensional $\mathbb{Q}_\ell$-vector spaces.

If T is a finite set of prime numbers, an lcc $\mathbb{A}^f_T$-sheaf consists of the giving, for each prime number ℓ, of an lcc $\mathbb{Z}_\ell$-sheaf if $\ell \notin T$, and an lcc $\mathbb{Q}_\ell$-sheaf if $\ell \in T$. For $T = \varnothing$, one speaks of lcc $\widehat{\mathbb{Z}}$-sheaves rather than of lcc $\mathbb{A}^f_\varnothing$-sheaves.

For general T, the category of lcc $\mathbb{A}^f_T$-sheaves is the inductive limit of the categories of lcc $\mathbb{A}^f_{T'}$-sheaves for $T' \subset T$ finite. One puts

$$\mathbb{Z}_\ell = \varprojlim_n \mathbb{Z}/(\ell^n), \qquad \mathbb{Q}_\ell = \mathbb{Z}_\ell \otimes \mathbb{Q}, \qquad \widehat{\mathbb{Z}} = (\mathbb{Z}_\ell)_\ell,$$

and uses the corresponding finite-adelic notation $\mathbb{A}^f_T$.

The stack of elliptic curves up to isogeny over S is the stack obtained from the stack of elliptic curves over S by formally inverting isogenies. One denotes by $\otimes \mathbb{Q}$ the functor associating to an elliptic curve its underlying elliptic curve up to isogeny. For S quasi-compact, one has

$$\mathrm{Hom}(E, F) \otimes \mathbb{Q} \xrightarrow{\sim} \mathrm{Hom}(E \otimes \mathbb{Q}, F \otimes \mathbb{Q}),$$

and, for S normal, every elliptic curve up to isogeny over S underlies an elliptic curve over S.

(3.2) Let $f : E \to S$ be an elliptic curve over a scheme S. One denotes by $T_\ell(E)$ the projective system of kernels E_{ℓ^n} of multiplication by ℓ^n on E, the transition maps from E_{ℓ^n} to E_{ℓ^m}, for $n \geq m$, being multiplication by ℓ^{n-m}. Proceeding in the same way for $\mathbb{G}_m$, one puts $T_\ell(\mathbb{G}_m) = \mathbb{Z}_\ell(1)$. If ℓ is invertible on S, then $T_\ell(E)$ and $\mathbb{Z}_\ell(1)$ are $\mathbb{Z}_\ell$-sheaves on S. One defines $T_\infty(E)$ to be the relative Lie algebra of E over S, that is, the invertible dual of the invertible sheaf ω of (2.1(a)).

Suppose S is of characteristic 0. One then defines the $\widehat{\mathbb{Z}}$-sheaf $T_f(E)$ over S to be the system of the $T_\ell(E)$, and puts

$$V_f(E) = T_f(E) \otimes \mathbb{A}_f.$$

If $u : E \to F$ is an isogeny, then u induces an isomorphism of $V_f(E)$ onto $V_f(F)$ and of $T_\infty(E)$ onto $T_\infty(F)$; the functors V_f and T_∞ therefore factor through the category of elliptic curves up to isogeny over S.

Proposition (3.3)**.** *Let S be a scheme of characteristic 0; let $E_1(S)$ be the category of elliptic curves over S; and let $E_2(S)$ be the category of triples composed of an elliptic curve up to isogeny E over S, a $\widehat{\mathbb{Z}}$-sheaf T which is a twisted form of $\widehat{\mathbb{Z}}^2$, and an isomorphism*

$$\beta : V_f(E) \xrightarrow{\sim} T \otimes \mathbb{A}_f.$$

Then the functor

$$I : E \longmapsto \big(E \otimes \mathbb{Q},\ T_f(E),\ V_f(E) \simeq T_f(E) \otimes \mathbb{A}_f\big)$$

from $E_1(S)$ to $E_2(S)$ is an equivalence of categories.

The question is local on S, which one may suppose quasi-compact. If $f : E \to F$ is a morphism of S-elliptic curves, and if f is an isogeny, then one has an exact sequence

$$0 \to T_f(E) \to T_f(F) \to \ker(f) \to 0. \tag{3.4}$$

The morphism f is divisible by n if and only if it kills the kernel E_n of multiplication by n, because the multiplication-by-n map $E/E_n \to E$ is an isomorphism. By (3.4), this holds if and only if $T_f(f)$ is divisible by n, and one deduces from this that $\mathrm{Hom}_S(E, F)$ is the subgroup of $\mathrm{Hom}_S(E \otimes \mathbb{Q}, F \otimes \mathbb{Q})$ consisting of the morphisms f such that $V_f(f)$ sends $T_f(E)$ into $T_f(F)$. The functor I is thus fully faithful.

Let $X \in \mathrm{Ob}(E_2(S))$. Locally on S, X is defined by an elliptic curve up to isogeny $E \otimes \mathbb{Q}$, and by a lattice T in $V_f(E)$ which, for almost all ℓ, coincides with $T_\ell(E)$. For $q \in \mathbb{Q}$, $(E \otimes \mathbb{Q}, T)$ is isomorphic to $(E \otimes \mathbb{Q}, qT)$, which allows us to suppose that $T_f(E) \subset T$. The quotient $K = T/T_f(E)$ is then canonically isomorphic to a finite subgroup of E, and X is the image of E/K under I (cf. 3.4). ☐

Proposition (3.5, Corollary)**.** *The functor $F_1(S)$, respectively $F_1'(S)$, which associates with each scheme S of characteristic 0 the set of isomorphism classes of elliptic curves over S equipped with an isomorphism*

$$\alpha : T_f(E) \xrightarrow{\sim} \widehat{\mathbb{Z}}^2$$

respectively also with an isomorphism

$$\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathbb{G}_a,$$

is isomorphic to the functor $F_2(S)$, respectively $F_2'(S)$, which associates with S the set of isomorphism classes of elliptic curves up to isogeny over S, equipped with an isomorphism

$$\beta : V_f(F) \xrightarrow{\sim} \mathbb{A}_f^2$$

respectively also with an isomorphism

$$\beta_\infty : T_\infty(F) \xrightarrow{\sim} \mathbb{G}_a.$$

Proposition (3.6)**.** *The functor F_1 (respectively F_1') is represented by a scheme M_∞ (respectively M_∞') over $\mathbb{Q}$.*

Let $n \geq 3$ be an integer. The functor which associates with each scheme S the set of isomorphism classes of elliptic curves, equipped with an isomorphism

$$\alpha_n : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$$

respectively also with $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_X$, is represented by an affine curve M_n (respectively by an affine surface M_n') over $\mathrm{Spec}(\mathbb{Z}[1/n])$. For $n \mid m$, the morphism from M_m to M_n defined by

$$(E, \alpha_m : E_m \xrightarrow{\sim} (\mathbb{Z}/m)^2) \longmapsto (E, (m/n)\alpha_m : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2)$$

is finite and étale over $\mathrm{Spec}(\mathbb{Z}[1/m])$, and one has

$$M_\infty = \varprojlim M_n.$$

One proceeds in the same way to represent F_1'. ☐

(3.7) The scheme M_∞ (respectively M'_∞) is equipped with a universal elliptic curve $f_\infty : E_\infty \to M_\infty$ and an isomorphism

$$\alpha : T_f(E_\infty) \xrightarrow{\sim} \widehat{\mathbb{Z}}^2,$$

respectively also with an isomorphism

$$\alpha_\infty : T_\infty(E'_\infty) \xrightarrow{\sim} \mathbb{G}_a.$$

Following (3.5), the scheme M_∞ (respectively M'_∞) represents the functor F_2 (respectively F'_2), which gives an evident left action of the adelic group $\mathrm{GL}_2(\mathbb{A}_f)$ on

$$(M_\infty, E_\infty \otimes \mathbb{Q}, \alpha \otimes \mathbb{A}_f^2)$$

respectively on

$$(M'_\infty, E'_\infty, \alpha \otimes \mathbb{A}_f^2, \alpha_\infty).$$

On the functor this action is given, for $g \in \mathrm{GL}_2(\mathbb{A}_f)$, by

$$g : (F, \beta : V_f(E) \xrightarrow{\sim} \mathbb{A}_f^2, \beta_\infty) \longmapsto (F, g \circ \beta : V_f(E) \xrightarrow{\sim} \mathbb{A}_f^2, \beta_\infty).$$

Shafarevich first noted this fact.

Let Y be a scheme over $\mathbb{C}$, equal to the projective limit of schemes Y_i of finite type over $\mathbb{C}$, with finite transition maps. The locally compact ringed space Y^{an}, equal to the projective limit of the Y_i^{an}, depends only on Y and not on its representation as a projective limit. If Y is a scheme over $\mathbb{Q}$, equal to the projective limit of schemes Y_i of finite type over $\mathbb{Q}$ with finite transition maps, one puts $Y^{\mathrm{an}} = (Y \otimes \mathbb{C})^{\mathrm{an}}$. This applies to M_∞ and M'_∞.

Proposition (3.8)**.** *One has canonically*

$$M'^{\mathrm{an}}_\infty \simeq \mathrm{Hom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2) / \mathrm{GL}_2(\mathbb{Q})$$

and

$$M^{\mathrm{an}}_\infty \simeq \mathbb{C}^\times \backslash \mathrm{Hom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2) / \mathrm{GL}_2(\mathbb{Q}),$$

and, less canonically,

$$M'^{\mathrm{an}}_\infty \simeq \mathrm{GL}_2(\mathbb{A}) / \mathrm{GL}_2(\mathbb{Q}), \qquad M^{\mathrm{an}}_\infty \simeq K_\infty \backslash \mathrm{GL}_2(\mathbb{A}) / \mathrm{GL}_2(\mathbb{Q}),$$

where K_∞ is the stabilizer of a chosen complex structure at the real place. These isomorphisms are compatible with the action of $\mathrm{GL}_2(\mathbb{A}_f)$*.*

The notion of elliptic curve up to isogeny, and its variants, extends to the complex analytic case. On the other hand, an isogeny $\varphi : E \to F$ induces an isomorphism φ^* between the local systems of rational cohomology of E and F, which allows us to define this last object for an elliptic curve up to isogeny. Let S be a complex analytic space. With the aid of (2.1), one sees that to give an elliptic curve up to isogeny over S is equivalent to giving an invertible sheaf T_∞, a local system $T_\mathbb{Q}$ of $\mathbb{Q}$-vector spaces, and a morphism $\alpha : T_\mathbb{Q} \to T_\infty$ inducing an isomorphism $T_\mathbb{Q} \otimes \mathbb{R} \to T_\infty$ point by point.

Let n be an integer and let K_n be the kernel of the natural mapping

$$\prod_\ell \mathrm{GL}_2(\mathbb{Z}_\ell) \to \mathrm{GL}_2(\mathbb{Z}/n).$$

Let G_1 be the functor which associates with S the set of isomorphism classes of elliptic curves $f : E \to S$ over S, equipped with an isomorphism

$$\varphi : \mathbb{Q}^2 \xrightarrow{\sim} T_\mathbb{Q}(E),$$

an isomorphism $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$, and an isomorphism $\alpha_n : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$. As in (3.3), one sees that G_1 is isomorphic to the functor G_2 which associates with S the set of isomorphism classes of elliptic curves up to isogeny E over S, equipped with $\varphi : \mathbb{Q}^2 \xrightarrow{\sim} T_\mathbb{Q}(E)$, $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$, and an isomorphism $V_f(E) \xrightarrow{\sim} \mathbb{A}_f^2$ given locally over S up to composition with an element of K_n.

Such an object is determined by the composite map φ' (given locally modulo K_n) obtained from

$$\varphi' : \mathbb{Q}^2 \xrightarrow{\varphi} T_\mathbb{Q}(E) \longrightarrow T_\infty(E) \times V_f(E) \xrightarrow{\sim} \mathcal{O}_S \times \mathbb{A}_f^2.$$

One has

$$E = \widehat{\mathbb{Z}}^2 \backslash \mathcal{O}_S \times \mathbb{A}_f^2 / \varphi'\big(\mathbb{Q}^2 \cap \varphi'^{-1}(T_\infty(E) \times T_f(E))\big),$$

so that, cf. 2.2, G_1 and G_2 are represented by

$$K_n \backslash \operatorname{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2).$$

Suppose still that $n \geq 3$, so that $\mathrm{GL}_2(\mathbb{Q})$ acts freely on the preceding space. The analytic space M_n^{an}, respectively $M_n'^{\mathrm{an}}$, represents the functor analogous in analytic geometry to the functor represented by M_n, respectively M_n', because this functor, say X, is representable and the arrow $X \to M_n^{\mathrm{an}}$, respectively $X \to M_n'^{\mathrm{an}}$, induces a bijection on sets of points with values in any finite-rank $\mathbb{C}$-algebra. Consequently one deduces from the preceding that

$$M_n'^{\mathrm{an}} \simeq K_n \backslash \operatorname{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2) / \mathrm{GL}_2(\mathbb{Q}).$$

Proceeding in the same way for M_n, one obtains the first assertion of (3.8) by passage to the limit on n.

A point x of

$$\operatorname{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2) / \mathrm{GL}_2(\mathbb{Q})$$

is identified with a lattice L_x of $\mathbb{C} \times \mathbb{A}_f^2$, and the corresponding curve is

$$E_x \simeq \widehat{\mathbb{Z}}^2 \backslash \mathbb{C} \times \mathbb{A}_f^2 / L_x,$$

equipped with

$$V_f(E) \simeq L_x \otimes \mathbb{A}_f \xrightarrow{\sim} \mathbb{A}_f^2.$$

The last assertion of (3.8) follows easily. $\square$

Denote by $f_n : E \to M_n$ the universal elliptic curve over M_n. The integer k being fixed, one makes the following definition.

Definition (3.9)**.** *One denotes by W (or by ${}_kW$ if confusion is possible) the $\mathbb{Q}$-vector space*

$$W = \varinjlim_n \widetilde{H}^1(M_n^{\mathrm{an}}, \operatorname{Sym}^k(R^1 f_* \mathbb{Q})).$$

This vector space does not depend on the universal elliptic curve, up to isogeny, $f_\infty : E \to M_\infty$, so that, by transport of structure, it is equipped with a left action of the adelic group $\mathrm{GL}_2(\mathbb{A}_f)$.

If ℓ is a prime number, the vector space $W_\ell = W \otimes \mathbb{Q}_\ell$ admits a purely algebraic definition, in terms of the ℓ-adic cohomology of the scheme over the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ obtained from M_n by extension of scalars:

$$W_\ell = \varinjlim_n \widetilde{H}^1(M_n \otimes \overline{\mathbb{Q}}, \operatorname{Sym}^k(R^1 f_{n*} \mathbb{Q}_\ell)). \tag{3.10}$$

Thus the Galois group of $\overline{\mathbb{Q}}$ over $\mathbb{Q}$ acts, by transport of structure, on W_ℓ and on the finite-level spaces.

Finally, the space M_n^{an} is the disjoint union of quotients of the Poincaré half-plane by congruence subgroups of $\mathrm{SL}_2(\mathbb{Z})$, so that, denoting by ω the invertible sheaf on M_n defined by E, Shimura's theory (2.10) gives

$$W_\infty = W \otimes \mathbb{C} = \varinjlim_n \left(H^0(M_n^{\mathrm{an}}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(M_n^{\mathrm{an}}, \Omega^1 \otimes \omega^k)} \right). \tag{3.11}$$

This decomposition of $W \otimes \mathbb{C}$ into two complex-conjugate subspaces, one of which is the space of all cusp forms of weight $k+2$ relative to a general congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$, is analogous to a Hodge decomposition, of type

$$(0, k+1) + (k+1, 0).$$

The action of the adelic group commutes with the action of the Galois group and respects the preceding decomposition.

While the ℓ-adic local system $R^1 f_* \mathbb{Q}_\ell$ is trivial on M_∞, I do not know whether there is a relation between W_ℓ and

$$\varinjlim_n \big(\widetilde{H}^1(M_n \otimes \overline{\mathbb{Q}}, \mathbb{Q}_\ell)\big) \otimes \operatorname{Sym}^k(\mathbb{Q}_\ell^2).$$

(3.12) Let $n \geq 3$ be an integer and let K_n be as in (3.8). Then one has

$$W^{K_n} = {}_nW.$$

This follows by passage to the limit and from the fact that, in rational cohomology, the cohomology of the quotient of a space by a finite group is obtained by taking the invariants of that group in the cohomology.

Put, for p prime,

$$W^{(p)} = W^{\mathrm{GL}_2(\mathbb{Z}_p)}.$$

By passage to the limit, one obtains

$$W^{(p)} = \varinjlim_{(n,p)=1} {}_nW.$$

On this cohomology space there again act:

(i) the subgroup $\prod_{\ell \neq p} \mathrm{GL}_2(\mathbb{Q}_\ell)$ of $\mathrm{GL}_2(\mathbb{A}_f)$, because this subgroup centralizes $\mathrm{GL}_2(\mathbb{Z}_p)$;

(ii) the Hecke algebra $\mathcal{H}(\mathrm{GL}_2(\mathbb{Q}_p), \mathrm{GL}_2(\mathbb{Z}_p))$, which is the algebra of integral measures on the discrete space $\mathrm{GL}_2(\mathbb{Q}_p)/\mathrm{GL}_2(\mathbb{Z}_p)$ invariant on the left by $\mathrm{GL}_2(\mathbb{Z}_p)$. This subalgebra of the group algebra of $\mathrm{GL}_2(\mathbb{Q}_p)$ acts on W and respects $W^{(p)}$. It acts already on each ${}_nW$ for n prime to p.

The Hecke algebra admits as a basis the measures associated with characteristic functions of the double cosets of $\mathrm{GL}_2(\mathbb{Z}_p)$ in $\mathrm{GL}_2(\mathbb{Q}_p)$, and one knows that

$$\mathcal{H}(\mathrm{GL}_2(\mathbb{Q}_p), \mathrm{GL}_2(\mathbb{Z}_p)) = \mathbb{Z}[T_p, R_p, R_p^{-1}],$$

where T_p and R_p are the double cosets of

$$\begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} p^{-1} & 0 \\ 0 & p^{-1} \end{pmatrix}.$$

(3.13) Let p be a prime number, let $n \geq 3$ be an integer prime to p, and let $F_{n,p}$ be the functor which associates to each scheme S the set of isomorphism classes of commutative diagrams of S-schemes

$$\begin{array}{ccccc} & & \mathbb{Z}/(n)^2 & & \\ & \alpha \nearrow & & \nwarrow \alpha' & \\ E_n & \longrightarrow & F_n & & \\ \cap & & \cap & & \\ E & \xrightarrow{\varphi} & F & & \end{array} \tag{3.14}$$

where φ is a p-isogeny between elliptic curves and α is an isomorphism. One denotes by q_1 and $q_2 : F_{n,p} \to M_n$ the morphisms of functors associating to a diagram (3.14) the subdiagrams (E, E_n, α) and (F, F_n, α').

Proposition (3.15). *The functor $F_{n,p}$ is represented by a scheme $M_{n,p}$, and the morphisms $q_1, q_2 : M_{n,p} \to M_n$ are finite.*

The automorphism σ of $F_{n,p}$ sending $\varphi : E \to F$ to ${}^t\varphi : F \to E$ exchanges q_1 and q_2; it suffices therefore to consider q_1. This morphism identifies $F_{n,p}$ with the functor of subgroups of order p of the universal elliptic curve E over M_n, so that, by the theory of Hilbert schemes, $F_{n,p}$ is representable and $M_{n,p}$ is proper over M_n. If s is a geometric point of M_n, $q_1^{-1}(s)$ is the set of subgroups of order p of E_s, and has $p+1$ elements if $\mathrm{char}(k(s)) \neq p$, and only one, the kernel of Frobenius, if $\mathrm{char}(k(s)) = p$. ☐

One can show that $M_{n,p}$ is regular, and that q_1 and q_2 are finite flat; we do not use this delicate result, but content ourselves here to note that over $\mathrm{Spec}(\mathbb{Z}[1/p])$, each q_i makes $M_{n,p}$ an étale covering of degree $p+1$ of M_n.

The morphisms q_i can be inserted into a commutative diagram

$$\begin{array}{ccccc} q_1^*E & \xrightarrow{\varphi} & q_2^*E & & \\ \swarrow & & \searrow & & \\ E & \xleftarrow{u} & M_{n,p} & \xrightarrow{v} & E \\ \downarrow f_n & & \downarrow q_1 \quad \downarrow q_2 & & \downarrow f_n \\ M_n & & M_n & & M_n \end{array} \tag{3.16}$$

where (φ, u, v) is part of the universal diagram (3.14).

One denotes by I_p the morphism from M_n to M_n corresponding to the morphism of functors $(E,\alpha)\mapsto(E,\alpha/p)$:

$$\begin{array}{ccc} E & \longrightarrow & E \\ \downarrow & & \downarrow \\ M_n & \xrightarrow{I_p} & M_n \end{array} \qquad I_p^*(E,\alpha)=(E,\alpha/p). \tag{3.17}$$

I_p^* is an automorphism of $\widetilde{H}^i(M_n^{\mathrm{an}},\mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}))$. It is tiresome, but routine, to prove the following.

Proposition (3.18)**.**

(i) The endomorphism T_p of ${}_nW$ is expressed, with the aid of (3.16), as the composite map

$$\widetilde{H}^1(M_n^{\mathrm{an}},\mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q}))\xrightarrow{q_2^*}\widetilde{H}^1(M_{n,p}^{\mathrm{an}},\mathrm{Sym}^k(R^1 v_*\mathbb{Q}))$$
$$\xrightarrow{\varphi^*}\widetilde{H}^1(M_{n,p}^{\mathrm{an}},\mathrm{Sym}^k(R^1 u_*\mathbb{Q}))\xrightarrow{q_{1*}}\widetilde{H}^1(M_n^{\mathrm{an}},\mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q})).$$

where q_{1*} is the trace morphism for the covering q_1.

(ii) Similarly, $R_p=p^kI_p^*$.

□

The distrustful reader could forget the adelic preliminaries and define T_p by (i). When $n=1$ or 2, one puts ${}_nW=W^{K_n}$, so that

$${}_1W={}_nW^{\mathrm{GL}_2(\mathbb{Z}/(n))}.$$

If S_{k+2} denotes the space of cusp forms, for the group $\mathrm{SL}_2(\mathbb{Z})$, of weight $k+2$, the Shimura isomorphism (3.11) induces an isomorphism

$${}_1^kW_\infty={}_1^kW\otimes\mathbb{C}=S_{k+2}\oplus\overline{S}_{k+2}.$$

It is tiresome, but routine, to prove the following.

Proposition (3.19)**.** *The endomorphism T_p of ${}_1^kW_\infty$ is identified, via the Shimura isomorphism, with the direct sum of the Hecke operator on S_{k+2}, including the factor p^{k-1}, and its conjugate.*

□

(3.20) One has canonically

$$\bigwedge^2 R^1 f_{n*}\mathbb{Z}_\ell\simeq R^2 f_{n*}\mathbb{Z}_\ell\simeq\mathbb{Z}_\ell(-1),$$

so that $\mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)$ is equipped with a bilinear form, symmetric for k even and alternating for k odd, with values in $\mathbb{Z}_\ell(-k)$. The form induced by tensoring with $\mathbb{Q}_\ell$ is nondegenerate.

If F is an lcc $\mathbb{Q}_\ell$-sheaf over a smooth scheme X of pure dimension n over an algebraically closed field k, then Poincaré duality gives

$$H^i(X,F)^\vee\simeq H_c^{2n-i}(X,\mathrm{Hom}(F,\mathbb{Q}_\ell(n))),$$
$$H_c^i(X,F)^\vee\simeq H^{2n-i}(X,\mathrm{Hom}(F,\mathbb{Q}_\ell(n))),$$
$$\widetilde{H}^i(X,F)^\vee\simeq\widetilde{H}^{2n-i}(X,\mathrm{Hom}(F,\mathbb{Q}_\ell(n))).$$

Taking $X=\overline{M}_n$ and $F=\mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q}_\ell)$ in these considerations, one defines a nondegenerate bilinear form ${}_n(\ ,\ )$ on ${}_n^kW_\ell$ with values in $\mathbb{Q}_\ell(-k-1)$. This form is symmetric for k odd and alternating for k even. This is the ℓ-adic analogue of the Peterson inner product. If $n\mid m$, and if the covering $\psi: M_m\to M_n$ is of degree d, then one has

$${}_m(\psi^*x,\psi^*y)=d\cdot{}_n(x,y).$$

4 The Congruence Formula

In this section we fix integers $k \geq 0$ and $n \geq 3$, and prime numbers p and ℓ. We suppose that p is prime to ℓ and n. We denote by $f_n : E \to M_n$ the universal elliptic curve over M_n, equipped with $\alpha : E_n \xrightarrow{\sim} \mathbb{Z}/(n)^2$.

Whatever the scheme Y, one denotes by a the unique morphism of Y to $\operatorname{Spec}(\mathbb{Z})$ or, if necessary, to a subscheme of $\operatorname{Spec}(\mathbb{Z})$. If Y is separated and of finite type over $\operatorname{Spec}(\mathbb{Z})$, and if F is a $\mathbb{Z}_\ell$- or $\mathbb{Q}_\ell$-sheaf on Y, one denotes by $R^i a_*(Y, F)$, respectively $R^i a_!(Y, F)$, respectively $R^i \tilde{a}(Y, F)$, the $\mathbb{Z}_\ell$- or $\mathbb{Q}_\ell$-sheaf over $\operatorname{Spec}(\mathbb{Z})$ that is the ith higher direct image of F under a, respectively the ith direct image with proper supports, respectively $\operatorname{Im}(R^i a_!(Y, F) \to R^i a_*(Y, F))$. For $m \in \mathbb{N}$, one puts

$$Y[1/m] = Y \times \operatorname{Spec}(\mathbb{Z}[1/m]).$$

Theorem (4.1, Igusa [1])**.** *The scheme M_n can be compactified to a scheme of curves M_n^* that is projective and smooth over $\operatorname{Spec}(\mathbb{Z}[1/n])$, such that $M_n^* \setminus M_n$ is an étale covering of $\operatorname{Spec}(\mathbb{Z}[1/n])$.*

The schemes M_n are formally smooth, thus smooth over $\operatorname{Spec}(\mathbb{Z})$. The modular invariant j of the universal curve over M_n is a morphism of M_n to the affine line $\mathbb{A}^1$ over $\operatorname{Spec}(\mathbb{Z}[1/n])$. The morphism j is finite, and is an étale covering away from the sections 0 and 1728 of $\mathbb{A}^1$; in fact:

(a) Two elliptic curves over an algebraically closed field with the same j-invariant are isomorphic (e.g. [8] 6.3), thus the geometric fibers of j are finite. The schemes M_n and $\mathbb{A}^1$ being smooth of the same dimension over $\operatorname{Spec}(\mathbb{Z})$, j is quasi-finite and flat.

(b) If E is an elliptic curve over the field of fractions K of a discrete valuation ring R, of j-invariant in R, and whose points of order n are rational over K, then E has good reduction. The valuative criterion for properness shows thus that j is proper.

(c) If E and F are two elliptic curves over a scheme S, of the same j-invariant, and if j and $j-1728$ are invertible, then the scheme $\operatorname{Isom}(S; E, F)$ of isomorphisms between E and F is étale over S ([8] 6.3). In the diagram

$$\begin{array}{ccc} \operatorname{Isom}(M_n \times_{\mathbb{A}^1} M_n; \operatorname{pr}_1^* E, \operatorname{pr}_2^* E) & \sim & M_n \times \operatorname{GL}_2(\mathbb{Z}/(n)) \\ \downarrow u & & \downarrow v \\ M_n \times_{\mathbb{A}^1} M_n & \xrightarrow{\operatorname{pr}_1} & M_n, \end{array}$$

where $j \neq 0, 1728$, u and v are étale surjective, thus pr_1 is étale and, by flat descent, j is étale.

The section at infinity of the projective line $\mathbb{P}^1 \supset \mathbb{A}^1$ over $\operatorname{Spec}(\mathbb{Z}[1/n])$ is a regular divisor whose generic point is of characteristic 0 in a regular scheme. It follows from a theorem of Abyankhar (see [5]) that along this divisor $j = \infty$, M_n is moderately (tamely?) ramified over $\mathbb{P}^1$, and that the normalization M_n^* of $\mathbb{P}^1$ in M_n satisfies (4.1). □

It follows from the same theorem that the lcc $\mathbb{Z}_\ell$-sheaves $R^i f_{n*}\mathbb{Z}_\ell$ on $M_n[1/\ell]$ are moderately ramified at infinity. Hence, from (4.1) and the specialization theorems for ℓ-adic cohomology (see [5]), it follows that $R^i a_*(M_n, \operatorname{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$, $R^i a_!(M_n, \operatorname{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$, and therefore $R^i \tilde{a}(M_n, \operatorname{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$ are lcc $\mathbb{Z}_\ell$-sheaves over $\operatorname{Spec}(\mathbb{Z}[1/n, 1/\ell])$, whose formation is compatible with all base changes.

Proposition (4.2, Corollary)**.** *The Galois module ${}_nW_\ell$ is the fiber of the lcc $\mathbb{Q}_\ell$-sheaf*

$$R^1 \tilde{a}(M_n, \operatorname{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)) \otimes \mathbb{Q}_\ell$$

over $\operatorname{Spec}(\mathbb{Z}[1/n, 1/\ell])$ at the geometric point $\overline{\mathbb{Q}}$. It is unramified away from n and ℓ.

□

Let the two commutative diagrams over $M_n \otimes \mathbb{F}_p$

$$\begin{array}{ccccc} & & \mathbb{Z}/(n)^2 & & \\ & \alpha \nearrow & & \nwarrow \alpha^{(p)} & \\ E_n & \xrightarrow{F} & E_n^{(p)} & & \\ \cap & & \cap & & \\ E & \xrightarrow{F} & E^{(p)} & & \end{array} \qquad \begin{array}{ccccc} & & \mathbb{Z}/(n)^2 & & \\ & p\alpha^{(p)} \nearrow & & \nwarrow \alpha & \\ E_n^{(p)} & \xrightarrow{V} & E_n & & \\ \cap & & \cap & & \\ E^{(p)} & \xrightarrow{V} & E & & \end{array}$$

be written more briefly

$$F : (E, \alpha) \to (E^{(p)}, \alpha^{(p)}) \qquad \text{and} \qquad V : (E^{(p)}, p\alpha^{(p)}) \to (E, \alpha),$$

where F is the Frobenius morphism and V, its transpose, is the Verschiebung. These diagrams define morphisms Φ_1 and Φ_2 of $M_n \otimes \mathbb{F}_p$ to $M_{n,p}$. These morphisms are finite, considered as sections of q_1 and q_2, and define a morphism

$$\Phi = \Phi_1 \coprod \Phi_2 : M_n \otimes \mathbb{F}_p \coprod M_n \otimes \mathbb{F}_p \to M_{n,p} \otimes \mathbb{F}_p.$$

Let Φ^h be the restriction of Φ to the opens M_n^h and $M_{n,p}^h$ of $M_n \otimes \mathbb{F}_p$ and $M_{n,p} \otimes \mathbb{F}_p$ corresponding to curves of nonzero Hasse invariant h.

Proposition (4.3)**.** Φ^h *is an isomorphism.*

Let $\varphi : E_1 \to E_2$ be a p-isogeny between elliptic curves of invertible Hasse invariant over a scheme S of characteristic p. At each geometric point of S, either the kernel $\ker(\varphi)$ of φ is étale over S, or its Cartier dual, isomorphic to $\ker({}^t\varphi)$, is étale over S. The property "$\ker(\varphi)$ is étale" is an open property, so that locally on S, either $\ker(\varphi)$ is purely infinitesimal, or $\ker({}^t\varphi)$ is. The only infinitesimal subgroup of order p of E_1 or E_2 being the kernel of Frobenius, in the first case φ is isomorphic to $F : E_1 \to E_1^{(p)}$, and in the second case ${}^t\varphi$ is isomorphic to $F : E_2 \to E_2^{(p)}$ and φ to $V : E_2^{(p)} \to E_2$. □

Proposition (4.4)**.**

(i) The scheme $M_{n,p}$ *is smooth over* $\mathrm{Spec}(\mathbb{Z})$ *away from points of characteristic* p *where* $h = 0$*.*

(ii) The morphisms q_1 *and* q_2 *induce finite flat morphisms* q_1' *and* q_2' *from the normalization* $M'_{n,p}$ *of* $M_{n,p}$ *to* M_n*.*

(iii) The morphism Φ *can be factored through a surjective morphism*

$$\Phi' : M_n \otimes \mathbb{F}_p \coprod M_n \otimes \mathbb{F}_p \to M'_{n,p} \otimes \mathbb{F}_p.$$

The automorphism σ of (3.15) exchanges φ and ${}^t\varphi$, so that it suffices to prove (i) at the points of characteristic p of $M_{n,p}$ where the kernel of φ is infinitesimal: there is no obstruction to infinitesimally lifting an elliptic curve and the infinitesimal part of the kernel of multiplication by p.

Where $p = h = 0$, the fiber of the finite morphism (3.15) $q_i : M_{n,p} \to M_n$ is reduced to a point, so that the open of smoothness of $M_{n,p}$ is dense in $M_{n,p}$ and $M'_{n,p}$ is everywhere of dimension 2. The scheme M_n being regular, following EGA 0_{IV} 16.5.1 and 17.3.5(ii), the morphism $q_i : M'_{n,p} \to M_n$ is flat. Finally, (iii) results from the fact that Φ is finite and $M_n \otimes \mathbb{F}_p$ is a normal curve. □

The Hecke endomorphism T_p of ${}_nW_\ell$, such as is made explicit in (3.18), is the tensorization with $\mathbb{Q}_\ell$ of the fiber at the geometric point $\overline{\mathbb{Q}}$ of $\mathrm{Spec}(\mathbb{Z}[1/n, 1/\ell])$ of the endomorphism, again denoted by T_p, of $R^1\widetilde{a}(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$ defined by the correspondence

$$\begin{array}{ccccc} q_1'^*E & \xrightarrow{\varphi} & q_2'^*E & & \\ \swarrow & & \searrow & & \\ E & \xleftarrow{u} & M'_{n,p} & \xrightarrow{v} & E \\ \downarrow f_n & & \downarrow q_1' \quad \downarrow q_2' & & \downarrow f_n \\ M_n & & M_n & & M_n, \end{array} \tag{4.5}$$

with

$$T_p = q'_{1*}\, \varphi^*\, q_2'^* \qquad \text{(cf. 3.18)}.$$

The endomorphisms R_p and I_p can be interpreted in the same way.

Lemma (4.6)**.** *Let* X, Y, Z_1*, and* Z_2 *be four schemes that are separated and of finite type over a noetherian scheme* S*, let* F *be a* $\mathbb{Z}_\ell$*-sheaf on* X*, let* G *be a* $\mathbb{Z}_\ell$*-sheaf on* Y*, and denote by* a *each of the structural maps from* X, Y, Z_1 *or* Z_2 *to* S*. Suppose we have the following commutative diagram of* S*-schemes and morphisms of sheaves:*

$$\begin{array}{ccc} y_1^*G & \xrightarrow{z_1} & x_1^*F \\ & & \\ Z_1 & \xrightarrow{f} & Z_2 \\ \downarrow x_1 & & \downarrow x_2 \\ X & & X \end{array} \qquad \begin{array}{ccc} y_2^*G & \xrightarrow{z_2} & x_2^*F \\ & & \\ Z_1 & \xrightarrow{f} & Z_2 \\ \downarrow y_1 & & \downarrow y_2 \\ & Y & \end{array}$$

Suppose that $f^*z_2 = z_1$*, that* y_1 *and* y_2 *are proper, that* x_1 *and* x_2 *are finite and flat, and that, for every geometric point* s *of* Z_2*, the multiplicity of* s *in its fiber* $x_2^{-1}(x_2(s))$ *is equal to the sum of the multiplicities in their fibers,* *for x_1, of the geometric points of Z_1 in the inverse image of s under f. Then the diagram*

$$\begin{array}{ccccccc}
R^i\widetilde{a}(Y,G) & \xrightarrow{y_1^*} & R^i\widetilde{a}(Z_1,G) & \xrightarrow{z_1} & R^i\widetilde{a}(Z_1,F) & \xrightarrow{x_{1*}} & R^i\widetilde{a}(X,F) \\
\| & & \uparrow f^* & & \uparrow f^* & & \| \\
R^i\widetilde{a}(Y,G) & \xrightarrow{y_2^*} & R^i\widetilde{a}(Z_2,G) & \xrightarrow{z_2} & R^i\widetilde{a}(Z_2,F) & \xrightarrow{x_{2*}} & R^i\widetilde{a}(X,F)
\end{array}$$

commutes.

This lemma results from the analogous lemmas for $R^ia_!$ and R^ia_*. The commutativity of the first squares is trivial. The last is rewritten

$$\begin{array}{ccccc}
R^i\widetilde{a}(Z_1,x_1^*F) & \xleftarrow{\sim} & R^i\widetilde{a}(X,x_{1*}x_1^*F) & \xrightarrow{\mathrm{Tr}} & R^i\widetilde{a}(X,F) \\
\uparrow & & \uparrow & & \| \\
R^i\widetilde{a}(Z_2,x_2^*F) & \xleftarrow{\sim} & R^i\widetilde{a}(X,x_{2*}x_2^*F) & \xrightarrow{\mathrm{Tr}} & R^i\widetilde{a}(X,F),
\end{array}$$

and one recalls the definition of the trace in verifying that the square

$$\begin{array}{ccc}
x_{1*}x_1^*F & \xrightarrow{\mathrm{Tr}} & F \\
\uparrow & & \| \\
x_{2*}x_2^*F & \xrightarrow{\mathrm{Tr}} & F
\end{array}$$

is commutative. ☐

(4.7) One denotes by $T_p/\mathbb{F}_p$ the endomorphism induced by T_p on the restriction to $\mathrm{Spec}(\mathbb{F}_p)$ of the lcc $\mathbb{Z}_\ell$-sheaf $R^1\widetilde{a}(M_n,\mathrm{Sym}^k R^1f_{n*}\mathbb{Z}_\ell)$. One has

$$R^1\widetilde{a}(M_n,\mathrm{Sym}^k R^1f_{n*}\mathbb{Z}_\ell)|\,\mathrm{Spec}(\mathbb{F}_p) \xrightarrow{\sim} R^1\widetilde{a}(M_n\otimes\mathbb{F}_p,\mathrm{Sym}^k R^1f_{n*}\mathbb{Z}_\ell);$$

the formation of the trace morphism for a finite flat morphism is compatible with change of base, so that $T_p/\mathbb{F}_p$ can be constructed, over the model (3.18), starting from the fiber over $\mathbb{F}_p$ of the correspondence (4.5). Lemma (4.6), applied to the commutative diagrams

$$\begin{array}{ccc}
M_n\otimes\mathbb{F}_p\coprod M_n\otimes\mathbb{F}_p & \xrightarrow{\Phi'} & M'_{n,p}\otimes\mathbb{F}_p \\
\downarrow & \swarrow q'_1 & \\
M_n\otimes\mathbb{F}_p & &
\end{array}
\qquad
\begin{array}{ccc}
M_n\otimes\mathbb{F}_p\coprod M_n\otimes\mathbb{F}_p & \xrightarrow{\Phi'} & M'_{n,p}\otimes\mathbb{F}_p \\
& q'_2 \searrow & \downarrow \\
& & M_n\otimes\mathbb{F}_p,
\end{array}$$

fournishes then a decomposition of $T_p/\mathbb{F}_p$ into the sum of endomorphisms defined by the two following correspondences:

$$(a)\qquad \begin{array}{ccccc}
& (E,\alpha) & \xrightarrow{F} & (E^{(p)},\alpha^{(p)}) = F^*(E,\alpha) & \\
& \swarrow & & \searrow & \\
(E,\alpha) & & M_n\otimes\mathbb{F}_p & & (E,\alpha) \\
& \searrow & & \swarrow F & \\
& M_n\otimes\mathbb{F}_p & & M_n\otimes\mathbb{F}_p &
\end{array}$$

where F is the absolute Frobenius. One recognizes in this correspondence the geometric Frobenius

$$(b)\qquad \begin{array}{ccccc}
& x^*(E,\alpha) = (E^{(p)},p\alpha^{(p)}) & \xrightarrow{V} & (E,\alpha) & \\
& \swarrow & & \searrow & \\
(E,\alpha) & & M_n\otimes\mathbb{F}_p & & (E,\alpha) \\
& \searrow x & & \swarrow & \\
& M_n\otimes\mathbb{F}_p & & M_n\otimes\mathbb{F}_p. &
\end{array}$$

The map x is the composite map $I_p^{-1}\circ F$:

$$\begin{array}{ccccc}
(E,\alpha) & \longleftarrow & (E,p\alpha) & \longleftarrow & (E^{(p)},p\alpha^{(p)}) \\
\downarrow & & \downarrow & & \downarrow \\
M_n\otimes\mathbb{F}_p & \xleftarrow{I_p^{-1}} & M_n\otimes\mathbb{F}_p & \xleftarrow{F} & M_n\otimes\mathbb{F}_p.
\end{array}$$

The corresponding endomorphism is thus the composite of

$$V : R^1\widetilde{a}(M_n \otimes \mathbb{F}_p, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)) \xrightarrow{V^*} R^1\widetilde{a}(M_n \otimes \mathbb{F}_p, \mathrm{Sym}^k(R^1 f^{(p)}_{n*}\mathbb{Z}_\ell)) \xrightarrow{\mathrm{Tr}_F} R^1\widetilde{a}(M_n \otimes \mathbb{F}_p, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$$

and of

$$I_p^* = \mathrm{Tr}_{I_p^{-1}} : \quad \text{endomorphism of } R^1\widetilde{a}(M_n \otimes \mathbb{F}_p, \mathrm{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell).$$

Proposition (4.8)**.** *One has $T_p/\mathbb{F}_p = F + I_p^* V$, and:*

(i) F can be identified with the inverse of the arithmetic Frobenius element φ_p of the Galois group $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ acting on $\widetilde{H}^1(M_n \otimes \overline{\mathbb{F}}_p, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$;

(ii) F and V are transposes of one another, relative to the scalar product (3.20);

(iii) $FV = VF = p^{k+1}$.

For the relation (i) between the geometric and arithmetic Frobenius, one is referred to the exposé of C. Houzel (SGA 5.XV). The composite VF is the composite of the homomorphisms obtained from the following maps:

$$\begin{array}{ccccc} E & \xleftarrow{F_E} & E^{(p)} & \xrightarrow{V_E} & E \\ \downarrow & & \downarrow & & \downarrow \\ M_n & \xrightarrow{F} & M_n & \xrightarrow{F} & M_n. \end{array}$$

Thus

$$VF = \mathrm{Tr}_F \circ F_E^* \circ V_E^* \circ F^*.$$

The map $F_E^* V_E^* = (F_E V_E)^* = (p \cdot 1_E)^*$ acts by multiplication by p^k on $\mathrm{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell$, so that

$$VF = p^k \,\mathrm{Tr}_F \circ F^* = p^k \cdot p = p^{k+1},$$

because $F : M_n \to M_n$ is of degree p.

By transport of structure, φ_p respects the scalar product (3.20) with values in $\mathbb{Q}_\ell(-k-1)$, a group on which φ_p acts by multiplication by p^{-k-1}. One has thus

$$(Fx, y) = p^{k+1}(\varphi_p F x, \varphi_p y) = (x, p^{k+1}F^{-1}y) = (x, Vy).$$

□

The following theorem, synonymous with (4.8), goes back to Eichler.

Theorem (4.9, The Congruence Formula)**.** *Let $K_{n,\ell}$ be the largest subextension of $\overline{\mathbb{Q}}$ that is unramified away from n and ℓ, let φ_p be a Frobenius element relative to p in $\mathrm{Gal}(K_{n,\ell}/\mathbb{Q})$, let F be the endomorphism φ_p^{-1} of ${}_nW_\ell$, and let V be the transpose of F relative to the scalar product (3.20). Then*

$$T_p = F + I_p^* V, \qquad FV = p^{k+1}, \qquad 1 - T_p X + pR_p X^2 = (1 - FX)(1 - I_p^* V X).$$

□

5 Weil implies Ramanujan

If p is a prime number and X a scheme over $\mathbb{F}_p$, then one denotes by $\overline{\mathbb{F}}_p$ an algebraic closure of $\mathbb{F}_p$, by $F : X \to X$ the geometric Frobenius endomorphism, and one puts $\overline{X} = X \otimes \overline{\mathbb{F}}_p$; ℓ always denotes a prime number different from p.

By the Weil conjectures one means to claim the following: let X be a projective smooth scheme over $\mathbb{F}_p$ and let ℓ be a prime number different from p. Then the eigenvalues of the endomorphism F^* of $H^i(\overline{X}, \mathbb{Q}_\ell)$ are algebraic integers all of whose complex conjugates have absolute value $p^{i/2}$.

With the hypotheses and notations of (4.9), one has, recalling that $(p, n) = 1$:

Theorem (5.1)**.** *If the Weil conjectures are true, then the absolute values of the endomorphism F of ${}^k_nW_\ell$ are algebraic integers, all of whose complex conjugates are of absolute value $p^{k+1/2}$.*

Assume the Weil conjectures.

Lemma (5.2, modulo Weil)**.** *Let X be a smooth scheme over $\mathbb{F}_p$, which can be represented as an open in a smooth projective scheme X^*. Then the absolute values of the endomorphism F^* of $\widetilde{H}^i(X, \mathbb{Q}_\ell)$ are algebraic integers of absolute value $p^{i/2}$.*

The natural map of $H^i_c(X, \mathbb{Q}_\ell)$ to $H^i(X, \mathbb{Q}_\ell)$ can be factored through $H^i(X^*, \mathbb{Q}_\ell)$:

$$H^i_c(X, \mathbb{Q}_\ell) \to H^i(X^*, \mathbb{Q}_\ell) \to H^i(X, \mathbb{Q}_\ell),$$

so that, as a $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$-module, $\widetilde{H}^i(X, \mathbb{Q}_\ell)$ is the quotient of a subobject of $H^i(X^*, \mathbb{Q}_\ell)$. $\square$

Lemma (5.3, modulo Weil)**.** *Let S be a smooth scheme over $\mathbb{F}_p$ and $f : A \to S$ an abelian scheme over S. Suppose that A can be represented as an open in a projective smooth scheme A^* over $\mathbb{F}_p$. Then the geometric Frobenius F^* of $\widetilde{H}^i(S, R^j f_* \mathbb{Q}_\ell)$ has eigenvalues that are algebraic integers of absolute value $p^{i+j/2}$.*

Let m be an integer > 1, and consider the Leray spectral sequences

$$E : E_2^{ij} = H^i(\overline{S}, R^j f_* \mathbb{Q}_\ell) \Longrightarrow H^{i+j}(\overline{A}, \mathbb{Q}_\ell), \qquad {}_cE : {}_cE_2^{ij} = H^i_c(\overline{S}, R^j f_* \mathbb{Q}_\ell) \Longrightarrow H^{i+j}_c(\overline{A}, \mathbb{Q}_\ell).$$

The endomorphism of multiplication by m, $\psi_m = m \cdot 1_A$, defines endomorphisms of E and ${}_cE$ that fit into a commutative diagram

$$\begin{array}{ccc} {}_cE & \longrightarrow & E \\ \psi_m^* \downarrow & & \psi_m^* \downarrow \\ {}_cE & \longrightarrow & E. \end{array}$$

ψ_m^* acts on $R^j f_* \mathbb{Q}_\ell$ by multiplication by m^j, so that ψ_m^* acts by multiplication by m^j on the terms ${}_cE_r^{ij}$ and E_r^{ij} of ${}_cE$ and E. The maps d_r for $r \geq 2$ commute with ψ_m^*, and send E_r^{ij}, respectively ${}_cE_r^{ij}$, to $E_r^{i'j'}$, respectively ${}_cE_r^{i'j'}$, with $j \neq j'$. These are nonzero, and E_2^{ij}, respectively ${}_cE_2^{ij}$, can be identified with a subspace of $H^{i+j}(\overline{A}, \mathbb{Q}_\ell)$, respectively $H_c^{i+j}(\overline{A}, \mathbb{Q}_\ell)$, where $\psi_m^* = m^j$. Consequently, $\widetilde{H}^i(S, R^j f_* \mathbb{Q}_\ell)$ is identified, as a Galois module, with a subspace of $\widetilde{H}^{i+j}(A, \mathbb{Q}_\ell)$ where $\psi_m^* = m^j$, and one applies (5.2). The trick used here is due to Lieberman. $\square$

Let $f_n : E \to M_n \otimes \mathbb{F}_p$ be the universal elliptic curve over $M_n \otimes \mathbb{F}_p$, and let $f_{n,k} : E_k \to M_n \otimes \mathbb{F}_p$ be its k-fold fiber product with itself. The Künneth formula shows that the $\mathbb{Q}_\ell$-sheaf $R^k f_{n,k*} \mathbb{Q}_\ell$ admits as a direct factor the k-fold tensor power of $R^1 f_{n*} \mathbb{Q}_\ell$; the latter in turn contains as a direct factor the $\mathbb{Q}_\ell$-sheaf $\mathrm{Sym}^k(R^1 f_{n*} \mathbb{Q}_\ell)$. Theorem 5.1 results thus from (5.3) and the following lemma.

Lemma (5.4)**.** *The scheme E_k is an open in a smooth projective scheme E_k^* over $\mathbb{F}_p$.*

Let E^* be the Néron minimal model of E over $M_n^* \otimes \mathbb{F}_p$ (4.1). The scheme E^* is smooth and projective over $\mathbb{F}_p$. Since $n \geq 3$ and the points of order n of E form a trivial covering of $M_n \otimes \mathbb{F}_p$, the Néron model is semistable, case a or b_m in Néron's classification. In particular, the projection $f_n : E^* \to M_n^*$ has only a finite number of nonsmooth points, and at these points f_n is nondegenerate, presenting an ordinary quadratic singularity.

Let E_k^{**} be the k-fold fiber product of E^* over M_n^*. To prove (5.4), it suffices to resolve the singularities of E_k^{**} without touching the open E_k. We first prove:

Lemma (5.5)**.** *Let V be the subvariety of affine space over a field k, with coordinates $(X_i)_{0 \leq i \leq r}$, $(Y_i)_{0 \leq i \leq r}$, and $(T_i)_{1 \leq i \leq s}$, with equation*

$$X_0 Y_0 = X_1 Y_1 = \cdots = X_r Y_r.$$

Let $\mathfrak{m}$ be the ideal of $\mathcal{O}_V$ generated by the monomials obtained from the monomial $\prod_{i=0}^r X_i^i$ by a permutation of the coordinates which respects the set of pairs $\{X_i, Y_i\}$, $0 \leq i \leq r$. Then $\mathfrak{m} = \mathcal{O}_V$ away from the singular locus of V, and the variety $\widetilde{V}$ obtained from V by blowing up $\mathfrak{m}$ is smooth over k.

The singular locus is the place where the four coordinates X_i, Y_i, X_j, Y_j for $i \neq j$ vanish. The open affine of $\widetilde{V}$ defined by the element $\prod_{i=0}^r X_i^i$ of the ideal $\mathfrak{m}$ of blowing up is the spectrum of the regular ring

$$k[Y_0/X_1, X_0/X_1, X_1/X_2, \ldots, X_{r-1}/X_r, X_r, T_1, \ldots, T_s]$$

(for the proof, note that $X_i/X_{i+1} = Y_{i+1}/Y_i$), and 5.5 results. $\square$

One still shows that, locally for the étale topology, the singularities of E_k^{**} are isomorphic to those of V for $r = k - 1$, and that this allows the definition on E_k^{**} of an ideal $\mathfrak{m}$ analogous to the ideal $\mathfrak{m}$ of (5.5). Blowing up this ideal, one obtains E_k^*. $\square$

An approximation to the following theorem has been proven by Ihara [2].

Theorem (5.6). *The Weil conjectures imply the Ramanujan conjecture.*

We note first that (5.1) is true for $n = 1$, because ${}^k_1W_\ell$ is the Galois submodule of ${}^k_mW_\ell$ invariant under $\mathrm{GL}_2(\mathbb{Z}/(m))$. On ${}^k_1W_\ell$, I_p^* induces the identity, and (4.8) reduces to

$$1 - T_pX + p^{k+1}X^2 = (1 - FX)(1 - VX).$$

The endomorphisms F and V are transposes of each other, so that

$$\det(1 - FX; {}^k_1W_\ell) = \det(1 - VX; {}^k_1W_\ell).$$

The action of T_p on ${}^k_1W_\ell$ is induced by its action on ${}^k_1W$, and is compatible with the decomposition of ${}^k_1W \otimes \mathbb{C}$ into the sum of the space S_{k+2} of cusp forms relative to $\mathrm{SL}_2(\mathbb{Z})$ of weight $k + 2$, and of the complex conjugate space. From the Hermitian property for T_p, for the Peterson inner product, and from (3.19), one deduces then that

$$\det(1 - T_pX + p^{k+1}X^2; {}^k_1W_\ell) = \det(1 - T_pX + p^{k+1}X^2; S_{k+2})^2,$$

and

$$\det(1 - T_pX + p^{k+1}X^2; S_{k+2})^2 = \det(1 - FX; {}^k_1W_\ell)^2,$$

so that

$$\det(1 - T_pX + p^{k+1}X^2; S_{k+2}) = \det(1 - FX; {}^k_1W_\ell). \tag{5.7}$$

Resume the notations of section 1 and make $k = 10$. By virtue of Hecke's theory and of (3.19), (5.7) can be rewritten

$$H_p(X) = \det(1 - FX; {}^{10}_1W_\ell),$$

and one applies (5.1). □

One verifies in the same way that the Weil conjectures imply Petersson's generalization of the Ramanujan conjecture.

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GRIFFITHS'S WORK

by Pierre DELIGNE

This talk contains part of Griffiths's fundamental results on families of Hodge structures. It contains none of his results on algebraic cycles.

1 Hodge structures

Let $H_{\mathbb{R}}$ be a real vector space of finite dimension.

Definition (1.1)**.** *A real Hodge structure on $H_{\mathbb{R}}$ is a bigrading of*

$$H_{\mathbb{C}} = H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}, \qquad H_{\mathbb{C}} = \bigoplus_{p,q} H^{pq},$$

such that H^{pq} and H^{qp} are complex conjugates.

The Hodge numbers of a real Hodge structure H are the integers

$$h^{pq} = \dim_{\mathbb{C}}(H^{pq}).$$

One says that H has weight n if $h^{pq} = 0$ for $p + q \neq n$. A variant of 1.1 is the following.

Definition (1.2)**.** *A real Hodge structure of weight n on $H_{\mathbb{R}}$ is a finite decreasing filtration F of $H_{\mathbb{C}}$ (the Hodge filtration) such that, for $p + q = n + 1$, one has*

$$F^p(H_{\mathbb{C}}) \oplus \overline{F^q(H_{\mathbb{C}})} \xrightarrow{\sim} H_{\mathbb{C}}. \tag{1.2.1}$$

One passes from 1.1 to 1.2 by putting

$$F^p(H_{\mathbb{C}}) = \sum_{p' \geq p} H^{p'q'} \tag{1.2.2}$$

and one passes from 1.2 to 1.1 by putting, for $p + q = n$,

$$H^{pq} = F^p(H_{\mathbb{C}}) \cap \overline{F^q(H_{\mathbb{C}})}. \tag{1.2.3}$$

1.3. Let $\mathbb{S}$ denote the real algebraic group obtained by Weil restriction of scalars, from $\mathbb{C}$ to $\mathbb{R}$, of the multiplicative group $\mathbb{G}_m$. One has $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$. There is a natural morphism w from $\mathbb{G}_m$ to $\mathbb{S}$ which, on real points, is the inclusion of $\mathbb{R}^*$ into $\mathbb{C}^*$; and there is a natural morphism t from $\mathbb{S}$ to $\mathbb{G}_m$ which, on real points, is the norm. The composite tw is squaring. A new variant of 1.1 is the following.

Definition (1.4)**.** *A real Hodge structure on $H_{\mathbb{R}}$ is an action, defined over $\mathbb{R}$, of the algebraic group $\mathbb{S}$ on $H_{\mathbb{R}}$.*

If z denotes the defining isomorphism $z : \mathbb{S}(\mathbb{R}) \xrightarrow{\sim} \mathbb{C}^*$, one passes from 1.1 to 1.4 by defining H^{pq} as the subspace of $H_{\mathbb{C}}$ on which $\mathbb{S}(\mathbb{R})$ acts by multiplication by $z^p \overline{z}^q$.

If $H_{\mathbb{R}}$ is endowed with a real Hodge structure, the Weil automorphism C is defined as multiplication by i^{p-q} on H^{pq}. This automorphism is real and is simply the action of the element i of $\mathbb{S}(\mathbb{R}) \subset \mathbb{C}^*$.

Definition (1.5)**.** *A polarization of a real Hodge structure, of weight n, H, is a bilinear form ψ on $H_{\mathbb{R}}$, invariant under the compact subgroup* $\mathrm{Ker}(t)$ *of $\mathbb{S}$, and such that the form $\psi(x, Cy)$ is symmetric and positive definite.*

From the identity $C^2 = (-1)^n$, one obtains

$$\psi(x, y) = (-1)^n \psi(x, C^2 y) = (-1)^n \psi(Cy, Cx) = (-1)^n \psi(y, x),$$

so that ψ is alternating or symmetric according to the parity of n. The variance property under $\mathbb{S}$ and the positivity property are written as follows:

$$\text{the orthogonal, for } \psi, \text{ of } F^p(H_{\mathbb{C}}) \text{ is } F^{n-p}(H_{\mathbb{C}}); \tag{1.5.1}$$

$$\text{for } x \neq 0 \text{ in } H^{pq}, \text{ one has } i^{p-q}\psi(x,\overline{x}) > 0. \tag{1.5.2}$$

These are the Riemann bilinear relations.

Definition (1.6)**.** *(i) A Hodge structure, of weight n, H, consists of a free $\mathbb{Z}$-module of finite type $H_{\mathbb{Z}}$ (the "integral lattice") and a real Hodge structure of weight n on*

$$H_{\mathbb{R}} = H_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{R}.$$

(ii) A polarization of a Hodge structure, of weight n, H, is a polarization ψ of the underlying real Hodge structure, taking integral values on the integral lattice $H_{\mathbb{Z}}$.

2 Hodge theory

2.1. Let $X \subset \mathbb{P}^r(\mathbb{C})$ be a nonsingular projective complex algebraic variety, pure of dimension d, and let $\eta \in H^2(X,\mathbb{Z})$ be the cohomology class of a hyperplane section of X, that is, the first Chern class of $\mathcal{O}(1)$.

The holomorphic de Rham complex $\Omega_X^{\bullet}$ is a resolution of the constant sheaf $\mathbb{C}$ (the holomorphic Poincaré lemma), whence a spectral sequence

$$E_1^{pq} = H^q(X, \Omega_X^p) \Rightarrow H^{p+q}(X,\mathbb{C}). \tag{2.1.1}$$

2.2. Hodge theory asserts that:

(A) the spectral sequence (2.1.1) degenerates: $E_1 = E_\infty$;

(B) the filtration of $H^n(X,\mathbb{C})$ to which it abuts is a Hodge structure of weight n, in the sense of 1.2, on $H^n(X,\mathbb{R})$.

By (A), one has

$$H^{pq} \simeq H^q(X, \Omega_X^p).$$

(C) For $n \leq d$, let $P_{\mathbb{Q}}^n$ denote the kernel of the iterated cup-product

$$\eta^{d-n+1} : H^n(X,\mathbb{Q}) \longrightarrow H^{2d-n+2}(X,\mathbb{Q})$$

(the primitive part of cohomology), and let $P_{\mathbb{Z}}^n$ be the intersection of $P_{\mathbb{Q}}^n$ with the image of $H^n(X,\mathbb{Z})$ in $H^n(X,\mathbb{Q})$. Since the class η is of type $(1,1)$, $P_{\mathbb{R}}^n = P_{\mathbb{Z}}^n \otimes \mathbb{R}$ is a Hodge substructure of $H^n(X,\mathbb{R})$. Up to a sign depending only on d and n, the form

$$\psi(x,y) = \int \eta^{d-n} \wedge x \wedge y$$

is a polarization of the Hodge structure P^n.

(D) The natural map

$$\bigoplus_{n-2k \leq d} \eta^k \wedge : \bigoplus P_{\mathbb{Q}}^{n-2k} \longrightarrow H^n(X,\mathbb{Q})$$

is an isomorphism (the Hodge–Lepage decomposition).

3 Families of Hodge structures

3.1. Let $f : X \to S$ be a projective and smooth morphism of analytic spaces, of pure relative dimension d. For simplicity, we suppose that S is nonsingular and that f is provided with a factorization through $\mathbb{P}_S^r = S \times \mathbb{P}^r(\mathbb{C})$. The fibers $X_s = f^{-1}(s)$ therefore form an analytic family, parametrized by S, of nonsingular subvarieties of $\mathbb{P}^r(\mathbb{C})$.

3.2. From the C^∞ point of view, f is a locally trivial fibration. The cohomology algebra of the fibers of f therefore forms a local system on S, denoted

$$R^*_{\mathbb{Z}}(f) = \sum R^n f_* \mathbb{Z}.$$

For every abelian sheaf F on S (for example $\mathbb{Q}$, $\mathbb{C}$, $\mathcal{O}$), put

$$R^n_F(f) = F \otimes R^n_{\mathbb{Z}}(f). \tag{3.2.1}$$

Since f is proper, one also has

$$R^n_F(f) \simeq R^n f_*(f^*F).$$

The cohomology classes of the hyperplane sections of the X_s define a locally constant section

$$\eta \in H^0(S, R^2_{\mathbb{Z}}(f)). \tag{3.2.2}$$

For $n \leq d$, the $P^n_{\mathbb{Z}}(X_s)$ therefore form a local subsystem $P^n_{\mathbb{Z}}(f)$ of $R^n_{\mathbb{Q}}(f)$. Putting $P^n_F(f) = F \otimes P^n_{\mathbb{Z}}(f)$, one has by definition

$$P^n_{\mathbb{Q}}(f) = \operatorname{Ker}\bigl(\eta^{d-n+1} \wedge : R^n_{\mathbb{Q}}(f) \to R^{2d-n+2}_{\mathbb{Q}}(f)\bigr), \tag{3.2.3}$$

$$P^n_{\mathbb{Z}}(f) = P^n_{\mathbb{Q}}(f) \cap \operatorname{Im}\bigl(R^n_{\mathbb{Z}}(f) \to R^n_{\mathbb{Q}}(f)\bigr). \tag{3.2.4}$$

The polarization form of 2.2(C),

$$\psi(x, y) = \pm \int_{X_s} \eta^{d-n} \wedge x \wedge y, \tag{3.2.5}$$

is locally constant on S, that is, it defines

$$\psi : P^n_{\mathbb{Z}}(f) \otimes P^n_{\mathbb{Z}}(f) \longrightarrow \mathbb{Z}. \tag{3.2.6}$$

3.3. For F a coherent analytic sheaf on S, $f'F$ will denote its inverse image as a sheaf, and

$$f^*F = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_S} f'F$$

its inverse image as a sheaf of modules. For $s \in S$, $F(s)$ denotes the $\mathcal{O}_{S,s}$-module of germs of sections of F at s; the fiber of F at s is the vector space

$$F_s = F(s) \otimes_{\mathcal{O}_{S,s}} \mathbb{C}.$$

Recall finally that the relative de Rham complex $\Omega^\bullet_{X/S}$ is the quotient of $\Omega^\bullet_X$ with components

$$\Omega^1_{X/S} = \Omega^1_X / f^*\Omega^1_S, \qquad \Omega^p_{X/S} = \bigwedge^p \Omega^1_{X/S}.$$

If $T^1_{X/S}$, the dual of $\Omega^1_{X/S}$, is the relative tangent bundle, then one has contraction pairings

$$\lrcorner : T^1_{X/S} \otimes \Omega^p_{X/S} \longrightarrow \Omega^{p-1}_{X/S}. \tag{3.3.1}$$

3.4. By the relative holomorphic Poincaré lemma, $\Omega^\bullet_{X/S}$ is a resolution of $f^*\mathcal{O}_S$. Hence one has a spectral sequence

$$E_1^{pq} = R^q f_* \Omega^p_{X/S} \Rightarrow R^n f_*(f^*\mathcal{O}_S) = R^n_{\mathcal{O}}(f). \tag{3.4.1}$$

It follows from 2.2(A) that the E_1^{pq} are locally free, that the spectral sequence (3.4.1) degenerates, $E_1 = E_\infty$, and that its formation commutes with arbitrary base change. The filtration of $R^n_{\mathcal{O}}(f)$ to which it abuts (the Hodge filtration) therefore induces on

$$R^n_{\mathcal{O}}(f)_s \simeq H^n(X_s, \mathbb{C})$$

the Hodge filtration of 2.2; the latter varies holomorphically with $s \in S$.

Definition (3.5)**.** *The Gauss–Manin connection on the holomorphic vector bundle $R^n_{\mathcal{O}}(f)$, identified here with its sheaf of holomorphic sections, is the holomorphic and integrable connection*

$$\nabla : R^n_{\mathcal{O}}(f) \longrightarrow \Omega^1_S \otimes R^n_{\mathcal{O}}(f)$$

whose horizontal local sections ($\nabla v = 0$) are the local sections of $R^n_{\mathbb{C}}(f)$.

Theorem (transversality 3.6)**.** *The Hodge filtration F on $R^n_{\mathcal{O}}(f)$ satisfies*

$$\nabla F^p(R^n_{\mathcal{O}}(f)) \subset \Omega^1_S \otimes F^{p-1}(R^n_{\mathcal{O}}(f)).$$

Proof (following Katz–Oda [10] and a personal communication from Katz). The question is local on S. Let $\mathcal{U} = (U_i)_{0 \le i \le N}$ be a finite covering of X by Stein opens. For $Q \subset [0, N]$, let

$$U_Q = \bigcap_{i \in Q} U_i,$$

let j_Q be the inclusion of U_Q in X, and put

$$f_*(U_Q, \Omega^p_{X/S}) = (f j_Q)_*(\Omega^p_{X/S} | U_Q).$$

We denote by $f_*(\mathcal{U}, \Omega^\bullet_{X/S})$ the double complex of sheaves on S with components

$$f_*(\mathcal{U}, \Omega^\bullet_{X/S})^{pq} = \bigoplus_{|Q|=q+1} f_*(U_Q, \Omega^p_{X/S}), \tag{3.6.1}$$

whose first differential is induced by the exterior differential and whose second differential is the Čech differential. The spectral sequence 3.4.1 is the spectral sequence of this double complex obtained from the filtration by the first degree.

Let v be a holomorphic vector field on S, and suppose that on each U_i a vector field v_i lifting v is given. We denote by $\theta(v_i)$ the endomorphism of $f_*(\mathcal{U}, \Omega^\bullet_{X/S})$ satisfying, for $Q = (i_1, \ldots, i_q)$ with $i_1 < \cdots < i_q$,

$$\theta(v_i) f_*(U_Q, \Omega^p_{X/S}) \subset f_*(U_Q, \Omega^p_{X/S}) \oplus \bigoplus_{i_0 < i_1} f_*(U_{\{i_0\} \cup Q}, \Omega^{p-1}_{X/S}),$$

and having as coordinates the Lie derivative

$$L_{v_{i_1}} : f_*(U_Q, \Omega^p_{X/S}) \longrightarrow f_*(U_Q, \Omega^p_{X/S})$$

and the contractions

$$(-1)^p (v_{i_1} - v_{i_0}) \lrcorner : f_*(U_Q, \Omega^p_{X/S}) \longrightarrow f_*(U_{\{i_0\} \cup Q}, \Omega^{p-1}_{X/S}).$$

A simple calculation shows that $\theta(v_i)$ commutes with d.

Lemma (3.7)**.** *On the cohomology sheaves $R^n_{\mathcal{O}}(f)$ of $f_*(\mathcal{U}, \Omega^\bullet_{X/S})$, the endomorphism $\theta(v_i)$ induces ∇_v.*

Let $\mathcal{C}^\infty$ be the sheaf of C^∞ functions on S. If $f_*(\mathcal{U}, \Omega^\bullet_{\infty X/S})$ is the C^∞ analogue of (3.6.1), constructed in terms of the complex of relative C^∞ differential forms with complex values on X, then

$$R^n_{\mathcal{C}^\infty}(f) = H^n(f_*(\mathcal{U}, \Omega^\bullet_{\infty X/S})).$$

Since the sheaves $\Omega^p_{\infty X/S}$ are soft, this formula remains valid for every covering $\mathcal{U}'$.

If $(v'_i)_{0 \le i \le N}$ are C^∞ liftings of v on the U_i, the construction 3.6 still defines endomorphisms $\theta(v'_i)$ of $R^n_{\mathcal{C}^\infty}(f)$. If (v'_i) and (v''_i) are two systems of liftings, a simple calculation shows that, at the level of complexes,

$$\theta(v'_i) - \theta(v''_i) = dH - Hd, \tag{3.7.1}$$

where the only nonzero coordinates of H are the contractions

$$v'_{i_1} - v''_{i_1} \lrcorner : f_*(U_Q, \Omega^p_{\infty X/S}) \longrightarrow f_*(U_Q, \Omega^{p-1}_{\infty X/S}),$$

$(Q = \{i_1, \ldots, i_q\},\ i_1 < \cdots < i_q)$.

Thus the endomorphism $\theta(v'_i)$ of $R^n_{\mathcal{C}^\infty}(f)$ is independent of the choice of the v'_i. Via the injection of $R^n_{\mathcal{O}}(f)$ into $R^n_{\mathcal{C}^\infty}(f)$, it is compatible with the endomorphism $\theta(v_i)$ of 3.7. It therefore remains only to verify that $\theta(v'_i) = \nabla_v$ on $R^n_{\mathcal{C}^\infty}(f)$. One verifies that $\theta(v'_i)$, which is independent of the v'_i, is also independent of $\mathcal{U}$. Taking $\mathcal{U} = \{X\}$ and a lift v' of v, one then has

$$f_*(\mathcal{U}, \Omega^\bullet_{\infty X/S}) = f_*(\Omega^\bullet_{\infty X/S}),$$

and $\theta(v')$ is the Lie derivative along v', visibly equal to ∇_v.

3.8. We finish the proof of 3.6. By construction, one has

$$\theta(v_i)\left(\bigoplus_{p'\geq p} f_*(\mathcal{U},\Omega^\bullet_{X/S})^{p'q'}\right) \subset \bigoplus_{p'\geq p-1} f_*(\mathcal{U},\Omega^\bullet_{X/S})^{p'q'}. \tag{3.8.1}$$

Since the spectral sequence (3.4.1) is obtained from the double complex $f_*(\mathcal{U},\Omega^\bullet_{X/S})$, it follows from 3.7 and (3.8.1) that

$$\theta(v_i)F^p(R^n_{\mathcal{O}}(f)) = \nabla_v F^p(R^n_{\mathcal{O}}(f)) \subset F^{p-1}(R^n_{\mathcal{O}}(f)). \tag{3.8.2}$$

Since this is true locally on S and for every v, it implies 3.6.

3.9. By the Leibniz formula

$$\nabla(fh) = df\cdot h + f\cdot\nabla h,$$

the map induced by ∇,

$$\mathrm{def}(\nabla) : \mathrm{Gr}^p_F(R^n_{\mathcal{O}}(f)) \longrightarrow \Omega^1_S\otimes\mathrm{Gr}^{p-1}_F(R^n_{\mathcal{O}}(f)),$$

is $\mathcal{O}$-linear. By 3.4.1 it is identified with

$$\mathrm{def}(\nabla) : R^q f_*\Omega^p_{X/S} \longrightarrow \Omega^1_S\otimes R^{q+1}f_*\Omega^{p-1}_{X/S}. \tag{3.9.1}$$

From the formula 3.7 for ∇_v one obtains the following.

Proposition (3.10)**.** *The homomorphism (3.9.1) is cup-product, via the pairing (3.3.1), with the Kodaira–Spencer class*

$$c\in\Omega^1_S\otimes R^1f_*(T^1_{X/S})$$

which expresses how X_s is deformed with s.

Definition (3.11)**.** *(i) A family of real Hodge structures, of weight n, H on S consists of*

a) a local system of real vector spaces $H_{\mathbb{R}}$ on S;

b) a finite holomorphic filtration F of the locally free analytic sheaf

$$H_{\mathcal{O}} = H_{\mathbb{R}}\otimes_{\mathbb{R}}\mathcal{O},$$

these data satisfying the following conditions:

(H.1) the natural connection ∇ on $H_{\mathcal{O}}$ is such that

$$\nabla F^p(H_{\mathcal{O}}) \subset \Omega^1_S\otimes F^{p-1}(H_{\mathcal{O}});$$

(H.2) at every point $s\in S$, F defines on $(H_{\mathbb{R}})_s$ a Hodge structure of weight n.

(ii) A polarization of H is a locally constant bilinear form ψ on $H_{\mathbb{R}}$ which at every point $s\in S$ induces a polarization of $(H_{\mathbb{R}})_s$.

3.12. A family of Hodge structures H of weight n on S is a local system of free $\mathbb{Z}$-modules of finite type $H_{\mathbb{Z}}$ on S, together with a family of real Hodge structures on $H_{\mathbb{R}} = \mathbb{R}\otimes H_{\mathbb{Z}}$. A polarization of H is a polarization ψ of $H_{\mathbb{R}}$ taking integral values on $H_{\mathbb{Z}}$.

With these definitions, one can reformulate 2.2(C) and 3.6 by saying that, under the hypotheses of 3.1, $P^n_{\mathbb{Z}}(f)$ is a polarized family of Hodge structures on S.

4 The regularity theorem

4.1. When one starts with a projective and smooth morphism $f : X\to S$ of schemes of finite type over $\mathbb{C}$, such that $f^{\mathrm{an}} : X^{\mathrm{an}}\to S^{\mathrm{an}}$ satisfies the hypotheses of 3.1, some of the objects constructed in no. 3 are obtained from purely algebraic objects by applying the functor of passage to the analytic category. They are:

a) the sheaves of modules $R^n_{\mathcal{O}}(f)$, their Hodge filtration, the spectral sequence 3.4.1, and the Gauss–Manin connection;

b) the algebra structure (cup-product) on $R^*_{\mathcal{O}}(f)$, and the "primitive part" $P^n_{\mathcal{O}}(f)$;

c) the product, by a suitable power of $2\pi i$, of the polarization form ψ.

By contrast, the "integral lattice" $R^*_{\mathbb{Z}}(f)$, and the real structure thereby obtained on the complex cohomology of the fibers of f, are of transcendental nature.

4.2. Indeed, let S be a smooth scheme of finite type over $\mathbb{C}$, and let X be a closed subscheme of $\mathbb{P}^r(\mathbb{C}) \times S$, such that the projection $f : X \to S$ is a smooth morphism, pure of relative dimension d.

By the relative variant of GAGA, for every coherent algebraic sheaf F on X one has

$$(R^n f_* F)^{\mathrm{an}} \xrightarrow{\sim} R^n f^{\mathrm{an}}_*(F^{\mathrm{an}}).$$

More generally, if $K^\bullet$ is a complex of coherent algebraic sheaves whose

differentials d_i are (algebraic) differential operators which are $f^{-1}\mathcal{O}_S$-linear, then the direct images in hypercohomology satisfy

$$R^n f_*(K^\bullet)^{\mathrm{an}} \xrightarrow{\sim} R^n f^{\mathrm{an}}_*(K^{\bullet\mathrm{an}})$$

(this case is reduced to the preceding one by one of the hypercohomology spectral sequences).

Take for $K^\bullet$ the relative algebraic de Rham complex $\Omega^\bullet_{X/S}$. One obtains (in the notation of 3.2)

$$\big(R^n f_* \Omega^\bullet_{X/S}\big)^{\mathrm{an}} \xrightarrow{\sim} R^n f^{\mathrm{an}}_*\big((\Omega^\bullet_{X/S})^{\mathrm{an}}\big) \simeq R^n f^{\mathrm{an}}_*\big(\Omega^\bullet_{X^{\mathrm{an}}/S^{\mathrm{an}}}\big) = R^n_{\mathcal{O}}(f).$$

Moreover, the spectral sequence (3.4.1) comes from a purely algebraic spectral sequence

$$E_1^{pq} = R^q f_* \Omega^p_{X/S} \Longrightarrow R^{p+q} f_* \Omega^\bullet_{X/S}. \tag{4.2.1}$$

According to 3.4, the E_1^{pq} are locally free (since the $E_1^{pq\,\mathrm{an}}$ are), the spectral sequence (4.2.1) degenerates, and its formation is compatible with every change of base. The Hodge filtration to which it abuts is therefore, ipso facto, algebraic.

The cup-product is defined in terms of the exterior product in $\Omega^\bullet_{X/S}$. The de Rham cohomology class of a hyperplane section can be defined algebraically (up to multiplication by a power of $2\pi i$), from the logarithmic differential

$$\mathrm{d}f/f : \mathcal{O}^*_X \longrightarrow \Omega^1_{X/S};$$

the primitive part $P^n_{\mathcal{O}}(f)$ and ψ are also of algebraic nature.

Finally, the construction 3.6–3.7 of the Gauss–Manin connection is transported without difficulty to the algebraic case (Katz–Oda [10]).

4.3. Let D be a Riemann surface isomorphic to the unit disk, let 0 be a point of D, put $D^* = D - \{0\}$, let V be a holomorphic vector bundle on D^*, let ∇ be a holomorphic connection on V, and let V_0 be the local system of horizontal sections of V. One has

$$V \simeq V_0 \otimes_{\mathbb{C}} \mathcal{O}.$$

The (commutative) fundamental group $\pi_1(D^*) \simeq \mathbb{Z}$ acts on V_0. We denote by T the monodromy transformation, i.e. the action on V_0 or on V of the positive generator of $\pi_1(D^*)$.

There exists U such that

$$T = \exp(2\pi i U).$$

Let $z : D \to \mathbb{C}$ be a coordinate, with $z(0) = 0$, and let $\widetilde{D}^*$ be the universal covering of D^* on which $\log z$ is defined. For v a section of V_0 on $\widetilde{D}^*$,

$$\exp(-\log z \cdot U)(v)$$

is the inverse image of a holomorphic section of V on D^*.

This construction defines an isomorphism

$$m_{U,z} : \mathcal{O} \otimes_{\mathbb{C}} H^0(\widetilde{D}^*, V_0) \xrightarrow{\sim} V,$$

and hence a natural extension V_U of V to D, depending only on U. A section v of V on D^* will be called ∇-moderate if, as a section of V_U, it is meromorphic at 0. This notion does not depend on the choice of U.

4.4. Let S be the complement, in a smooth projective curve $\overline{S}$, of a finite set T, and let V be an algebraic vector bundle on S. An algebraic connection ∇ on V is called regular if, for every $t \in T$, the meromorphic sections of V in a neighborhood of t are ∇-moderate.

Theorem (4.5)**.** *For $f : X \to S$ a projective and smooth morphism of schemes, the Gauss–Manin connection on $R^n f_* \Omega^\bullet_{X/S}$ is regular.*

This theorem is proved in [2]. A completely different proof, proceeding by arithmetic methods, is due to Katz [9].

4.6. Let D^* be as in 4.3, let H be a family of real Hodge structures on D^*, let T be the monodromy transformation of $H_{\mathbb{C}}$, let U be an endomorphism of $H_{\mathbb{C}}$ such that

$$T = \exp(2\pi i U),$$

and let $H_{\mathcal{O},U}$ be the corresponding locally free extension of $H_{\mathcal{O}}$ to D.

One says that H is regular at 0 if the Hodge filtration of $H_{\mathcal{O}}$ extends to $H_{\mathcal{O},U}$. This condition does not depend on U. Theorem 4.5 has the following corollary, whose conclusion is purely analytic.

Theorem (Corollary 4.7)**.** *Under the hypotheses of 4.5, the families of Hodge structures $R^n(f)$ on S^{an} are regular in a neighborhood of every point $t \in T$.*

One may hope that, in fact, every polarized family of Hodge structures on D^* is automatically regular.

5 Modular spaces

5.1. Let G be a real algebraic group, let $C \in G(\mathbb{R})$, and let V be a real representation of G. A C-polarization of V is a bilinear form Ψ on V, invariant under G, and such that $\Psi(x, Cy)$ is symmetric and positive definite.

Lemma (5.2)**.** *If G is connected (by which one means that $G(\mathbb{C})$ is connected), the following conditions are equivalent:*

(i) G admits a representation $\rho : G \to \mathrm{GL}(V)$, with finite kernel $\mathrm{Ker}(\rho)$, which is C-polarizable.

(ii) Every representation of G is C-polarizable.

(iii) G is reductive and $\mathrm{ad}\, C$ is a Cartan involution of G.

The proof is left to the reader.

Remark 5.3. There exist in $G(\mathbb{R})$ elements C satisfying the conditions of 5.2 if and only if G is reductive and admits a compact maximal torus. In this case:

a) such an element C is contained in a single maximal compact subgroup, namely its centralizer;

b) for G adjoint, the correspondence

$$C \longmapsto (\text{centralizer of } C)$$

is a bijection between the set of these elements and the set of maximal compact subgroups of $G(\mathbb{R})$.

5.4. Let G be a real algebraic group, endowed with homomorphisms

$$\mathbb{G}_m \xrightarrow{w} G \xrightarrow{t} \mathbb{G}_m \tag{5.4.1}$$

such that w is central and $tw(x) = x^2$. A representation $\rho : G \to \mathrm{GL}(V)$ will be called homogeneous of weight n if $\rho w(x) = x^n$.

By a morphism from $\mathbb{S}$ to G we shall always mean a homomorphism $h : \mathbb{S} \to G$ making the diagram

$$\begin{array}{ccccc} \mathbb{G}_m & \xrightarrow{w} & \mathbb{S} & \xrightarrow{t} & \mathbb{G}_m \\ \| & & \downarrow h & & \| \\ \mathbb{G}_m & \xrightarrow{w} & G & \xrightarrow{t} & \mathbb{G}_m \end{array}$$

commutative.

If $h : \mathbb{S} \to G$ is a morphism and $\rho : G \to \mathrm{GL}(V)$ a real representation of G, we shall denote by h_V the real Hodge structure (1.4) $\rho \circ h$ on V. For ρ homogeneous of weight n, a polarization of V is a polarization Ψ of (V, h_V) which satisfies

$$\Psi(gx, gy) = t(g)^n \Psi(x, y). \tag{5.4.2}$$

We shall say that h is positive if every representation of G is polarizable.

5.5. The group G, endowed with (5.4.1), is uniquely determined by $G^0 = \mathrm{Ker}(t)$, endowed with the central element, of order dividing 2,

$$\varepsilon = w(-1).$$

Indeed G is identified with the quotient of $\mathbb{G}_m \times G^0$ by the subgroup generated by $(-1, \varepsilon)$.

From this point of view, a morphism $h : \mathbb{S} \to G$ is identified with

$$h^0 : \mathbb{S}^0 \to G^0$$

such that $h^0(-1) = \varepsilon$. A homogeneous representation of weight n of G is identified with a representation ρ of G^0 such that $\rho(\varepsilon) = (-1)^n$, and a polarization of this representation is identified with an $h^0(i)$-polarization (5.1). According to 5.2, if G^0 is connected, h is a positive morphism if and only if G is reductive and $\mathrm{ad}\, h^0(i)$ is a Cartan involution of G^0.

5.6. We shall henceforth suppose that G^0 is reductive and connected.

Let h be a morphism from $\mathbb{S}$ to G. For $\rho : G \to \mathrm{GL}(V)$ a representation of G, denote by F_h the Hodge filtration of $V_{\mathbb{C}}$ deduced from h_V. For the adjoint representation $\mathrm{Lie}(G)$,

$$F_h^0(\mathrm{Lie}(G)_{\mathbb{C}}) = \bigoplus_{p \geq 0} \mathrm{Lie}(G_{\mathbb{C}})^{p,-p} \tag{5.6.1}$$

is the Lie algebra of a parabolic subgroup $P(h)$ of $G_{\mathbb{C}}$. For every representation $\rho : G \to \mathrm{GL}(V)$ of G, $P(h)$ respects the Hodge filtration F_h of $V_{\mathbb{C}}$; if $\mathrm{Ker}(\rho) \cap G^0$ is central, then $P(h)$ is exactly the subgroup of $G_{\mathbb{C}}$ which respects this filtration.

Finally, denote by $H(h)$ the centralizer of h in $G(\mathbb{R})$.

5.7. Let X be a conjugacy class of morphisms from $\mathbb{S}$ to G. We shall denote by $\check{X}$ the set of conjugates in $G(\mathbb{C})$ of $P(h)$, for $h \in X$.

Lemma (5.8). *The $G(\mathbb{R})$-equivariant map $P : h \mapsto P(h)$ identifies X with an open subset of the flag space $\check{X}$.*

Let $h \in X$ and let $\rho : G \to \mathrm{GL}(V)$ be a homogeneous representation of G such that $\mathrm{Ker}(\rho) \cap G^0$ is finite. The set X is identified with the set of Hodge structures on V, conjugate under $G(\mathbb{R})$ to h_V, while $\check{X}$ is identified with the set of filtrations on $V_{\mathbb{C}}$, conjugate under $G(\mathbb{C})$ to F_h. Since a homogeneous Hodge structure is determined by its Hodge filtration, the map P is injective. It is identified with the map

$$G(\mathbb{R})/H(h) \longrightarrow G(\mathbb{C})/P(h)$$

whose differential at the origin is

$$\mathrm{Lie}(G)/\mathrm{Lie}(H) \longrightarrow \mathrm{Lie}(G_{\mathbb{C}})/\mathrm{Lie}(P(h)),$$

i.e.

$$\mathrm{Lie}(G)/(F^0 \cap \overline{F^0})(\mathrm{Lie}(G)) \longrightarrow F^{-\infty}/F^0(\mathrm{Lie}(G_{\mathbb{C}})).$$

This differential is bijective, whence 5.8.

We have seen in passing that

$$H(h) = G(\mathbb{R}) \cap P(h). \tag{5.8.1}$$

5.9. Let V be a real representation of G, let $V(\check{X},\mathbb{R})$ be the constant sheaf on $\check{X}$ with value V, put

$$V(\check{X},\mathbb{C}) = \mathbb{C}\otimes V(\check{X},\mathbb{R}), \qquad V(\check{X},\mathcal{O}) = \mathcal{O}\otimes_{\mathbb{C}} V(\check{X},\mathbb{C}),$$

and let ∇ be the natural integrable connection of $V(\check{X},\mathcal{O})$. The $G(\mathbb{R})$-equivariant map $h\mapsto h_V$ from X to the Hodge structures on V extends uniquely to a $G(\mathbb{C})$-equivariant, and hence holomorphic, map from $\check{X}$ to the filtrations of $V_{\mathbb{C}}$. This map corresponds to a filtration F of $V(\check{X},\mathcal{O})$, the Hodge filtration. The system $(V(\check{X},\mathcal{O}),\nabla,F)$ is $G(\mathbb{C})$-equivariant. The real structure $V(\check{X},\mathbb{R})$ of $V(\check{X},\mathbb{C})$ is $G(\mathbb{R})$-equivariant. Finally, at $h\in X$, F induces on $V_{\mathbb{C}}$ the Hodge filtration F_h.

5.10. The action of $G(\mathbb{C})$ on $\check{X}$ defines a morphism

$$\mathrm{Lie}(G)(\check{X},\mathcal{O}) \longrightarrow T_{\check{X}}, \tag{5.10.1}$$

where $T_{\check{X}}$ is the tangent bundle. At $h\in X$, the kernel of (5.10.1) is

$$\mathrm{Lie}(P(h)) = F_h^0(\mathrm{Lie}(G_{\mathbb{C}})) \qquad (5.6),$$

so that (5.10.1) factors through a $G(\mathbb{C})$-equivariant isomorphism

$$F^{-\infty}/F^0\big(\mathrm{Lie}(G)(\check{X},\mathcal{O})\big) \xrightarrow{\sim} T_{\check{X}}. \tag{5.10.2}$$

Let $T^1_{\check{X}}$ denote the holomorphic subbundle of $T_{\check{X}}$ which is the image of $F^{-1}(\mathrm{Lie}(G)(\check{X},\mathcal{O}))$. It is easily verified that if v is a local section of $T^1_{\check{X}}$, then

$$\nabla_v\big(F^p(V(\check{X},\mathcal{O}))\big) \subset F^{p-1}(V(\check{X},\mathcal{O})),$$

whatever the representation $\rho: G\to \mathrm{GL}(V)$ may be. The converse is true when $\mathrm{Ker}(\rho)\cap G^0$ is central.

5.11. Suppose now that X is a conjugacy class of positive morphisms from $\mathbb{S}$ to G. For $h\in X$, put $C_h = h(i)$, and denote by $K(h)$ the centralizer of C_h in $G(\mathbb{R})$. The group

$$K^0(h) = K(h)\cap G^0(\mathbb{R})$$

is compact. Let V be a representation of G and let Ψ be a G-invariant bilinear form on V. If Ψ is a polarization of (V,h_V) for one $h\in X$, then Ψ is a polarization of (V,h_V) for every $h\in X$. Indeed, putting

$$\Phi_h(x,y) = \Psi(x, C_h y),$$

one has

$$\Phi_{gh} = g(\Phi_h). \tag{5.11.1}$$

We shall then say that Ψ is a polarization of V. By hypothesis, every representation of G is polarizable.

5.12. Let Ψ be a polarization of the adjoint representation of G on $\mathrm{Lie}(G)$. By 5.10.2, Ψ induces, at each point $h\in X$, a positive definite Hermitian form on

$$(T_X)_h = F_h^{-\infty}/F_h^0(\mathrm{Lie}(G)_{\mathbb{C}}) \simeq \bigoplus_{p<0}\mathrm{Lie}(G)^{p,-p}. \tag{5.12.1}$$

The polarization Ψ therefore defines a Hermitian structure, invariant under $G(\mathbb{R})$, on X.

If Z is the center of G, one already has

$$F^{-\infty}/F^0\big(\mathrm{Lie}(G/Z)(\check{X},\mathcal{O})\big) \xrightarrow{\sim} T_{\check{X}}, \tag{5.12.2}$$

and a polarization Ψ of $\mathrm{Lie}(G/Z)$ already defines a structure of Hermitian variety on X. One may take for Ψ the negative of the Killing form of G/Z.

Put

$$\begin{cases} (T_X^v)_h = \bigoplus_{p<0}(\mathrm{Lie}(G/Z))^{2p,-2p}, \\ (T_X^h)_h = \bigoplus_{p<0}(\mathrm{Lie}(G/Z))^{2p-1,-2p+1}. \end{cases} \tag{5.12.3}$$

Thus one obtains an orthogonal C^∞ decomposition of the tangent bundle into two complex subbundles. One has

$$(T_X^1)_h \subset (T_X^h)_h \qquad (h\in X). \tag{5.12.4}$$

5.13. The subgroup $K(h)_{\mathbb{C}} \cap P(h)$ of $K(h)_{\mathbb{C}}$ is parabolic, so that $K(h)_{\mathbb{C}} \subset K(h) \cdot P(h)$ and $K(h) \cdot h = K(h)_{\mathbb{C}} \cdot h$ is a compact complex analytic subvariety of X. These varieties are the fibers of the C^∞, $G(\mathbb{R})$-equivariant fibration

$$X = G/H(h) \longrightarrow G/K(h).$$

It is easily verified that T_X^v is the tangent bundle to this fibration.

5.14. Recall that on a holomorphic Hermitian bundle F there exists a unique Hermitian connection ∇ such that $\nabla'' = \partial''$. The curvature R of the bundle is, by definition, the curvature of this connection. It is a form of type $(1,1)$ with values in the Lie algebra of the unitary group of the bundle.

If F is the constant bundle defined by a vector space F_0, and if the Hermitian form is written

$$\Phi(x,y) = \Phi_0(hx,y), \tag{5.14.1}$$

then, for $x : S \to F_0$ a C^∞ section of F, one has

$$\nabla x = \partial x + h^{-1}\partial' h\, x. \tag{5.14.2}$$

It follows that, for u and v two tangent fields,

$$R(u,v) = \partial_u(h^{-1}\partial'_v h) - \partial_v(h^{-1}\partial'_u h) - h^{-1}\partial'_{[u,v]}h + [h^{-1}\partial'_u h, h^{-1}\partial'_v h]. \tag{5.14.3}$$

5.15. Let F be the tangent bundle of a Hermitian complex analytic variety. For u a tangent vector to S, denote by $u_{1,0}$ and $u_{0,1} = \overline{u_{1,0}}$ its components of type $(1,0)$ and $(0,1)$ in the complexification of the tangent bundle. The holomorphic curvature of S in the direction u is the real number

$$R(u) = \Phi(R(u_{1,0}, u_{0,1})(u), u)/\Phi(u,u)^2. \tag{5.15.1}$$

If X is endowed with the Hermitian structure 5.12, one has the fundamental result ([8], §9):

Theorem (5.16)**.** *There exists $\lambda < 0$ such that $R(u) \leq \lambda$ in every direction belonging to T_X^h.*

Proof (a sketch). By homogeneity it is enough to prove 5.16 at a point $e \in X$. Put $H = H(e)$, $P = P(e)$, let N^+ be the unipotent radical of P, and let N^- be the conjugate complex subgroup whose Lie algebra is

$$\mathfrak{n}^- = \bigoplus_{p<0} \mathrm{Lie}(G)^{p,-p}.$$

In the calculations which follow, near e, we identify X with N^- by the maps

$$X = G/H \hookrightarrow G(\mathbb{C})/P \longleftrightarrow N^-.$$

Left translations on N^- permit one to identify the tangent bundle with $\mathfrak{n}^-$. We shall denote by t^+, t^0, and t^- the orthogonal projections of $\mathrm{Lie}(G)_{\mathbb{C}}$ onto $\mathfrak{n}^+$, $\mathrm{Lie}(H_{\mathbb{C}})$, and $\mathfrak{n}^- = \mathrm{Lie}(N^-)$, respectively. For $g = np$ ($g \in G(\mathbb{R})$, $n \in N^-$, $p \in P$), the differential, at e, of the action of g on X is identified with

$$t^- \operatorname{ad} p : \mathfrak{n}^- \longrightarrow \mathfrak{n}^-.$$

The Hermitian structure is therefore written

$$\Phi_n(x,y) = \Phi_e(t^- \operatorname{ad} p^{-1} x, t^- \operatorname{ad} p^{-1} y).$$

Denoting by Ψ the polarization form and by C the Cartan involution associated to e, one has

$$\begin{aligned}\Phi_n(x,y) &= \Psi(t^- \operatorname{ad} p^{-1} x,\ t^- C \operatorname{ad} \overline{p}^{-1}\, \overline{y})\\ &= \Psi(t^- \operatorname{ad} p^{-1} x,\ \operatorname{ad} C(\overline{p})^{-1} \cdot C\overline{y})\\ &= \Psi(\operatorname{ad} C(\overline{p}) \cdot t^- \cdot \operatorname{ad} p^{-1} \cdot x,\ C\overline{y})\\ &= \Phi_e(\operatorname{ad} C(\overline{p}) \cdot t^- \cdot \operatorname{ad} p^{-1} \cdot x,\ y).\end{aligned}$$

One verifies that, for $n = \exp(u)$ with $u \sim 0$, to third order one has

$$p = \exp\bigl(\overline{u} - \tfrac{1}{2}t^0[u,\overline{u}] - t^+[u,\overline{u}]\bigr)$$

and

$$h = \operatorname{ad} C(\overline{p})\, t^- \ \operatorname{ad} p^{-1} = \exp(H),$$

for

$$H = t^- \circ \operatorname{ad}(Cu - \overline{u} + [u,\overline{u}]) - \tfrac{1}{2}[t^- \operatorname{ad} Cu, t^- \operatorname{ad}\overline{u}].$$

Under the hypotheses of 5.14, if $h = \exp(H)$ and if $H = 0$ at $s_0 \in S$, one deduces from 5.14.3 that, at s_0,

$$\begin{aligned} R(u,v) = (\partial_u\partial'_v - \partial_v\partial'_u)H - \tfrac{1}{2}[\partial''_u H, \partial'_v H] + \tfrac{1}{2}[\partial''_v H, \partial'_u H] \\ + (\partial'_u\partial'_v - \partial'_v\partial'_u - \partial'_{[u,v]})H. \end{aligned}$$

Applying this formula, one sees that the curvature is given, at e, by

$$R(u,v)_e = S(u,\overline{v}) - S(v,\overline{u}),$$

with

$$\begin{aligned} S(u,\overline{v}) &= -t^- \circ \operatorname{ad}[u,\overline{v}] + \tfrac{1}{2}[t^- \operatorname{ad} Cu, t^- \operatorname{ad}\overline{v}] + \tfrac{1}{2}[-t^- \operatorname{ad}\overline{v}, t^- \operatorname{ad} Cu] \\ &= [t^- \operatorname{ad} Cu, t^- \operatorname{ad}\overline{v}] - t^- \circ \operatorname{ad}[u,\overline{v}]. \end{aligned}$$

Whatever $u, v \in \mathfrak{n}^-$ may be, if u_{10} and u_{01} are the components of type (10) and (01) of u, as in 5.15, one has, thanks to the invariance of Ψ,

$$\begin{aligned} \Phi(R(u_{10},u_{01})(v),v) &= \Psi(S(u,\overline{u})(v), C\overline{v}) \\ &= \Psi\bigl([Cu, t^-[\overline{u},v]] - t^-[\overline{u},[Cu,v]] - t^-[[u,\overline{u}],v], C\overline{v}\bigr) \\ &= -\Psi\bigl(t^-[\overline{u},v],[Cu,C\overline{v}]\bigr) + \Psi\bigl([Cu,v],[\overline{u},C\overline{v}]\bigr) - \Psi\bigl([u,\overline{u}],[v,C\overline{v}]\bigr). \end{aligned}$$

For $u = v \in T^h_X$, one has $Cu = -u$ and

$$\begin{aligned} \Phi(R(u_{10},u_{01})(u),u) &= \Psi\bigl(t^-[u,\overline{u}],[u,\overline{u}]\bigr) + \Psi\bigl([u,\overline{u}],[v,\overline{v}]\bigr) \\ &= -\Phi\bigl(t^-[u,\overline{u}], t^-[u,\overline{u}]\bigr) - \Phi\bigl([u,\overline{u}],[u,\overline{u}]\bigr) < 0, \end{aligned}$$

which proves 5.16.

A generalization of Schwarz's lemma allows one to deduce from 5.16 (see [1], III, no. 10):

Corollary 5.17. Let $f : D \to X$ be a holomorphic map from the unit disk into X. Endow D with the hyperbolic metric d for which it has curvature -1, and X with the metric 4.12. Then, if $f(D)$ is tangent to the T^h_X, one has

$$d(f(x), f(y)) \le |\lambda|\, d(x,y).$$

6 The monodromy theorem, after A. Borel

6.1. Let S be a connected complex analytic variety, endowed with a base point $s_o \in S$, and let $p : \widetilde{S} \to S$ be its universal covering. Let H be a polarized family of Hodge structures of weight n on S. Denote by H_o the polarized Hodge structure which is the fiber of H at s_o.

The polarization Ψ is a bilinear form on $H_{o\mathbb{Q}}$. Denote by G^o the algebraic group, defined over $\mathbb{Q}$, which is the neutral component of the group of automorphisms of $(H_{o\mathbb{Q}}, \Psi)$; by $\varepsilon \in G^o$ the homothety of ratio $(-1)^n$; by G the group deduced, by the procedure 5.5, from (G^o, ε); and by Γ the discrete group of automorphisms of $(H_{o\mathbb{Z}}, \Psi)$. One has

$$T : \pi_1(S, s_o) \longrightarrow \Gamma. \tag{6.1.1}$$

6.2. For each point $s \in \widetilde{S}$, one has a natural isomorphism

$$H_{o\mathbb{Z}} \simeq H_{p(s),\mathbb{Z}},$$

and the Hodge structure $H_{p(s)}$ is identified with a Hodge structure on $H_{o,\mathbb{Z}}$, defined by $h(s) : \mathbb{S} \to G_{\mathbb{R}}$. The morphisms $h(s)$ are positive and mutually conjugate, because they form a connected continuous family and G is reductive. If X is the corresponding modular space, defined in § 5, one therefore has a map

$$h : \widetilde{S} \longrightarrow X.$$

The family H is the inverse image by h of the system of Hodge structures on X defined in (5.9) by the natural representation of weight n of G in H_o.

The morphism h is equivariant, via (5.5.1), for the actions of $\pi_1(S, s_o)$ and Γ on $\widetilde{S}$ and X.

By 3.4 and 3.6, the map h is holomorphic and $\mathrm{Im}(dh) \subset T^1_X$ (cf. 5.10).

6.3. Suppose that S is the punctured unit disk

$$D^* = \{z \mid 0 < |z| < 1\}$$

and take for the universal covering of D^* the Poincaré half-plane Y, endowed with

$$\exp(2\pi i y) : Y \longrightarrow D^*.$$

If $T \in \Gamma$ is the monodromy transformation (4.3) of $H_{\mathbb{Z}}$, the map h satisfies

$$h(y+1) = Th(y). \tag{6.3.1}$$

Choose $h_o \in X$. If $h(y) = g_y h_o$, the distance in X, for a metric (5.12), satisfies

$$d(h(y+1), h(y)) = d(Tg_y h_o, g_y h_o) = d(g_y^{-1} T g_y h_o, h_o). \tag{6.3.2}$$

By 5.17, if Y is endowed with its natural hyperbolic distance, the map h decreases distances (suitably normalized). In Y, one has

$$d(y, y+1) \sim (\mathrm{Im}(y))^{-1}$$

and therefore

$$d(g_y^{-1} T g_y h_o, h_o) \longrightarrow 0 \qquad \text{for } \mathrm{Im}(y) \to \infty. \tag{6.3.3}$$

The same argument applies to T^2; moreover, $T^2 \in G(\mathbb{R})$, and (6.2.3) implies that a limit of conjugates of T^2 lies in the compact subgroup $H(h_o)$ of $G(\mathbb{R})$. The eigenvalues of T^2 (and of T) are therefore all of absolute value 1. These eigenvalues form a set of algebraic integers stable under conjugation. Now,

Lemma (6.4)**.** *If all the complex conjugates of an algebraic integer α have absolute value 1, then α is a root of unity.*

The eigenvalues of T are therefore roots of unity, which proves

Theorem (Monodromy theorem 6.5 (A. Borel))**.** *Let H be a polarized family of Hodge structures on the punctured disk D^*. There exist integers N and M such that the monodromy transformation T satisfies*

$$(T^N - I)^M = 0.$$

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WORKS OF SHIMURA[1]

by Pierre DELIGNE

Let X/Γ be the quotient of a symmetric Hermitian domain X by a discrete arithmetic group Γ (assumed to be defined by congruence conditions, see 1.7). By Baily and Borel, X/Γ is a complex algebraic variety. In the papers [10] to [14], Shimura shows, among other things, for many of these varieties, that they can be defined over number fields which he makes explicit. This is the subject of the present report.

To obtain clean statements, one is forced to use adelic language. This leads one to consider, for a given "congruence condition", a scheme over $\mathbb{C}$, a disjoint sum of finitely many varieties of the type X/Γ (see §§ 1 and 2). In many cases, this scheme can be defined over a number field E, independent of the congruence condition under consideration. Its geometric connected components, however, will be defined over class fields of E. This phenomenon, although essential, will be neglected in the rest of this introduction.

In a small number of cases, X/Γ can be interpreted as the set of isomorphism classes of complex abelian varieties, endowed with some additional algebraic structures (polarizations, endomorphisms, structures on the points of order n). The solution, over $\mathbb{Q}$, of a corresponding moduli problem then provides a scheme M over an explicit number field F, whose complex points are X/Γ. We shall call M a "model" of X/Γ. The fundamental case is that of polarized abelian varieties of the principal series, endowed with a level-n structure (1.6, 1.11, 4.16, 4.17 and 4.21).

In other cases, X/Γ can be interpreted as the set of isomorphism classes of complex abelian varieties A endowed with some structures as above, and with some integral cohomology classes of type (p,p)

$$a_i \in H^{2p}(A \times \cdots \times A, \mathbb{C}) \qquad \text{(cf. Mumford [7]).}$$

In this case, the idea will be to construct a model M of X/Γ as a subscheme of a model M' already constructed for a variety X'/Γ'. Roughly speaking, one first constructs in M' the points of M corresponding to abelian varieties of C.M. type (= maximum of complex multiplication). One then defines M as the closure of the set of these points. This construction rests on the detailed knowledge of abelian varieties of C.M. type provided by [16].

In the cases considered above, one has much information on the fields of definition of the points of M corresponding to abelian varieties of C.M. type. This makes it possible to give partial solutions to Hilbert's twelfth problem (cf. the introduction of [15] and 3.15, 3.16).

Looking at what these cases have in common, one arrives at the notion of a "canonical model" (§ 3). One shows that there is at most one "canonical model" of the X/Γ (5.5). Descent techniques then make it possible to construct some canonical models which seem far from being solutions of moduli problems ([12] [13] [14] and [5]). For the description of these "strange models", one is referred to §§ 5 and 6.

We shall denote by $\mathbb{Z}/n(1)$ both the group scheme μ_n of nth roots of unity and the group of nth roots of unity in a fixed algebraically closed field $\bar{k}$. For $\bar{k} = \mathbb{C}$, the exponential map identifies $\mathbb{Z}/n(1)$ with $\mathbb{Z}/n$. The $\mathbb{Z}/n(1)$ form a projective system: the transition maps are

$$\varphi_{n,m} : x \longmapsto x^{n/m}.$$

We put

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \simeq \prod_p \mathbb{Z}_p, \qquad \widehat{\mathbb{Z}}(1) = \varprojlim_n \mathbb{Z}/n\mathbb{Z}(1),$$

$$\mathbb{A}^f = \mathbb{Q} \otimes \widehat{\mathbb{Z}} \simeq {\prod_p}' \mathbb{Q}_p \quad \text{(restricted product)}, \qquad \mathbb{A}^f(1) = \mathbb{Q} \otimes \widehat{\mathbb{Z}}(1) = \mathbb{A}^f \otimes_{\widehat{\mathbb{Z}}} \widehat{\mathbb{Z}}(1),$$

and we denote by $\mathbb{A}$ the ring of adeles of $\mathbb{Q}$.

We shall use the following notation:

[1] Text written in March 1971

$\pi_0(X)$ (for X a topological space): the set of connected components of X. In what follows, $\pi_0(X)$, endowed with the quotient topology from X, will be discrete or compact totally disconnected.

G^0: the connected component of the identity of a topological group G (or of a pointed topological space).

$G(K)$, G_K, $G \otimes_F K$: for G a scheme over F (for instance an algebraic group over F) and K an F-algebra, $G(K)$ denotes the set of K-valued points of G, and G_K or $G \otimes_F K$ denotes the scheme over K deduced from G by extension of scalars.

E^*: for E an algebra of finite rank over a field F, E^* denotes the algebraic group over F of invertible elements of E. For E a number field and $F = \mathbb{Q}$, $E^*(\mathbb{A})/E^*(\mathbb{Q})$ is the idele class group of E.

We shall say that an algebraic group G over $\mathbb{Q}$ satisfies the Hasse principle if

$$H^1(\mathbb{Q}, G) = \operatorname{dfn} H^1(\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), G(\overline{\mathbb{Q}}))$$

injects into the product, over all places, of the $H^1(\mathbb{Q}_v, G_{\mathbb{Q}_v})$. We shall use the following theorems.

(0.1) Hasse principle. A simply connected semisimple group, without a factor of type E_8, satisfies the Hasse principle [3].

(0.2) A simply connected semisimple group G over a non-archimedean local field K satisfies $H^1(K, G) = 0$ (a proof independent of the classification is announced in Bruhat–Tits [1]).

(0.3) Strong approximation theorem. Let G be a simply connected semisimple group over a global field K. If G is K-simple, and if v is a place of K such that $G(K_v)$ is non-compact, then $G(K_v)G(K)$ is dense in $G(\mathbb{A})$ (for a proof independent of the classification, see Platonov [9]).

(0.4) Real approximation theorem. Let G be a connected (linear) algebraic group over $\mathbb{Q}$. Then $G(\mathbb{Q})$ is dense in $G(\mathbb{R})$ (to prove this, one reduces easily to the case of tori, which is a corollary of ([4] 5.1)).

§ 1. The adelic language.

1.1. Let G be a reductive group over $\mathbb{Q}$. We denote by G' the derived group of the neutral component of G, by C the center of this neutral component, and we put $T = G/G'$. There are exact sequences of algebraic groups

$$0 \longrightarrow C \longrightarrow G \longrightarrow G/C \longrightarrow 0, \tag{1.1.1}$$

$$0 \longrightarrow G' \longrightarrow G \xrightarrow{\nu} T \longrightarrow 0, \tag{1.1.2}$$

and the canonical maps from C to T and from G' to G/C are isogenies with kernel $C \cap G'$.

1.2. Let $\mathbb{S}$ be the real algebraic group of invertible elements of the $\mathbb{R}$-algebra $\mathbb{C}$. It is also the real algebraic group obtained, by Weil restriction of scalars from $\mathbb{C}$ to $\mathbb{R}$, from the multiplicative group $\mathbb{G}_m$. One has $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$, and $\mathbb{S}_{\mathbb{C}} \simeq \mathbb{G}_{m,\mathbb{C}} \times \mathbb{G}_{m,\mathbb{C}}$. The group $\operatorname{Hom}(\mathbb{S}_{\mathbb{C}}, \mathbb{G}_{m,\mathbb{C}})$ of characters of $\mathbb{S}_{\mathbb{C}}$ has as a basis the characters z and $\bar{z}$ such that the composite

$$\mathbb{C}^* = \mathbb{S}(\mathbb{R}) \longrightarrow \mathbb{S}(\mathbb{C}) \xrightarrow{z,\bar{z}} \mathbb{C}^*$$

is respectively the identity and complex conjugation.

Let V be a real vector space. For any representation $h : \mathbb{S} \to \operatorname{GL}(V)$, one denotes by V^{pq} the subspace of $V_{\mathbb{C}}$ on which $\mathbb{S}$ acts by the character $z^p \bar{z}^q$. The V^{pq} form a bigrading of $V_{\mathbb{C}}$, and the construction $h \mapsto V^{pq}$ identifies the representations of $\mathbb{S}$ in $\operatorname{GL}(V)$ with Hodge bigradings of $V_{\mathbb{C}}$, i.e. with bigradings of $V_{\mathbb{C}}$ such that $V^{pq} = \overline{V^{qp}}$. We shall denote by $F(h)$ the Hodge filtration of $V_{\mathbb{C}}$:

$$F(h)^p = \bigoplus_{p' \geq p} V^{p'q'}.$$

1.3. We denote by $r : \mathbb{G}_{m,\mathbb{C}} \to \mathbb{S}_{\mathbb{C}}$ the $\mathbb{C}$-homomorphism such that

$$(z^p \bar{z}^q) \circ r = (x \longmapsto x^p),$$

and by $w : \mathbb{G}_{m,\mathbb{R}} \to \mathbb{S}$ the $\mathbb{R}$-homomorphism such that

$$(z^p \bar{z}^q) \circ w = (x \longmapsto x^{p+q}).$$

For a representation $h : \mathbb{S} \to \mathrm{GL}(V)$ and $v^{pq} \in V^{pq}$, one has

$$hr(x)\, v^{pq} = x^p v^{pq} \quad \text{and} \quad hw(x)\, v^{pq} = x^{p+q} v^{pq}.$$

The composite $w : \mathbb{R}^* = \mathbb{G}_m(\mathbb{R}) \xrightarrow{w} \mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ is the evident inclusion.

1.4. Let $h : \mathbb{S} \to G_{\mathbb{R}}$ be a homomorphism. For any representation $\rho : G \to \mathrm{GL}(V)$ of G on a $\mathbb{Q}$-vector space V, ρh defines a bigrading of $V_{\mathbb{C}}$ (1.2). If V is faithful, and if G is the subgroup of $\mathrm{GL}(V)$ fixing some tensors s_i, the construction $h \mapsto (V^{pq})$ identifies the homomorphisms h with Hodge bigradings of $V_{\mathbb{C}}$ such that the s_i are of type $(0,0)$.

1.5. Let $h_o : \mathbb{S} \to G_{\mathbb{R}}$ be a homomorphism satisfying the following conditions (cf. [7] and [2]; (1.5.2) is motivated by Griffiths' transversality theorem).

(1.5.1) The image of $h_o w : \mathbb{G}_m \to G_{\mathbb{R}}$ is central. We shall call $h_o w$ the weight of h_o;

(1.5.2) the Hodge bigrading of the complexification $\mathrm{Lie}(G)_{\mathbb{C}}$ of the space of the adjoint representation (1.2) is of type $(-1,1) + (0,0) + (1,-1)$;

(1.5.3) the automorphism $\mathrm{ad}\, h_o(i)$ of G, an involution by (1.5.1), induces a Cartan involution of G'.

Let K_∞ be the centralizer of h_o in $G(\mathbb{R})$; it contains the center of $G(\mathbb{R})$, and $K_\infty \cap G'(\mathbb{R})^0$ is a maximal compact subgroup of $G'(\mathbb{R})^0$, identical with the centralizer in $G'(\mathbb{R})^0$ of $h_o(i)$. The space $K_\infty \backslash G(\mathbb{R})$ is identified with the set X of conjugates of h_o by an element of $G(\mathbb{R})$. One checks that it has one and only one complex structure such that, for every representation $\rho : G \to \mathrm{GL}(V)$, the filtration $F(\rho h)$ of $V_{\mathbb{C}}$ depends holomorphically on $h \in X$. This structure is invariant on the right under $G(\mathbb{R})$, and the connected components of X are symmetric Hermitian domains. We shall denote by X^0 the connected component of h_o.

1.6. Example I. Let $\mathrm{Gp}(V)$ be the group of symplectic similitudes of a real vector space V endowed with a non-degenerate alternating form ψ. There is then one and only one conjugacy class X of homomorphisms $h : \mathbb{S} \to \mathrm{Gp}(V)$ satisfying conditions (1.5.1) and of weight -1, i.e. of weight

$$hw : \mathbb{G}_m \to \mathrm{Gp}(V) : x \longmapsto \text{homothety of ratio } x^{-1}.$$

The space X is identified with the sum of two Siegel half-spaces.

Construction 1.5 identifies the $h \in X$ with decompositions

$$V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1}$$

of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$, such that $V^{pq} = \overline{V^{qp}}$, such that $V^{0,-1}$ is totally isotropic for ψ, and such that, if C denotes the endomorphism of V which induces multiplication by i^{p-q} on V^{pq}, the symmetric form $\psi(x, Cy) = \psi(x, h(i)y)$ on V is definite (> 0 or < 0).

1.7. Let Γ be an arithmetic subgroup of $G(\mathbb{Q})$. We are interested in the case where Γ is defined by congruence conditions. If $V_{\mathbb{Z}}$ is an integral lattice in a faithful representation V of G, this means that there exists n such that Γ contains, with finite index, the group $\Gamma_n \subset G(\mathbb{Q})$ of $\gamma \in G(\mathbb{Q})$ such that $\gamma \in G(\mathbb{R})^0$, $\gamma V_{\mathbb{Z}} = V_{\mathbb{Z}}$, and $\gamma \equiv \mathrm{Id} \pmod{n}$. By Baily and Borel, the quotient X^0/Γ is endowed with a natural structure of quasi-projective scheme over $\mathbb{C}$. Let K_n be the compact open subgroup of $G(\mathbb{A}^f)$ formed by the $k = (k_p)$ such that

$$k_p \cdot V_{\mathbb{Z}} \otimes \mathbb{Z}_p = V_{\mathbb{Z}} \otimes \mathbb{Z}_p \quad \text{and} \quad k_p \equiv 1 \pmod{n} \quad \text{for } p \mid n.$$

The group Γ_n is then the intersection, inside $G(\mathbb{A})$, of $G(\mathbb{Q})$ and $G(\mathbb{R})^0 \times K_n$.

1.8. Let K be a compact open subgroup of $G(\mathbb{A}^f)$. The space

$$K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}) = K\backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q}) = X \times (K\backslash G(\mathbb{A}^f))/G(\mathbb{Q})$$

is then a sum of a finite family of spaces of the type considered in 1.7. It is therefore a quasi-projective scheme over $\mathbb{C}$. We shall denote it by ${}_K M_{\mathbb{C}}(G, h_o)$ (or simply ${}_K M_{\mathbb{C}}(G)$ or ${}_K M_{\mathbb{C}}$).

For K becoming smaller and smaller, these schemes form a projective system of schemes with finite and surjective transition morphisms

$$ {}_K M_{\mathbb{C}}(G, h_o) \qquad (K \to 0). \tag{1.8.1}$$

We shall denote by $M_{\mathbb{C}}(G, h_o)$ (or simply $M_{\mathbb{C}}(G)$, or $M_{\mathbb{C}}$) the quasi-compact separated scheme (not of finite type for $G \neq \{1\}$) which is the projective limit of (1.8.1)

$$M_{\mathbb{C}}(G, h_o) = \varprojlim_K K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}). \tag{1.8.2}$$

It must be considered as an avatar of the projective system (1.8.1). The group $G(\mathbb{A}^f)$ acts on $M_{\mathbb{C}}$, or, if one prefers, on the pro-object defined by the projective system (1.8.1), but of course not on the individual ${}_K M_{\mathbb{C}}$ (cf. 3.2). One has

$$K\backslash M_{\mathbb{C}} = {}_K M_{\mathbb{C}},$$

so that

$$M_{\mathbb{C}} = \varprojlim_K K\backslash M_{\mathbb{C}} \qquad (K \text{ compact open in } G(\mathbb{A}^f)). \tag{1.8.3}$$

This is expressed by saying that $G(\mathbb{A}^f)$ acts continuously on $M_{\mathbb{C}}$.

The following construction will be useful for interpreting the ${}_K M_{\mathbb{C}}$ as moduli schemes.

1.9. Let H be an algebraic group over $\mathbb{Q}$ and V a faithful representation of H. We define an H-structure $\bar{\iota}$ on a vector space W to be an isomorphism from V to W, given modulo $H(\mathbb{Q})$: $\bar{\iota} \in \operatorname{Isom}(V, W)/H(\mathbb{Q})$. If $H(\mathbb{Q})$ is the group of automorphisms of a structure s of species Σ on V, then an H-structure on W is identified with a structure t of the same species on W, isomorphic to s (with $\bar{\iota} = \operatorname{Isom}((V, s), (W, t))$). Let W be endowed with an H-structure $\bar{\iota}$. The algebraic group $\iota H \iota^{-1} \subset \mathrm{GL}(W)$ ($\iota \in \bar{\iota}$) depends only on $\bar{\iota}$. It is denoted $H^{\bar{\iota}}$. Let A be a $\mathbb{Q}$-algebra. The set $\operatorname{Isom}_{\bar{\iota}}(W \otimes A, V \otimes A)$ of permissible isomorphisms from $W \otimes A$ to $V \otimes A$ is the set of isomorphisms k such that, for $\iota \in \bar{\iota}$, one has $k\iota \in H(A)$.

1.10. Let (G, h_o) be as in 1.5, let V be a faithful linear representation of G, let K be a compact open subgroup of $G(\mathbb{A}^f)$, and consider objects H of the following type: H consists of

(a) a vector space $H_{\mathbb{Q}}$ over $\mathbb{Q}$, endowed with a G-structure $\bar{\iota}$ (1.9);

(b) a homomorphism $h : \mathbb{S} \to G_{\mathbb{R}}^{\bar{\iota}}$ such that, for $\iota \in \bar{\iota}$, $\iota^{-1} h \iota : \mathbb{S} \to G_{\mathbb{R}}$ is conjugate to h_o;

(c) a class $\bar{k} \in K\backslash \operatorname{Isom}_{\bar{\iota}}(H \otimes \mathbb{A}^f, V \otimes \mathbb{A}^f)$ (1.9).

1.11. Example I (continued from 1.6). Let V be a $\mathbb{Q}$-vector space endowed with a non-degenerate alternating form ψ, let G be the group of symplectic similitudes of V, let $V_{\mathbb{Z}}$ be an integral lattice on which ψ has integral values and discriminant 1, and let h_o be as in 1.6. Take

$$K = \prod_\ell \mathrm{Gp}(V_{\mathbb{Z}} \otimes \mathbb{Z}_\ell).$$

A datum (a) is interpreted as a vector space H of the same dimension as V, endowed with an alternating form ψ given up to a rational factor. A datum (b) is interpreted as a Hodge bigrading of $H_{\mathbb{C}}$, of type 1.6. A datum (c) is interpreted as an integral lattice $H_{\mathbb{Z}}$ of H, on which a rational multiple of ψ has integral values and discriminant 1: $\bar{k}$ is the set of k such that, for every ℓ, $V_{\mathbb{Z}} \otimes \mathbb{Z}_\ell = k_\ell(H_{\mathbb{Z}} \otimes \mathbb{Z}_\ell)$. One can normalize ψ by the conditions of having integral values and discriminant 1 on $H_{\mathbb{Z}}$, and of satisfying

$$\psi(x, Cx) > 0 \qquad (\text{cf. } 1.6).$$

Let n be an integer, and let K_n be the subgroup of K formed by the k such that $k \equiv 1 \pmod{n}$. For K_n, a datum (c) is interpreted as $H_{\mathbb{Z}}$ as above, plus a symplectic similitude

$$H_{\mathbb{Z}}/nH_{\mathbb{Z}} \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}.$$

1.12. Under the hypotheses of 1.10, the points of ${}_KM_{\mathbb{C}}(G,h_o)$ are in bijective correspondence with the isomorphism classes of objects H of type 1.10. Indeed, let

$$H = (H_{\mathbb{Q}}, \bar{\iota}, h, \bar{k}).$$

To H endowed with $\iota \in \bar{\iota}$, one associates the morphism $\iota^{-1}h\iota : \mathbb{S} \to G_{\mathbb{R}}$, an element of $X \simeq K_\infty\backslash G(\mathbb{R})$, and $\bar{k}$ in $K\backslash G(\mathbb{A}^f)$, hence in total an element of $K\backslash X \times G(\mathbb{A}^f)$; to H alone, one associates only an element of

$${}_KM_{\mathbb{C}}(G,h_o) = K\backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q}) = K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}),$$

which uniquely determines the isomorphism class of H.

In certain cases (see 4.11), to give an object H as above amounts to giving an abelian variety endowed with some supplementary structures. One then obtains an interpretation of ${}_KM_{\mathbb{C}}(G)$ (and of $M_{\mathbb{C}}(G)$) as the coarse moduli scheme for this type of abelian variety.

1.13. Let G^i $(i = 1, 2)$ be two reductive groups, let $h^i : \mathbb{S} \to G^i_{\mathbb{R}}$ be homomorphisms satisfying the conditions of 1.5, and let X^i be the set of conjugates of h^i by an element of $G^i(\mathbb{R})$. Let $G = G^1 \times G^2$, let $h : \mathbb{S} \to G_{\mathbb{R}}$ be the morphism (h^1, h^2), and let $X = X^1 \times X^2$ be the set of conjugates of h. In the evident way one defines an isomorphism

$$M_{\mathbb{C}}(G^1,h^1) \times M_{\mathbb{C}}(G^2,h^2) = M_{\mathbb{C}}(G,h). \tag{1.13.1}$$

For K^i compact open in $G^i(\mathbb{A}^f)$ and $K = K^1 \times K^2$, one likewise has

$${}_{K^1}M_{\mathbb{C}}(G^1,h^1) \times {}_{K^2}M_{\mathbb{C}}(G^2,h^2) = {}_KM_{\mathbb{C}}(G,h). \tag{1.13.2}$$

1.14. Let G^1, G^2, h^1, h^2 be as in 1.13. We shall denote by $u : (G^1,h^1) \to (G^2,h^2)$ a homomorphism $u : G^1 \to G^2$ such that $uX^1 \subset X^2$. Such a homomorphism defines

$$u : M_{\mathbb{C}}(G^1,h^1) \longrightarrow M_{\mathbb{C}}(G^2,h^2). \tag{1.14.1}$$

For K^1 compact open in $G^1(\mathbb{A}^f)$, with $u(K^1) \subset K^2$, it defines

$$u(K^1,K^2) : {}_{K^1}M_{\mathbb{C}}(G^1,h^1) \longrightarrow {}_{K^2}M_{\mathbb{C}}(G^2,h^2). \tag{1.14.2}$$

Proposition 1.15. With the notation of 1.14, suppose that G^1 is a subgroup of G^2. Then, for every compact open subgroup K^1 of $G^1(\mathbb{A}^f)$, there exists a compact open subgroup $K^2 \supset K^1$ of $G^2(\mathbb{A}^f)$ such that $u(K^1,K^2)$ identifies ${}_{K^1}M_{\mathbb{C}}(G^1,h^1)$ with a closed subscheme of ${}_{K^2}M_{\mathbb{C}}(G^2,h^2)$.

If the K^i are small enough that the map from $X^i \times (K^i\backslash G^i(\mathbb{A}^f))$ to ${}_{K^i}M_{\mathbb{C}}(G^i,h^i)$ is étale, then the map $u(K^1,K^2)$ is finite and unramified. It suffices to treat this case, and to show that $u(K^1,K^2)$ is injective for $K^2 \supset K^1$ sufficiently small. The graph of the equivalence relation

$$u(K^1,K^2)(x) = u(K^1,K^2)(y)$$

is a closed subscheme of $({}_{K^1}M_{\mathbb{C}}(G^1,h^1))^2$, decreasing with K^2, hence stationary for $K^2 \supset K^1$ sufficiently small. It is enough to show that

$$u(K^1) : {}_{K^1}M_{\mathbb{C}}(G^1,h^1) \longrightarrow \varprojlim_{K^2 \supset K^1} {}_{K^2}M_{\mathbb{C}}(G^2,h^2) = \mathrm{dfn}\, {}_{K^1}M_{\mathbb{C}}(G^2,h^2)$$

is injective. One has

$${}_{K^1}M_{\mathbb{C}}(G^1,h^1) = K^1\backslash M_{\mathbb{C}}(G^1,h^1), \qquad {}_{K^1}M_{\mathbb{C}}(G^2,h^2) = K^1\backslash M_{\mathbb{C}}(G^2,h^2),$$

so it is enough to prove the

Variant 1.15.1. $u : M_{\mathbb{C}}(G^1,h^1) \longrightarrow M_{\mathbb{C}}(G^2,h^2)$ is injective.

This assertion follows easily from the following two lemmas.

Lemma 1.15.2. Let $U^i \subset C^i(\mathbb{Q})$ be the group of units of the centre of G^i, and put

$$G^i(\mathbb{Q})^\wedge = \varprojlim_n \big(G^i(\mathbb{Q})/(U^i)^n\big).$$

Then

$$M_{\mathbb{C}}(G^i, h^i) = K^i_\infty \backslash G^i(\mathbb{A})/G^i(\mathbb{Q})^\wedge.$$

This is a corollary of Chevalley's theorem, according to which the $(U^i)^n$ are congruence subgroups of U^i.

Lemma 1.15.3. The map

$$u : G^1(\mathbb{A}^f)/G^1(\mathbb{Q})^\wedge \longrightarrow G^2(\mathbb{A}^f)/G^2(\mathbb{Q})^\wedge$$

is injective.

By left translation it is enough to show that $u^{-1}(\{e\}) = \{e\}$. Consider the diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & G^1(\mathbb{Q})^\wedge & \longrightarrow & G^2(\mathbb{Q})^\wedge & \longrightarrow & (G^2/G^1)(\mathbb{Q})^\wedge \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & G^1(\mathbb{A}^f) & \longrightarrow & G^2(\mathbb{A}^f) & \longrightarrow & (G^2/G^1)(\mathbb{A}^f) \\
 & & \downarrow & & \downarrow & & \\
 & & G^1(\mathbb{A}^f)/G^1(\mathbb{Q})^\wedge & \longrightarrow & G^2(\mathbb{A}^f)/G^2(\mathbb{Q})^\wedge. & &
\end{array}$$

where

$$(G^2/G^1)(\mathbb{Q})^\wedge = \varprojlim_n (G^2/G^1)(\mathbb{Q})/(U^2/U^1)^n.$$

§ 2. Connected components.

2.1. Let G be a connected reductive group over $\mathbb{Q}$. We use the notation of 1.1 and assume that G' admits no non-trivial invariant subgroup H (defined over $\mathbb{Q}$) such that $H(\mathbb{R})$ is compact. This hypothesis does not exclude the possibility that $G'_{\mathbb{R}}$ has compact factors. We denote by $\widetilde{G}'$ the simply connected covering of the derived group G'.

Proposition 2.2. (i) The group $G(\mathbb{A}^f)$ acts transitively on the set

$$\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$$

of connected components of $G(\mathbb{A})/G(\mathbb{Q})$.

(ii) The group $\widetilde{G}'(\mathbb{A})$ acts trivially on $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$.

(iii) The quotient of $G(\mathbb{A})$ by the image of $\widetilde{G}'(\mathbb{A})$ is a commutative group.

Assertion (i) follows from (0.4):

$$G(\mathbb{A}) = G(\mathbb{A}^f)G(\mathbb{R})^0 G(\mathbb{Q}).$$

Since $\widetilde{G}'$ is simply connected, $\widetilde{G}'(\mathbb{R})$ is connected. By (0.3), $\widetilde{G}'(\mathbb{R})\widetilde{G}'(\mathbb{Q})$ is dense in $\widetilde{G}'(\mathbb{A})$, so that $\widetilde{G}'(\mathbb{A})/\widetilde{G}'(\mathbb{Q})$ is connected. For $x \in G(\mathbb{A})$ one has $\widetilde{G}'(\mathbb{A})x = x\widetilde{G}'(\mathbb{A})$; hence the image of $\widetilde{G}'(\mathbb{A})x$ in $G(\mathbb{A})/G(\mathbb{Q})$ is an image of $\widetilde{G}'(\mathbb{A})/\widetilde{G}'(\mathbb{Q})$: it is connected, and (ii) follows.

Let Z be the centre of $\widetilde{G}'$. Assertion (iii) follows from the fact that the commutator map

$$(x, y) \longmapsto xyx^{-1}y^{-1} : G \times G \longrightarrow G$$

has a factorization

$$G \times G \longrightarrow G/C \times G/C \longleftarrow \widetilde{G}'/Z \times \widetilde{G}'/Z \longrightarrow \widetilde{G}' \longrightarrow G.$$

Definition 2.3. We denote by $\pi(G)$ the commutative quotient of $G(\mathbb{A})$, or of $G(\mathbb{A}^f)$, which acts faithfully on $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$.

In other words, $\pi(G)$ is $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$, endowed with its "evident" commutative group structure. We recall the following theorem.

Theorem 2.4. Under the hypotheses of 2.1, if G' is simply connected, then the canonical homomorphism

$$\pi(G) = \pi_0(G(\mathbb{A})/G(\mathbb{Q})) \longrightarrow \pi_0(T(\mathbb{A})/T(\mathbb{Q})) \tag{2.4.1}$$

is bijective.

To prove the surjectivity of (2.4.1), by 2.2(i) applied to T it is enough to show that $G(\mathbb{A}^f)$ maps onto $T(\mathbb{A}^f)$. Since the kernel G' of $\nu : G \to T$ is connected, one knows that, for almost all p, $\nu(G(\mathbb{Z}_p)) = T(\mathbb{Z}_p)$; this formula has a meaning for almost all p, and is a corollary of Lang's theorem applied to G' reduced modulo p. It is therefore enough to prove that, for every p,

$$\nu(G(\mathbb{Q}_p)) = T(\mathbb{Q}_p),$$

which follows from (0.2) applied to G'.

Theorem 2.4 is equivalent to the following more concrete statement.

Variant 2.5. For every compact open subgroup K of $G(\mathbb{A}^f)$, the map

$$\nu : \pi_0(K\backslash G(\mathbb{A})/G(\mathbb{Q})) \longrightarrow \pi_0(\nu(K)\backslash T(\mathbb{A})/T(\mathbb{Q})) \tag{2.5.1}$$

is bijective.

Indeed,

$$\pi_0(G(\mathbb{A})/G(\mathbb{Q})) = \varprojlim_K \pi_0(K\backslash G(\mathbb{A})/G(\mathbb{Q})),$$

and similarly for T. Also,

$$\pi_0(K\backslash G(\mathbb{A})/G(\mathbb{Q})) = G(\mathbb{R})^0\backslash K\backslash G(\mathbb{A})/G(\mathbb{Q}).$$

We show that if $x, y \in G(\mathbb{A})$ have the same image in the right-hand side of (2.5.1), then they have the same image in the left-hand side. Left translation by y^{-1}, which replaces K by $y^{-1}Ky$, allows us to suppose that $y = e$. Modifying x by an element of $G(\mathbb{R})^0 \times K$, we may suppose that $\nu(x) = \tau$ with $\tau \in T(\mathbb{Q})$. By the Hasse principle (0.1) for G', the equation $\nu(y) = \tau$ has a solution in $G(\mathbb{Q})$. Correcting x by y, we may suppose that $x \in G'(\mathbb{A})$. By 2.2(ii), the image of x in $G(\mathbb{A})/G(\mathbb{Q})$ is in the neutral component, and a fortiori x and y have the same image in the left-hand side of 2.5.1.

This proof seems to use, via (0.1), that G' has no factor of type E_8. In fact, the class in $H^1(\mathbb{Q}, G')$ which obstructs the equation $\nu(y) = \tau$ comes from the centre of G', and the E_8 factors, being adjoint, count for nothing.

Proposition 2.6. One has

$$K_\infty = C(\mathbb{R})\big(K_\infty \cap G'(\mathbb{R})^0\big).$$

The group $K_\infty \cap G'(\mathbb{R})^0$ is connected, and

$$\pi_0(K_\infty) \xrightarrow{\sim} \mathrm{Im}\big(\pi_0(C(\mathbb{R})) \longrightarrow \pi_0(G(\mathbb{R}))\big).$$

Let $h' : \mathbb{S} \to G_\mathbb{R} \to (G/C)_\mathbb{R}$. The centralizer of h' in $(G/C)(\mathbb{R})$ is connected, because it is a compact group whose complexification is connected, as the centralizer of a torus. In particular, the image of K_∞ in $G/C(\mathbb{R})$ lies in the neutral component, the image of $G'(\mathbb{R})^0$, and

$$K_\infty = C(\mathbb{R})\big(K_\infty \cap G'(\mathbb{R})^0\big).$$

The group $K_\infty \cap G'(\mathbb{R})^0$ is connected as a maximal compact subgroup of a connected group. This proves 2.6.

2.7. The image of $\pi_0(K_\infty)$ in $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$ depends only on the conjugacy class of h, and

$$\pi_0(M_\mathbb{C}(G,h)) = \varprojlim_K \pi_0(K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}))$$

is identified with $\pi(G)/\pi_0(K_\infty)$.

In particular, if G' is simply connected, one has

$$\pi_0(M_\mathbb{C}(G,h)) \xrightarrow{\sim} \pi_0(T(\mathbb{A})/T(\mathbb{Q}))/\pi_0(K_\infty). \tag{2.7.1}$$

More concretely, for every compact open subgroup K of $G(\mathbb{A}^f)$, $\pi_0({}_K M_\mathbb{C}(G,h))$ is then a principal homogeneous space under

$$\nu(K_\infty \times K)\backslash T(\mathbb{A})/T(\mathbb{Q}).$$

§ 3. Models.

Let G be a reductive group over $\mathbb{Q}$, let $h : \mathbb{S} \to G_{\mathbb{R}}$ satisfy the conditions of 1.5, and let E be a field endowed with a complex embedding $\rho : E \to \mathbb{C}$. Put $M_{\mathbb{C}}(G) = M_{\mathbb{C}}(G, h)$.

Definition 3.1. A model over E of $M_{\mathbb{C}}(G)$ consists of

(a) a scheme M over E, endowed with a continuous action of $G(\mathbb{A}^f)$;

(b) an isomorphism m from $M \otimes_{E,\rho} \mathbb{C}$ to $M_{\mathbb{C}}(G)$, compatible with the action of $G(\mathbb{A}^f)$.

Let F be a finite extension of E, endowed with a complex embedding extending that of E. If $M_E(G,h)$ is a model of $M_{\mathbb{C}}(G,h)$ over E, we denote by $M_F(G,h)$ the model $M_E(G,h) \otimes_E F$ of $M_{\mathbb{C}}(G,h)$ over F.

Remark 3.2. Giving a scheme M over E, endowed with a continuous action of $G(\mathbb{A}^f)$, is equivalent to giving

(a) for every compact open subgroup K of $G(\mathbb{A}^f)$, a scheme ${}_K M$ over E;

(b) for compact open subgroups K, L of $G(\mathbb{A}^f)$ and for $x \in G(\mathbb{A}^f)$ such that $xKx^{-1} \subset L$, a homomorphism

$$J_{L,K}(x) : {}_K M \longrightarrow {}_L M,$$

these homomorphisms satisfying

$$(\alpha) \quad J_{M,L}(y) J_{L,K}(x) = J_{M,K}(yx), \tag{3.2α}$$

$$(\beta) \quad J_{K,K}(x) = \mathrm{Id} \qquad \text{if } x \in K, \tag{3.2β}$$

$$(\gamma) \quad \text{for } K \text{ normal in } L,\ J \text{ defines an action of } L/K \text{ on } {}_K M, \text{ and } J_{L,K}(e) \text{ defines } (L/K)\backslash {}_K M \xrightarrow{\sim} {}_L M. \tag{3.2γ}$$

One has

$$M = \varprojlim_K {}_K M, \tag{3.2.1}$$

and

$${}_K M = K \backslash M. \tag{3.2.2}$$

3.3. Let $M(G)$ be a model over E of $M_{\mathbb{C}}(G)$. For K compact open in $G(\mathbb{A}^f)$, put ${}_K M(G) = K\backslash M(G)$. The ${}_K M(G)$ are quasi-projective schemes over E, not necessarily geometrically connected. The structural isomorphism m induces isomorphisms

$${}_K M(G) \otimes_{E,\rho} \mathbb{C} \xrightarrow{\sim} {}_K M_{\mathbb{C}}(G). \tag{3.3.1}$$

3.4. Suppose that G satisfies the hypotheses of 2.1, and let (M, m) be a model of $M_{\mathbb{C}}(G,h)$ over E. Let $\overline{E}$ be the algebraic closure of E in $\mathbb{C}$. Since M is defined over E, the Galois group $\mathrm{Gal}(\overline{E}/E)$ acts on the profinite set

$$\pi_0(M \otimes_E \overline{E}) = \varprojlim_K \pi_0({}_K M \otimes_E \overline{E}) \xrightarrow{\sim}_m \pi_0(M_{\mathbb{C}}(G,h)).$$

The group $G(\mathbb{A}^f)$ acts on $\pi_0(M \otimes_E \overline{E})$ through its quotient $\pi(G)/\pi_0(K_\infty)$, and $\pi_0(M \otimes_E \overline{E})$ is a principal homogeneous space under the commutative group $\pi(G)/\pi_0(K_\infty)$ (2.6). The action of $\mathrm{Gal}(\overline{E}/E)$ commutes with that of $G(\mathbb{A}^f)$, hence with that of $\pi(G)/\pi_0(K_\infty)$. The action of an element σ of $\mathrm{Gal}(\overline{E}/E)$ can therefore only be the translation defined by an element $\lambda(\sigma)$ of $\pi(G)/\pi_0(K_\infty)$, and λ is a homomorphism

$$\lambda_M : \mathrm{Gal}(\overline{E}/E) \longrightarrow \pi(G)/\pi_0(K_\infty). \tag{3.4.1}$$

Assume that E is a number field. Class field theory identifies the largest abelian quotient of $\mathrm{Gal}(\overline{E}/E)$ with $\pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q}))$, and identifies (3.4.1) with

$$\lambda_M : \mathrm{Gal}(\overline{E}/E)^{\mathrm{ab}} = \pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q})) \longrightarrow \pi(G)/\pi_0(K_\infty). \tag{3.4.2}$$

In the special case where G' is simply connected, the homomorphism (3.4.2) is identified, by (2.7.1), with

$$\lambda_M : \pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q})) \longrightarrow \pi_0(T(\mathbb{A})/T(\mathbb{Q}))/\pi_0(K_\infty). \tag{3.4.3}$$

Definition 3.5. The homomorphisms (3.4.1), (3.4.2), and (3.4.3) are called the reciprocity law of the model M. If G' is simply connected, and if λ_M comes from a homomorphism of algebraic groups over $\mathbb{Q}$ from E^* to T, the latter will also be called the reciprocity law of M.

3.6. There is a rule, applicable to each of the models constructed by Shimura, for determining E and the reciprocity law λ_M from G and h. The field E, for arbitrary G, and λ_M, for simply connected G', are described as follows.

3.7. The composite homomorphism hr (1.3)

$$hr : \mathbb{G}_{m,\mathbb{C}} \xrightarrow{r} \mathbb{S}_{\mathbb{C}} \xrightarrow{h} G_{\mathbb{C}}$$

is a homomorphism over $\mathbb{C}$ between algebraic groups defined over $\mathbb{Q}$. The subfield E of $\mathbb{C}$ is the field of definition $E(G,h)$ of the conjugacy class of this homomorphism.

For G semisimple adjoint, the field $E(G,h)$ can be described as follows. Let Δ be the Dynkin diagram of $G_{\mathbb{C}}$, and let $|\Delta|$ be the set of vertices of Δ. The Galois group $\mathrm{Gal}(\mathbb{C}/\mathbb{Q})$ acts on Δ.

Let H be a maximal torus of $G_{\mathbb{C}}$, endowed with a system of simple roots $(\alpha_i)_{i\in|\Delta|}$. The α_i identify H with $\mathbb{G}_m^{\Delta}$. For every homomorphism $u : \mathbb{G}_{m,\mathbb{C}} \to G_{\mathbb{C}}$, there is then a unique homomorphism $u' : \mathbb{G}_{m,\mathbb{C}} \to H = \mathbb{G}_{m,\mathbb{C}}^{\Delta}$,

$$x \longmapsto (x^{n_i})_{i\in|\Delta|} \qquad (n_i \geq 0),$$

which is conjugate to u. The construction $u \mapsto \eta(u) = (n_i)_{i\in|\Delta|}$ identifies the set of conjugacy classes of maps from $\mathbb{G}_{m,\mathbb{C}}$ to $G_{\mathbb{C}}$ with the set of functions from $|\Delta|$ to $\mathbb{N}$. It follows that $E(G,h)$ is defined by the subgroup of $\mathrm{Gal}(\mathbb{C}/\mathbb{Q})$ which stabilizes $\eta(hr)$.

Proposition 3.8. Let

$$G/C = \prod_{i=1}^{g} G_i$$

be the decomposition of the adjoint group G/C into $\mathbb{Q}$-simple factors, and let h_i be the composite

$$h_i : \mathbb{S} \xrightarrow{h} G_{\mathbb{R}} \longrightarrow (G_i)_{\mathbb{R}}.$$

(i) The subfield $E(G,h)$ of $\mathbb{C}$ is the compositum of the subfields $E(T,\nu h)$ and $E(G_i,h_i)$ $(1 \leq i \leq g)$ of $\mathbb{C}$.

(ii) If the opposition involution of the Dynkin diagram of G_i respects $\eta(h_i r)$, then $E(G_i,h_i)$ is totally real. Otherwise, $E(G_i,h_i)$ is a totally imaginary quadratic extension of a totally real number field.

Assertion (i) is easy to check, and it is enough to prove (ii) for G $\mathbb{Q}$-simple and adjoint. Let Δ be its Dynkin diagram. The hypotheses 1.5 imply that $G_{\mathbb{R}}$ has a compact maximal torus; this implies that complex conjugation acts on Δ by the opposition involution ι. This involution is in the centre of the automorphism group of Δ.

Let E' be the Galois extension of $\mathbb{Q}$ defined by the subgroup of $\mathrm{Gal}(\mathbb{C}/\mathbb{Q})$ which acts trivially on Δ. Then the image c of complex conjugation is central in $\mathrm{Gal}(E'/\mathbb{Q})$; hence E' is either totally real or a totally imaginary quadratic extension of a totally real field. Finally, c is the identity on $E(G,h) \subset E'$ if and only if ι respects $\eta(hr)$, and the assertion follows.

3.9. We use the notation of 3.6. Suppose that G' is simply connected, that E contains $E(G,h)$, and that the complex embedding of E induces that of $E(G,h)$. The composite morphism

$$r''(h) : \mathbb{G}_{m,\mathbb{C}} \xrightarrow{r} \mathbb{S}_{\mathbb{C}} \xrightarrow{h} G_{\mathbb{C}} \xrightarrow{\nu} T_{\mathbb{C}}$$

depends only on the conjugacy class of hr. It is therefore defined over E, i.e. it is obtained by extension of scalars from E to $\mathbb{C}$ from a homomorphism of algebraic groups over E, again denoted

$$r''(h) : \mathbb{G}_m \longrightarrow T_E.$$

Apply Weil restriction of scalars, and let $r'(h) : E^* \to T$ be the composite

$$E^* = \mathrm{Res}_{E/\mathbb{Q}}\,\mathbb{G}_m \xrightarrow{\mathrm{Res}_{E/\mathbb{Q}}(r''(h))} \mathrm{Res}_{E/\mathbb{Q}}(T_E) \xrightarrow{N_{E/\mathbb{Q}}} T.$$

The reciprocity law of M is the inverse $r(G,h)$ of $r'(h)$:

$$r(G,h) = r'(h)^{-1} : E^* \longrightarrow T. \tag{3.9.1}$$

3.10. Example II. Let H be a torus and let $h : \mathbb{S} \to H_{\mathbb{R}}$. The conditions of 1.5 are automatically satisfied. The ${}_K M_{\mathbb{C}}(H,h)$ are finite sets. A finite reduced scheme over the field $E(H,h)$ (3.7) is the same as a Galois set. It follows that, up to unique isomorphism, there is a unique model of $M_{\mathbb{C}}(H,h)$ over $E(H,h)$, with reciprocity law given by 3.9.1. It is denoted $M(H,h)$.

3.11. Let F be a finite extension of E, endowed with a complex embedding extending that of E. The diagram

$$\begin{tikzcd} F^* \arrow[r, "N_{F/E}"] \arrow[dr] & E^* \arrow[d] \\ & T \end{tikzcd}$$

is commutative. If $M_E(G,h)$ is a model over E of $M_{\mathbb{C}}(G,h)$ whose reciprocity law is (3.9.1), it follows that $M_F(G,h)$ (3.1) again has (3.9.1) as its reciprocity law.

3.12. Let $u : (G_1,h_1) \to (G_2,h_2)$ be as in 1.14, and let $M^i(G_i,h_i)$ be a model of $M_{\mathbb{C}}(G_i,h_i)$ over E_i. Let E be the compositum of the subfields E_1 and E_2 of $\mathbb{C}$. With the notation of 3.1 one has

$$M^i(G_i,h_i) \otimes_{E_i} E = M_E(G_i,h_i) \qquad (i = 1,2).$$

It therefore makes sense to ask whether

$$u : M_{\mathbb{C}}(G_1,h_1) \longrightarrow M_{\mathbb{C}}(G_2,h_2)$$

is defined over E.

Definition 3.13. Let G be a connected reductive group over $\mathbb{Q}$ satisfying the condition of 2.1, let $h : \mathbb{S} \to G_{\mathbb{R}}$ be a homomorphism satisfying the conditions of 1.5, and let E be an extension of $E(G,h)$ endowed with a complex embedding extending that of $E(G,h)$. A model $M_E(G,h)$ of $M_{\mathbb{C}}(G,h)$ over E is called weakly canonical if, for every torus $u : H \to G$ in G, endowed with $h' : \mathbb{S} \to H_{\mathbb{R}}$ such that uh' is the conjugate of h by an element of $G(\mathbb{R})$, the morphism (1.14)

$$M_{\mathbb{C}}(H,h') \longrightarrow M_{\mathbb{C}}(G,h)$$

is defined, in the sense of (3.10) and (3.12), over the compositum $E(H,h') \cdot E \subset \mathbb{C}$. A model $M_E(G,h)$ is called canonical if it is weakly canonical and if $E = E(G,h)$.

3.14. Translation. Let M be a canonical model of $M_{\mathbb{C}}(G,h)$, with reciprocity law λ_M. If G' is simply connected, then λ_M is given by 3.9.1 (5.6). For every compact open subgroup K of $G(\mathbb{A}^f)$, again denote by $\nu(K)$ the image of K in $\pi(G)/\pi_0(K_\infty)$, and let $E(K)$ be the extension of $E(G,h)$ defined by the idèle class group $\lambda_M^{-1}(\nu(K))$. One verifies that $E(K)$ is the field of definition of the neutral component ${}_K M^0_{\mathbb{C}}$ of ${}_K M_{\mathbb{C}}$, considered as a subscheme of ${}_K M_{\mathbb{C}} = {}_K M \otimes_E \mathbb{C}$.

By definition, ${}_K M^0_{\mathbb{C}}$ is obtained by extension of scalars from $E(K)$ to $\mathbb{C}$ from a subscheme ${}_K M'$ of ${}_K M \otimes_E E(K)$. The scheme ${}_K M'/E(K)$ is geometrically connected over $E(K)$, and one can also describe ${}_K M'$ as the union of those connected components, over E, of ${}_K M$ which, after extension of scalars from E to $\mathbb{C}$, contain the origin:

$$\begin{tikzcd} {}_K M^0_{\mathbb{C}} \arrow[r, phantom, "\subset"] \arrow[d] & {}_K M_{\mathbb{C}} \arrow[d] \\ \operatorname{Spec} E(K) \arrow[r] \arrow[d] & \operatorname{Spec} \mathbb{C} \\ \operatorname{Spec} E \arrow[r, phantom, "="] & \operatorname{Spec} E. \end{tikzcd}$$

Shimura usually states his results in terms of the system of varieties ${}_K M'/E(K)$ and the homomorphisms of type 2.2 between them. For a typical statement, see [14, p. 146]. For the translation, cf. [14, 2.7].

3.15. For $u : (H, h') \longrightarrow (G, h)$ as in 3.13, the image of $M_{\mathbb{C}}(H, h')$ in ${}_K M_{\mathbb{C}}(G, h)$ is a finite set. The hypothesis is that this part of ${}_K M_{\mathbb{C}}(G, h)$ be defined over $E(H, h')$, that its points be defined over an abelian extension of $E(H, h')$, and that $\mathrm{Gal}(\overline{E(H, h')}/E(H, h'))$ permute them in a prescribed manner.

The points of ${}_K M_{\mathbb{C}}(G, h)$ lying in the image of some $M_{\mathbb{C}}(H, h')$ will be called *special*.

Suppose that a projective embedding

$$q : {}_K M^0_{\mathbb{C}} \longrightarrow \mathbb{P}^r$$

is known, and is defined over $E(K)$. It may be interpreted as

$$\widetilde{q} : X^0 \longrightarrow \mathbb{P}^r$$

(cf. 1.7, 1.8). If $uh' \in X$ lies in X^0, then the coordinates of $\widetilde{q}(uh')$ generate, over $E(K)E(H, h')$, an abelian extension of $E(H, h')$ which can be made explicit as a class field.

3.16. The classical example of such a situation is provided by the moduli scheme of elliptic curves, corresponding to $G = \mathrm{GL}_2$, $K = \mathrm{GL}_2(\widehat{\mathbb{Z}})$, h as in 1.6, and by those of its points defined by elliptic curves with complex multiplication.

Let j be the modular invariant, viewed as a function on the Poincaré half-plane X^0. If $\tau \in X^0$ belongs to an imaginary quadratic field K, put

$$L(\tau) = \mathbb{Z} \oplus \mathbb{Z}\tau \subset \mathbb{C}, \qquad \mathcal{O}(\tau) = \{\, k \in K \mid kL(\tau) \subset L(\tau) \,\}.$$

The field $K(j(\tau))$ is the abelian extension of K whose Galois group is the ideal class group of $\mathcal{O}(\tau)$. If σ in Galois corresponds to an invertible module L, and if

$$L(\tau') \simeq L(\tau) \otimes_{\mathcal{O}(\tau)} L,$$

then $j(\tau) = j(\tau')^{\sigma}$.

For more unusual explicit examples, see [15], pp. 45–46.

§4. Abelian varieties

4.1. Let k be an algebraically closed field of characteristic 0. Let A be an abelian variety over k. For $n \in \mathbb{N}^+$, denote by A_n the kernel of multiplication by n. The A_n form a projective system

$$\varphi_{nm,n} : A_{nm} \longrightarrow A_n, \qquad x \longmapsto x^m,$$

and we put

$$\widehat{T}(A) = \varprojlim_n A_n, \qquad \widehat{V}(A) = \mathbb{A}^f \otimes_{\widehat{\mathbb{Z}}} \widehat{T}(A).$$

The $\widehat{\mathbb{Z}}$-module $\widehat{T}(A)$ is the product of the Tate modules, or Weil representations, $T_\ell(A)$. If $k = \mathbb{C}$, then

$$\widehat{T}(A) = \widehat{\mathbb{Z}} \otimes H_1(A, \mathbb{Z}), \tag{4.1.1}$$

$$\widehat{V}(A) = \mathbb{A}^f \otimes H_1(A, \mathbb{Q}). \tag{4.1.2}$$

4.2. The category of *abelian varieties up to isogeny* over k is the category whose objects are the abelian varieties over k and in which

$$\mathrm{Hom}_{\mathrm{iso}}(A, B) = \mathrm{Hom}(A, B) \otimes \mathbb{Q}.$$

We denote by $A \otimes \mathbb{Q}$ the abelian variety up to isogeny underlying an abelian variety A. Every additive functor from the category of abelian varieties to a $\mathbb{Q}$-linear additive category factors through the functor $A \mapsto A \otimes \mathbb{Q}$. Thus the contravariant functor “dual variety” $A \mapsto A^\vee$ extends to abelian varieties up to isogeny. The functor $A \mapsto \widehat{V}(A)$ extends in the same way. Moreover, if A_0 is an abelian variety up to isogeny, it is equivalent to give B together with an isomorphism $B \otimes \mathbb{Q} = A_0$, or to give $\widehat{T}(B) \subset \widehat{V}(A_0)$.

4.3. Let A be an abelian variety over k. Let $\mathrm{NS}(A)$ denote the group of algebraic-equivalence classes of invertible sheaves on A. An *effective polarization*, respectively a *polarization*, of A is defined to be an element of $\mathrm{NS}(A)$, respectively of $\mathrm{NS}(A)\otimes\mathbb{Q}$, a positive multiple of which is defined by a projective embedding of A. A *homogeneous polarization* is an element of $(\mathrm{NS}(A)\otimes\mathbb{Q})/\mathbb{Q}^*$ which is the class of a polarization.

Recall that $\mathrm{NS}(A)\otimes\mathbb{Q}$ is identified, by a bijection $p\mapsto p'$, with the set of elements of $\mathrm{Hom}(A,A^\vee)\otimes\mathbb{Q}$ equal to their transpose. An isomorphism

$$u\in\mathrm{Hom}(A\otimes\mathbb{Q},B\otimes\mathbb{Q})$$

induces isomorphisms from $\mathrm{Hom}(A,A^\vee)\otimes\mathbb{Q}$ to $\mathrm{Hom}(B,B^\vee)\otimes\mathbb{Q}$, and from $\mathrm{NS}(A)\otimes\mathbb{Q}$ to $\mathrm{NS}(B)\otimes\mathbb{Q}$, and carries polarizations to polarizations. This makes it possible to speak of a polarization of an abelian variety up to isogeny.

4.4. A polarization p of A defines a positive involution on $\mathrm{End}(A)\otimes\mathbb{Q}$,

$$u\longmapsto p'^{-1}u^*p',$$

which depends only on the homogeneous polarization $\mathbb{Q}^*p$.

Let F be a product of totally real fields, and let $\rho : F\longrightarrow\mathrm{End}(A)\otimes\mathbb{Q}$. Let $\mathrm{NS}_\rho(A)\subset\mathrm{NS}(A)$ be the set of those p such that

$$p'\rho(f)=\rho(f)^*p'\qquad(f\in F).$$

The vector space $\mathrm{NS}_\rho(A)\otimes\mathbb{Q}$ is endowed, for $f\in F$, with an action of F:

$$(f.p)'=p'\rho(f)=\rho(f)^*p'.$$

A *weak polarization* of A, relative to ρ, F, is an element of

$$(\mathrm{NS}_\rho(A)\otimes\mathbb{Q})/F^*$$

which is the class of a polarization. The set of weak polarizations of A depends only on $A\otimes\mathbb{Q}$. The restriction to the centralizer of F of the involution of $\mathrm{End}(A)\otimes\mathbb{Q}$ defined by a polarization $p\in\mathrm{NS}_\rho(A)$ depends only on the weak polarization F^*p.

4.5. If $A_0^\vee$ is the dual of an abelian variety up to isogeny A_0, then $\widehat{V}(A_0)$ and $\widehat{V}(A_0^\vee)$ are in duality with values in $\mathbb{A}^f(1)$. If p is a polarization of A_0, the *polarization form*

$$\psi_p(x,y)=\langle x,p'(y)\rangle$$

is a nondegenerate alternating form on the free $\mathbb{A}^f$-module $\widehat{V}(A_0)$, with values in $\mathbb{A}^f(1)$.

The reminders 4.1 and 4.5 extend to abelian schemes over an arbitrary base.

4.6. Now take $k=\mathbb{C}$. The additive functor $A\mapsto H_1(A,\mathbb{Q})$ extends to abelian varieties up to isogeny over $\mathbb{C}$ (4.2). For an abelian variety up to isogeny A_0, $H_1(A_0,\mathbb{Q})$ and $H_1(A_0^\vee,\mathbb{Q})$ are in duality. Via 4.1.2 and the isomorphism of $\mathbb{A}^f(1)$ with $\mathbb{A}^f$ defined by the exponential, this duality is compatible with the one considered in 4.5. The alternating form

$$\psi_p(x,y)=\langle x,p'(y)\rangle$$

on $H_1(A_0,\mathbb{Q})$ defined by a polarization p of A_0 is again called the polarization form.

The Hodge structure of $H_1(A_0,\mathbb{Q})$ is the homomorphism

$$h:\mathbb{S}\longrightarrow\mathrm{GL}(H_1(A_0,\mathbb{Q}))_\mathbb{R}$$

such that $\mathbb{S}$ acts on $H_1(A_0,\mathbb{Q})\otimes\mathbb{C}$ through the characters z^{-1} and $\bar{z}^{-1}$, and such that $F(h)^\circ$ (1.5) is the kernel of the exponential

$$H_1(A_0,\mathbb{Q})\otimes\mathbb{C}\longrightarrow\mathrm{Lie}(A).$$

Theorem 4.7 (Riemann). The constructions

$$(A,p) \longmapsto (H_1(A,\mathbb{Q}), \psi_p, H_1(A,\mathbb{Z}), h),$$

$$(V,\psi,V_{\mathbb{Z}},h) \longmapsto F(h)^\circ \backslash (V\otimes\mathbb{C})/V_{\mathbb{Z}}$$

establish an equivalence between

(a) polarized abelian varieties over $\mathbb{C}$;

(b) systems consisting of a vector space V over $\mathbb{Q}$, a nondegenerate alternating form ψ on V, an integral lattice $V_{\mathbb{Z}}$ in V, and a homomorphism h from $\mathbb{S}$ into the group $\mathrm{Gp}(V)_{\mathbb{R}}$ of symplectic similitudes of $V\otimes\mathbb{R}$, of type 1.6 and such that

$$\psi(x,h(i)x) > 0 \qquad (x\in V\otimes\mathbb{R},\ x\neq 0).$$

Variant 4.8. To give an abelian variety up to isogeny A over $\mathbb{C}$, endowed with a homogeneous polarization, is equivalent to giving a vector space over $\mathbb{Q}$, endowed with a nondegenerate alternating form ψ given up to scalar, together with $h:\mathbb{S}\longrightarrow \mathrm{Gp}(V)$ of type 1.6.

4.9. Let L be a semisimple algebra with involution over $\mathbb{Q}$, and let V be a vector space over $\mathbb{Q}$, endowed with a faithful L-module structure and with a nondegenerate alternating form ψ such that

$$\psi(\ell x,y)=\psi(x,\ell^* y). \tag{4.9.1}$$

Let G denote the algebraic group over $\mathbb{Q}$ of L-linear symplectic similitudes of V. The group $G(\mathbb{Q})$ is the set of $g\in\mathrm{GL}_L(V)$ for which there exists $\mu(g)\in\mathbb{Q}^*$ satisfying

$$\psi(gx,gy)=\mu(g)\psi(x,y). \tag{4.9.2}$$

Suppose given a homomorphism $h_0:\mathbb{S}\longrightarrow G_{\mathbb{R}}$ such that, via h_0, $\mathbb{S}$ acts on $V_{\mathbb{C}}$ through the characters z^{-1} and $\bar{z}^{-1}$ and such that the form $\psi(x,h_0(i)y)$ is symmetric and positive definite. The conditions (1.5.1) to (1.5.3) are then satisfied by (G,h_0), and the involution of L is positive.

4.10. Apply (1.11) to (G,h_0) and to the representation V of G. A G-structure $\bar{\xi}$ on a vector space H is interpreted, by 1.8, as the datum on H of an L-module structure and of an alternating form ψ, given up to scalar, the vector space H, endowed with these structures, being isomorphic to V. Let $\bar{\xi}$ be a G-structure on H. A datum 1.9 (b), $h:\mathbb{S}\to G^1_{\mathbb{R}}$ on H, then defines, by 4.8 applied to (H,ψ,h), an abelian variety up to isogeny A, endowed with a homogeneous polarization $\bar{p}$, such that H is $H_1(A,\mathbb{Q})$, h is the Hodge structure of $H_1(A,\mathbb{Q})$, and ψ is the polarization form. Moreover, the L-module structure on H comes from $\rho:L\to\mathrm{End}(A)$, and H, endowed with $\bar{\xi}$ and h, is determined by $(A,\bar{p},\rho)$.

For a triple $(A,\bar{p},\rho)$ to arise from such an H, it is necessary and sufficient that

$$H_1(A,\mathbb{Q}),\ \text{endowed with its } L\text{-module structure and with the polarization form given up to scalar, be isomorphic to } V; \tag{4.10.1}$$

$$\text{for an isomorphism } t:V\longrightarrow H_1(A,\mathbb{Q}),\ t^{-1}h_1t \text{ be conjugate to } h_0. \tag{4.10.2}$$

Let t be the map from L to $\mathbb{C}$ given by

$$t(\ell)=\mathrm{Tr}(\ell;V_{\mathbb{C}}/F^0(h_0)). \tag{4.10.3}$$

Scholium 4.11. Let K be a compact open subgroup of $G(\mathbb{A}^f)$. The points of ${}_KM_{\mathbb{C}}(G,h_0)$ correspond bijectively to isomorphism classes of abelian varieties up to isogeny A, endowed with

(a) $\rho:L\to\mathrm{End}(A)$ such that

$$\mathrm{Tr}(\rho(\ell),\mathrm{Lie}(A))=t(\ell) \qquad (\ell\in L),$$

(b) a homogeneous polarization $\bar{p}$ which induces the given involution of L,

(c) a class modulo K, $\bar{k}$, of L-linear symplectic similitudes

$$k:\widehat{V}(A)\xrightarrow{\sim} V\otimes\mathbb{A}^f,$$

and such that

$$(*)\qquad \text{the conditions (4.10.1) and (4.10.2) are satisfied.}$$

One applies 1.11 and 4.10, noting that a datum 1.9 (c) is identified with a datum (c), and that the conditions in (b) and (a) result respectively from (4.10.1) and (4.10.2).

4.12. Let K be a compact open subgroup of $G(\mathbb{A}^f)$, let $V_{\mathbb{Z}}$ be an integral lattice in V such that $V_{\widehat{\mathbb{Z}}} = V_{\mathbb{Z}} \otimes \widehat{\mathbb{Z}}$ is K-invariant, let $L_{\mathbb{Z}}$ be an order in L such that $L_{\mathbb{Z}} V_{\mathbb{Z}} \subset V_{\mathbb{Z}}$, let $\psi_{\mathbb{Z}}$ be a positive rational multiple of the alternating form of V, with integral values on $V_{\mathbb{Z}}$, and let $V'_{\mathbb{Z}}$ be the largest lattice in V such that $\psi_{\mathbb{Z}}(V_{\mathbb{Z}}, V'_{\mathbb{Z}}) \subset \mathbb{Z}(1)$. Let n be an integer, and put

$$K_n = \{\, g \in G(\mathbb{A}^f) \mid (g-1)V'_{\widehat{\mathbb{Z}}} \subset nV_{\widehat{\mathbb{Z}}} \,\}.$$

Assume that n is large enough that $K \supset K_n$.

If A is as in 4.11, then $k^{-1}(V_{\widehat{\mathbb{Z}}}) \subset \widehat{V}(A)$ does not depend on $k \in \bar{k}$, and defines an abelian variety B with $B \otimes \mathbb{Q} \simeq A$ and

$$\widehat{T}(B) = k^{-1}(V_{\widehat{\mathbb{Z}}}) \subset \widehat{V}(A).$$

The action of L on A induces an action of $L_{\mathbb{Z}}$ on B. There exists a unique effective polarization p of B, with $p \in \bar{p}$, such that $k^{-1}(V'_{\widehat{\mathbb{Z}}})$ is the largest "lattice" $\widehat{T}'(B) \supset \widehat{T}(B)$ satisfying

$$\psi_p(\widehat{T}(B), \widehat{T}'(B)) \subset \widehat{\mathbb{Z}}(1).$$

Finally, the set $\bar{k}$ of L-linear symplectic isomorphisms from $\widehat{T}(B)$ to $V_{\widehat{\mathbb{Z}}}$ is the inverse image of its image $\bar{k}_n$ in $\mathrm{Isom}(B_n, V_{\mathbb{Z}}/nV_{\mathbb{Z}})$.

This again permits the points of ${}_K M_{\mathbb{C}}(G, h_0)$ to be interpreted as corresponding to isomorphism classes of systems $(B, p, \rho, \bar{k}_n)$ consisting of

(a) a polarized abelian variety (B, p), with complex multiplication by $L_{\mathbb{Z}}$, satisfying 4.11 (a) and (b);

(b) a class modulo K/K_n, $\bar{k}_n$, of isomorphisms

$$k_n : B_n \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}},$$

which can be lifted to L-linear symplectic isomorphisms

$$k : \widehat{T}(B) \longrightarrow V_{\widehat{\mathbb{Z}}},$$

and which satisfy (4.10.1) and (4.10.2).

4.13. Let F denote the algebra of invariants under $*$ in the center of L. It is a product of totally real fields. There exists a unique F-bilinear alternating form ψ_F on V such that

$$\mathrm{Tr}_{F/\mathbb{Q}}\, \psi_F = \psi.$$

Let G_1 denote the group of L-linear symplectic similitudes of ψ_F: $G_1(\mathbb{Q})$ is the set of $g \in \mathrm{GL}_L(V)$ for which there exists $\mu(g) \in F^*$ satisfying

$$\psi_F(gx, gy) = \mu(g)\psi_F(x, y), \tag{4.13.1}$$

that is,

$$\psi(gx, gy) = \psi(\mu(g)x, y). \tag{4.13.2}$$

As above, one verifies the following variant.

Variant 4.14. Let K be a compact open subgroup of $G_1(\mathbb{A}^f)$. The points of ${}_K M_{\mathbb{C}}(G_1, h_0)$ correspond bijectively to isomorphism classes of abelian varieties up to isogeny A, endowed with

(a) $\rho : L \longrightarrow \mathrm{End}(A)$ as in 4.11,

(b) a weak polarization, relative to F, $\bar{p}$, which induces the given involution of L,

(c) a class modulo K of L-linear symplectic similitudes

$$k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$$

for the $F \otimes \mathbb{A}^f$-bilinear forms on the two sides,

and such that

$$(*) \qquad \text{conditions analogous to (4.10.1) and (4.10.2) are satisfied.}$$

4.15. One can make 4.11 and 4.14 more precise by showing that ${}_K M_{\mathbb{C}}(G, h)$ is the coarse moduli scheme of the evident functor on schemes of finite type over $\mathbb{C}$.

4.16. Example I (continuation of 1.6, 1.11). Place ourselves in the special case of 4.9 where $L = \mathbb{Q}$ and $\dim(V) = 2g$. Then $G = \mathrm{Gp}(V)$, the group of symplectic similitudes. For $h_0 : \mathbb{S} \to G_{\mathbb{R}}$ of type 1.6, one has $E(\mathrm{Gp}, h_0) = \mathbb{Q}$. Let $V_{\mathbb{Z}}$, n, and K_n be as in 1.11. For every scheme S, let $F(S)$ be the set of isomorphism classes of principally polarized abelian schemes A/S, endowed with a symplectic similitude

$$k_n : A_n \xrightarrow{\sim} (V_{\mathbb{Z}}/nV_{\mathbb{Z}})_S.$$

For $n \geq 3$, the classified objects no longer have automorphisms and the functor F is represented by a $\mathbb{Q}$-scheme ${}_{K_n}M$ [6]. From 1.11, 4.11, and 4.12, one has

$$m_n : {}_{K_n}M(\mathbb{C}) \xrightarrow{\sim} {}_{K_n}M_{\mathbb{C}}(\mathrm{Gp}, h_0).$$

In the language of 3.1, this gives:

Proposition 4.17. The $\mathbb{Q}$-scheme

$$M(\mathrm{Gp}, h_0) = \varprojlim_{K_n} {}_{K_n}M$$

is a model over $\mathbb{Q} = E(\mathrm{Gp}, h_0)$ of $M_{\mathbb{C}}(\mathrm{Gp}, h_0)$.

4.18. Place ourselves in the special case of 4.9 where V is a monogenic L-module. The groups G and G_1 are then contained in the center of L^*.

Let $\bar{E}$ be an algebraic closure of $E(G, h)$ and let M^+ be the set of isomorphism classes $[A, \rho, k, \bar{p}]$ of objects $(A, \rho, k, \bar{p})$ consisting of

(a) an abelian variety up to isogeny $A/\bar{E}$, endowed with a homogeneous polarization $\bar{p}$ and with $\rho : L \to \mathrm{End}(A)$ such that, with the notation of 4.10.3, for $\ell \in L$ one has

$$\mathrm{Tr}(\rho(\ell), \mathrm{Lie}(A)) = t(\ell) \in E(G, h);$$

(b) an $L \otimes \mathbb{A}^f$-linear isomorphism

$$k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f.$$

The abelian variety A is therefore of CM type. We refer to [16] for the proof of the following fundamental theorem.

Theorem 4.19 (Shimura–Taniyama). The Galois group $\mathrm{Gal}(\bar{E}/E(G, h))$ acts on M^+ through its largest abelian quotient

$$\pi_0\big(E(G, h)^*(\mathbb{A})/E(G, h)^*(\mathbb{Q})\big).$$

For $e \in E(G, h)^*(\mathbb{A})$, with image $\varphi(e)$ in abelianized Galois and with component e^f in $E(G, h)^*(\mathbb{A}^f)$, one has

$$\varphi(e)([A, \rho, k, \bar{p}]) = [A, \rho, r(G, h)(e^f)k, \bar{p}]. \tag{4.19.1}$$

Let τ be an embedding of $\bar{E}$ into $\mathbb{C}$ extending that of $E(G, h)$, let K be a compact open subgroup of $G(\mathbb{A}^f)$, and let ${}_K M(G, h)(\bar{E})$ be the set of isomorphism classes of objects $(A, \rho, \bar{k}, \bar{p})$ where

(a) A, ρ, and $\bar{p}$ are as above;

(b) $\bar{k}$ is a class modulo K of $L \otimes \mathbb{A}^f$-linear symplectic similitudes

$$k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f;$$

(c) the vector space $H_1(A_{\mathbb{C}}, \mathbb{Q})$, where $A_{\mathbb{C}}$ is defined via τ, endowed with the action of L and with the polarization form given up to scalar, is isomorphic to V.

By 4.19, condition (c) is independent of the choice of τ, so that the finite set ${}_K M(G,h)(\bar{E})$ is endowed with an action of $\mathrm{Gal}(\bar{E}/E(G,h))$; as a Galois set, it is identified with the set of $\bar{E}$-points of a finite étale $E(G,h)$-scheme ${}_K M(G,h)$. By 4.14, ${}_K M(G,h)(\mathbb{C}) = G(\mathbb{R})XK\backslash G(\mathbb{A})/G(\mathbb{Q})$, and by 4.19 one has:

Proposition 4.20. The $E(G,h)$-scheme

$$M(G,h) = \varprojlim_K {}_K M(G,h)$$

is a canonical model of $M_{\mathbb{C}}(G,h)$.

Theorem 4.21. The model $M(\mathrm{Gp}, h_0)$ constructed in 4.17 is a canonical model.

Let (V,ψ) be as in 4.16, and let L, $\rho : L \to \mathrm{End}(V)$, G, and $h : \mathbb{S} \to G_{\mathbb{R}}$ be as in 4.18. We have

$$u : (G,h) \longrightarrow (\mathrm{Gp}, h_0).$$

Let $K \subset G(\mathbb{A}^f)$ be such that $u(K) \subset K_n$ (4.16). If $\bar{E}$ is an algebraic closure of $E(G,h)$, define

$$u : {}_K M(G,h)(\bar{E}) \longrightarrow {}_{K_n} M(\mathrm{Gp}, h_0)(\bar{E})$$

by "forgetting the complex multiplication", associating to $[A, \rho, \bar{k}, \bar{p}]$ the triple consisting of

(a) the abelian variety B, endowed with $B \otimes \mathbb{Q} \simeq A$ and such that

$$\widehat{T}(B) = k^{-1}(V_{\widehat{\mathbb{Z}}}) \subset \widehat{V}(A) \qquad (k \in \bar{k});$$

(b) the polarization $\bar{p}$ of B;

(c) the isomorphism

$$k : B_n \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}} \qquad (k \in \bar{k}).$$

The maps u define a morphism of models

$$u : M(G,h) \longrightarrow M(\mathrm{Gp}, h_0) \otimes E(G,h).$$

The proof is completed by showing ([8]) that for every

$$u : (H, h') \longrightarrow (\mathrm{Gp}, h_0)$$

as in 3.13, there exists $v : (G,h) \longrightarrow (\mathrm{Gp}, h_0)$ as above, for a suitable L, such that $uh' = vh$.

Variant 4.22. Let F be a totally real number field, let $n = [F : \mathbb{Q}]$, and let $\mathrm{Gp}_{2g}^{(F)}$ be the group of symplectic similitudes of a symplectic F-vector space of dimension $2g$. Let

$$h_0 : \mathbb{S} \longrightarrow \mathrm{Gp}_{2g,\mathbb{R}}^{(F)} \simeq \mathrm{Gp}_{2g}(\mathbb{R})^n$$

be as in 1.6. One can construct a canonical model of $M_{\mathbb{C}}(\mathrm{Gp}_{2g}^{(F)}, h_0)$ from moduli of weakly polarized abelian schemes.

§5. Construction techniques

Definition 3.13 is usable only because there are "many" special points.

Theorem 5.1. Let G and h be as in 3.13. For every finite extension F of $E(G,h)$, there exists $(H, h', u : H \hookrightarrow G)$ as in 3.13 such that the extension $E(H,h')$ of $E(G,h)$ is linearly disjoint from F.

Let

$$Y = \mathrm{Hom}(\mathbb{G}_m, G)/G$$

be the scheme of conjugacy classes of morphisms from $\mathbb{G}_m$ to G. The Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on the infinite discrete set $Y(\mathbb{C})$ of conjugacy classes of morphisms from $\mathbb{G}_{m\mathbb{C}}$ to $G_{\mathbb{C}}$. Let Y_o be the finite subscheme of Y such that $Y_o(\mathbb{C})$ is the Galois orbit of the class of hr. It is the spectrum of $E(G,h)$.

Let $V' = \mathrm{Lie}(G)$, regarded as an affine space over $\mathbb{Q}$, and let V be the open subscheme of regular elements of the Lie algebra. For v in V, let T_v denote the centralizer of v, a maximal torus. Let W be the moduli scheme of maximal tori T of G, endowed with

$$s : \mathbb{G}_m \longrightarrow T, \qquad v \in \mathrm{Lie}(T),$$

such that

(a) v is regular in $\mathrm{Lie}(G)$;

(b) the class of $s : \mathbb{G}_m \to G$ lies in Y_o.

The morphism

$$f : W \longrightarrow V, \qquad (T, s, v) \longmapsto v$$

is finite étale and surjective; note that $T = T_v$. Let p be the morphism from W to Y_o,

$$p : (T, s, v) \longmapsto \text{the class of } s.$$

Thus one has the diagram

$$\begin{array}{ccc} W & \xrightarrow{f} & V \\ {\scriptstyle p}\downarrow & & \downarrow \\ \mathrm{Spec}(E(G,h)) & \longrightarrow & \mathrm{Spec}(\mathbb{Q}). \end{array} \tag{5.1.1}$$

Lemma 5.1.2. The scheme W is irreducible, and the geometric fibers of p are geometrically irreducible.

Since Y_o is irreducible, being the spectrum of an extension of $\mathbb{Q}$, it is enough to prove the second assertion. It follows from the following facts:

(a) since G is connected, the scheme over $\mathbb{C}$ of homomorphisms from $\mathbb{G}_{m\mathbb{C}}$ to $G_{\mathbb{C}}$ conjugate to a given homomorphism is irreducible;

(b) for a given $i : \mathbb{G}_{m\mathbb{C}} \to G_{\mathbb{C}}$, by the theorem on conjugacy of maximal tori applied to the connected centralizer of $i(\mathbb{G}_{m\mathbb{C}})$, the scheme of maximal tori of $G_{\mathbb{C}}$ containing $i(\mathbb{G}_{m\mathbb{C}})$ is irreducible.

Let $U \subset V(\mathbb{R})$ be the set of $v \in V(\mathbb{R})$ such that $(T_v/C)(\mathbb{R})$ is compact. For the usual topology, U is open.

For $v \in U$ and $w \in W(\mathbb{C})$ such that $f(w) = v \in V(\mathbb{R}) \subset V(\mathbb{C})$, there exists a unique homomorphism

$$h(w) : \mathbb{S} \longrightarrow T_v$$

such that $h(w) \circ r = s_w$: after extension of scalars to $\mathbb{C}$,

$$h(w) = s_w \circ z \cdot \overline{s_w} \circ \overline{z}.$$

Let $v \in U$. By the theorem on conjugacy of maximal compact subgroups in $G/C(\mathbb{R})^\circ$, there exists $g \in G(\mathbb{R})$, even $g \in G'(\mathbb{R})^\circ$, such that

$$gT_vg^{-1} \subset K_\infty.$$

Since T_v is a maximal torus, gT_vg^{-1} contains the center of K_∞. It follows that there exists $w \in W(\mathbb{C})$ such that $f(w) = v$ and such that $h(w)$ is conjugate to h.

The proof of 5.1 is concluded by applying to the diagram (5.1.1) and to U the following variant of Hilbert's irreducibility theorem, for whose proof I am indebted to J.-P. Serre.

Lemma 5.13. Let E be a finite extension of $\mathbb{Q}$, and let

$$\begin{array}{ccc} W & \xrightarrow{f} & V \\ {\scriptstyle p}\downarrow & & \downarrow \\ \mathrm{Spec}(E) & \longrightarrow & \mathrm{Spec}(\mathbb{Q}) \end{array}$$

be a commutative diagram in which

(i) V is a Zariski open subset of an affine space over $\mathbb{Q}$;

(ii) the geometric fibers of p are irreducible, and W is reduced;

(iii) f is quasi-finite and dominant.

Then, for every open set $U \subset V(\mathbb{R})$ in the usual topology and every extension F of E, there exists

$$v \in V(\mathbb{Q}) \cap U$$

such that $f^{-1}(v)$ is the spectrum of an extension of E linearly disjoint from F.

Let $W' = W \otimes_E F$. The hypotheses of 4.12 are again satisfied for W'/F and

$$f' : W' \longrightarrow W \longrightarrow V,$$

and one must construct $v \in V(\mathbb{Q}) \cap U$ such that

$$f'^{-1}(v) = f^{-1}(v) \otimes_E F$$

is the spectrum of a field. This reduces us to the case $E = F$, which follows from Hilbert's irreducibility theorem; cf. Lang, *Diophantine Geometry*, chap. VIII.

Proposition 5.2. For $x \in M_{\mathbb{C}}(G, h)$ and K compact open in $G(\mathbb{A}^f)$, the image in ${}_K M_{\mathbb{C}}(G, h)$ of $G(\mathbb{A}^f) \cdot x \subset M_{\mathbb{C}}(G, h)$ is dense.

This is a consequence of (0.4). The reader may verify it.

Proposition 5.3. Let E be a field endowed with a complex embedding ρ, let X be a scheme over E, and let Y be a reduced closed subscheme of $X_{\mathbb{C}}$. Suppose that there exist finite extensions E_i of E in $\mathbb{C}$, schemes $M_{i,\alpha}/E_i$, and E_i-morphisms

$$v_{i,\alpha} : M_{i,\alpha} \longrightarrow X \otimes_E E_i,$$

or, writing

$$v_i : \coprod_\alpha M_{i,\alpha} \longrightarrow X \otimes_E E_i,$$

such that $E = \bigcap_i E_i$,

$$v_i\Big((\coprod_\alpha M_{i,\alpha}) \otimes_{E_i} \mathbb{C}\Big) \subset Y,$$

and

$$v_i\Big((\coprod_\alpha M_{i,\alpha}) \otimes_{E_i} \mathbb{C}\Big)$$

is dense in Y. Then Y is defined over E.

Corollary 5.4. Let $u : (G^1, h^1) \to (G^2, h^2)$ be as in 1.14. Let $M_E(G^1, h^1)$ and $M_E(G^2, h^2)$ be weakly canonical models of the $M_{\mathbb{C}}(G^i, h^i)$ over a number field E, with $E(G^i, h^i) \subset E \subset \mathbb{C}$. Then

$$v : M_{\mathbb{C}}(G^1, h^1) \longrightarrow M_{\mathbb{C}}(G^2, h^2)$$

is defined over E, cf. 3.12.

Let $K^i \subset G^i(\mathbb{A}^f)$ be compact open subgroups such that $v(K^1) \subset K^2$. We prove that

$$v(K^1, K^2) : {}_{K^1} M_{\mathbb{C}}(G^1, h^1) \longrightarrow {}_{K^2} M_{\mathbb{C}}(G^2, h^2)$$

is defined over E. Let $u : (H, h') \to (G^1, h^1)$ be as in 3.13 and set $F = E \cdot E(H, h')$. For $g \in G^1(\mathbb{A}^f)$, one has F-morphisms

$$\begin{aligned} a(g): \quad & M_F(H, h') \longrightarrow M_F(G^1, h^1) \xrightarrow{g} M_F(G^1, h^1) \longrightarrow {}_{K^1}M_F(G^1, h^1), \\ b(g): \quad & M_F(H, h') \longrightarrow M_F(G^2, h^2) \xrightarrow{v(g)} M_F(G^2, h^2) \longrightarrow {}_{K^2}M_F(G^2, h^2), \end{aligned}$$

and, after extension of scalars to $\mathbb{C}$,

$$v(K^1, K^2) \circ a(g) = b(g).$$

By 5.1 and 5.2, one can apply 5.3 to

$$X = {}_{K^1}M_E(G^1, h^1) \times {}_{K^2}M_E(G^2, h^2),$$

to the graph Y of v, to the $E_i = E \cdot E(H, h')$ for variable (H, h', u), and to

$$(a(g), b(g)) : M_{E_i}(H, h') \longrightarrow X_{E_i}.$$

It follows that the graph of v is defined over E.

In the special case $(G^1, h^1) = (G^2, h^2)$ and $u = \mathrm{Id}$, 5.4 says:

Corollary 5.5. Up to unique isomorphism, there exists at most one weakly canonical model of $M_{\mathbb{C}}(G, h)$ over E, for $E(G, h) \subset E \subset \mathbb{C}$.

Corollary 5.6. If G' is simply connected, the reciprocity law of a weakly canonical model of $M_{\mathbb{C}}(G, h)$ over E, for $E(G, h) \subset E \subset \mathbb{C}$, is given by 3.9.1.

This is the same proof as in the special case $v : (G, h) \to (T, vh)$ of 5.4.

Corollary 5.7. Let $v : (G^1, h^1) \hookrightarrow (G^2, h^2)$ be as in 1.15. If $M_{\mathbb{C}}(G^2, h^2)$ admits a weakly canonical model over E, with

$$E(G^2, h^2) \subset E(G^1, h^1) \subset E \subset \mathbb{C},$$

then $M_{\mathbb{C}}(G^1, h^1)$ admits a weakly canonical model over E.

Let K^i be a compact open subgroup of $G^i(\mathbb{A}^f)$. Using 1.15, we are reduced to showing only that, if $v(K^1) \subset K^2$ and

$${}_{K^1}M_{\mathbb{C}}(G^1, h^1) \subset {}_{K^2}M_{\mathbb{C}}(G^2, h^2),$$

then the subscheme ${}_{K^1}M_{\mathbb{C}}(G^1, h^1)$ of

$${}_{K^2}M_{\mathbb{C}}(G^2, h^2) = {}_{K^2}M_E(G^2, h^2) \otimes_E \mathbb{C}$$

is defined over E.

Let $u : (H, h') \to (G^1, h^1)$ be as in 3.13 and, for $g \in G^1(\mathbb{A}^f)$, let $a(g)$ be the composite homomorphism, defined over $F = E \cdot E(H, h')$:

$$a(g) : M_F(H, h') \longrightarrow M_F(G^1, h^1) \xrightarrow{v(g)} M_F(G^2, h^2) \longrightarrow {}_{K^2}M_F(G^2, h^2).$$

By 5.1 and 5.2, one concludes by applying 5.3 to

$$X = {}_{K^2}M_E(G^2, h^2), \qquad Y = {}_{K^1}M_{\mathbb{C}}(G^1, h^1),$$

to the $E_i = E \cdot E(H, h')$ for variable (H, h', u) and to the

$$a(g) : M_{E_i}(H, h') \longrightarrow X_{E_i}.$$

Example 5.8. For (G, h_0) as in 4.9, one constructs, by 5.7 and 4.21, a canonical model

$$M(G^\circ, h_0) \subset M(\mathrm{Gp}(V, \psi), h_0) \otimes_{\mathbb{Q}} E(G^\circ, h^\circ).$$

Variant 5.9. Let (G_1, h_0) be such that, with the notation of 4.22, there exists

$$u : (G_1, h_0) \hookrightarrow (\mathrm{Gp}_{2g}^{(F)}, h_0),$$

for example (G_1°, h_0) with (G_1, h_0) as in 4.13. An application of 5.7 gives a canonical model of $M(G_1, h_0)$.

Proposition 5.10. Let (G, h) be as in 3.13, let

$$E_i : \ E(G,h) \subset E_i \subset \mathbb{C}$$

be number fields, let E be the intersection of the E_i, and let $M_{E_i}(G,h)$ be a weakly canonical model of $M_{\mathbb{C}}(G,h)$ over E_i. Then there exists a unique model $M_E(G,h)$ of $M_{\mathbb{C}}(G,h)$ over E such that, for every i, the model $M_E(G,h) \otimes_E E_i$ over E_i is isomorphic to $M_{E_i}(G,h)$.

In view of the uniqueness assertion, one may suppose the E_i finite in number. Let F be the extension of E in $\mathbb{C}$ that they generate, and let $\mathfrak{E}$ be the set of subextensions of F containing one of the E_i. Using 5.5, one obtains for every $F' \in \mathfrak{E}$ a model $M_{F'}(G,h)$, and for every inclusion $F' \subset F''$, $F', F'' \in \mathfrak{E}$, an isomorphism of models

$$\alpha(F'', F') : M_{F'}(G,h) \otimes_{F'} F'' \longrightarrow M_{F''}(G,h).$$

Let $u : (H, h') \to (G, h)$ be as in 3.13, with $E(H, h')$ linearly disjoint from F over $E(G,h)$, cf. 5.1. Put

$$E_1 = E(H,h') \cdot E, \qquad F_1' = E_1 \cdot F' = E(H,h') \cdot F' \qquad (F' \in \mathfrak{E}).$$

Let $g \in G(\mathbb{A}^f)$ and let $K \subset G(\mathbb{A}^f)$ be a compact open subgroup. Since $M_{F'}(G,h)$ is weakly canonical, one has an F_1'-morphism

$$a'(g) : M_{F_1'}(H,h') \longrightarrow M_{F_1'}(G,h) \xrightarrow{g} M_{F_1'}(G,h) \longrightarrow {}_K M_{F_1'}(G,h).$$

Since $F_1' = E_1 \otimes_E F'$, one may "interpret" this as a morphism of F'-schemes

$$a(g) : M_{E_1}(H,h') \otimes_E F' \longrightarrow {}_K M_{F'}(G,h).$$

The $a(g)$ satisfy

$$\alpha(F'', F')\, a(g)_{F''} = a(g).$$

One concludes by 5.2 and by the following lemma, applied to

$$X^1 = \coprod_{g \in G(\mathbb{A}^f)} M_{E_1}(H, h').$$

Its proof is left to the reader as an exercise.

Lemma 5.10.1. Let F/E be a finite extension of fields and let $\mathfrak{E}$ be a set of subfields of F, whose intersection is reduced to E, such that

$$F \in \mathfrak{E}, \qquad F' \subset F'' \subset F \Longrightarrow F'' \in \mathfrak{E}.$$

(i) The functor which sends a scheme X/E to the system $\mathcal{X}$ of schemes $X_{F'}/F'$, for $F' \in \mathfrak{E}$, and of isomorphisms

$$X_{F'} \otimes_{F'} F'' \xrightarrow{\sim} X_{F''} \qquad (F' \subset F'',\ F', F'' \in \mathfrak{E})$$

is fully faithful.

(ii) For a system

$$\mathcal{X} = \{X(F')/F' \ (F' \in \mathfrak{E});\ \alpha(F'', F') : X(F') \otimes_{F'} F'' \xrightarrow{\sim} X(F'')\}$$

to come from a scheme X/E, it is enough that the $X(F')/F'$ be quasi-projective and that there exist a scheme X^1/E and a morphism

$$u : X^1_{\mathfrak{E}} \longrightarrow \mathcal{X}$$

such that the $u(F') : X^1_{F'} \to X(F')$ have dense image.

Remark 5.10.2. Under the hypotheses of 5.10, for $u : (H, h') \hookrightarrow (G, h)$ as in 3.13, the field of definition of

$$u : M_{\mathbb{C}}(H, h') \longrightarrow M_{\mathbb{C}}(G, h)$$

is contained in the extension

$$\bigcap_i E(H, h') \cdot E_i$$

of $E(H, h') \cdot E$.

Proposition 5.11. Let $h : \mathbb{S} \to G_{\mathbb{R}}$ be as in 3.13 and let $\delta : \mathbb{S} \to C^{\circ}_{\mathbb{R}}$, where C is the center of G. If

$$E(G, h),\ E(C, \delta) \subset E \subset \mathbb{C}$$

and if $M_{\mathbb{C}}(G, h)$ admits a weakly canonical model over E, then $M_{\mathbb{C}}(G, h\delta)$ also admits one.

Let K be a compact open subgroup of $G(\mathbb{A}^f)$ and let $L = K \cap C^{\circ}(\mathbb{A}^f)$. Over $\mathbb{C}$, the morphism

$$u : (G \times C^{\circ}, (h, \delta)) \longrightarrow (G, h\delta), \qquad (g, c) \longmapsto gc,$$

induces morphisms

$$u_K : {}_K M_{\mathbb{C}}(G, h) \times {}_L M_{\mathbb{C}}(C^{\circ}, \delta) \longrightarrow {}_K M_{\mathbb{C}}(G, h\delta).$$

Let d be the morphism

$$d : C^{\circ} \longrightarrow G \times C^{\circ}, \qquad c \longmapsto (c, c^{-1}).$$

The subgroup $d(C^{\circ}(\mathbb{A}^f))$ of $G \times C^{\circ}(\mathbb{A}^f)$ acts on

$$M_{\mathbb{C}}(G, h) \times M_{\mathbb{C}}(C^{\circ}, \delta)$$

through the quotient $\pi_0(C^{\circ}(\mathbb{A}))$ of $C^{\circ}(\mathbb{A}^f)$, and u_K induces an isomorphism

$$\bar{u}_K : \big({}_K M_{\mathbb{C}}(G, h) \times {}_L M_{\mathbb{C}}(C^{\circ}, \delta)\big) / \big(L \backslash \pi_0(C^{\circ}(\mathbb{A}))\big) \xrightarrow{\sim} {}_K M_{\mathbb{C}}(G, h\delta).$$

Define ${}_K M_E(G, h\delta)$ to be the quotient

$$\big({}_K M_E(G, h) \times {}_L M_E(C^{\circ}, \delta)\big) / \big(L \backslash \pi_0(C^{\circ}(\mathbb{A}))\big).$$

One verifies that this gives a weakly canonical model. By construction, one has

$$M_E(G, h) \times_E M_E(C^{\circ}, \delta) \longrightarrow M_E(G, h\delta). \tag{5.11.1}$$

Conjecture 5.12. For every (G, h_0) as in 3.13, there exists a canonical model of $M_{\mathbb{C}}(G, h_0)$.

Suppose, this being a typical case, that G is $\mathbb{Q}$-simple and adjoint. Let $\widetilde{G}$ be the universal covering of G and let ω be a weight of $\widetilde{G}_{\mathbb{C}}$. If, with the notation of 3.7, the conjugacy class of $h_0 r$ is described by integers $(n_i)_{i \in |\Delta|}$, and if

$$\omega = \sum m_i \omega_i,$$

put

$$\langle \omega, h_0 r \rangle = \sum n_i m_i.$$

Up to now one has been able to attack 5.12 only when $\widetilde{G}$ admits a nontrivial representation V such that all irreducible components V' of $V_{\mathbb{C}}$ satisfy:

$(*)$ as ω runs through the weights of V', $\langle \omega, h_0 r \rangle$ takes at most two values.

Let σ be the opposition involution. Condition $(*)$ also means that the dominant weight ω of V' satisfies

$$\langle \omega, h_0 r \rangle + \langle \sigma\omega, h_0 r \rangle = 0 \text{ or } 1.$$

There is no such representation V if G is exceptional, in which case the noncompact factors of $G_{\mathbb{R}}$ are of type $E_{6(-14)}$ or $E_{7(-25)}$, or of trialitarian type D_4, nor if G is of type D_n and $G_{\mathbb{R}}$ has noncompact factors of both types $D_n^{\mathbb{H}}$ and $D_n^{\mathbb{R},2}$.

§6. Strange models

6.1. Let F be a totally real number field, and let

$$ {}_F\Sigma : \quad 0 \longrightarrow {}_F G' \longrightarrow {}_F G \longrightarrow \mathbb{G}_m \longrightarrow 0 \tag{6.1.1} $$

be a form over F of the exact sequence

$$ {}_F\Sigma_0 : \quad 0 \longrightarrow \mathrm{Sp}_{2n} \longrightarrow \mathrm{Gp}_{2n} \xrightarrow{\mu} \mathbb{G}_m \longrightarrow 0, \tag{6.1.2} $$

where Gp denotes a group of symplectic similitudes.

Assume that, for every real place τ of F, $\Sigma \otimes_{F,\tau} \mathbb{R}$ is of one of the following types:

(a) ${}_F\Sigma \otimes_{F,\tau} \mathbb{R}$ is isomorphic to ${}_F\Sigma_0 \otimes_{F,\tau} \mathbb{R}$, or

(b) $G' \otimes_{F,\tau} \mathbb{R}$ is compact.

Let $\tau_1, \ldots, \tau_g$ denote the real places of F, and suppose that the places of type (a) are $\tau_1, \ldots, \tau_r$, with $r > 0$.

6.2. The group ${}_F G$ may be described as follows:

(a) Choose a quaternion algebra B over F, indefinite at the places τ_i for $i \leq r$, and definite for $i > r$. Let $x \mapsto \overline{x}$ denote its canonical involution.

(b) Choose a free B-module V, endowed with a nondegenerate F-bilinear form Φ, symmetric and such that

$$ \Phi(bx, y) = \Phi(x, \overline{b}\, y). $$

(c) Assume that, for $i > r$, the form deduced from Φ by extension of scalars through $\tau_i : F \to \mathbb{R}$ is positive definite.

(d) Take ${}_F G$ to be the group of B-linear similitudes of Φ; in other words, the elements $g \in {}_F G(F)$ are the $g \in \mathrm{GL}_B(V)$ for which there exists $\mu(g) \in F^*$ satisfying

$$ \Phi(gx, gy) = \mu(g)\Phi(x, y). $$

6.3. Let

$$ \Sigma : \quad 0 \longrightarrow G' \longrightarrow G \longrightarrow F^* \longrightarrow 0 $$

denote the exact sequence of algebraic groups over $\mathbb{Q}$ deduced from ${}_F\Sigma$ by Weil restriction of scalars from F to $\mathbb{Q}$. We have

$$ G_{\mathbb{R}} = \prod_{\tau} {}_F G \otimes_{F,\tau} \mathbb{R} \qquad (\tau \text{ a real place}). \tag{6.3.1} $$

A morphism $h : \mathbb{S} \to G_{\mathbb{R}}$ is defined by its coordinates

$$ h_i : \mathbb{S} \longrightarrow {}_F G \otimes_{F,\tau_i} \mathbb{R}. $$

By 1.6, there exists a unique conjugacy class of morphisms h such that, for $i \leq r$, h_i is of type 1.6, and for $i > r$, h_i is trivial. Let h_0 be in this class.

The field $E(G, h_0)$ is the totally real subfield of $\mathbb{C}$ generated by the elements

$$ \sum_{i=1}^{r} \tau_i(f) \qquad (f \in F). $$

Theorem 6.4. With the preceding notation, $M_{\mathbb{C}}(G, h_0)$ admits a canonical model $M(G, h_0)$.

Let Z be a totally imaginary quadratic extension of F. Let

$$ V' = V \otimes_F Z, \qquad L = B \otimes_F Z, $$

and let G' be the algebraic subgroup of $\mathrm{GL}_L(V')$ generated by G and Z^*:

$$0 \longrightarrow F^* \xrightarrow{(f,f^{-1})} G \times Z^* \xrightarrow{q} G' \longrightarrow 0. \tag{6.4.1}$$

Let $x \mapsto \widetilde{x}$ be the involution of L given by the tensor product of the involutions of B and Z. Up to a factor in F, there exists on Z a unique alternating F-bilinear form ψ_Z. If ${}_F\psi$ is the alternating F-bilinear form on V' which is the tensor product of Φ and ψ_Z, then

$${}_F\psi(\ell x, y) = {}_F\psi(x, \widetilde{\ell}\, y).$$

Put

$$\psi(x,y) = \mathrm{Tr}_{F/\mathbb{Q}}\big({}_F\psi(x,y)\big).$$

Choose, for every real place of F, a complex embedding of Z inducing it. Then

$$Z^*(\mathbb{R}) \simeq \prod_{i=1}^{g} \mathbb{C}^*.$$

Let h_Z be the homomorphism from $\mathbb{S}$ into $Z^*_{\mathbb{R}}$ with coordinates

$$h_i : \mathbb{S}(\mathbb{R}) = \mathbb{C}^* \longrightarrow \mathbb{C}^* \qquad (1 \le i \le g)$$

equal to $z \mapsto 1$ for $i \le r$, and to $z \mapsto z^{-1}$ for $i > r$.

For every $h : \mathbb{S} \to G_{\mathbb{R}}$ as in 6.3, put

$$h' = q \circ (h \times h_Z) : \mathbb{S} \longrightarrow G'_{\mathbb{R}}.$$

Lemma 6.4.2. There exists $b \in L$ such that, for a suitable choice of ψ_Z, the form

$$\psi\big(x, b\, h'_0(i) y\big) \qquad \text{on } V' \otimes_{\mathbb{Q}} \mathbb{R}$$

is symmetric and positive definite.

This is verified after extension of scalars from $\mathbb{Q}$ to $\mathbb{R}$.

This lemma and 5.9 make it possible to construct a canonical model of $M(G', h'_0)$, since G' is contained in the group of symplectic similitudes of ${}_F\psi(x, by)$. Applying 5.11 and 5.7, one obtains a weakly canonical model of $M_{\mathbb{C}}(G, h_0)$ over the extension $E(Z^*, h_Z)$ of $E(G, h_0)$. The conclusion follows from 5.10, 5.10.2 and the following lemma, which can be deduced from 5.13.

Lemma 6.5. Let F be a totally real number field, endowed with a set S of real embeddings. Let F^* be the number field corresponding to the subgroup of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which stabilizes S. For Z a totally imaginary quadratic extension of F, endowed with a set T of complex embeddings, one above each element of S, let likewise $Z^* \supset F^*$ be the field of invariants of the stabilizer of T. Then, for every extension E of F^*, there exist Z and T such that Z^* is linearly disjoint from E.

Remark 6.6. The canonical map 1.13, 1.14

$$M_{\mathbb{C}}(G, h_0) \times M_{\mathbb{C}}(Z^*, h_Z) \longrightarrow M_{\mathbb{C}}(G', h'_0) \tag{6.6.1}$$

is interpreted as follows. Each conjugate of h defines on $V_{\mathbb{C}}$ a

a Hodge bigrading invariant under F:

$$V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1} \oplus V^{0,0} \qquad (\overline{V^{pq}} = V^{qp}).$$

One should beware that the grading, by weight, of $V_{\mathbb{C}}$ by

$$V^n = \sum_{p+q=n} V^{pq}$$

is not defined over $\mathbb{Q}$ if $r \neq g$.

Similarly, $Z_{\mathbb{C}}$ is bigraded:

$$Z_{\mathbb{C}} = Z^{-1,0} \oplus Z^{0,-1} \oplus Z^{0,0}.$$

The miracle is that the $\mathbb{Q}$-vector space $V' = V \otimes_F Z$, endowed with the tensor-product Hodge bigrading

$$V'_{\mathbb{C}} = V'^{-1,0}_{\mathbb{C}} \oplus V'^{0,-1}_{\mathbb{C}} \qquad (!)$$

is the H_1 of an abelian variety.

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HODGE THEORY I

by PIERRE DELIGNE

We propose to give a heuristic dictionary between statements in ℓ-adic cohomology and statements in Hodge theory. This dictionary has, in particular, [3] and Grothendieck's conjectural theory of motives [2] as sources. Up to now, it has mainly served to formulate conjectures in Hodge theory, and it has sometimes suggested a proof of them.

Definition 1.1. A mixed Hodge structure H consists of

(a) a finitely generated $\mathbb{Z}$-module $H_{\mathbb{Z}}$ (the "integral lattice");

(b) a finite increasing filtration W on $H_{\mathbb{Q}} = H_{\mathbb{Z}} \otimes \mathbb{Q}$ (the "weight filtration");

(c) a finite decreasing filtration F on $H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$ (the "Hodge filtration").

These data are subject to the axiom:

There exists on $\mathrm{Gr}_W(H_{\mathbb{C}})$ a unique bigrading by subspaces $H^{p,q}$ such that

(i)

$$\mathrm{Gr}^n_W(H_{\mathbb{C}}) = \bigoplus_{p+q=n} H^{p,q};$$

(ii) the filtration F induces on $\mathrm{Gr}_W(H_{\mathbb{C}})$ the filtration

$$\mathrm{Gr}_W(F)^p = \bigoplus_{p' \geq p} H^{p',q'};$$

(iii) $\overline{H^{p,q}} = H^{q,p}$.

A morphism $f : H \to H'$ is a homomorphism $f_{\mathbb{Z}} : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ such that $f_{\mathbb{Q}} : H_{\mathbb{Q}} \to H'_{\mathbb{Q}}$ and $f_{\mathbb{C}} : H_{\mathbb{C}} \to H'_{\mathbb{C}}$ are respectively compatible with the filtrations W and F.

The Hodge numbers of H are the integers

$$h^{pq} = \dim H^{p,q} = h^{qp}. \tag{1.2}$$

One says that H is pure of weight n if $h^{pq} = 0$ for $p + q \neq n$ (i.e. if $\mathrm{Gr}^i_W(H) = 0$ for $i \neq n$). One also says that H is a Hodge structure of weight n.

The Tate Hodge structure $\mathbb{Z}(1)$ is the Hodge structure of weight -2, purely of type $(-1,-1)$, for which $\mathbb{Z}(1)_{\mathbb{C}} = \mathbb{C}$ and $\mathbb{Z}(1)_{\mathbb{Z}} = 2\pi i\mathbb{Z} = \mathrm{Ker}(\exp : \mathbb{C} \to \mathbb{C}^*) \subset \mathbb{C}$. Put $\mathbb{Z}(n) = \mathbb{Z}(1)^{\otimes n}$.

One can show that mixed Hodge structures form an abelian category. If $f : H \to H'$ is a morphism, then $f_{\mathbb{Q}}$ and $f_{\mathbb{C}}$ are strictly compatible with the filtrations W and F ([1], 2.3.5).

2. Let A be a normal integral ring of finite type over $\mathbb{Z}$, let K be its field of fractions, and let $\overline{K}$ be an algebraic closure of K. Let K_{nr} be the largest subextension of $\overline{K}$ unramified at every prime ideal of A. It is known, or we define,

$$\pi_1(\mathrm{Spec}(A), \overline{K}) = \mathrm{Gal}(K_{nr}/K).$$

For every closed point x of $\mathrm{Spec}(A)$, defined by a maximal ideal m_x of A, the residue field $k_x = A/m_x$ is finite; the point x defines a conjugacy class of "Frobenius substitutions" $\varphi_x \in \pi_1(\mathrm{Spec}(A), \overline{K})$. Put $q_x = \# k_x$ and $F_x = \varphi_x^{-1}$.

Let K be a field of finite type over the prime field of characteristic p, let $\overline{K}$ be an algebraic closure of K, let ℓ be a prime number $\neq p$, and let H be a finite type $\mathbb{Z}_\ell$-module (or $\mathbb{Q}_\ell$-module) endowed with a continuous action

ρ of $\mathrm{Gal}(\overline{K}/K)$. In what follows we shall always suppose that there exists A as above, with ℓ invertible in A, such that ρ factors through

$$\pi_1(\mathrm{Spec}(A), \overline{K}) = \mathrm{Gal}(K_{nr}/K).$$

We shall say that H is pure of weight n if, for every closed point x of a nonempty open subset of $\mathrm{Spec}(A)$, the eigenvalues α of F_x acting on H are algebraic integers all of whose complex conjugates have absolute value

$$|\alpha| = q_x^{n/2}.$$

Principle 2.1. If the Galois module H "comes from algebraic geometry", there exists on $H_{\mathbb{Q}_\ell} = H \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ a unique increasing filtration W (the "weight filtration"), invariant under Galois, such that $\mathrm{Gr}_n^W(H)$ is pure of weight n.

One may think that $\mathrm{Gr}_n^W(H)$ is moreover semisimple.

When resolution of singularities is available, one can often give a conjectural definition of W, whose correctness follows from the Weil conjectures [5] (cf. 6).

Let μ be the subgroup of $\overline{K}^*$ formed by the roots of unity. The Tate module $\mathbb{Z}_\ell(1)$, defined by

$$\mathbb{Z}_\ell(1) = \mathrm{Hom}(\mathbb{Q}_\ell/\mathbb{Z}_\ell, \mu),$$

is pure of weight -2. Put $\mathbb{Z}_\ell(n) = \mathbb{Z}_\ell(1)^{\otimes n}$.

It is trivial that every morphism $f : H \to H'$ is strictly compatible with the weight filtration.

Principle 2.1 agrees with the fact that every extension of $\mathbb{G}_m$ ("weight -2") by an abelian variety ("weight $-1 > -2$") is trivial.

3. Translation. The Galois modules which appear in ℓ-adic cohomology have, over $\mathbb{C}$, mixed Hodge structures as analogues. In addition one has the dictionary

pure module of weight n	Hodge structure of weight n
weight filtration	weight filtration
homomorphism compatible with Galois	morphism
Tate module $\mathbb{Z}_\ell(1)$	Tate Hodge structure $\mathbb{Z}(1)$.

4. Let X be a complex algebraic variety (= a scheme of finite type over $\mathbb{C}$, which we shall suppose separated). There exists a subfield K of $\mathbb{C}$, of finite type over $\mathbb{Q}$, such that X can be defined over K (i.e. arises by extension of scalars from K to $\mathbb{C}$ from a K-scheme X'). Let $\overline{K}$ be the algebraic closure of K in $\mathbb{C}$. The Galois group $\mathrm{Gal}(\overline{K}/K)$ then acts on the ℓ-adic cohomology groups $H^*(X, \mathbb{Z}_\ell)$; one has

$$H^*(X(\mathbb{C}), \mathbb{Z}) \otimes \mathbb{Z}_\ell = H^*(X, \mathbb{Z}_\ell) = H^*(X'_{\overline{K}}, \mathbb{Z}_\ell).$$

According to 3, one should expect the cohomology groups $H^n(X(\mathbb{C}), \mathbb{Z})$ to carry natural mixed Hodge structures. This can be proved (see [1], 3.2.5, for the case where X is smooth; the proof is algebraic, starting from classical Hodge theory [6]). For X projective and smooth, the Weil conjectures imply that $H^n(X, \mathbb{Z}_\ell)$ is pure of weight n, while classical Hodge theory endows $H^n(X, \mathbb{Z})$ with a Hodge structure of weight n. For every morphism $f : X \to Y$ and for K large enough, $f^* : H^*(Y, \mathbb{Z}_\ell) \to H^*(X, \mathbb{Z}_\ell)$ commutes with Galois (by transport of structure); likewise, $f^* : H^*(Y, \mathbb{Z}) \to H^*(X, \mathbb{Z})$ is a morphism of mixed Hodge structures. For X smooth, the cohomology class in $H^{2n}(X, \mathbb{Z}_\ell(n))$ of an algebraic cycle of codimension n, Z, defined over K, is Galois-invariant, i.e. defines

$$c(Z) \in \mathrm{Hom}_{\mathrm{Gal}}(\mathbb{Z}_\ell(-n), H^{2n}(X, \mathbb{Z}_\ell)).$$

Similarly, the cohomology class $c(Z) \in H^{2n}(X(\mathbb{C}), \mathbb{Z})$ is purely of type (n, n), i.e. corresponds to

$$c(Z) \in \mathrm{Hom}_{\mathrm{H.M.}}(\mathbb{Z}(-n), H^{2n}(X(\mathbb{C}), \mathbb{Z})).$$

5. If $f : H \to H'$ is a Galois-compatible morphism between $\mathbb{Q}_\ell$-vector spaces of different weights, then $f = 0$. Likewise, if $f : H \to H'$ is a morphism of pure mixed Hodge structures of different weights, then f is torsion. A more useful remark is the following.

Scholium 5.1. Let H and H' be Hodge structures of weights n and n', with $n > n'$. Let $f : H_{\mathbb{Q}} \to H'_{\mathbb{Q}}$ be a homomorphism such that $f : H_{\mathbb{C}} \to H'_{\mathbb{C}}$ respects F. Then $f = 0$.

6. Let X be a projective smooth variety over $\mathbb{C}$, let $D = \sum_1^n D_i$ be a normal-crossings divisor in X, a sum of smooth divisors, and let j be the inclusion into X of $U = X - D$. For $Q \subset [1, n]$, put

$$D_Q = \bigcap_{i \in Q} D_i.$$

In ℓ-adic cohomology, one has canonically

$$R^q j_* \mathbb{Z}_\ell = \bigoplus_{\#Q = q} \mathbb{Z}_\ell(-q)_{D_Q}, \tag{6.1}$$

and the Leray spectral sequence for j is

$$E_2^{pq} = \bigoplus_{\#Q=q} H^p(D_Q, \mathbb{Q}_\ell) \otimes \mathbb{Z}_\ell(-q) \Rightarrow H^{p+q}(U, \mathbb{Q}_\ell). \tag{6.2}$$

According to the Weil conjectures [5], $H^p(D_Q, \mathbb{Q}_\ell)$ is pure of weight p, so that E_2^{pq} is pure of weight $p+2q$. As a quotient of a subobject of E_2^{pq}, E_r^{pq} is also pure of weight $p+2q$. By 5, $d_r = 0$ for $r \geq 3$, since the weights $p+2q$ and $p+2q-r+2$ of E_r^{pq} and $E_r^{p+r,q-r+1}$ are different. Hence $E_3^{pq} = E_\infty^{pq}$. Up to reindexing, the weight filtration of $H^*(U, \mathbb{Q}_\ell)$ is the abutment of (6.2):

$$\mathrm{Gr}_n^W(H^k(U, \mathbb{Q}_\ell)) = E_3^{2k-n, n-k}. \tag{6.3}$$

7. In integral cohomology, for the usual topology, the Leray spectral sequence for j is

$${}'E_2^{pq} = \bigoplus_{\#Q=q} H^p(D_Q, \mathbb{Z}) \Rightarrow H^{p+q}(U, \mathbb{Z}). \tag{7.1}$$

Since every D_Q is a nonsingular projective variety, ${}'E_2^{pq}$ is endowed with a Hodge structure of weight p. Put $E_2^{pq} = {}'E_2^{pq} \otimes \mathbb{Z}(-q)$ (a Hodge structure of weight $p+2q$). As an abelian group, $E_2^{pq} = {}'E_2^{pq}$; it is useful to regard (7.1) as a spectral sequence with initial term E_2^{pq}. According to 3, one should expect

$$d_2 : E_2^{pq} \to E_2^{p+2,q-1}$$

to be a morphism of Hodge structures. This is proved by interpreting d_2 as a Gysin morphism. Thus E_3^{pq} is endowed with a Hodge structure of weight $p+2q$. According to 3, one expects that, modulo torsion, the spectral sequence (6.4) degenerates at E_3 ($E_3 = E_\infty$), and that the vanishing of the d_r ($r \geq 3$) is an application of 5.1. This program is carried out in [1] 3.2. There the weight filtration of $H^*(U, \mathbb{Q})$ is defined as the abutment of (7.1), up to the reindexing (6.3).

In fact, in order to endow cohomology groups H^* with a mixed Hodge structure, the key point up to now has always been to find a spectral sequence E abutting to H^* such that the ℓ-adic analogue of E_2^{pq} is conjecturally pure (of weight $p+2q$); E_2^{pq} must then carry a natural Hodge structure (of weight $p+2q$), and the filtration W is the abutment of E.

8. Let $\mathrm{Spec}(V)$ be the spectrum of a henselian discrete valuation ring (a henselian trait), with fraction field K and residue field k of finite type over the prime field of characteristic p. Let $\overline{K}$ be an algebraic closure of K and let H be a finite-dimensional vector space over $\mathbb{Q}_\ell$ ($\ell \neq p$), on which $\mathrm{Gal}(\overline{K}/K)$ acts continuously. According to Grothendieck, it is known ([4], appendix) that a finite-index subgroup of the inertia group I acts unipotently. Replacing V by a finite extension, one reduces to the case in which the action of all of I is unipotent (the semistable case); it then factors through the largest pro-ℓ quotient I_ℓ of I, canonically isomorphic to $\mathbb{Z}_\ell(1)$.

Principle 8.1. In the semistable case, if the Galois module H "comes from algebraic geometry", there exists a unique increasing filtration W of H (the "weight filtration") such that I acts trivially on $\mathrm{Gr}_n^W(H)$ and such that $\mathrm{Gr}_n^W(H)$, as a Galois module under $\mathrm{Gal}(\bar{k}/k) \simeq \mathrm{Gal}(\overline{K}/K)/I$, is pure of weight n.

Compare 2.1 and the appendix of [4].

When resolution of singularities is available, one can sometimes give a conjectural definition of W, whose validity follows from the Weil conjectures. With the help of resolution and Weil, it is often easy to show that in any case H admits a dévissage into Galois modules (under $\mathrm{Gal}(\bar{k}/k)$) that are pure.

Suppose H is semistable. For $T \in I_\ell$, define $\log T$ as the finite sum

$$-\sum_{n>0} (\mathrm{Id} - T)^n/n.$$

The map $(T, x) \mapsto \log T(x)$ is identified with a homomorphism

$$M : \mathbb{Z}_\ell(1) \otimes H \longrightarrow H. \tag{8.2}$$

Since $\mathbb{Z}_\ell(1)$ has weight -2, one necessarily has (cf. 5)

$$M(\mathbb{Z}_\ell(1) \otimes W_n(H)) \subset W_{n-2}(H) \tag{8.3}$$

and M induces

$$\mathrm{Gr}(M) : \mathbb{Z}_\ell(1) \otimes \mathrm{Gr}_n^W(H) \longrightarrow \mathrm{Gr}_{n-2}^W(H). \tag{8.4}$$

8.5. If X is a nonsingular projective variety over an algebraically closed field k_0, define

$$L : \mathbb{Z}_\ell(-1) \otimes H^*(X, \mathbb{Z}_\ell) \longrightarrow H^*(X, \mathbb{Z}_\ell)$$

to be cup product with the cohomology class of a hyperplane section. One will note a formal analogy between L and M: just as M is defined by an action of $\mathbb{Z}_\ell(1)$, one may regard L as defined by an action of $\mathbb{Z}_\ell(-1)$; L raises degree by 2, and $\mathrm{Gr}\, M$ (8.4) lowers it by 2.

9. Let D be the unit disk, $D^* = D - \{0\}$, and let X

$$\begin{array}{ccc} X & \subset & \mathbb{P}^r(\mathbb{C}) \times D \\ {}_f\searrow & & \swarrow^{\mathrm{pr}_2} \\ & D & \end{array}$$

be a family of projective varieties parametrized by D, with f proper and $f|D^*$ smooth. Keep the notation of 8, and recall that in the analogy between a henselian trait and a small neighborhood of 0 in the complex line, one has the following dictionary (note that the spectrum of the ring of germs of holomorphic functions at 0 is a henselian trait):

D	$\mathrm{Spec}(V)$
D^*	$\mathrm{Spec}(K)$
a universal covering $\widetilde{D}^*$ of D^*	$\mathrm{Spec}(\overline{K})$
fundamental group $\pi_1(D^*)$	inertia group I
(with $\pi_1(D^*) = \mathbb{Z} \simeq \mathbb{Z}(1)_{\mathbb{Z}}$)	(with $I_\ell = \mathbb{Z}_\ell(1)$)
X	projective scheme X over $\mathrm{Spec}(V)$
$X^* = f^{-1}(D^*)$	X_K
$\widetilde{X} = X \times_D \widetilde{D}^*$	$X_{\overline{K}}$
local system $R^i f_* \mathbb{Z}\|D^*$	Galois module $H^i(X_{\overline{K}}, \mathbb{Z}_\ell)$
$H^i(\widetilde{X}, \mathbb{Z})$	$H^i(X_{\overline{K}}, \mathbb{Z}_\ell)$.

Note that $\widetilde{X}$ is homotopy equivalent to each of the fibers $X_t = f^{-1}(t)$ ($t \in D^*$): $H^i(X_K, \mathbb{Z}_\ell)$ again has $H^i(X_t, \mathbb{Z})$ as analogue, and the action of I corresponds to the monodromy transformation T.

Here again, it is known that a finite-index subgroup of $\pi_1(D^*)$ acts unipotently on $H^i(\widetilde{X}, \mathbb{Q}) = H^i(X_t, \mathbb{Q})$. Place ourselves in the semistable case where all of $\pi_1(D^*)$ acts unipotently (this amounts to replacing D by a finite covering), and let T be the action of the canonical generator of $\pi_1(D^*)$.

By 3 and 8, one expects $H^i(\widetilde{X},\mathbb{Q}) \simeq H^i(X_t,\mathbb{Q})$ to be endowed with an increasing filtration W, $\mathrm{Gr}_n^W(H^i(\widetilde{X},\mathbb{Q}))$ to be endowed with a Hodge structure of weight n, $\log T(W_n) \subset W_{n-2}$, and $\log T$ to induce a morphism of Hodge structures

$$M_n : \mathbb{Z}(-1) \otimes \mathrm{Gr}_n^W(H^i) \longrightarrow \mathrm{Gr}_{n-2}^W(H^i).$$

One would moreover like (8.2), and not only (8.3) and (8.4), to have an analogue.

In fact, for each vector u in the tangent space to D at $\{0\}$, one succeeds in defining a mixed Hodge structure $\mathcal{H}_u$ on $H^i(\widetilde{X},\mathbb{Z})$. The filtration W and the Hodge structures on the $\mathrm{Gr}_n^W(H^i)$ are independent of u, and the dependence of $\mathcal{H}_u$ on u is expressed simply in terms of T. In analogy with (8.2), one finds that, for every u, $\log T$ induces a homomorphism of mixed Hodge structures

$$M : \mathbb{Z}(1) \otimes H^i(\widetilde{X},\mathbb{Z}) \longrightarrow H^i(\widetilde{X},\mathbb{Z}).$$

Finally, the analogy 8.5 is not misleading (but here the fact that $f|D^*$ is supposed proper and smooth is no doubt essential). One proves that

$$(\log T)^k : \mathrm{Gr}_{n+k}^W(H^n(\widetilde{X},\mathbb{Q})) \longrightarrow \mathrm{Gr}_{n-k}^W(H^n(\widetilde{X},\mathbb{Q}))$$

is an isomorphism for every k (cf. [6], IV 6, cor. to th. 5). This characterizes the filtration W. Up to now, an analogue of Hodge's positivity theorem (cf. [6], IV 7, cor. to th. 7) is available only in very special cases. One hopes that the mixed structures $\mathcal{H}_u$ determine, as $t \to 0$, the asymptotic behavior of the family of pure structures $H^i(X_t,\mathbb{Z})$ $(t \in D^*)$.

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HODGE THEORY, II

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CONTENTS

Introduction

According to Hodge, the cohomology space $H^n(X,\mathbb{C})$ of a compact Kähler variety X is endowed with a *Hodge structure* of weight n, that is, with a natural bigrading

$$H^n(X,\mathbb{C}) = \bigoplus_{p+q=n} H^{pq}$$

satisfying $\overline{H^{pq}} = H^{qp}$. We show here that the complex cohomology of a nonsingular algebraic variety, not necessarily compact, is endowed with a somewhat more general kind of structure, which presents $H^n(X,\mathbb{C})$ as a successive extension of Hodge structures of decreasing weights, lying between $2n$ and n, whose Hodge numbers $h^{pq} = \dim H^{pq}$ are zero for $p > n$ or $q > n$.

The reader will find in [14] an exposition of the yoga underlying this construction.

The proof, essentially algebraic, rests on the one hand on Hodge theory and on the other hand on Hironaka's resolution of singularities, which permits one, via a spectral sequence, to *express* the cohomology of a nonsingular quasi-projective algebraic variety in terms of the cohomology of nonsingular projective varieties.

Section 1 contains, besides a few elementary routines on filtrations assembled for the reader's convenience, two key results:

a) Theorem (1.2.10), which will be used only through its corollary (2.3.5), giving the fundamental properties of mixed Hodge structures;

b) the *lemma on two filtrations* (1.3.16).

Section 2 recalls Hodge theory and introduces mixed Hodge structures. The heart of the paper is no. 3.2, which defines the mixed Hodge structure on $H^n(X,\mathbb{C})$ and establishes certain degeneracies of spectral sequences.

Section 4 gives various applications, all deduced from (4.1.1) and from the theory of the (K/k)-trace for Hodge structures which follows from it (4.1.2). The principal ones are (4.2.6) and (4.4.15).

[1]Work presented as a doctoral thesis at the University of Orsay.

1. Filtrations

1.1. Filtered objects

(1.1.1) Let $\mathcal{A}$ be an abelian category. We shall have to consider filtrations of type $\mathbb{Z}$, usually finite, on the objects of $\mathcal{A}$.

Definition (1.1.2). A decreasing, respectively increasing, filtration F of an object A of $\mathcal{A}$ is a family $(F^n(A))_{n\in\mathbb{Z}}$, respectively $(F_n(A))_{n\in\mathbb{Z}}$, of subobjects of A, satisfying

$$\forall n, m \qquad n \leq m \Rightarrow F^n(A) \supset F^m(A)$$

(resp. $\forall n, m \quad n \leq m \Rightarrow F_n(A) \subset F_m(A)$).

A filtered object is an object endowed with a filtration. When no confusion is possible, the same letter will often denote filtrations on different objects of $\mathcal{A}$.

If F is a decreasing, respectively increasing, filtration on A, put $F^{\infty}(A) = 0$ and $F^{-\infty}(A) = A$ (resp. $F_{-\infty}(A) = 0$ and $F_{\infty}(A) = A$).

The shifted filtrations of a decreasing filtration W are defined by

$$W[n]^p(A) = W^{n+p}(A).$$

(1.1.3) If F is a decreasing, respectively increasing, filtration of A, then the $F_n(A) = F^{-n}(A)$, respectively the $F^n(A) = F_{-n}(A)$, form an increasing, respectively decreasing, filtration of A. Thus in principle it is enough to consider decreasing filtrations; unless explicitly stated otherwise, the word *filtration* will henceforth mean *decreasing filtration*.

(1.1.4) A filtration F of A is called finite if there exist n and m such that $F^n(A) = A$ and $F^m(A) = 0$.

(1.1.5) A morphism from a filtered object (A, F) to a filtered object (B, F) is a morphism f from A to B satisfying

$$f(F^n(A)) \subset F^n(B) \qquad (n \in \mathbb{Z}).$$

The filtered objects, respectively the finitely filtered objects, of $\mathcal{A}$ form an additive category in which finite inductive and projective limits exist, hence kernels, cokernels, images, and coimages of a morphism exist.

A morphism $f : (A, F) \to (B, F)$ is called strict, or strictly compatible with the filtrations, if the canonical arrow from $\mathrm{Coim}(f)$ to $\mathrm{Im}(f)$ is an isomorphism of filtered objects (cf. (1.1.11)).

(1.1.6) Let $^\circ$ be the contravariant identity functor from $\mathcal{A}$ to the dual category $\mathcal{A}^\circ$. If (A, F) is a filtered object of $\mathcal{A}$, the objects $(A/F^{1-n}(A))^\circ$ are identified with subobjects of A°. The filtration on A° dual to F is defined by

$$F^n(A^\circ) = (A/F^{1-n}(A))^\circ.$$

The bidual of (A, F) is identified with (A, F). This construction identifies the dual of the category of filtered objects of $\mathcal{A}$ with the category of filtered objects of $\mathcal{A}^\circ$.

(1.1.7) If (A, F) is a filtered object of $\mathcal{A}$, its associated graded object is the object of $\mathcal{A}^{\mathbb{Z}}$ defined by

$$\mathrm{Gr}^n(A) = F^n(A)/F^{n+1}(A).$$

The convention (1.1.6) is justified by the simple formula

$$\mathrm{Gr}^n(A^\circ) = \mathrm{Gr}^{-n}(A)^\circ,$$

which is verified on the autodual diagram

$$\begin{tikzcd}
A/F^n(A) & A/F^{n+1}(A) \arrow[l] & 0 \arrow[l] & \\
& A \arrow[ul] \arrow[u] \arrow[dl] \arrow[d] & \mathrm{Gr}^n(A) \arrow[u] \arrow[dr] & \\
F^{n+1}(A) \arrow[r] & F^n(A) & & 0
\end{tikzcd} \tag{1.1.7.1}$$

(1.1.8) Let (A, F) be a filtered object and let $j : X \hookrightarrow A$ be a subobject of A. The filtration induced on X by F is the unique filtration on X such that j is strictly compatible with the filtrations; one has

$$F^n(X) = j^{-1}(F^n(A)) = X \cap F^n(A).$$

Dually, the quotient filtration on A/X, the unique filtration such that $p : A \to A/X$ is strictly compatible with the filtrations, is given by

$$F^n(A/X) = p(F^n(A)) \simeq (X + F^n(A))/X \simeq F^n(A)/(X \cap F^n(A)).$$

Lemma (1.1.9). If X and Y are two subobjects of A, with $X \subset Y$, then on $Y/X \simeq \mathrm{Ker}(A/X \to A/Y)$ the quotient filtration from Y coincides with the filtration induced from A/X. In the diagram

$$\begin{tikzcd}
& A/Y & A/X \arrow[l] \\
& A \arrow[u] \arrow[ur] & Y/X \arrow[u] \\
X \arrow[r] \arrow[ur] & Y \arrow[u] \arrow[ur] &
\end{tikzcd}$$

the arrows are strict.

(1.1.10) The filtration of (1.1.9) on Y/X will be called the filtration induced by that of A. By (1.1.9), its definition is autodual.

In particular, if $\Sigma : A \xrightarrow{f} B \xrightarrow{g} C$ is a 0-sequence and if B is filtered, then

$$H(\Sigma) = \mathrm{Ker}(g)/\,\mathrm{Im}(f) = \mathrm{Ker}(\mathrm{Coker}(f) \to \mathrm{Coim}(g))$$

is endowed with a canonical induced filtration.

The reader will verify the following.

Proposition (1.1.11).

(i) Let $f : (A, F) \to (B, F)$ be a morphism of finitely filtered objects. For f to be strict, it is necessary and sufficient that the sequence

$$0 \to \mathrm{Gr}(\mathrm{Ker}(f)) \to \mathrm{Gr}(A) \to \mathrm{Gr}(B) \to \mathrm{Gr}(\mathrm{Coker}(f)) \to 0$$

be exact.

(ii) Let $\Sigma : (A, F) \to (B, F) \to (C, F)$ be a 0-sequence of strict morphisms. Then one has canonically

$$H(\mathrm{Gr}(\Sigma)) \simeq \mathrm{Gr}(H(\Sigma)).$$

In particular, if Σ is exact in $\mathcal{A}$, then $\mathrm{Gr}(\Sigma)$ is exact in $\mathcal{A}^{\mathbb{Z}}$.

In a category of modules, to say that a morphism $f : (A, F) \to (B, F)$ is strict means that every $b \in B$ of filtration $\geq n$ ($b \in F^n(B)$) which is in the image of A is already in the image of $F^n(A)$:

$$f(F^n(A)) = f(A) \cap F^n(B).$$

(1.1.12) If $\otimes : \mathcal{A}_1 \times \cdots \times \mathcal{A}_n \to \mathcal{B}$ is a multiadditive right exact functor, and if A_i is a finitely filtered object of $\mathcal{A}_i$ $(1 \leq i \leq n)$, one defines a filtration on $\bigotimes_{i=1}^n A_i$ by

$$F^k\left(\bigotimes_{i=1}^n A_i\right) = \sum_{\sum k_i = k} \operatorname{Im}\left(\bigotimes_{i=1}^n F^{k_i}(A_i) \to \bigotimes_{i=1}^n A_i\right)$$

(sum of subobjects).

Dually, if H is left exact, put

$$F^k(H(A_i)) = \bigcap_{\sum k_i = k} \operatorname{Ker}\left(H(A_i) \to H(A_i/F^{k_i}(A_i))\right).$$

For exact H, the two definitions are equivalent. These definitions are extended to functors contravariant in certain variables by (1.1.6). In particular, for the left exact functor Hom, one puts

$$F^n(\operatorname{Hom}(A,B)) = \{f : A \to B \mid \forall p,\ f(F^p(A)) \subset F^{p+n}(B)\}.$$

Thus

$$\operatorname{Hom}((A,F),(B,F)) = F^0(\operatorname{Hom}(A,B)).$$

Under the preceding hypotheses, one has evident morphisms

$$\bigotimes_{i=1}^n \operatorname{Gr}(A_i) \to \operatorname{Gr}\left(\bigotimes_{i=1}^n A_i\right)$$

and

$$\operatorname{Gr} H(A_i) \to H(\operatorname{Gr}(A_i)).$$

For exact H, these are mutually inverse isomorphisms.

These constructions are compatible with the composition of functors, in a sense which we leave to the reader not to make explicit.

1.2. Opposite filtrations

(1.2.1) Let A be an object of $\mathcal{A}$ endowed with two filtrations F and G. By definition, $\operatorname{Gr}_F^p(A)$ is a quotient of a subobject of A, and as such is endowed with a filtration induced by G (1.1.10). Passing to the associated graded object, one defines a bigraded object $(\operatorname{Gr}_G^n \operatorname{Gr}_F^p(A))_{n,p\in\mathbb{Z}}$. By a lemma of Zassenhaus, $\operatorname{Gr}_G^n \operatorname{Gr}_F^p(A)$ and $\operatorname{Gr}_F^p \operatorname{Gr}_G^n(A)$ are canonically isomorphic: if one defines the induced filtrations (1.1.10) as quotient filtrations of filtrations induced on a subobject, one has

$$\operatorname{Gr}_G^n \operatorname{Gr}_F^p(A) \simeq \frac{F^p(A) \cap G^n(A)}{(F^{p+1}(A) \cap G^n(A)) + (F^p(A) \cap G^{n+1}(A))}$$

$$\simeq \frac{G^n(A) \cap F^p(A)}{(G^{n+1}(A) \cap F^p(A)) + (G^n(A) \cap F^{p+1}(A))} \simeq \operatorname{Gr}_F^p \operatorname{Gr}_G^n(A).$$

(1.2.2) Let H be a third filtration of A. It induces a filtration on $\operatorname{Gr}_F(A)$, and hence on $\operatorname{Gr}_G \operatorname{Gr}_F(A)$. It also induces a filtration on $\operatorname{Gr}_F \operatorname{Gr}_G(A)$. One should note that these filtrations do not in general correspond under the isomorphism (1.2.1). In the expression $\operatorname{Gr}_H \operatorname{Gr}_G \operatorname{Gr}_F(A)$, therefore, G and H play symmetric roles, but F and G do not.

Definition (1.2.3). Two finite filtrations F and $\overline{F}$ on A are called n-opposite if

$$\operatorname{Gr}_F^p \operatorname{Gr}_{\overline{F}}^q(A) = 0 \qquad \text{for } p + q \neq n.$$

(1.2.4) If $A^{p,q}$ is a bigraded object of $\mathcal{A}$ such that:

a) $A^{p,q} = 0$ except for finitely many pairs (p, q);

b) $A^{p,q} = 0$ for $p + q \neq n$,

then one defines two finite n-opposite filtrations of $A = \sum_{p,q} A^{p,q}$ by putting

$$F^p(A) = \sum_{p' \geq p} A^{p',q'}, \tag{1.2.4.1}$$

$$\overline{F}^q(A) = \sum_{q' \geq q} A^{p',q'}. \tag{1.2.4.2}$$

One has

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = A^{p,q}. \tag{1.2.4.3}$$

Conversely:

Proposition (1.2.5).

(i) Let F and $\overline{F}$ be two finite filtrations on A. For F and $\overline{F}$ to be n-opposite, it is necessary and sufficient that

$$\forall p, q \qquad p + q = n + 1 \Rightarrow F^p(A) \oplus \overline{F}^q(A) \xrightarrow{\sim} A.$$

(ii) If F and $\overline{F}$ are n-opposite, and if one puts

$$\begin{cases} A^{p,q} = 0 & \text{for } p + q \neq n, \\ A^{p,q} = F^p(A) \cap \overline{F}^q(A) & \text{for } p + q = n, \end{cases}$$

then A is the direct sum of the $A^{p,q}$, and F and $\overline{F}$ are deduced from the bigrading $A^{p,q}$ of A by the procedure (1.2.4).

Proof. (i) The condition $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q > n$ means that

$$F^p \cap \overline{F}^q = (F^{p+1} \cap \overline{F}^q) + (F^p \cap \overline{F}^{q+1}) \qquad (p + q > n).$$

By hypothesis, $F^p \cap \overline{F}^q$ is zero for $p+q$ sufficiently large; by descending induction one deduces that the condition $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q > n$ is equivalent to the condition $F^p(A) \cap \overline{F}^q(A) = 0$ for $p + q > n$. Dually ((1.1.6), (1.1.7), (1.1.10)) the condition $\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q(A) = 0$ for $p + q < n$ is equivalent to $A = F^p(A) + \overline{F}^q(A)$ for $(1 - p) + (1 - q) > -n$, that is, for $p + q < n + 1$, and (i) follows.

(ii) If F and $\overline{F}$ are n-opposite, we prove by descending induction on p that

$$\bigoplus_{p' \geq p} A^{p',q'} \xrightarrow{\sim} F^p(A). \tag{1.2.5.1}$$

For $F^p(A) = 0$, the assertion is evident. The decomposition $A = F^{p+1}(A) \oplus \overline{F}^{n-p}(A)$ induces on $F^p(A) \supset F^{p+1}(A)$ a decomposition

$$F^p(A) = F^{p+1}(A) \oplus (F^p(A) \cap \overline{F}^{n-p}(A)),$$

and the induction concludes the proof. For p sufficiently small, $F^p(A) = A$. By (1.2.5.1), the $A^{p,q}$ therefore form a bigrading of A, and F satisfies (1.2.4.1). That $\overline{F}$ satisfies (1.2.4.2) follows by symmetry.

(1.2.6) The constructions (1.2.4) and (1.2.5) establish quasi-inverse equivalences of categories between objects of $\mathcal{A}$ endowed with two finite n-opposite filtrations and bigraded objects of $\mathcal{A}$ of the type considered in (1.2.4).

Definition (1.2.7). Three finite filtrations W, F, and $\overline{F}$ on an object A of $\mathcal{A}$ are called opposite if

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^n(A) = 0 \qquad \text{for } p+q+n \neq 0.$$

This condition is symmetric in F and $\overline{F}$. It means that F and $\overline{F}$ induce on $W^n(A)/W^{n+1}(A)$ two $(-n)$-opposite filtrations. Put

$$A^{p,q} = \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^{-p-q}(A),$$

whence decompositions (1.2.4), (1.2.5)

$$W^n(A)/W^{n+1}(A) = \bigoplus_{p+q=-n} A^{p,q} \tag{1.2.7.1}$$

which make $\mathrm{Gr}_W(A)$ into a bigraded object.

Lemma (1.2.8). Let W, F, and $\overline{F}$ be three finite opposite filtrations on A, and let σ be a sequence $(p_i, q_i)_{i\geq 0}$ of pairs of integers satisfying:

a) $p_i \leq p_j$ and $q_i \leq q_j$ for $i \geq j$;

b) for $i > 0$, $p_i + q_i = p_0 + q_0 - i + 1$.

Put $p = p_0$, $q = q_0$, $n = -p - q$, and

$$A_\sigma = \left(\sum_{0\leq i}(W^{n+i}(A) \cap F^{p_i}(A))\right) \cap \left(\sum_{0\leq i}(W^{n+i}(A) \cap \overline{F}^{q_i}(A))\right).$$

Then the projection from $W^n(A)$ to $\mathrm{Gr}_W^n(A)$ induces an isomorphism

$$A_\sigma \xrightarrow{\sim} A^{p,q} \subset \mathrm{Gr}_W^n(A).$$

We prove by induction on k the following assertion:

$$(*_k)$$

the projection from $W^n(A)/W^{n+k}(A)$ to $\mathrm{Gr}_W^n(A)$ induces an isomorphism from

$$\frac{\left(\left(\sum_{i<k}(W^{n+i}(A) \cap F^{p_i}(A))\right) + W^{n+k}(A)\right) \cap \left(\left(\sum_{i<k}(W^{n+i}(A) \cap \overline{F}^{q_i}(A))\right) + W^{n+k}(A)\right)}{W^{n+k}(A)}$$

to $A^{p,q} \subset \mathrm{Gr}_W^n(A)$.

For $k = 1$, this is exactly the definition of $A^{p,q}$. By (1.2.5)(i), one has

$$F^{p_k}(\mathrm{Gr}_W^{n+k}(A)) \oplus \overline{F}^{q_k}(\mathrm{Gr}_W^{n+k}(A)) \xrightarrow{\sim} \mathrm{Gr}_W^{n+k}(A). \tag{1.2.8.1}$$

Put

$$\begin{aligned}
B &= \sum_{i<k}(W^{n+i}(A) \cap F^{p_i}(A)),\\
C &= \sum_{i<k}(W^{n+i}(A) \cap \overline{F}^{q_i}(A)),\\
B' &= (W^{n+k}(A) \cap F^{p_k}(A)) + W^{n+k+1}(A),\\
C' &= (W^{n+k}(A) \cap \overline{F}^{q_k}(A)) + W^{n+k+1}(A),\\
D &= W^{n+k}(A),\\
E &= W^{n+k+1}(A).
\end{aligned}$$

Formula (1.2.8.1) becomes

$$B' + C' = D,$$

and

$$B' \cap C' = E.$$

Moreover, since $p_k \leq p_i$ $(i \leq k)$,

$$B \cap D \subset F^{p_k}(A) \cap W^{n+k}(A) \subset B'$$

and since $q_k \leq q_i$ $(i \leq k)$,

$$C \cap D \subset \overline{F}^{q_k}(A) \cap W^{n+k}(A) \subset C'.$$

The assertion $(*_{k+1})$ therefore follows from $(*_k)$ and from the following lemma.

Lemma (1.2.9). Let B, C, B', C', D, and E be subobjects of A. Assume that

$$B' + C' = D, \qquad B' \cap C' = E,$$

$$B \cap D \subset B', \qquad C \cap D \subset C'.$$

Then

$$((B + B') \cap (C + C'))/E \xrightarrow{\sim} ((B + D) \cap (C + D))/D.$$

To prove surjectivity, one writes

$$\begin{aligned}((B + B') \cap (C + C')) + D &= (((B + B') \cap (C + C')) + B') + C' \\ &= ((B + B') \cap (C + C' + B')) + C' \\ &= (B + B' + C') \cap (C + C' + B') \\ &= (B + D) \cap (C + D).\end{aligned}$$

To prove injectivity, one writes

$$(B + B') \cap (C + C') \cap D = ((B + B') \cap D) \cap ((C + C') \cap D).$$

Since $B' \subset D$, one has

$$(B + B') \cap D = (B \cap D) + B' = B';$$

similarly,

$$(C + C') \cap D = C',$$

and

$$(B + B') \cap (C + C') \cap D = B' \cap C' = E.$$

Finally, the proof of (1.2.8) is completed by observing that (1.2.8) is equivalent to $(*_k)$ for large k.

Theorem (1.2.10). Let $\mathcal{A}$ be an abelian category, and provisionally denote by $\mathcal{A}'$ the category of objects of $\mathcal{A}$ endowed with three opposite filtrations W, F, and $\overline{F}$. The morphisms in $\mathcal{A}'$ are the morphisms in $\mathcal{A}$ compatible with the three filtrations.

(i) $\mathcal{A}'$ is an abelian category.

(ii) The kernel, respectively cokernel, of an arrow $f : A \to B$ in $\mathcal{A}'$ is the kernel, respectively cokernel, of f in $\mathcal{A}$, endowed with the filtrations induced from those of A, respectively quotient filtrations from those of B.

(iii) Every morphism $f : A \to B$ in $\mathcal{A}'$ is strictly compatible with the filtrations W, F, and $\overline{F}$; the morphism $\mathrm{Gr}_W(f)$ is compatible with the bigradings of $\mathrm{Gr}_W(A)$ and $\mathrm{Gr}_W(B)$; the morphisms $\mathrm{Gr}_F(f)$ and $\mathrm{Gr}_{\overline{F}}(f)$ are strictly compatible with the filtration induced by W.

(iv) The functors *forget the filtrations*, Gr_W, Gr_F, $\mathrm{Gr}_{\overline{F}}$, and

$$\mathrm{Gr}_W \mathrm{Gr}_F \simeq \mathrm{Gr}_F \mathrm{Gr}_W, \quad \mathrm{Gr}_W \mathrm{Gr}_{\overline{F}} \simeq \mathrm{Gr}_{\overline{F}} \mathrm{Gr}_W, \quad \mathrm{Gr}_F \mathrm{Gr}_{\overline{F}} \simeq \mathrm{Gr}_{\overline{F}} \mathrm{Gr}_F$$

from $\mathcal{A}'$ to $\mathcal{A}$ are exact.

Let $\sigma_0(p,q)$ and $\sigma_1(p,q)$ denote the sequences

$$\sigma_0(p,q) = (p,q), (p,q), (p,q-1), (p,q-2), (p,q-3), \ldots$$

$$\sigma_1(p,q) = (p,q), (p,q), (p-1,q), (p-2,q), (p-3,q), \ldots$$

and, with the notation of (1.2.8), put

$$A_i^{p,q} = A_{\sigma_i(p,q)} \qquad (i = 0, 1).$$

If $f : A \to B$ is compatible with W, F, and $\overline{F}$, then

$$f(A_i^{p,q}) \subset B_i^{p,q} \qquad (i = 0, 1). \tag{1.2.10.1}$$

Assertion (iii) therefore follows from the following lemma.

Lemma (1.2.11). The $A_i^{p,q}$ form a bigrading of A. One has

$$W^n(A) = \sum_{n+p+q \leq 0} A_i^{p,q} \qquad (i = 0, 1), \tag{1.2.11.1}$$

$$F^p(A) = \sum_{p' \geq p} A_0^{p',q}, \tag{1.2.11.2}$$

$$\overline{F}^q(A) = \sum_{q' \geq q} A_1^{p,q'}. \tag{1.2.11.3}$$

By symmetry, it is enough to prove the assertions for $i = 0$. Put $A_0 = \bigoplus A_0^{p,q}$, and define on A_0 filtrations W and F by the formulae of (1.2.11). The canonical map i from A_0 to A is compatible with the filtrations W and F. Moreover, by (1.2.8), $\mathrm{Gr}_W(i)$ is an isomorphism, and it induces isomorphisms of graded objects

$$\sum_{p+q=-n} A_0^{p,q} \xrightarrow{\sim} \mathrm{Gr}_W^{-n}(A) = \sum_{p+q=-n} A^{p,q}. \tag{1.2.11.4}$$

The morphism i is therefore an isomorphism, and the $A_0^{p,q}$ form a bigrading of A. Formula (1.2.11.1) then expresses that $\mathrm{Gr}_W(i)$ is an isomorphism. By (1.2.11.4), $\mathrm{Gr}_F \mathrm{Gr}_W(i)$ is an isomorphism, hence so are $\mathrm{Gr}_W \mathrm{Gr}_F(i)$ and $\mathrm{Gr}_F(i)$. Formula (1.2.11.2) follows.

(1.2.12) We prove (1.2.10). Let $f : A \to B$ be in $\mathcal{A}'$ and endow $K = \mathrm{Ker}(f)$ with the filtrations induced from those of A. By (1.2.11), $\mathrm{Gr}_W(K) \to \mathrm{Gr}_W(A)$; moreover, the filtration F, respectively $\overline{F}$, of K induces on $\mathrm{Gr}_W(K)$ the inverse-image filtration of the filtration F on $\mathrm{Gr}_W(A)$. Finally, the subobject $\mathrm{Gr}_W(K)$ of $\mathrm{Gr}_W(A)$ is compatible with the bigrading of $\mathrm{Gr}_W(A)$:

$$\mathrm{Gr}_W(K) = \bigoplus_{p,q} (\mathrm{Gr}_W(K) \cap A^{p,q}).$$

It follows that

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^n(K) \hookrightarrow \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^n(A);$$

the filtrations W, F, and $\overline{F}$ of K are therefore opposite, and K is a kernel of f in $\mathcal{A}'$. Together with the dual result, this proves (ii).

If f is an arrow of $\mathcal{A}'$, the canonical morphism from $\mathrm{Coim}(f)$ to $\mathrm{Im}(f)$ is an isomorphism in $\mathcal{A}$; by (iii), it is also an isomorphism in $\mathcal{A}'$, which is therefore abelian.

The functor *forget the filtrations* is exact by (ii). The exactness of the other functors in (iv) follows at once from (ii), (iii), and (1.1.11)(i) or (ii).

(1.2.13) Let A be an object of $\mathcal{A}$ endowed with a finite increasing filtration W, and with two finite decreasing filtrations F and $\overline{F}$. The construction (1.1.3) associates to $W_\bullet$ a decreasing filtration $W^\bullet$. We shall say that the filtrations $W_\bullet$, F, and $\overline{F}$ are opposite if the filtrations $W^\bullet$, F, and $\overline{F}$ are opposite, that is, if for every n the filtrations induced by F and $\overline{F}$ on

$$\mathrm{Gr}_n^W(A) = W_n(A)/W_{n-1}(A)$$

are n-opposite.

Theorem (1.2.10) carries over trivially to this variant.

1.3. The lemma on two filtrations

(1.3.1) Let K be a differential complex of objects of $\mathcal{A}$, endowed with a filtration F. This filtration is called biregular if it induces on each component of K a finite filtration. Recall the definition of the terms $E_r^{pq}(K,F)$, or simply E_r^{pq}, of the spectral sequence defined by F. Put

$$Z_r^{pq} = \mathrm{Ker}\left(d : F^p(K^{p+q}) \to K^{p+q+1}/F^{p+r}(K^{p+q+1})\right),$$

and define B_r^{pq} dually by the formula

$$K^{p+q}/B_r^{pq} = \mathrm{coker}\left(d : F^{p-r+1}(K^{p+q-1}) \to K^{p+q}/F^{p+1}(K^{p+q})\right).$$

These formulae still make sense for $r=\infty$.

One should note that the use made here of the notation B_r^{pq} is different from that of Godement [TF].

By definition one has:

$$\begin{aligned} E_r^{pq} &= \mathrm{Im}\left(Z_r^{pq} \to K^{p+q}/B_r^{pq}\right), && (1.3.1.1)\\ &= Z_r^{pq}/(B_r^{pq} \cap Z_r^{pq}), && (1.3.1.2)\\ &= \mathrm{Ker}\left(K^{p+q}/B_r^{pq} \to K^{p+q}/(Z_r^{pq}+B_r^{pq})\right). && (1.3.1.3)\end{aligned}$$

One can also write

$$\begin{aligned} B_r^{pq} \cap Z_r^{pq} &= (dF^{p-r+1} + F^{p+1}) \cap (d^{-1}F^{p+r} \cap F^p)\\ &= (dF^{p-r+1} \cap F^p) + F^{p+1} \cap d^{-1}F^{p+r}, && (1.3.1.4)\end{aligned}$$

since $dF^{p-r+1} \subset d^{-1}F^{p+r}$ and $F^{p+1} \subset F^p$.

For $r<\infty$, the E_r form a complex graded by the degree $p-r(p+q)$, and E_{r+1} is expressed as the cohomology of this complex:

$$E_{r+1}^{p,q} = H\big(E_r^{p-r,q+r-1} \xrightarrow{d_r} E_r^{p,q} \xrightarrow{d_r} E_r^{p+r,q-r+1}\big). \qquad (1.3.1.5)$$

For $r=0$ one has

$$E_0^{**} = \mathrm{Gr}_F(K^*). \qquad (1.3.1.6)$$

Proposition (1.3.2). Let K be a complex endowed with a biregular filtration F. The following conditions are equivalent:

(i) the spectral sequence defined by F degenerates: $E_1 = E_\infty$;

(ii) the morphisms $d : K^i \to K^{i+1}$ are strictly compatible with the filtrations.

We check this when $\mathcal{A}$ is a category of modules. For fixed p and q, the hypothesis that the arrows d_r with source $E_r^{p,q}$ are zero for $r \geq 1$ means that, if $x \in F^p(K^{p+q})$ satisfies $dx \in F^{p+r}(K^{p+q+1})$, then there exists y in K^{p+q} such that $dy=0$ and such that x and y have the same image in $E_r^{p,q}$. Modifying y by a boundary and putting $z = x-y$, one obtains

$$\forall x \in F^p(K^{p+q})\ \big(dx \in F^{p+1}(K^{p+q+1}) \Rightarrow \exists z \in F^{p+1}(K^{p+q}),\ dz = dx\big),$$

or, in other words,

$$F^{p+1}(K^{p+q+1}) \cap dF^p(K^{p+q}) = dF^{p+1}(K^{p+q}). \qquad (1)$$

If this condition holds for all p and q, then by induction on r one has

$$F^{p+r} \cap dF^p = dF^{p+r},$$

which, for $p+r$ large, becomes

$$F^p \cap dK = dF^p. \tag{2}$$

Assertion (2) trivially implies (i), and is equivalent to (ii). This proves (1.3.2).

(1.3.3) If (K, F) is a filtered complex, we shall denote by $\mathrm{Dec}(K)$ the complex K endowed with the shifted filtration

$$\mathrm{Dec}(F)^p K^n = Z_1^{p+n,-p}.$$

This filtration is compatible with the differentials:

$$dZ_1^{p+n,-p} \subset F^{p+n+1}(K^{n+1}) \cap \mathrm{Ker}(d) \subset Z_\infty^{p+n+1,-p} \subset Z_1^{p+n+1,-p}.$$

Since

$$Z_1^{p+1+n,-p-1} \subset F^{p+1+n}(K^n) \subset B_1^{p+n,-p} \subset Z_1^{p+n,-p}, \tag{1.3.3.1}$$

the evident map from $Z_1^{p+n,-p}/Z_1^{p+1+n,-p-1}$ to $Z_1^{p+n,-p}/B_1^{p+n,-p}$ is a morphism

$$u : E_0^{p,n-p}(\mathrm{Dec}\, K) \longrightarrow E_1^{p+n,-p}(K). \tag{1.3.3.2}$$

Proposition (1.3.4).

(i) The morphisms (1.3.3.2) form a morphism of graded complexes from $E_0(\mathrm{Dec}\, K)$ to $E_1(K)$.

(ii) This morphism induces an isomorphism on cohomology.

(iii) Step by step, via (1.3.1.5), it induces isomorphisms of graded complexes

$$E_r(\mathrm{Dec}(K)) \simeq E_{r+1}(K) \qquad (r \geq 1).$$

Proof. Let F' be the filtration of K defined by

$$F'^p(K^n) = \mathrm{Dec}(F)^{p-n}(K^n) = Z_1^{p,n-p}.$$

There are trivial isomorphisms, compatible with the d_r and with (1.3.1.5),

$$E_r^{p,n-p}(\mathrm{Dec}\, K) = E_r^{p+n,-p}(K, F'). \tag{1.3.4.1}$$

The map u is deduced from (1.3.4.1) and from the identity morphism

$$(K, F') \longrightarrow (K, F).$$

This proves (i), and it remains to verify that, for $r \geq 2$,

$$E_r^{p,q}(K, F') \simeq E_r^{p,q}(K, F).$$

Indeed one has

$$Z_r^{p,q}(K, F') = Z_r^{p,q}(K, F) \qquad (r \geq 1),$$

and

$$Z_r^{p,q}(K, F') \cap B_r^{p,q}(K, F') = Z_r^{p,q}(K, F) \cap B_r^{p,q}(K, F) \qquad (r \geq 2),$$

and then applies (1.3.1.2).

(1.3.5) The construction (1.3.3) is not self-dual. The dual construction consists in defining

$$\mathrm{Dec}^*(F)^p K^n = B_1^{p+n-1,-p+1}.$$

One then has morphisms

$$E_0^{p,n-p}(\mathrm{Dec}\, K) \longrightarrow E_1^{p+n,-p}(K) \longrightarrow E_0^{p,n-p}(\mathrm{Dec}^*\, K)$$

and, for $r \geq 1$, isomorphisms

$$E_r^{p,n-p}(\mathrm{Dec}\, K) \simeq E_{r+1}^{p+n,-p}(K) \simeq E_r^{p,n-p}(\mathrm{Dec}^*\, K).$$

Recall that a morphism of complexes is called a quasi-isomorphism if it induces an isomorphism on cohomology.

Definition (1.3.6).

(i) A morphism $f : (K, F) \to (K', F')$ of filtered complexes with biregular filtration is a filtered quasi-isomorphism if $\mathrm{Gr}_F(f)$ is a quasi-isomorphism, i.e. if the $E_1^{p,q}(f)$ are isomorphisms.

(ii) A morphism $f : (K, F, W) \to (K', F', W')$ between biregular bifiltered complexes is a bifiltered quasi-isomorphism if $\mathrm{Gr}_F \mathrm{Gr}_W(f)$ is a quasi-isomorphism.

(1.3.7) Let K be a differential complex of objects of $\mathcal{A}$, endowed with two filtrations F and W. Let $E_r^{p,q}$ be the spectral sequence defined by W. The filtration F induces several filtrations on the $E_r^{p,q}$, which we shall compare.

(1.3.8) Formula (1.3.1.2) identifies $E_r^{p,q}$ with a quotient of a subobject of K^{p+q}. Thus the term $E_r^{p,q}$ receives a filtration F_d induced by F, called the first direct filtration.

(1.3.9) Dually, formula (1.3.1.3) identifies $E_r^{p,q}$ with a subobject of a quotient of K^{p+q}, and hence gives a new filtration F_{d*} induced by F, the second direct filtration.

Lemma (1.3.10). On E_0 and E_1 one has $F_d = F_{d*}$. For $r = 0$ or 1, one has $B_r \subset Z_r$, and one applies (1.1.9).

(1.3.11) Formula (1.3.1.5) identifies $E_{r+1}^{p,q}$ with a quotient of a subobject of $E_r^{p,q}$. The recurrent filtration F_r of the $E_r^{p,q}$ is defined by the following conditions:

(i) on $E_0^{p,q}$, $F_r = F_d = F_{d*}$;

(ii) on $E_{r+1}^{p,q}$, the recurrent filtration is the one induced by the recurrent filtration of $E_r^{p,q}$.

(1.3.12) The definitions (1.3.8) and (1.3.9) still make sense for $r = \infty$. If the filtration of K is biregular, the direct filtrations of $E_\infty^{p,q}$ coincide with those of $E_r^{p,q} = E_\infty^{p,q}$ for r large enough, and the recurrent filtration of $E_\infty^{p,q}$ is defined by requiring it to coincide with that of $E_r^{p,q}$ for r large enough.

The filtrations F and W each induce a filtration of $H^*(K)$, and $E_\infty^{p,q} = \mathrm{Gr}_W^p(H^{p+q}(K))$. The filtration F of $H^*(K)$ therefore induces a new filtration on $E_\infty^{p,q}$.

Proposition (1.3.13).

(i) For the first direct filtration, the morphisms d_r are compatible with filtrations. If $E_{r+1}^{p,q}$ is regarded as a quotient of a subobject of $E_r^{p,q}$, then the first direct filtration on $E_{r+1}^{p,q}$ is finer than the filtration induced by the first direct filtration on $E_r^{p,q}$.

(ii) Dually, the morphisms d_r are compatible with the second direct filtration, and the second direct filtration on $E_{r+1}^{p,q}$ is less fine than the filtration induced by that of $E_r^{p,q}$.

(iii) $F_d(E_r^{p,q}) \subset F_r(E_r^{p,q}) \subset F_{d*}(E_r^{p,q})$.

(iv) On $E_\infty^{p,q}$, the filtration induced by the filtration F of $H^*(K)$ is finer than the first direct filtration and less fine than the second.

The first assertion is evident, the second is its dual, and the third follows by induction. The first assertion of (iv) is easy to check, and the second is dual to it.

(1.3.14) Denote by $\mathrm{Dec}(K)$, respectively by $\mathrm{Dec}^*(K)$, the complex K endowed with the filtrations $\mathrm{Dec}(W)$ and F, respectively $\mathrm{Dec}^*(W)$ and F.

It is clear from (1.3.4.1) that the isomorphism (1.3.4) transforms the first direct filtration of $E_r(\mathrm{Dec}\, K)$ into the first direct filtration of $E_{r+1}(K)$, for $r \geq 1$. The dual isomorphism (1.3.5) transforms the second direct filtration of $E_r(\mathrm{Dec}^*\, K)$ into the second direct filtration of $E_{r+1}(K)$.

Lemma (1.3.15). If the filtration F is biregular, and if, on the $\mathrm{Gr}_W^p(K)$, the morphisms d are strictly compatible with the filtration induced by F, then:

(i) the morphism (1.3.3.2) of graded complexes filtered by F

$$u : \mathrm{Gr}_{\mathrm{Dec}(W)}(K) \longrightarrow E_1(K, W)$$

is a filtered quasi-isomorphism;

(ii) dually, the morphism (1.3.5)

$$u : E_1(K, W) \longrightarrow \mathrm{Gr}_{\mathrm{Dec}^*(W)}(K)$$

is a filtered quasi-isomorphism.

By duality it is enough to prove (i). By (1.3.3) and (1.3.4), the complex $E_1(K, W)$, filtered by F, is a quotient of the filtered complex $\mathrm{Gr}_{\mathrm{Dec}(W)}(K)$. Let U be the filtered kernel complex; it is acyclic by (1.3.4) (ii). The long exact cohomology sequence associated with the exact sequence of complexes

$$0 \to \mathrm{Gr}_F^p(U) \to \mathrm{Gr}_F^p(\mathrm{Gr}_{\mathrm{Dec}(W)}(K)) \to \mathrm{Gr}_F^p(E_1(K, W)) \to 0$$

shows that u is a filtered quasi-isomorphism if and only if $\mathrm{Gr}_F^p(U)$ is an acyclic complex. By (1.3.2), and because U is acyclic, this amounts to requiring that the differentials of U be strictly compatible with the filtration F. From (1.3.3.1) one obtains that U is the sum, over p, of the complexes

$$(U^p)^n = B_1^{p+n,-p}/Z_1^{p+1+n,-p-1},$$

endowed with the filtration induced by F.

Each differential d of each of the complexes U^p is included in a commutative diagram of filtered objects of the following type, where, for simplicity, the total or complementary degree has been omitted:

$$\begin{array}{ccccc}
B^p/Z^{p+1} & \xrightarrow{d} & & & B^{p+1}/Z^{p+1} \\
\uparrow & & \mathrm{Coim}(d) & \xrightarrow{(5)} & \mathrm{Im}(d) \ \downarrow \\
W^{p+1}/Z^{p+1} & \xrightarrow{(3)} & & & B^{p+1}/W^{p+2} \\
\uparrow & & (2) & & \downarrow \\
W^{p+1}/W^{p+2} & \xrightarrow{(1)} & & & W^{p+1}/W^{p+2}
\end{array}$$

By hypothesis, the morphism (1) is strict. Since the square (2) is just the canonical decomposition of (1), the arrow (3) is a filtered isomorphism. The arrows of the trapezium (4) are isomorphisms; hence they are filtered isomorphisms, because (3) is one. That (5) is a filtered isomorphism means that d is strict. This proves (1.3.15).

Theorem (1.3.16). Let K be a complex endowed with two filtrations W and F, with F biregular. Let $r_0 \geq 0$ be an integer, and suppose that, for $0 \leq r < r_0$, the differentials of the graded complex $E_r(K, W)$ are strictly compatible with the filtration F_r. Then, for $r \leq r_0 + 1$, one has

$$F_d = F_r = F_{d*} \quad \text{on } E_r^{p,q}.$$

The theorem is proved by induction on r_0. For $r_0 = 0$ the hypothesis is empty, and one applies (1.3.10) and (1.3.13) (iii). For $r_0 \geq 1$, by the induction hypothesis, one has $F_d = F_r = F_{d*}$ on $E_r^{p,q}$ for $r \leq r_0$.

By (1.3.15), the morphism $u : E_0(\mathrm{Dec}\, K) \to E_1(K)$ is a filtered quasi-isomorphism. It therefore induces a filtered isomorphism from $H^*(\mathrm{Dec}\, K)$ to $H^*(E_1(K))$:

$$u : (E_1(\mathrm{Dec}\, K), F_r) \xrightarrow{\sim} (E_2(K), F_r).$$

Step by step, it follows that the canonical isomorphism from $E_s(\mathrm{Dec}\, K)$ to $E_{s+1}(K)$ is a filtered isomorphism for the recurrent filtration. On $E_1(\mathrm{Dec}(K))$, $F_r = F_d$ (1.3.10), and we already know (1.3.14) that u' is a filtered isomorphism

$$u' : (E_1(\mathrm{Dec}\, K), F_d) \xrightarrow{\sim} (E_2(K), F_d).$$

Thus on $E_2(K)$ one has $F_d = F_r$. Together with the dual result, this proves (1.3.16) for $r_0 = 1$.

Assume now that $r_0 \geq 2$. Then the arrows d_1 of $E_1(K)$ are strictly compatible with the filtrations; hence so are the arrows d_0 of $E_0(\operatorname{Dec} K)$, since u induces an isomorphism of spectral sequences, and one applies the criterion (1.3.2).

For $0 < s < r_0 - 1$, the isomorphism $(E_s(\operatorname{Dec} K), F_r) \simeq (E_{s+1}(K), F_r)$ shows that the d_s are strictly compatible with the recurrent filtrations. By the induction hypothesis one therefore has $F_d = F_r$ on $E_s(\operatorname{Dec} K)$ for $s \leq r_0$. The isomorphism $(E_s(\operatorname{Dec} K), F_d) \simeq (E_{s+1}(K), F_d)$ (1.3.13) then shows that $F_d = F_r$ on $E_r(K)$ for $r \leq r_0 + 1$. Together with the dual result, this proves (1.3.16).

Corollary (1.3.17). Under the general hypotheses of (1.3.16), suppose that, for every r, the differentials d_r are strictly compatible with the recurrent filtrations of the E_r. Then, on E_∞, the filtrations F_d, F_r, and F_{d*} coincide, and coincide with the filtration induced by the filtration F of $H^*(K)$.

This follows at once from (1.3.16) and (1.3.13) (iv).

1.4. Hypercohomology of filtered complexes

In this section we recall some standard constructions in hypercohomology. We do not use the language of derived categories, which would be the most natural here.

Throughout this section, by a "complex" we mean a complex bounded below.

(1.4.1) Let T be a left exact functor from an abelian category $\mathcal{A}$ to an abelian category $\mathcal{B}$. Assume that every object of $\mathcal{A}$ injects into an injective object; the derived functors $R^iT : \mathcal{A} \to \mathcal{B}$ are therefore defined. An object A of $\mathcal{A}$ will be called acyclic for T if $R^iT(A) = 0$ for $i > 0$.

(1.4.2) Let (A, F) be an object with a finite filtration, and let TF be the filtration of TA by its subobjects TF^pA. If $\operatorname{Gr}_F^p(A)$ is T-acyclic, then the $F^p(A)$ are T-acyclic as successive extensions of T-acyclic objects. The image under T of the sequence

$$0 \to F^{p+1}(A) \to F^p(A) \to \operatorname{Gr}_F^p(A) \to 0$$

is therefore exact, and

$$\operatorname{Gr}_{TF}^p TA \xrightarrow{\sim} T \operatorname{Gr}_F^p A. \tag{1.4.2.1}$$

(1.4.3) Let A be an object with two finite filtrations F and W such that $\operatorname{Gr}_F^p \operatorname{Gr}_W^q A$ is T-acyclic. The objects $\operatorname{Gr}_F^p A$ and $\operatorname{Gr}_W^q A$ are then T-acyclic, as are the $F^p(A) \cap W^q(A)$. The sequences

$$0 \to T(F^p \cap W^{q+1}) \to T(F^p \cap W^q) \to T\big((F^p \cap W^q)/(F^p \cap W^{q+1})\big) \to 0$$

are therefore exact, and $T(F^p(\operatorname{Gr}_W^q A))$ is the image in $T(\operatorname{Gr}_W^q A)$ of $T(F^p \cap W^q)$. The diagram

$$\begin{tikzcd}
T(F^p \cap W^q) \arrow[r] \arrow[d] & T(F^p \operatorname{Gr}_W^q A) \arrow[r] & T \operatorname{Gr}_W^q A \arrow[d] \\
TF^p \cap TW^q \arrow[rr] & & \operatorname{Gr}_{TW}^q TA
\end{tikzcd}$$

then shows that the isomorphism (1.4.2.1) relative to W carries the filtration $\operatorname{Gr}_{TW}(TF)$ to the filtration $T(\operatorname{Gr}_W(F))$.

(1.4.4) Let K be a complex of objects of $\mathcal{A}$. The hypercohomology objects $R^iT(K)$ are computed as follows:

a) choose a quasi-isomorphism $i : K \to K'$ such that the components of K' are acyclic for T. For example, one may take for K' the simple complex associated with a Cartan–Eilenberg injective resolution of K;

b) put

$$R^iT(K) = H^i(T(K')).$$

One checks that $R^iT(K)$ does not depend on the choice of K', depends functorially on K, and that a quasi-isomorphism $f : K_1 \to K_2$ induces isomorphisms

$$R^iT(f) : R^iT(K_1) \xrightarrow{\sim} R^iT(K_2).$$

(1.4.5) Let F be a biregular filtration of K. A filtered T-acyclic resolution of K is a filtered quasi-isomorphism $i : K \to K'$ from K to a biregular filtered complex such that the $\mathrm{Gr}^p_F(K'^n)$ are acyclic for T. If K' is such a resolution, the K'^n are acyclic for T and the filtered complex $T(K')$ defines a spectral sequence

$$E_1^{p,q} = R^{p+q}T(\mathrm{Gr}^p_F K) \Rightarrow R^{p+q}T(K).$$

This is independent of the choice of K'. It is called the hypercohomology spectral sequence of the filtered complex K. It depends functorially on K, and a filtered quasi-isomorphism induces an isomorphism of spectral sequences.

The differentials d_1 of this spectral sequence are the connecting morphisms defined by the short exact sequences

$$0 \to \mathrm{Gr}^{p+1}_F K \to F^p K/F^{p+2}K \to \mathrm{Gr}^p_F K \to 0.$$

(1.4.6) Let K be a complex. Denote by $\tau_{\leq p}(K)$ the following subcomplex:

$$\tau_{\leq p}(K)^n = \begin{cases} K^n, & n < p, \\ \mathrm{Ker}(d), & n = p, \\ 0, & n > p. \end{cases}$$

The filtration, called canonical, of K by the $\tau_{\leq p}(K)$ is obtained by shifting the trivial filtration G for which $G^0(K) = K$ and $G^1(K) = 0$. For the canonical filtration one has

$$E_1^{p,q} = \begin{cases} 0, & p+q \neq -p, \\ H^{-p}, & p+q = -p. \end{cases}$$

A quasi-isomorphism $f : K \to K'$ is automatically a filtered quasi-isomorphism for the canonical filtrations.

(1.4.7) The subcomplexes $\sigma_{\geq p}(K)$ of K,

$$\sigma_{\geq p}(K)^n = \begin{cases} 0, & n < p, \\ K^n, & n \geq p, \end{cases}$$

define a biregular filtration, the stupid filtration of K.

The hypercohomology spectral sequences attached to the stupid or canonical filtrations of K are the two hypercohomology spectral sequences of K.

Example (1.4.8). Let $f : X \to Y$ be a continuous map between topological spaces, and let $\mathcal{F}$ be an abelian sheaf on X. Let $\mathcal{F}^*$ be a resolution of $\mathcal{F}$ by f_*-acyclic sheaves. Then $R^i f_* \mathcal{F} \simeq \mathcal{H}^i(f_*\mathcal{F}^*)$. Take as T the functor $\Gamma(Y, \cdot)$. The hypercohomology spectral sequence of the complex $f_*\mathcal{F}^*$, endowed with its canonical filtration (1.4.6),

$$E_1^{p,q} = H^{2p+q}(Y, R^{-p} f_*\mathcal{F}) \Rightarrow H^{p+q}(X, \mathcal{F}),$$

is, up to renumbering, the Leray spectral sequence for f and $\mathcal{F}$.

(1.4.9) Let (K, W, F) be a biregular bifiltered complex. To this complex one associates:

a) a spectral sequence

$$_W E_1^{p,n-p} = H^n(\mathrm{Gr}^p_W K) \Rightarrow H^n(K),$$

whose differentials $_W d_1$ are the connecting morphisms deduced from the short exact sequences

$$0 \to \mathrm{Gr}^{p+1}_W K \to W^p(K)/W^{p+2}(K) \to \mathrm{Gr}^p_W K \to 0;$$

b) an analogous spectral sequence for the filtration F;

c) exact squares

$$
\begin{tikzcd}
0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & & \\
0 \arrow[r] & \mathrm{Gr}_F^{p+1}\mathrm{Gr}_W^{q+1} K \arrow[r] \arrow[d] & F^p/F^{p+2}(\mathrm{Gr}_W^{q+1} K) \arrow[r] \arrow[d] & \mathrm{Gr}_F^{p}\mathrm{Gr}_W^{q+1} K \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{Gr}_F^{p+1}(W^q/W^{q+2}K) \arrow[r] \arrow[d] & F^p/F^{p+2}(W^q/W^{q+2}K) \arrow[r] \arrow[d] & \mathrm{Gr}_F^{p}(W^q/W^{q+2}K) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{Gr}_F^{p+1}\mathrm{Gr}_W^{q} K \arrow[r] \arrow[d] & F^p/F^{p+2}\,\mathrm{Gr}_W^{q} K \arrow[r] \arrow[d] & \mathrm{Gr}_F^{p}\mathrm{Gr}_W^{q} K \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

The exterior rows and columns of this square define connecting morphisms

$$_{F,W}d_1 : H^n \,\mathrm{Gr}_F^p \mathrm{Gr}_W^q K \longrightarrow H^{n+1}\,\mathrm{Gr}_F^{p+1}\mathrm{Gr}_W^q K,$$

$$_{W,F}d_1 : H^n \,\mathrm{Gr}_F^p \mathrm{Gr}_W^q K \longrightarrow H^{n+1}\,\mathrm{Gr}_F^{p}\mathrm{Gr}_W^{q+1} K.$$

These morphisms satisfy

$$_{F,W}d_1 \circ {}_{W,F}d_1 + {}_{W,F}d_1 \circ {}_{F,W}d_1 = 0, \qquad {}_{F,W}d_1^2 = 0, \qquad {}_{W,F}d_1^2 = 0.$$

The morphisms $_{F,W}d_1$ are the d_1 morphisms of the spectral sequences $E^{(q)}$, with term $E_1^{p,n-p-q}$ equal to

$$E_1^{p,n-p-q} = H^n(\mathrm{Gr}_F^p \mathrm{Gr}_W^q K) \Rightarrow H^n(\mathrm{Gr}_W^q K) = {}_W E_1^{q,n-q}, \tag{1.4.9.1}$$

defined by the filtered complex $\mathrm{Gr}_W^q(K)$. This spectral sequence abuts to the filtration induced by F on $H^* \,\mathrm{Gr}_W^q K$. Similarly, the $_{W,F}d_1$ are the d_1 of spectral sequences with the same initial terms

$$E_1^{p,q,n-p-q} = H^n(\mathrm{Gr}_F^p \mathrm{Gr}_W^q K) \Rightarrow H^n(\mathrm{Gr}_F^p K). \tag{1.4.9.2}$$

$$
\begin{tikzcd}
 & H^n \,\mathrm{Gr}_W^q K \arrow[dr, "(_W d_1)"] & \\
H^n \,\mathrm{Gr}_F^p \mathrm{Gr}_W^q K \arrow[ur, "(_{F,W} d_1)"] \arrow[dr, "(_{W,F} d_1)"'] & & H^n K \\
 & H^n \,\mathrm{Gr}_F^p K \arrow[ur, "(_F d_1)"'] &
\end{tikzcd}
$$

These constructions are symmetric in F and W, via the isomorphism

$$\mathrm{Gr}_F^p \mathrm{Gr}_W^q \simeq \mathrm{Gr}_W^q \mathrm{Gr}_F^p .$$

(1.4.10) One may also interpret the $_{W,F}d_1$ as the initial morphisms of a morphism of spectral sequences (1.4.9.2), abutting to $_W d_1$. Indeed, let C^q be the cone of the morphism

$$W^q(K)/W^{q+2}(K) \longrightarrow \mathrm{Gr}_W^q(K).$$

In the diagram

$$\Sigma : \mathrm{Gr}_W^{q+1}(K)[1] \longrightarrow C^q \longleftarrow \mathrm{Gr}_W^q(K),$$

u is a quasi-isomorphism, and one has

$$_W d_1 = H(u)^{-1} \circ H(i).$$

In fact, u is even a filtered quasi-isomorphism for F, and the preceding construction defines a morphism from the spectral sequence defined by $(\mathrm{Gr}_W^q(K), F)$ to the one defined by $(\mathrm{Gr}_W^{q+1}(K)[1], F)$, a morphism which abuts to $_W d_1$. The initial term of this morphism, deduced from $\mathrm{Gr}_F(\Sigma)$, is precisely $_{W,F}d_1$.

(1.4.11) These constructions pass unchanged to hypercohomology. Let K be a complex endowed with two biregular filtrations F and W.

A bifiltered T-acyclic resolution of K is a bifiltered quasi-isomorphism $i : K \to K'$ such that the $\mathrm{Gr}_F^p \mathrm{Gr}_W^q(K'^n)$ are T-acyclic. Such resolutions always exist. In the special case where $\mathcal{A}$ is the category of sheaves of A-modules on a topological space X, and where T is the functor Γ from $\mathcal{A}$ to A-modules, one example of a bifiltered T-acyclic resolution of K is the simple complex associated with the Godement double-resolution complex $\mathcal{C}^*(K)$ of K, filtered by the $\mathcal{C}^*(F^p(K))$ and the $\mathcal{C}^*(W^q(K))$. Since $\mathcal{C}^*$ is exact, one indeed has

$$\mathrm{Gr}_F^p \mathrm{Gr}_W^q(\mathcal{C}^*(K)) \simeq \mathcal{C}^*(\mathrm{Gr}_F^p \mathrm{Gr}_W^q(K)).$$

No other case will be needed here.

If K' is a bifiltered T-acyclic resolution of K, the complex TK' is filtered by the TF^pK' and the TW^qK' (1.4.3). Moreover, $\mathrm{Gr}_W^q(K')$ is a filtered T-acyclic resolution, for F, of $\mathrm{Gr}_W^q(K)$, $\mathrm{Gr}_F^p(K')$ is a filtered T-acyclic resolution, for W, of $\mathrm{Gr}_F^p(K)$, $\mathrm{Gr}_F^p \mathrm{Gr}_W^q(K')$ is a T-acyclic resolution of $\mathrm{Gr}_F^p \mathrm{Gr}_W^q(K)$, and

$$T\,\mathrm{Gr}_F^p K' \simeq \mathrm{Gr}_F^p TK' \quad \text{(as a } W\text{-filtered complex)},$$

$$T\,\mathrm{Gr}_W^q K' \simeq \mathrm{Gr}_W^q TK' \quad \text{(as an } F\text{-filtered complex)},$$

$$T\,\mathrm{Gr}_F^p \mathrm{Gr}_W^q K' \simeq \mathrm{Gr}_F^p \mathrm{Gr}_W^q TK'.$$

Lemma (1.4.12). Under the hypotheses of (1.4.11):

(i) The initial terms of the hypercohomology spectral sequences

$$_WE_1^{p,n-p} = R^nT(\mathrm{Gr}_W^p K) \Rightarrow R^nT(K) \tag{1}$$

$$_FE_1^{p,n-p} = R^nT(\mathrm{Gr}_F^p K) \Rightarrow R^nT(K) \tag{2}$$

are the abutments of the hypercohomology spectral sequences of the filtered complexes $\mathrm{Gr}_W^q K$ and $\mathrm{Gr}_F^p K$, with E_1 term given by

$$E_1^{p,n-p-q} = R^nT(\mathrm{Gr}_F^p \mathrm{Gr}_W^q K) \Rightarrow {_WE_1^{q,n-q}}, \tag{3}$$

$$E_1^{q,n-p-q} = R^nT(\mathrm{Gr}_F^p \mathrm{Gr}_W^q K) \Rightarrow {_FE_1^{p,n-p}}. \tag{4}$$

(ii) The filtration of $_WE_1^{q,n-q}$, the abutment of the spectral sequence (3), is the filtration of $_WE_1^{q,n-q}(TK')$ induced by the filtration F of TK'.

(iii) The differentials of the complexes $\mathrm{Gr}_W^q(T(K'))$ are strictly compatible with the filtration F if and only if the hypercohomology spectral sequences (3) degenerate at E_1.

(iv) The morphisms d_1 of the spectral sequence (4) are the initial terms of degree-one morphisms of spectral sequences (3) abutting to the morphisms d_1 of the spectral sequence (1).

Assertions (i) and (iv) follow from (1.4.9) and (1.4.10) applied to TK', through the isomorphisms (1.4.11). Assertion (ii) is then immediate from the definition of the recurrent filtration F, identical with the direct filtrations by (1.3.10) and (1.3.13) (iii), and assertion (iii) follows from (1.3.2).

2. Hodge structures

2.1. Pure structures

(2.1.1) In what follows, $\mathbb{C}$ will denote an algebraic closure of $\mathbb{R}$: we do not assume that a root i of the equation $x^2 + 1 = 0$ has been chosen. The theory will be invariant under complex conjugation (cf. (2.1.14)).

(2.1.2) Let S denote the real algebraic group $\mathbb{C}^*$, obtained by Weil restriction of scalars from $\mathbb{C}$ to $\mathbb{R}$ of the group $\mathbb{G}_m$:

$$S = \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m, \qquad S(\mathbb{R}) = \mathbb{C}^*.$$

The group S is a torus, i.e. it is connected and of multiplicative type. It is therefore described by the finitely generated free abelian group

$$X(S) = \mathrm{Hom}(S_{\mathbb{C}}, \mathbb{G}_m) = \mathrm{Hom}(S, \mathbb{G}_m)(\mathbb{C})$$

of its complex characters, endowed with the action of $\mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \mathbb{Z}/(2)$.

The group $X(S)$ has generators z and $\bar{z}$, inducing respectively the identity and complex conjugation:

$$\mathbb{C}^* = S(\mathbb{R}) \longrightarrow S(\mathbb{C}) \longrightarrow \mathbb{G}_m(\mathbb{C}) = \mathbb{C}^*.$$

Complex conjugation exchanges z and $\bar{z}$.

(2.1.3) There is a canonical map

$$w : \mathbb{G}_m \longrightarrow S. \tag{2.1.3.1}$$

which, on real points, induces the inclusion of $\mathbb{R}^*$ into $\mathbb{C}^*$. One has

$$zw = \bar{z}w = \mathrm{Id}\,. \tag{2.1.3.2}$$

There is also a map

$$N : S \longrightarrow \mathbb{G}_m \tag{2.1.3.3}$$

which, on real points, is identified with the norm $N_{\mathbb{C}/\mathbb{R}} : \mathbb{C}^* \to \mathbb{R}^*$. One has

$$N = z\bar{z} \tag{2.1.3.4}$$

and

$$N \circ w = (x \mapsto x^2). \tag{2.1.3.5}$$

Definition (2.1.4). A real Hodge structure is a finite-dimensional real vector space V endowed with an action of the real algebraic group S.

(2.1.5) By the general theory of groups of multiplicative type, giving a real Hodge structure on V is equivalent to giving a bigrading

$$V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V = \bigoplus_{p,q} V^{pq}$$

such that $\overline{V^{pq}} = V^{qp}$. The action of S and the bigrading determine one another by the condition:

$$\text{on } V^{pq}, \quad S \text{ acts by multiplication by } z^p \bar{z}^q. \tag{2.1.5.1}$$

(2.1.6) Let $(\mathbb{C}, \times)$ be the multiplicative monoid, and put

$$\overline{S} = \prod_{\mathbb{C}/\mathbb{R}} (\mathbb{C}, \times).$$

If V is a real vector space, giving an action of $\overline{S}$ on V is equivalent to giving on V a bigrading such that $\overline{V^{pq}} = V^{qp}$ and $V^{pq} = 0$ for $p < 0$ or $q < 0$.

(2.1.7) Let V be a real Hodge structure, defined by a representation σ of S and by a bigrading V^{pq}. The grading of $V_{\mathbb{C}}$ by

$$V_{\mathbb{C}}^n = \sum_{p+q=n} V^{pq}$$

is then defined over $\mathbb{R}$. It is called the weight grading. On $V^n = V \cap V_{\mathbb{C}}^n$, the representation σw of $\mathbb{G}_m$ is multiplication by x^n.

We shall say that V is of weight n if $V^{pq} = 0$ for $p + q \neq n$, i.e. if σw is multiplication by x^n.

(2.1.8) Let V be a real Hodge structure. The Hodge filtration on $V_{\mathbb{C}}$ is defined by

$$F^p(V_{\mathbb{C}}) = \sum_{p' \geq p} V^{p'q'}.$$

By (1.2.6), one has:

Proposition (2.1.9). Let n be an integer. The construction (2.1.8) establishes an equivalence of categories between:

a) the category of real Hodge structures of weight n;

b) the category of pairs consisting of a finite-dimensional real vector space V and a filtration F on $V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$ which is n-opposed to its complex conjugate $\overline{F}$.

Definition (2.1.10). A Hodge structure H of weight n consists of:

a) a finitely generated $\mathbb{Z}$-module $H_{\mathbb{Z}}$, the "integral lattice";

b) a real Hodge structure of weight n on $H_{\mathbb{R}} = \mathbb{R} \otimes_{\mathbb{Z}} H_{\mathbb{Z}}$.

(2.1.11) A morphism $f : H \to H'$ is a homomorphism $f : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ such that $f_{\mathbb{R}} : H_{\mathbb{R}} \to H'_{\mathbb{R}}$ is compatible with the action of S, that is, such that $f_{\mathbb{C}}$ is compatible with the bigrading, or with the Hodge filtration.

Hodge structures of weight n form an abelian category. If H has weight n and H' has weight n', one defines a Hodge structure $H \otimes H'$ of weight $n + n'$ by the formulas:

a) $(H \otimes H')_{\mathbb{Z}} = H_{\mathbb{Z}} \otimes H'_{\mathbb{Z}}$;

b) the action of S on $(H \otimes H')_{\mathbb{R}} = H_{\mathbb{R}} \otimes H'_{\mathbb{R}}$ is the tensor product of the actions of S on $H_{\mathbb{R}}$ and $H'_{\mathbb{R}}$.

The bigrading, respectively the Hodge filtration, of $(H \otimes H')_{\mathbb{C}} = H_{\mathbb{C}} \otimes H'_{\mathbb{C}}$ is the tensor product of the bigradings, respectively of the Hodge filtrations (cf. (1.1.12)), of $H_{\mathbb{C}}$ and $H'_{\mathbb{C}}$.

Similarly one defines the Hodge structure $\mathrm{Hom}(H, H')$, of weight $n' - n$, the Hodge structures $\bigwedge^p H$, of weight pn, and the dual Hodge structure H^*.

The preceding internal Hom and the group of homomorphisms are related by the following remark.

Remark (2.1.11.1). $\mathrm{Hom}(H, H')$ is the subgroup of

$$\mathrm{Hom}(H, H')_{\mathbb{Z}} = \mathrm{Hom}_{\mathbb{Z}}(H_{\mathbb{Z}}, H'_{\mathbb{Z}})$$

formed by the elements of type $(0, 0)$.

The actions of S on $H_{\mathbb{R}}$, $H'_{\mathbb{R}}$, and $\mathrm{Hom}(H, H')_{\mathbb{R}} = \mathrm{Hom}(H_{\mathbb{R}}, H'_{\mathbb{R}})$ are indeed related by

$$s(f(x)) = s(f)(s(x)).$$

For f to be of type $(0, 0)$, i.e. invariant under S, therefore means that it commutes with the action of S.

(2.1.12) For A a noetherian subring of $\mathbb{R}$, an A-Hodge structure of weight n is defined as consisting of a finite-type A-module H_A and of a real Hodge structure of weight n on $H_{\mathbb{R}} = \mathbb{R} \otimes_A H_A$. These definitions are used mainly for $A = \mathbb{Q}$. A Hodge A-structure consists of a finite-type A-module H_A and a real Hodge structure on $H_A \otimes_A \mathbb{R}$, such that the weight grading is defined over the field of fractions of A.

Definition (2.1.13). The Tate Hodge structure $\mathbb{Z}(1)$ is the Hodge structure of weight -2, of rank 1, purely of bidegree $(-1, -1)$, with integral lattice $2\pi i\mathbb{Z} \subset \mathbb{C}$.

The action of S is therefore multiplication by the inverse of the norm (2.1.3.3). For $n \in \mathbb{Z}$, define $\mathbb{Z}(n)$ to be the n-th tensor power of $\mathbb{Z}(1)$: $\mathbb{Z}(n)$ is the Hodge structure of weight $-2n$, rank 1, purely of bidegree $(-n, -n)$, with integral lattice $(2\pi i)^n\mathbb{Z} \subset \mathbb{C}$. The action of S is multiplication by $N(x)^{-n}$.

(2.1.14) The choice in $\mathbb{C}$ of a solution i of the equation $x^2+1=0$ determines on every complex variety X purely of dimension n an orientation $\mathrm{or}_i(X)$. When i is changed to $-i$, one has

$$\mathrm{or}_{-i}(X) = (-1)^n \,\mathrm{or}_i(X).$$

The choice of i also defines an element C of order 4 in $S(\mathbb{R})$: the image of i under the isomorphism $S(\mathbb{R}) \simeq \mathbb{C}^*$.

Finally it defines an isomorphism between $\mathbb{Z}$ and the integral lattice of $\mathbb{Z}(n)$: multiplication by $(2\pi i)^n$.

Whenever i, an orientation of X, C, or an identification $\mathbb{Z} \simeq \mathbb{Z}(n)_{\mathbb{Z}}$ occurs in a formula, it will in principle be understood that they are subordinated to the same choice of i, and that changing i to $-i$ would give an equivalent definition or formula.

Definition (2.1.15). A polarization of a Hodge structure H of weight n is a homomorphism

$$(x,y) : H \otimes H \longrightarrow \mathbb{Z}(-n)$$

such that the real bilinear form $(2\pi i)^n(x, Cy)$ on $H_{\mathbb{R}}$ is symmetric and positive definite.

(2.1.16) The real Tate Hodge structure is the real Hodge structure $\mathbb{R}(1)$ underlying $\mathbb{Z}(1)$. Similarly one defines $\mathbb{R}(n)$ underlying $\mathbb{Z}(n)$. A polarization of a real Hodge structure H of weight n is a homomorphism

$$(x,y) : H \otimes H \longrightarrow \mathbb{R}(-n)$$

such that the real bilinear form $(2\pi i)^n(x, Cy)$ on $H_{\mathbb{R}}$ is symmetric and positive definite. A polarization is entirely determined by the positive-definite quadratic form $(2\pi i)^n(x, Cy)$ on $H_{\mathbb{R}}$, subject only to the condition that it be invariant under the compact subtorus of S, the kernel of N.

One has

$$(x,y) = (Cx, Cy) = (y, C^2x) = (-1)^n(y,x).$$

The form (x,y) is therefore symmetric or alternating according to the parity of n.

(2.1.17) The reader will generalize these definitions to A-Hodge structures of weight n (2.1.12).

2.2. Hodge theory

(2.2.1) Let X be a compact Kähler variety, for example smooth projective. By the holomorphic Poincaré lemma, the De Rham complex $\Omega^\bullet_X$ is a resolution of the constant sheaf $\mathbb{C}$. Thus one has an isomorphism

$$H^*(X,\mathbb{C}) \simeq \mathbb{H}^*(X, \Omega^\bullet_X),$$

and the bête filtration of $\Omega^\bullet_X$ (1.4.5) defines the hypercohomology spectral sequence

$$E_1^{pq} = H^q(X, \Omega^p_X) \Longrightarrow H^{p+q}(X,\mathbb{C}), \tag{2.2.1.1}$$

whose abutment is the Hodge filtration of $H^r(X,\mathbb{C})$.

By Hodge theory [9], [12], one has:

(A) The spectral sequence (2.2.1.1) degenerates: $E_1 = E_\infty$.

(B) The Hodge filtration of $H^n(X,\mathbb{C})$ is n-opposed to the complex conjugate filtration.

(2.2.2) Let V be a local system of real vector spaces on X, i.e. a sheaf of $\mathbb{R}$-vector spaces locally isomorphic to a constant sheaf $\mathbb{R}^n$. Suppose that on V there exists a bilinear form

$$Q : V \otimes V \longrightarrow \mathbb{R}$$

which is locally constant and definite. For X connected, this is the case if V is defined by a representation of a finite quotient of the fundamental group of X.

The assertions (A) and (B) above remain valid exactly as stated, without any change in the proofs, for cohomology with coefficients in $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$: the spectral sequence

$$E_1^{pq} = H^q(X, \Omega_X^p(V)) \Longrightarrow H^{p+q}(X, V_{\mathbb{C}})$$

deduced from the De Rham resolution of $V_{\mathbb{C}}$ by $\Omega_X^\bullet(V_{\mathbb{C}})$ degenerates and abuts to a filtration on $H^n(X, V_{\mathbb{C}})$ which is n-opposed to the complex conjugate filtration.

The vector space $H^n(X, V)$ is therefore endowed with a canonical real Hodge structure of weight n.

(2.2.3) It is shown in [2] that the statements (2.2.1) remain valid for X a complete nonsingular algebraic variety, not necessarily Kähler. The proof in loc. cit., based on a reduction to the projective case via Chow's lemma and resolution of singularities, extends to the setting (2.2.2).

(2.2.4) Let $\mathcal{L}$ be an invertible sheaf. Here are two ways to define the class $c_1(\mathcal{L})$.

(2.2.4.1) The sheaf $\mathcal{L}$ defines an element c in $H^1(X, \mathcal{O}^*)$. Its image under $df/f : \mathcal{O}^* \to \Omega_X^1$ lies in $H^1(X, \Omega_X^1)$. More precisely, df/f defines a morphism of complexes

$$d\log : \mathcal{O}^*[-1] \longrightarrow [\,0 \to \Omega_X^1 \to \Omega_X^2 \to \cdots] = \sigma_{\geq 1}(\Omega_X^\bullet).$$

This complex maps to $\Omega_X^\bullet$, whence

$$d\log : \mathcal{O}^*[-1] \longrightarrow \Omega_X^\bullet,$$

and the image of c under $d\log$ lies in $H^2(\Omega_X^\bullet)$. This construction would still make sense for an algebraic variety over an arbitrary field k. For $k = \mathbb{C}$, one also has

$$H^2(X, \mathbb{C}) \simeq H^2(X, \Omega_X^\bullet),$$

and hence a class

$$c_1'(\mathcal{L}) \in H^2(X, \mathbb{C}).$$

(2.2.4.2) The exponential exact sequence

$$0 \to \mathbb{Z}(1) \to \mathcal{O} \to \mathcal{O}^* \to 0$$

defines a homomorphism

$$\partial : H^1(X, \mathcal{O}^*) \longrightarrow H^2(X, \mathbb{Z}(1)),$$

hence a class

$$c_1''(\mathcal{L}) \in H^2(X, \mathbb{Z}(1)).$$

If i is chosen, this class is identified with

$$c_1'''(\mathcal{L}) \in H^2(X, \mathbb{Z}),$$

and for $\mathcal{L} = \mathcal{O}(D)$, $c_1'''(\mathcal{L})$ is just the integral cohomology class defined by D, the orientations being defined by i.

(2.2.5) We prove that:

$$-\alpha c_1''(\mathcal{L}) = c_1'(\mathcal{L}), \tag{2.2.5.1}$$

for α the natural injection of $\mathbb{Z}(1)$ into $\mathbb{C}$, and that

$$-\beta c_1'''(\mathcal{L}) = \frac{1}{2\pi i} c_1'(\mathcal{L}), \tag{2.2.5.2}$$

for β the natural injection of $\mathbb{Z}$ into $\mathbb{C}$.

This is verified by contemplating the following diagram of complexes, in which $\simeq$ denotes a quasi-isomorphism, and whose upper rectangle is anticommutative up to homotopy:

$$\begin{tikzcd}
\mathbb{C} \arrow[r, "\simeq"] \arrow[d, equal] & \Omega_X^\bullet & \sigma_{\geq 1}\Omega^\bullet \arrow[l, dashed] \arrow[d, equal] \\
\mathbb{C} & {[\mathbb{C} \to \mathcal{O}]} \arrow[l] \arrow[r, "\simeq"'] \arrow[d] & \sigma_{\geq 1}\Omega^\bullet \arrow[d, "d\log"] \\
\mathbb{Z}(1) \arrow[u] & {[\mathbb{Z}(1) \to \mathcal{O}]} \arrow[l] \arrow[r, "\simeq"'] & \mathcal{O}^*[-1].
\end{tikzcd}$$

(2.2.6) Let X be a nonsingular projective variety purely of dimension n. A choice of i defines an orientation of X and an isomorphism $\mathbb{Z}(1) \simeq \mathbb{Z}$. The corresponding trace morphism

$$H^{2n}(X, \mathbb{Z}(n)) \longrightarrow \mathbb{Z}$$

which follows from it does not depend on the choice of i.

According to Hodge, for $i \leq n$ the morphism

$$L^{n-i} = c_1''(\mathcal{O}(1))^{n-i} : H^i(X, \mathbb{Z}) \longrightarrow H^{2n-i}(X, \mathbb{Z}(n-i))$$

is an isomorphism and, combined with Poincaré duality

$$H^i(X, \mathbb{Z}) \otimes H^{2n-i}(X, \mathbb{Z}(n-i)) \longrightarrow H^{2n}(X, \mathbb{Z}(n-i)) \longrightarrow \mathbb{Z}(-i),$$

it furnishes a polarization on the primitive part $\operatorname{Ker}(L^{n-i+1})$ of $H^i(X, \mathbb{Z})$. It follows that the rational Hodge structures $H^i(X, \mathbb{Q})$ are polarizable.

2.3. Mixed structures

Definition (2.3.1). A mixed Hodge structure H consists of:

a) a finitely generated $\mathbb{Z}$-module $H_{\mathbb{Z}}$, the "integral lattice";

b) a finite increasing filtration W of $H_{\mathbb{Q}} = \mathbb{Q} \otimes_{\mathbb{Z}} H_{\mathbb{Z}}$, called the weight filtration;

c) a finite decreasing filtration F of $H_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{Z}} H_{\mathbb{Z}}$, called the Hodge filtration.

It is required that, on $H_{\mathbb{C}}$, the filtration $W_{\mathbb{C}}$ deduced from W by extension of scalars, the filtration F, and its complex conjugate $\overline{F}$ form a system $(W_{\mathbb{C}}, F, \overline{F})$ of three opposed filtrations ((1.2.7) and (1.2.13)).

(2.3.2) Let W again denote the filtration of $H_{\mathbb{Z}}$ which is the inverse image of the filtration W of $H_{\mathbb{Q}}$. The axiom of mixed Hodge structures means that, for every n, the filtration F induces on $\mathbb{C} \otimes_{\mathbb{Z}} \operatorname{Gr}_n^W(H_{\mathbb{Z}})$ a filtration n-opposed to its complex conjugate. By (2.1.9), $\operatorname{Gr}_n^W(H_{\mathbb{Z}})$ is endowed with a Hodge structure of weight n, with Hodge filtration induced by F.

Example (2.3.3). If H is a Hodge structure of weight n, one defines a mixed Hodge structure with the same integral lattice and the same Hodge filtration by putting

$$W_i(H_{\mathbb{Q}}) = 0 \quad \text{for } i < n, \qquad W_i(H_{\mathbb{Q}}) = H_{\mathbb{Q}} \quad \text{for } i \geq n.$$

(2.3.4) A morphism $f : H \to H'$ of mixed Hodge structures is a homomorphism $f : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ compatible with the filtrations W and F, and therefore compatible with $\overline{F}$. One immediately deduces from (1.2.10) the following theorem.

Theorem (2.3.5).

(i) The category of mixed Hodge structures is abelian.

(ii) The kernel, respectively cokernel, of a morphism $f : H \to H'$ has as integral lattice the kernel, respectively cokernel, K of $f : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$, while $K \otimes \mathbb{Q}$ and $K \otimes \mathbb{C}$ are endowed with the induced, respectively quotient, filtrations from the filtrations W and F of $H_{\mathbb{Q}}$ and $H_{\mathbb{C}}$, respectively of $H'_{\mathbb{Q}}$ and $H'_{\mathbb{C}}$.

(iii) Every morphism $f : H \to H'$ is strictly compatible with the filtration W of $H_{\mathbb{Q}}$ and $H'_{\mathbb{Q}}$, and with the filtration F of $H_{\mathbb{C}}$ and $H'_{\mathbb{C}}$. It induces morphisms of $\mathbb{Q}$-Hodge structures

$$\operatorname{Gr}_n^W(f) : \operatorname{Gr}_n^W(H_{\mathbb{Q}}) \longrightarrow \operatorname{Gr}_n^W(H'_{\mathbb{Q}})$$

and morphisms strictly compatible with the filtration induced by $W_{\mathbb{C}}$:

$$\operatorname{Gr}_F^p(f) : \operatorname{Gr}_F^p(H_{\mathbb{C}}) \longrightarrow \operatorname{Gr}_F^p(H'_{\mathbb{C}}).$$

(iv) The functor Gr_n^W is an exact functor from the category of mixed Hodge structures to the category of $\mathbb{Q}$-Hodge structures of weight n.

(v) The functor Gr_F^p is exact.

(2.3.6) Let H be a mixed Hodge structure. The $W_n(H_{\mathbb{Z}})$, endowed with the filtrations induced by W and F, then form mixed Hodge substructures $W_n(H)$ of H. The quotient $W_n(H)/W_{n-1}(H)$ is identified with $\mathrm{Gr}_n^W(H_{\mathbb{Z}})$ endowed with its Hodge structure (2.3.2), (2.3.3). This Hodge structure will be denoted by $\mathrm{Gr}_n^W(H)$.

(2.3.7) Put

$$H^{pq} = \mathrm{Gr}_F^p \,\mathrm{Gr}_{\overline{F}}^q \,\mathrm{Gr}_{p+q}^W(H_{\mathbb{C}}) = \big(\mathrm{Gr}_F^p \,\mathrm{Gr}_{p+q}^W(H)\big)^{p,q}.$$

The Hodge numbers of H are the integers

$$h^{pq} = \dim_{\mathbb{C}} H^{pq}.$$

Thus the Hodge number h^{pq} of H is the Hodge number h^{pq} of the Hodge structure $\mathrm{Gr}_{p+q}^W(H)$.

(2.3.8) A $\mathbb{Q}$-mixed Hodge structure H is defined as consisting of a finite-dimensional vector space $H_{\mathbb{Q}}$ over $\mathbb{Q}$, a finite increasing filtration W of $H_{\mathbb{Q}}$, and a finite decreasing filtration F of $H_{\mathbb{C}}$, the filtrations $W_{\mathbb{C}}$, F, and $\overline{F}$ being opposed. Theorem (2.3.5) generalizes trivially to this variant.

3. Hodge theory of nonsingular algebraic varieties

3.1. Logarithmic poles and residues

(3.1.1) We recall some classical properties of "logarithmic poles" of holomorphic differential forms. The reader will find proofs in [3], II, (3.1) to (3.7), for example.

(3.1.2) A divisor Y in a smooth complex analytic variety X will be said to have normal crossings if the inclusion of Y into X is locally isomorphic to the inclusion of a union of coordinate hyperplanes in $\mathbb{C}^n$; this does not imply that Y is a union of smooth divisors. Let Y be a normal-crossings divisor in X, and let j be the inclusion of $X^* = X - Y$ in X. Let $\Omega_X^1\langle Y\rangle$ denote the locally free $\mathcal{O}$-submodule of $j_*\Omega_{X^*}^1$ generated by Ω_X^1 and by the dz_i/z_i, where z_i is a local equation of a local irreducible component of Y. The sheaf $\Omega_X^p\langle Y\rangle$ of differential p-forms on X with logarithmic pole along Y is, by definition, the locally free subsheaf $\bigwedge^p \Omega_X^1\langle Y\rangle$ of $j_*\Omega_{X^*}^p$.

Proposition (3.1.3).

(i) A section α of $j_*\Omega_{X^*}^p$ belongs to $\Omega_X^p\langle Y\rangle$ if and only if α and $d\alpha$ have at worst simple poles along the divisor Y.

(ii) The $\Omega_X^p\langle Y\rangle$ form the smallest subcomplex of $j_*\Omega_{X^*}^\bullet$, stable under exterior product, containing $\Omega_X^\bullet$, and containing the logarithmic differential df/f of every local section, meromorphic along Y, of $j_*\mathcal{O}_{X^*}^*$.

The complex $\Omega_X^\bullet\langle Y\rangle$ is called the logarithmic De Rham complex of X along Y. By (3.1.3) (ii), this complex is contravariant in the pair (X, X^*).

(3.1.4) Locally on X, Y is a union of smooth divisors Y_i, and Y^n, respectively $\widetilde{Y}^n$, denotes the union, respectively the disjoint sum, of the intersections n at a time of the Y_i. The Y^n glue to a subspace Y^n of X, and the $\widetilde{Y}^n$ glue to the normalization of Y^n. One has $\widetilde{Y}^0 = Y^0 = X$ and puts $\widetilde{Y} = \widetilde{Y}^1$.

The two-element set of orientations of a finite set E with n elements is defined to be the set of generators of $\bigwedge^n \mathbb{Z}^E$. For $n \geq 2$, this set is the set of conjugacy classes, under the alternating group, of total orderings of E.

If to each point y of $\widetilde{Y}^n$ one associates the set of the n local components of Y which contain the image in X of a neighborhood of y in $\widetilde{Y}^n$, then one defines on $\widetilde{Y}^n$ a local system E^n of n-element sets. The local system of orientations of these sets is a torsor under $\mathbb{Z}/(2)$. Through the inclusion of $\mathbb{Z}/(2)$ into $\mathbb{C}^*$, this torsor defines a rank-one complex local system ε^n on $\widetilde{Y}^n$, endowed with an isomorphism

$$(\varepsilon^n)^{\otimes 2} \simeq \mathbb{C}.$$

One has

$$\varepsilon^n \simeq \bigwedge^n \mathbb{C}^{E^n}.$$

Locally on $\widetilde{Y}^n$, ε^n is endowed with two opposite isomorphisms $\pm\alpha : \varepsilon^n \xrightarrow{\sim} \mathbb{C}$. Put

$$\varepsilon^n_{\mathbb{Z}} = \alpha^{-1}\big((2\pi i)^{-n}\mathbb{Z}\big).$$

The local system ε^n, endowed with $\varepsilon^n_{\mathbb{Z}}$, should be regarded as a twisted form of $\mathbb{Z}(-n)$ on $\widetilde{Y}^n$.

Let ε^n_X, respectively $(\varepsilon^n_X)_{\mathbb{Z}}$, denote the direct image of ε^n, respectively of $\varepsilon^n_{\mathbb{Z}}$, under the map from $\widetilde{Y}^n$ to X. One has

$$\varepsilon^n_X \simeq \bigwedge^n \varepsilon^1_X \qquad (n \geq 0). \tag{3.1.4.1}$$

If Y is a union of distinct smooth divisors $(Y_i)_{i\in I}$, the choice of a total order on I trivializes the ε^n.

(3.1.5) Let $W_m(\Omega^k_X\langle Y\rangle)$ denote the submodule of $\Omega^k_X\langle Y\rangle$ formed by linear combinations of products

$$\alpha \wedge \frac{dt_{i(1)}}{t_{i(1)}} \wedge \cdots \wedge \frac{dt_{i(m)}}{t_{i(m)}} \qquad (m \leq n),$$

with α holomorphic and with the $t_{i(j)}$ local equations of distinct local components Y_j of Y. The weight filtration of $\Omega^\bullet_X\langle Y\rangle$ is the increasing filtration by the subcomplexes $W_m(\Omega^\bullet_X\langle Y\rangle)$. One has

$$W_m(\Omega^k_X\langle Y\rangle) \wedge W_n(\Omega^\ell_X\langle Y\rangle) \subset W_{m+n}(\Omega^{k+\ell}_X\langle Y\rangle). \tag{3.1.5.1}$$

Letting i_n denote the map from $\widetilde{Y}^n$ to X, one checks that the correspondence

$$\alpha \wedge \frac{dt_{i(1)}}{t_{i(1)}} \wedge \cdots \wedge \frac{dt_{i(n)}}{t_{i(n)}} \longmapsto (\alpha \mid Y_{i(1)} \cap \cdots \cap Y_{i(n)}) \otimes (\text{orientation } i(1), \ldots, i(n))$$

defines isomorphisms of complexes

$$\operatorname{Res} : \operatorname{Gr}^W_n(\Omega^\bullet_X\langle Y\rangle) \xrightarrow{\sim} i_{n*}\Omega^\bullet_{\widetilde{Y}^n}(\varepsilon^n)[-n], \tag{3.1.5.2}$$

the "Poincaré residue".

(3.1.6) The following interpretation, in terms of (3.1.5.2), of the Leray spectral sequence for the inclusion of X^* in X was pointed out to me by N. Katz. It will make it possible to verify a point which I had initially considered evident ((3.2.5), (ii), first part).

(3.1.7) Every point of X admits a fundamental system of Stein open neighborhoods whose trace on X^* is Stein. For $\mathcal{F}$ a coherent analytic sheaf on X^*, one therefore has $R^i j_*\mathcal{F} = 0$ for $i > 0$. The De Rham complex $\Omega^\bullet_{X^*}$ is therefore a resolution of the constant sheaf $\mathbb{C}$ by sheaves acyclic for the functor j_*. Hence

$$H^*(X^*, \mathbb{C}) \simeq \mathbb{H}^*(X^*, \Omega^\bullet_{X^*}) \simeq \mathbb{H}^*(X, j_*\Omega^\bullet_{X^*}), \tag{3.1.7.1}$$

and the Leray spectral sequence for the morphism j is identified with the hypercohomology spectral sequence for $j_*\Omega^\bullet_{X^*}$, corresponding to the filtration τ by the subcomplexes $\tau_{\leq n}(j_*\Omega^\bullet_{X^*})$ (1.4.6).

Proposition (3.1.8). The morphisms of filtered complexes

$$(\Omega^\bullet_X\langle Y\rangle, W) \xleftarrow{\alpha} (\Omega^\bullet_X\langle Y\rangle, \tau) \xrightarrow{\beta} (j_*\Omega^\bullet_{X^*}, \tau)$$

are filtered quasi-isomorphisms. They define an isomorphism between the Leray spectral sequence for j in complex cohomology and the hypercohomology spectral sequence of the filtered complex $(\Omega^\bullet_X\langle Y\rangle, W)$ on X.

By (1.4.5) and (3.1.7), it suffices to prove the first assertion. In [3], II, (6.9), or in [1], one finds the proof that β is a quasi-isomorphism, hence a filtered quasi-isomorphism. One can also compute directly the cohomology sheaves of the two sides: those of $\Omega^\bullet_X\langle Y\rangle$ are determined by (3.1.5.2), whereas those of $j_*\Omega^\bullet_{X^*}$ are the $R^i j_*\mathbb{C}$, which can be computed topologically.

For $n \geq p$ one has

$$W_n(\Omega^p_X\langle Y\rangle) = \Omega^p_X\langle Y\rangle,$$

so that α is a morphism from $\Omega^\bullet_X\langle Y\rangle$, endowed with τ, to $\Omega^\bullet_X\langle Y\rangle$, endowed with the decreasing filtration associated with W (1.1.3). By (3.1.5.2), one has

$$\mathcal{H}^i(\mathrm{Gr}^W_n(\Omega^\bullet_X\langle Y\rangle)) = \begin{cases} 0, & i \neq n, \\ \varepsilon^n_X, & i = n. \end{cases} \tag{3.1.8.1}$$

From the first line of this formula one deduces that α is a filtered quasi-isomorphism. This proves (3.1.8).

By (3.1.7), the isomorphism (3.1.8.1) defines an isomorphism

$$R^n j_*\mathbb{C} \simeq \mathcal{H}^n(j_*\Omega^\bullet_{X^*}) \simeq \mathcal{H}^n(\Omega^\bullet_X\langle Y\rangle) \simeq \varepsilon^n_X. \tag{3.1.8.2}$$

The isomorphisms (3.1.4.1) correspond, via (3.1.8.2), to the cup product.

Proposition (3.1.9). The canonical morphism from $R^n j_*\mathbb{Z}$ to $R^n j_*\mathbb{C}$ identifies, via (3.1.8.2), the sheaf $R^n j_*\mathbb{Z}$ with $(\varepsilon^n_X)_{\mathbb{Z}}$ (3.1.4).

The question is local on X. One can therefore suppose that X is an open polydisc D^m, with

$$D = \{z \in \mathbb{C} \mid |z| < 1\},$$

and that

$$Y = \bigcup_{k=1}^r Y_k, \qquad Y_k = \mathrm{pr}_k^{-1}(0).$$

The fiber at 0 of $R^n j_*\mathbb{Z}$ is then the integral cohomology of

$$X^* = D^{*r} \times D^{m-r}, \qquad D^* = \{z \in \mathbb{C} \mid 0 < |z| < 1\}.$$

The space X^* has the homotopy type of a torus; its cohomology is therefore torsion-free, and the cup product defines isomorphisms

$$\bigwedge^n (R^1 j_*\mathbb{Z})_0 \xrightarrow{\sim} (R^n j_*\mathbb{Z})_0.$$

It is therefore enough to prove (3.1.9) for $n = 1$.

The integral homology $H_1(X^*)$ is generated by the loops γ_k winding around the various Y_k. One has

$$\int_{\gamma_k} \frac{dz_k}{z_k} = \pm 2\pi i;$$

the integral cohomology is therefore generated by the $\frac{1}{2\pi i}\frac{dz_k}{z_k}$, and this proves (3.1.9).

(3.1.10) Let $\mathcal{F}$ be a coherent analytic sheaf on X^*, given as the restriction to X^* of a coherent analytic sheaf, again denoted $\mathcal{F}$, on X. The meromorphic direct image $j^m_*\mathcal{F}$ of $\mathcal{F}$ is the inductive limit

$$j^m_*\mathcal{F} = \varinjlim_n \mathcal{F}(nY).$$

Locally on X, Y is the sum of a finite family $(Y_i)_{i\in I}$ of smooth divisors, and the filtration by order of pole P on $j^m_*\mathcal{O}_{X^*}$ is defined by the formula

$$P^p(j^m_*\mathcal{O}_{X^*}) = \sum_{\mathbf{n}\in A_p} \mathcal{O}_X\Big(\sum_i (n_i+1)Y_i\Big), \tag{3.1.10.1}$$

where

$$A_p = \{(n_i)_{i\in I} \mid \sum_i n_i \leq -p \ \text{ and } \ \forall i,\ n_i \geq 0\}.$$

This construction globalizes and provides on $j^m_*\mathcal{O}_{X^*}$ an exhaustive filtration such that $P^p = 0$ for $p > 0$.

The filtration by order of pole of the complex $j^m_*\Omega^\bullet_{X^*} = j^m_*\mathcal{O}_{X^*} \otimes \Omega^\bullet_X$ is the filtration

$$P^p(j^m_*\Omega^k_{X^*}) = P^{p-k}(j^m_*\mathcal{O}_{X^*}) \otimes \Omega^k_X. \tag{3.1.10.2}$$

The filtration P induces on the subcomplex $\Omega^\bullet_X\langle Y\rangle$ of $j^m_*\Omega^\bullet_{X^*}$ the bête filtration $\sigma_{\geq p}(\Omega^\bullet_X\langle Y\rangle)$, a filtration also called the Hodge filtration F.

Proposition (3.1.11). The inclusion morphism

$$(\Omega^\bullet_X\langle Y\rangle, F) \longrightarrow (j^m_* \Omega^\bullet_{X^*}, P)$$

is a filtered quasi-isomorphism.

This statement was suggested to me by [4]. A proof appears in [3], II, (3.13).

3.2. Mixed Hodge theory

Recall that from now on, by a scheme we mean a scheme of finite type over $\mathbb{C}$, and by a sheaf on S a sheaf on S^{an}.

(3.2.1) Let X be a smooth separated scheme. By Nagata [11], X is a Zariski open subset of a complete scheme $\overline{X}$. By Hironaka [8], one can take $\overline{X}$ smooth, and such that $Y = \overline{X} - X$ is a normal-crossings divisor.

The reader who would like to avoid the reference to Nagata may suppose X quasi-

projective. The smooth completion $\overline{X}$ can then be chosen projective and such that Y is a union of smooth divisors. When one restricts oneself to such compactifications, Hodge theory is needed only in its standard form (2.2.1).

(3.2.2) By (3.1.7) and (3.1.8), one has

$$H^*(X, \mathbb{C}) \simeq H^*(\overline{X}, \Omega^\bullet_{\overline{X}}\langle Y\rangle).$$

The Hodge filtration F on the complex $\Omega^\bullet_{\overline{X}}\langle Y\rangle$ is defined to be the filtration $F^p = \sigma_{\geq p}$ by bête truncations (1.4.5). Thus $\Omega^\bullet_{\overline{X}}\langle Y\rangle$ carries two filtrations: F and W (3.1.5).

(3.2.3) We shall use the existence of bifiltered resolutions

$$i : \Omega^\bullet_{\overline{X}}\langle Y\rangle \longrightarrow K^\bullet$$

such that the $\mathrm{Gr}^p_F \mathrm{Gr}^W_n(K^j)$ are sheaves acyclic for the functor Γ:

$$H^i(\overline{X}, \mathrm{Gr}^p_F \mathrm{Gr}^W_n(K^j)) = 0 \qquad \text{for } i > 0.$$

Here are two ways to construct them:

a) One may take for $K^\bullet$ the canonical Godement resolution

$$\mathcal{C}^\bullet(\Omega^\bullet_{\overline{X}}\langle Y\rangle),$$

filtered by the $\mathcal{C}^\bullet(W_n(\Omega^\bullet_{\overline{X}}\langle Y\rangle))$ and the $\mathcal{C}^\bullet(F^p(\Omega^\bullet_{\overline{X}}\langle Y\rangle))$. This is a bifiltered resolution because $\mathcal{C}^\bullet$ is an exact functor.

b) One may take for $K^\bullet$ the d''-resolution of $\Omega^\bullet_{\overline{X}}\langle Y\rangle$. Let $\mathcal{E}^{p,q}_X$ be the sheaf of C^∞ forms of type (p,q); then $K^\bullet$ is the simple complex associated with the double complex of the

$$\Omega^p_{\overline{X}}\langle Y\rangle \otimes_{\Omega^0_X} \mathcal{E}^{0,q}_X,$$

a subcomplex of the $j_*\mathcal{E}^{p,q}_{X^*}$. This complex is filtered by

$$F^p(\Omega^\bullet_{\overline{X}}\langle Y\rangle) \otimes \mathcal{E}^{0,*}_X \quad \text{and by} \quad W_n(\Omega^\bullet_{\overline{X}}\langle Y\rangle) \otimes \mathcal{E}^{0,*}_X.$$

To prove that this is a bifiltered resolution, one uses the fact that the sheaf $\mathcal{E}^0_X$ of complex-valued C^∞ functions on X is flat over $\mathcal{O}$ (a corollary of Malgrange's C^∞ preparation theorem). The sheaves $\mathrm{Gr}_F \mathrm{Gr}_W(K^\bullet)$ are fine, because they are sheaves of modules over the soft sheaf $\mathcal{E}^0_X$.

(3.2.4) With the notation of (3.2.3), the complex cohomology of X appears as the cohomology of the bifiltered complex $\Gamma(\overline{X}, K^\bullet)$. Thus there are two spectral sequences abutting to $H^*(X, \mathbb{C})$. With the notation of (3.1.4), they are

$$ {}_W E_1^{pq} = H^{p+q}(\overline{X}, \varepsilon_{\overline{X}}^{-p}[p]) = H^{2p+q}(\widetilde{Y}^{-p}, \varepsilon^{-p}) \Longrightarrow H^*(X, \mathbb{C}), \tag{3.2.4.1} $$

$$ {}_F E_1^{pq} = H^q(\overline{X}, \Omega^p_{\overline{X}}\langle Y\rangle) \Longrightarrow H^{p+q}(X, \mathbb{C}). \tag{3.2.4.2} $$

The first of these, up to the renumbering

$$ {}_W E_1^{pq} \longmapsto E_2^{2p+q,-p}, $$

is just the Leray spectral sequence of the inclusion j.

Theorem (3.2.5).

(i) On the terms ${}_W E_r^{pq}$ of the spectral sequence (3.2.4.1), the first direct filtration, the second direct filtration, and the recurrent filtration defined by F coincide.

(ii) The filtration on $H^n(X, \mathbb{C})$ which is the abutment of the spectral sequence ${}_W E$ is deduced from a filtration W of $H^n(X, \mathbb{Q})$. Neither it, nor the filtration F which is the abutment of the spectral sequence ${}_F E$, depends on the chosen compactification $\overline{X}$ of X or on the choice of $K^\bullet$.

(iii) The filtrations $W[n]$ (1.1.2) and F define on $H^n(X, \mathbb{Z})$ a mixed Hodge structure, functorial in X.

By (3.1.7), the spectral sequence ${}_W E$ is the Leray spectral sequence for j, up to renumbering. It is therefore obtained by tensoring with $\mathbb{C}$ a spectral sequence of $\mathbb{Q}$-vector spaces, and the first assertion of (ii) is true. Complex conjugation acts on the ${}_W E_r$; it can be computed via (3.1.10).

Lemma (3.2.6). The hypercohomology spectral sequences of the filtered complexes

$$ \mathrm{Gr}_n^W(\Omega^\bullet_{\overline{X}}\langle Y\rangle), $$

endowed with the filtration induced by the Hodge filtration, degenerate at E_1.

Define Y^n and $\widetilde{Y}^n$ as in (3.1.4), and let $i_n : \widetilde{Y}^n \to X$. By (3.1.5.2), one has

$$ \mathrm{Gr}_n^W(\Omega^\bullet_{\overline{X}}\langle Y\rangle) \simeq i_{n*}\Omega^\bullet_{\widetilde{Y}^n}(\varepsilon^n)[-n]. $$

Moreover, the Hodge filtration induces the bête filtration (1.4.5), so that the spectral sequence (3.2.6) is obtained, by translations in the degrees, from the classical spectral sequence

$$ E_1^{pq} = H^q(\widetilde{Y}^n, \Omega^p_{\widetilde{Y}^n}(\varepsilon^n)) \Longrightarrow H^{p+q}(\widetilde{Y}^n, \varepsilon^n). $$

If $\overline{X}$ is projective and Y is a union of smooth divisors, then $\widetilde{Y}^n$ is projective, ε^n is a trivial local system, and classical Hodge theory (2.2.1) gives the degeneration (3.2.6). For the general case, one must refer to [2] (see (2.2.2), (2.2.3)).

Hodge theory further gives that the Hodge filtration on $H^k(\widetilde{Y}^n, \varepsilon^n)$ is k-opposed to its complex conjugate, where complex conjugation is defined in terms of $\varepsilon^n_{\mathbb{Z}}$ (3.1.4). Taking account here of the translations in the degrees, one has:

Lemma (3.2.7). The filtration on

$$ {}_W E_1^{-n,k+n} = H^k(\overline{X}, \mathrm{Gr}_n^W(\Omega^\bullet_{\overline{X}}\langle Y\rangle)) \simeq H^{k-n}(\widetilde{Y}^n, \varepsilon^n), $$

which is the abutment of the spectral sequence (3.2.6), is $(k+n)$-opposed to its complex conjugate.

Lemma (3.2.8). The differentials d_1 of the spectral sequence ${}_WE$ are strictly compatible with the filtration F.

On the terms E_1, there is only one filtration induced by F to consider ((1.3.10) and (1.3.13), (iii)), and d_1 is compatible with this filtration ((1.3.13), (i)). This filtration is the abutment of the spectral sequence (3.2.6) ((1.4.8), (iii)). By (3.2.7), the arrow d_1

$$d_1 : H^k(\overline{X}, \mathrm{Gr}_n^W(\Omega^\bullet\langle Y\rangle)) \longrightarrow H^{k+1}(\overline{X}, \mathrm{Gr}_{n-1}^W(\Omega^\bullet\langle Y\rangle))$$

that is,

$$d_1 : H^{k-n}(\widetilde{Y}^n, \varepsilon^n) \longrightarrow H^{k-n+2}(\widetilde{Y}^{n-1}, \varepsilon^{n-1}) \tag{3.2.8.1}$$

is compatible with filtrations that are $(k+n)$-opposed to their complex conjugates. Since d_1 commutes with complex conjugation, it respects the bigrading, and hence the filtration defined by F and $\overline{F}$; this proves (3.2.8). Moreover, the cohomology of the complex E_1 will still be bigraded.

Lemma (3.2.9). On ${}_WE_r^{pq}$, the recurrent filtration F is q-opposed to its complex conjugate.

We prove by induction on r that:

Lemma (3.2.10). For $r > 0$, the differentials d_r of the spectral sequence ${}_WE$ are strictly compatible with the recurrent filtration F. For $r \geq 2$, they are zero.

For $r = 0$, respectively $r = 1$, one applies (3.2.6) and (1.4.8), (iii), respectively (3.2.8). For $r \geq 2$, it suffices to prove that $d_r = 0$. By induction and by (1.3.16), one may suppose that on the terms ${}_WE_s$ ($s \geq r+1$) one has $F_d = F_r = F_\infty$, and that ${}_WE_r = {}_WE_2$. By (1.3.13), (i), d_r is therefore compatible with the filtration F_r.

On ${}_WE_r^{pq} = {}_WE_2^{pq}$, the filtration F_r is q-opposed to its complex conjugate. The morphism

$$d_r : {}_WE_r^{pq} \longrightarrow {}_WE_r^{p+r,q-r+1}$$

therefore satisfies, for $r - 1 > 0$,

$$\begin{aligned} d_r({}_WE_r^{pq}) &= d_r\left(\sum_{a+b=q} \left(F^a({}_WE_r^{pq}) \cap \overline{F}^b({}_WE_r^{pq})\right)\right) \\ &\subset \sum_{a+b=q} \left(F^a({}_WE_r^{p+r,q-r+1}) \cap \overline{F}^b({}_WE_r^{p+r,q-r+1})\right) = 0. \end{aligned}$$

This proves (3.2.10), which by (1.3.16) implies (3.2.5) (i).

By (1.3.17), the filtration of ${}_WE_\infty^{pq}$ induced by the filtration F of $H^{p+q}(X, \mathbb{C})$ is q-opposed to its complex conjugate. Since $q = -p + (p+q)$, this proves the first part of (3.2.5) (iii).

(3.2.11) We prove (ii) and (iii), which will complete the proof.

A. Independence of the choice of $K^\bullet$. The filtrations F and W of $H^*(X, \mathbb{C})$ are the abutments of the hypercohomology spectral sequences of $\Omega^\bullet_{\overline{X}}\langle Y\rangle$ for the filtrations F and W. These entire spectral sequences do not depend on the choice of $K^\bullet$.

B. Functoriality. Let $f : X_1 \to X_2$ be a morphism of schemes. Suppose given a morphism of smooth compactifications

$$\begin{tikzcd} X_1 \arrow[r, "f"] \arrow[d, "j_1"'] & X_2 \arrow[d, "j_2"] \\ \overline{X}_1 \arrow[r, "\overline{f}"'] & \overline{X}_2, \end{tikzcd} \tag{3.2.11.1}$$

where $Y_i = \overline{X}_i - X_i$ are normal-crossings divisors. The canonical morphism (see (3.1.3)) from $f^*\Omega^\bullet_{\overline{X}_2}\langle Y_2\rangle$ to $\Omega^\bullet_{\overline{X}_1}\langle Y_1\rangle$ is then a morphism of bifiltered complexes; on hypercohomology it induces a morphism compatible with F and W, and therefore

$$f^* : H^n(X_2, \mathbb{Z}) \longrightarrow H^n(X_1, \mathbb{Z})$$

is a morphism of mixed Hodge structures, for the structures defined by the compactifications $\overline{X}_i$.

C. Independence of the compactification. With the notation of B, if f is an isomorphism, then f^* is a bijective morphism of mixed Hodge structures, hence an isomorphism (2.3.5).

If $\overline{X}_1$ and $\overline{X}_2$ are two smooth compactifications of X, with $Y_i = \overline{X}_i - X_i$ a normal-crossings divisor, there exists a third smooth compactification $\overline{X}$, with $Y = \overline{X} - X$ a normal-crossings divisor, fitting into a commutative diagram

$$\begin{tikzcd} & X \arrow[dl] \arrow[d] \arrow[dr] & \\ \overline{X}_1 & \overline{X} \arrow[l] \arrow[r] & \overline{X}_2. \end{tikzcd}$$

Namely, one takes for $\overline{X}$ a resolution of the singularities of the closure of the diagonal image of X in $\overline{X}_1 \times \overline{X}_2$. The identity map from $H^n(X, \mathbb{Z})$, endowed with the mixed Hodge structure defined by $\overline{X}_1$, to $H^n(X, \mathbb{Z})$, endowed with that defined by $\overline{X}_2$, is therefore a composition of two isomorphisms.

To finish the proof of (ii) and (iii), one observes that every morphism f fits into a diagram (3.2.11.1): choose compactifications $\overline{X}'_1$ and $\overline{X}_2$ of X_1 and X_2, and then take for $\overline{X}_1$ a resolution of the singularities of the closure of the image of X_1 in $\overline{X}'_1 \times \overline{X}_2$.

Definition (3.2.12). The mixed Hodge structure on the cohomology of a smooth separated algebraic variety is the mixed Hodge structure (3.2.5), (iii).

Corollary (3.2.13). With the preceding notation:

(i) The spectral sequence (3.2.4.1) degenerates at E_2, i.e. the Leray spectral sequence for the inclusion $j : X \hookrightarrow \overline{X}$ degenerates at E_3 ($E_3 = E_\infty$).

(ii) The spectral sequence (3.2.4.2)

$$_F E_1^{pq} = H^q(\overline{X}, \Omega^p_{\overline{X}}\langle Y \rangle) \Longrightarrow H^{p+q}(X, \mathbb{C})$$

degenerates at E_1.

(iii) The spectral sequence defined by the sheaf $\Omega^\bullet_{\overline{X}}\langle Y \rangle$, endowed with the filtration W:

$$E_1^{-n,k+n} = H^k(\overline{X}, \mathrm{Gr}^W_n(\Omega^\bullet_{\overline{X}}\langle Y \rangle)) \simeq H^{k-n}(\widetilde{Y}^n, \Omega^\bullet_{\widetilde{Y}^n}(\varepsilon^n)) \Longrightarrow H^k(\overline{X}, \Omega^\bullet_{\overline{X}}\langle Y \rangle)$$

degenerates at E_2.

Assertion (i) was proved in (3.2.10).

Consider the four spectral sequences of (1.4.12), relative to the bifiltered complex $\Omega^\bullet_{\overline{X}}\langle Y \rangle$. By (i), one has

$$\sum_n \dim H^n(X, \mathbb{C}) = \sum_{p,q} \dim {}_W E_2^{p,q}.$$

The terms $_W E_1^{-n,k+n}$ are, for fixed n, the abutment of a spectral sequence degenerate at E_1 (3.2.6), with initial terms $E_1^{p,n-p}$ (notation of (1.4.12)), the initial terms of spectral sequence (iii). Since $_W d_1$ is strictly compatible with the filtration F that is the abutment of this spectral sequence (3.2.7), and is (1.4.12) the abutment of a morphism between these spectral sequences, beginning with the differentials of (iii), one has

$$\mathrm{Gr}^p_F({}_W E_2^{n,k-n}) \simeq H^*(E_1^{p,n-1,k-n-p} \to E_1^{p,n,k-n-p} \to E_1^{p,n+1,k-n-p})$$

and $\mathrm{Gr}^*_F({}_W E_2^{**})$ is the sum of the terms E_∞^{**} of the spectral sequences (iii). Therefore

$$\sum_n \dim H^n(X, \mathbb{C}) = \sum \dim E_\infty^{**}. \tag{3.2.13.1}$$

On the other hand,

$$\sum_n \dim H^n(X, \mathbb{C}) \leq \sum \dim {}_F E_1^{**},$$

with equality if and only if the spectral sequence (ii) degenerates at E_1, and

$$\sum \dim {}_F E_1^{**} \leq \sum \dim E_\infty^{**},$$

with equality if and only if the spectral sequences (iii) degenerate at E_2. Comparing with (3.2.13.1), one obtains (3.2.13).

Corollary (3.2.14). Let ω be a meromorphic differential p-form on $\overline{X}$, holomorphic on X and having at worst logarithmic poles along Y. Then the restriction $\omega|X$ of ω to X is closed, and if the cohomology class in $H^p(X, \mathbb{C})$ defined by ω is zero, then $\omega = 0$.

This is the special case ${}_F E_1^{p0} = {}_F E_\infty^{p0}$ of (3.2.13), (ii).

Corollary (3.2.15).

(i) If X is a complete smooth algebraic variety, the mixed Hodge structure on $H^n(X, \mathbb{Z})$ is the classical Hodge structure of weight n.

(ii) The Hodge numbers h^{pq} of the mixed Hodge structure of $H^n(X, \mathbb{Z})$, for X smooth algebraic, can be nonzero only for $p \leq n$, $q \leq n$, and $p + q \geq n$.

Assertion (i) is clear. To prove (ii), one observes that, for Y a union of smooth divisors, the rational Hodge structure $\mathrm{Gr}^W_{n+k}(H^n(X, \mathbb{Q}))$ is a quotient of a subobject of a Hodge structure of weight $n + k$, namely

$$H^{n-k}(\widetilde{Y}^k, \mathbb{Q}) \otimes \mathbb{Q}(-k).$$

(3.2.16) Let X be a smooth separated scheme. It is known that X admits smooth compactifications $\overline{X}$, and that the subgroup of $H^n(X, \mathbb{Z})$ which is the image of $H^n(\overline{X}, \mathbb{Z})$ is independent of the choice of $\overline{X}$ (cf. [6], (9.1) to (9.4)).

Corollary (3.2.17). Under the hypotheses (3.2.16), the image of $H^n(\overline{X}, \mathbb{Q})$ in $H^n(X, \mathbb{Q})$ is $W_n(H^n(X, \mathbb{Q}))$, where W denotes the weight filtration (3.2.12).

One may suppose that $\overline{X} - X$ is a normal-crossings divisor. The assertion then follows from the fact that $W[-n]$ is the abutment of the Leray spectral sequence for the inclusion $j : X \hookrightarrow \overline{X}$.

Corollary (3.2.18). Let f be a morphism from a proper smooth scheme Y to a smooth scheme X admitting a smooth compactification $\overline{X}$:

$$Y \xrightarrow{f} X \xrightarrow{j} \overline{X}.$$

Then the groups $H^n(X, \mathbb{Q})$ and $H^n(\overline{X}, \mathbb{Q})$ have the same image in $H^n(Y, \mathbb{Q})$.

Since f^* and $(jf)^*$ are strictly compatible with the weight filtration ((3.2.5), (iii)), it is enough to prove that $\mathrm{Gr}^W_m(f^*)$ and $\mathrm{Gr}^W_m((jf)^*)$ have the same image in $\mathrm{Gr}^W_m H^n(Y, \mathbb{Q})$. By (3.2.17) and (3.2.15), $\mathrm{Gr}^W_n(j^*)$ is an isomorphism, while $\mathrm{Gr}^W_m(f^*) = 0$ for $m \neq n$ because $\mathrm{Gr}^W_m H^n(Y, \mathbb{Q}) = 0$ for $m \neq n$.

Remark (3.2.19). By (3.1.11) and (1.4.5), under the hypotheses of (3.2.5), the Hodge filtration on $H^n(X, \mathbb{C})$ is the abutment of the hypercohomology spectral sequence of the complex $j_*^m \Omega^\bullet_{X^*}$, endowed with the filtration by order of pole (3.1.10): this spectral sequence coincides with (3.2.4.2).

4. Applications and complements

4.1. The fixed-part theorem

Theorem (4.1.1). Let S be a smooth separated scheme, and let $f : X \to S$ be a proper smooth morphism.

(i) In rational cohomology, the Leray spectral sequence

$$E_2^{pq} = H^p(S, R^q f_* \mathbb{Q}) \Longrightarrow H^{p+q}(X, \mathbb{Q})$$

degenerates ($E_2 = E_\infty$).

(ii) If $\overline{X}$ is a nonsingular compactification of X, the canonical morphism

$$H^n(\overline{X}, \mathbb{Q}) \longrightarrow H^0(S, R^n f_* \mathbb{Q})$$

is surjective.

For f projective and smooth, assertion (i) is proved in [2]. The proof of [2] is rewritten more readably in [5]; it does not use the smoothness of S.

Fix n, and prove that (i) implies (ii). We reduce to the case where S is connected and nonempty. If $s \in S$ is a point of S, the local system $R^n f_* \mathbb{Q}$ is entirely described by its fiber $(R^n f_* \mathbb{Q})_s$ at s, and by the action of the fundamental group $\pi = \pi_1(S, s)$ on this fiber. One has

$$H^0(S, R^n f_* \mathbb{Q}) \simeq [(R^n f_* \mathbb{Q})_s]^\pi.$$

If $X_s = f^{-1}(s)$, the composite arrow

$$H^0(S, R^n f_* \mathbb{Q}) \longrightarrow (R^n f_* \mathbb{Q})_s \simeq H^n(X_s, \mathbb{Q})$$

is therefore injective. Consider the arrows

$$H^n(\overline{X}, \mathbb{Q}) \xrightarrow{a} H^n(X, \mathbb{Q}) \xrightarrow{b} H^0(S, R^n f_* \mathbb{Q}) \xrightarrow{c} H^n(X_s, \mathbb{Q}).$$

The arrows cb and cba have the same image by (3.2.18). The arrow b is an edge homomorphism of the Leray spectral sequence, hence is surjective by hypothesis. Since c is injective, b and ba have the same image and ba is surjective. This proves (ii) for f projective. We now deduce the general case. Again reduce to the case where S is connected and nonempty.

By Chow's lemma and resolution of singularities, there exists a quasi-projective smooth scheme X' and a projective birational morphism

$$p : X' \longrightarrow X.$$

There then exists, by Bertini or Sard, a nonempty Zariski open subset S_1 of S such that

$$X'_1 = (fp)^{-1}(S_1)$$

is smooth over S_1. Finally let $X_1 = f^{-1}(S_1)$, let $\overline{X}$ be a smooth compactification of X, and let $\overline{X}'_1$ be a smooth compactification of X'_1 which dominates $\overline{X}$:

$$\begin{tikzcd}
\overline{X} & & \overline{X}'_1 \arrow[ll, "\overline{p}"'] \\
X \arrow[u] & X_1 \arrow[l] & X'_1 \arrow[l, "p"'] \arrow[u] \\
S \arrow[u, "f"] & S_1 \arrow[l, "i"'] \arrow[u, "f_1"] \arrow[r, equal] & S_1 \arrow[u, "f'_1"']
\end{tikzcd}$$

If $s \in S_1$, the morphism $i_* : \pi_1(S_1, s) \to \pi_1(S, s)$ is surjective because the topological codimension of $S - S_1$ is ≥ 2. Therefore

$$H^0(S, R^n f_* \mathbb{Q}) \simeq H^0(S_1, R^n f_{1*} \mathbb{Q}),$$

and it suffices to prove that

$$H^n(\overline{X}, \mathbb{Q}) \longrightarrow H^0(S_1, R^n f_{1*} \mathbb{Q})$$

is surjective.

The vertical arrows of the diagram

$$\begin{tikzcd}
H^n(\overline{X}, \mathbb{Q}) \arrow[r] \arrow[d, "p^*"'] & H^n(X_1, \mathbb{Q}) \arrow[r] \arrow[d, "p^*"'] & H^0(S_1, R^n f_{1*} \mathbb{Q}) \arrow[d, "p^*"] \\
H^n(\overline{X}'_1, \mathbb{Q}) \arrow[r] & H^n(X'_1, \mathbb{Q}) \arrow[r] & H^0(S_1, R^n f'_{1*} \mathbb{Q})
\end{tikzcd}$$

admit as left inverses the Gysin morphisms $\overline{p}_*$, p_*, and p_*, defined by Poincaré duality as transposes of the arrows analogous to the preceding ones in cohomology with proper support. Moreover, the diagram

$$\begin{tikzcd}
H^n(\overline{X}, \mathbb{Q}) \arrow[r, "u"] \arrow[d, "p^*"'] & H^0(S_1, R^n f_{1*} \mathbb{Q}) \arrow[d, "p^*"] \\
H^n(\overline{X}'_1, \mathbb{Q}) \arrow[r, "v"'] & H^0(S_1, R^n f'_{1*} \mathbb{Q})
\end{tikzcd}$$

is commutative. Thus u is a direct factor of v. The latter is surjective, since f'_1 is projective, and hence u is also surjective.

We now prove (i) in the general case. We reduce to the case where f is purely of relative dimension n. Let $f \times f$ be the projection of $X \times_S X$ onto S.

Let δ be the image of the cohomology class of the diagonal of $X \times_S X$ in

$$H^0(S, R^n(f \times f)_*\mathbb{Q}).$$

By Künneth,

$$R^n(f \times f)_*\mathbb{Q} \simeq \sum_{p+q=n} (R^p f_*\mathbb{Q} \otimes R^q f_*\mathbb{Q}).$$

Let δ_{pq}, for $p+q=n$, denote the components of δ in this decomposition, and let $\widetilde{\delta}_{pq}$ be classes in $H^n(X \times_S X)$ whose images are the δ_{pq}.

The $\widetilde{\delta}_{pq}$ define in the derived category $D^+(S)$ homomorphisms

$$\delta_p : Rf_*\mathbb{Q} \longrightarrow Rf_*\mathbb{Q}$$

such that $\mathcal{H}^q(\delta_p)$ is zero for $p \neq q$, and is the identity for $p = q$. By [2], one therefore has in $D^+(S)$

$$Rf_*\mathbb{Q} \simeq \sum_p R^p f_*\mathbb{Q}[-p], \tag{4.1.1.1}$$

and the Leray spectral sequence degenerates.

Corollary (4.1.2). Let $f : X \to S$ be a proper smooth morphism whose target is a reduced connected separated scheme S. Let $(R^n f_*\mathbb{Q})^0$ be the largest constant local subsystem of $R^n f_*\mathbb{Q}$, with fiber $H^0(S, R^n f_*\mathbb{Q})$. Then, for each $s \in S$, $(R^n f_*\mathbb{Q})^0_s$ is a Hodge substructure of

$$(R^n f_*\mathbb{Q})_s \simeq H^n(X_s, \mathbb{Q}),$$

and the Hodge structure induced on $H^0(S, R^n f_*\mathbb{Q})$ is independent of s.

Let $s \in S$ and $X_s = f^{-1}(s)$. If S is smooth, and if $\overline{X}$ is a smooth compactification of X, then by (4.1.1) the subspace $(R^n f_*\mathbb{Q})^0_s$ of $(R^n f_*\mathbb{Q})_s$ is the image of $H^n(\overline{X}, \mathbb{Q})$. Since the restriction map

$$H^n(\overline{X}, \mathbb{Q}) \longrightarrow H^n(X_s, \mathbb{Q})$$

is a morphism of Hodge structures, its image is a Hodge substructure, and the Hodge structure induced on $H^0(S, R^n f_*\mathbb{Q})$, a quotient of that of $H^n(\overline{X}, \mathbb{Q})$, is independent of s.

In the general case, (4.1.2) also means that if a is a global section of $R^n f_*\mathbb{C}$, then its components a^{pq} of type (p,q), *a priori* only continuous sections of the complex vector bundle defined by $R^n f_*\mathbb{C}$, are in fact locally constant, i.e. are sections of $R^n f_*\mathbb{C}$. This is so because a^{pq} is continuous and locally constant on the dense open smoothness locus of S, by what precedes.

(4.1.3) In the case where S is compact, a generalization of (4.1.2) is proved analytically in Griffiths [5].

As corollaries of (4.1.2), let us cite:

(4.1.3.1) (Griffiths [5]) Let $f : X \to S$ be a proper smooth morphism as in (4.1.2). If a global section a of $R^n f_*\mathbb{C}$ is of Hodge type (p,q) at one point, then a is of type (p,q) everywhere. In particular, if $n = 2$ and if, at one point s, a is the cohomology class of a divisor of X_s, then a is at every point the cohomology class of a divisor and, for S smooth, is even defined by a divisor D on X.

(4.1.3.2) (Grothendieck [7]) Let S be a reduced connected scheme of finite type over $\mathbb{C}$, and let $f_1 : X_1 \to S$ and $f_2 : X_2 \to S$ be two abelian schemes over S. If a morphism

$$u : R^1 f_{2*}\mathbb{Z} \longrightarrow R^1 f_{1*}\mathbb{Z}$$

comes at one point s of S from a morphism of abelian varieties

$$u_s : (X_1)_s \longrightarrow (X_2)_s,$$

then u comes from one, and only one, morphism of abelian schemes

$$v : X_1 \longrightarrow X_2.$$

(4.1.3.3) (Cf. Katz [10]) Let S be a smooth connected scheme, let s be a point of S, let $f : X \to S$ be a proper smooth morphism, and let P be a direct factor of the local system $R^i f_* \mathbb{Q}$ which is pointwise a Hodge substructure. The following conditions are equivalent:

a) The Hodge structure of P is locally constant.

b) The representation P_s of $\pi_1(S, s)$ factors through a finite quotient of $\pi_1(S, s)$.

c) There exists a nonempty finite étale covering $u : S' \to S$ such that $u^* P$ is a constant family of Hodge structures.

4.2. The semisimplicity theorem

(4.2.1) Let S be a topological space. A continuous family of Hodge structures on S consists of:

a) a local system $H_{\mathbb{Z}}$ of finite-type $\mathbb{Z}$-modules on S;

b) for every point $s \in S$, a Hodge structure on the fiber $(H_{\mathbb{Z}})_s$, this structure varying continuously with s.

A continuous family H of Hodge structures on S is said to have weight n if the fibers H_s ($s \in S$) have weight n.

Similarly, a continuous family of $\mathbb{Q}$-Hodge structures is defined as a local system of $\mathbb{Q}$-vector spaces endowed at each point with a continuously varying $\mathbb{Q}$-Hodge structure.

A polarization of a continuous family H of $\mathbb{Q}$-Hodge structures of weight n is a morphism of local systems from $H_{\mathbb{Q}} \otimes H_{\mathbb{Q}}$ into the constant local system $\mathbb{Q}(-n)_{\mathbb{Q}}$, which at every point $s \in S$ defines a polarization of H_s.

(4.2.2) Suppose S is connected, and let $\mathcal{C}$ be a strictly full subcategory of the category of continuous families of $\mathbb{Q}$-Hodge structures on S. We shall consider the following conditions:

(4.2.2.1) $\mathcal{C}$ is stable under direct factors, direct sums, and tensor products; the constant Tate families $\mathbb{Q}(n)$ ($n \in \mathbb{Z}$) belong to $\mathcal{C}$.

(4.2.2.2) Every homogeneous Hodge structure, i.e. one of some weight n, in $\mathcal{C}$ is polarizable.

(4.2.2.3) For every $H \in \operatorname{Ob} \mathcal{C}$, there exists a local system $H'_{\mathbb{Z}}$ of free $\mathbb{Z}$-modules on S such that $H'_{\mathbb{Z}} \otimes \mathbb{Q} \simeq H_{\mathbb{Q}}$.

(4.2.2.4) For every H in $\mathcal{C}$, the largest constant local subsystem H' of H is a constant family of Hodge substructures of H.

Lemma (4.2.3). If $\mathcal{C}$ satisfies the conditions (4.2.2.1) and (4.2.2.2), then:

(i) $\mathcal{C}$ is a semisimple abelian subcategory of the abelian category of continuous families of $\mathbb{Q}$-Hodge structures on S.

(ii) If $H \in \operatorname{Ob} \mathcal{C}$, then its dual H^* and the $\bigwedge^p H$ belong to $\mathcal{C}$; if $H_1, H_2 \in \operatorname{Ob} \mathcal{C}$, then $\operatorname{Hom}(H_1, H_2) \in \operatorname{Ob} \mathcal{C}$.

If $H \in \operatorname{Ob} \mathcal{C}$ and H_1 is a subobject of H, in the category of continuous families of $\mathbb{Q}$-Hodge structures, we prove that H_1 is a direct factor of H in that category. We may suppose H homogeneous. If ψ is a polarization form for H, the orthogonal complement of H_1 for ψ is indeed a subobject of H complementary to H_1. This proves (i).

If $H \in \operatorname{Ob} \mathcal{C}$ has weight n, a polarization of H defines an isomorphism between H^* and $H \otimes \mathbb{Q}(n)$. By (4.2.2.1), one therefore has $H^* \in \operatorname{Ob} \mathcal{C}$. For arbitrary H in $\mathcal{C}$, decomposing H into its homogeneous components, one has

$$H^* = \bigoplus_n (H^n)^*,$$

whence again $H^* \in \operatorname{Ob} \mathcal{C}$. Finally, $\bigwedge^p H$ is a direct factor in $\otimes^p H$, and $\operatorname{Hom}(H_1, H_2) \simeq H_1^* \otimes H_2$.

Let $f : X \to S$ be a proper smooth morphism of reduced schemes. The sheaf $R^n f_*\mathbb{Q}$ is then a local system, and for $s \in S$,

$$(R^n f_*\mathbb{Q})_s \simeq H^n(X_s, \mathbb{Q})$$

is endowed with a $\mathbb{Q}$-Hodge structure. This varies continuously with s.

Definition (4.2.4). Let S be a smooth connected scheme. A continuous family H of $\mathbb{Q}$-Hodge structures on S^{an} will be called algebraic if there exists a nonempty Zariski open subset U of S, an integer k, and a projective smooth morphism $f : X \to U$ such that $H|U$ is a direct factor of $R^k f_*\mathbb{Q}(k)$.

Proposition (4.2.5).

(i) The category of algebraic continuous families of $\mathbb{Q}$-Hodge structures on S satisfies the conditions of (4.2.2).

(ii) If a continuous family H of Hodge structures on S is such that its restriction to a dense Zariski open subset U of S is algebraic, then H is algebraic.

(iii) If $f : X \to S$ is proper and smooth, then $R^i f_*\mathbb{Q}$ is algebraic.

Assertion (ii) is evident. We prove (i). Let $\mathcal{C}_0$ be the set of continuous families of Hodge structures on S which are of the form $R^i f_*\mathbb{Q}$, for f projective and smooth. Then:

a) By the Künneth formula,

$$R(f \times g)_*\mathbb{Q} \simeq Rf_*\mathbb{Q} \otimes Rg_*\mathbb{Q},$$

$\mathcal{C}_0$ is stable under tensor product.

b) $\mathcal{C}_0$ is stable under direct sum:

$$R(f \amalg g)_*\mathbb{Q} \simeq Rf_*\mathbb{Q} \oplus Rg_*\mathbb{Q}.$$

c) The homogeneous components of every $H \in \mathcal{C}_0$ are polarizable (cf. (2.2.6)).

d) The objects $H \in \mathcal{C}_0$ satisfy (4.2.2.3), since

$$Rf_*\mathbb{Q} \simeq Rf_*\mathbb{Z} \otimes \mathbb{Q}.$$

e) The objects $H \in \mathcal{C}_0$ satisfy (4.2.2.4), by (4.1.2).

Let $\mathcal{C}_1$ be the set of direct factors of objects of $\mathcal{C}_0$. Then $\mathcal{C}_1$ is stable under tensor product, direct sum, and direct factor, and satisfies (4.2.2.2) to (4.2.2.4). Moreover $\mathbb{Q}(-1)$ belongs to $\mathcal{C}_1$, since it is of the form $R^2 f_*\mathbb{Q}$ for the projective bundle $f : \mathbb{P}^1_S \to S$. The category $\mathcal{C}_2$ of the $H \otimes \mathbb{Q}(k)$ ($k \in \mathbb{N}$) therefore satisfies (4.2.2.1) to (4.2.2.4).

To deduce (i), it suffices to note that if H is a continuous family of Hodge structures, and U is a dense Zariski open subset of S, then a subobject, a polarization, or an integral lattice of $H|U$ extends uniquely to H.

We prove (iii). If $f : X \to S$ is projective and smooth, then there exist a dense Zariski open subset U of S and a commutative diagram

$$\begin{tikzcd} X' \arrow[r, "p"] \arrow[dr, "f'"'] & X|U \arrow[d, "f|U"] \\ & U \end{tikzcd}$$

with f' projective and smooth and p birational and surjective (apply Chow's lemma and resolution of singularities to the generic fiber of f). The restriction map

$$p^* : (Rf_*\mathbb{Q})|U \longrightarrow Rf'_*\mathbb{Q}$$

is then a direct injection of continuous families of Hodge structures, with left inverse the Gysin morphism p_*; hence $Rf_*\mathbb{Q}$ is algebraic.

Theorem (4.2.6). [2] Let S be a connected, locally connected and locally simply connected topological space, endowed with a base point s. Let $\mathcal{C}$ be a category of continuous families of Hodge structures on S satisfying the conditions (4.2.2). Then, if $H \in \mathrm{Ob}\,\mathcal{C}$, the representation of $\pi_1(S,s)$ on the fiber $(H_{\mathbb{Q}})_s$ is semisimple.

Let $H \in \mathrm{Ob}\,\mathcal{C}$. The local system $H_{\mathbb{Q}}$ defines a complex local system $H_{\mathbb{C}} = H_{\mathbb{Q}} \otimes \mathbb{C}$. For every complex local system V on S, denote by V^c the complex vector bundle which it defines, identified with the sheaf of its continuous sections.

By definition, the group S (2.1.2) acts on $H^c_{\mathbb{C}}$. A subbundle of $H^c_{\mathbb{C}}$ will be called horizontal, or locally constant, if it is defined by a local subsystem of $H_{\mathbb{C}}$.

Lemma (4.2.7). Let V be a rank-one local subsystem of $H_{\mathbb{C}}$. Suppose that a tensor power $V^{\otimes n}$ ($n \geq 1$) of V is a trivial local system, i.e. that $\pi_1(S,s)$ acts on V_s through a finite, necessarily cyclic, group. Then, for $t \in S$, tV^c is again locally constant.

For tV^c to be locally constant, it suffices that

$$(tV^c)^{\otimes n} = tV^{c\otimes n} \subset H^c_{\mathbb{C}}$$

be locally constant. But $V^{\otimes n}$ is generated by a global horizontal section v, and by hypothesis tv is again horizontal (4.2.2.4).

We proceed by induction on $\dim(H_{\mathbb{Q}})_s$. We may suppose H homogeneous and nonzero. Let d be the minimal dimension of the nonzero complex local subsystems of $H_{\mathbb{C}}$. The sum W of all local subsystems of $H_{\mathbb{C}}$ of dimension d, automatically simple, is "defined over $\mathbb{Q}$", i.e. is of the form $W_{\mathbb{Q}} \otimes \mathbb{C}$ for $W_{\mathbb{Q}}$ a local subsystem of $H_{\mathbb{Q}}$. By construction, W_s is a semisimple complex $\pi_1(S,s)$-module, hence $(W_{\mathbb{Q}})_s$ is a semisimple $\pi_1(S,s)$-module over $\mathbb{Q}$.

Let $H_{\mathbb{Z}}$ be a local system of free $\mathbb{Z}$-modules such that $H_{\mathbb{Z}} \otimes \mathbb{Q} \simeq H_{\mathbb{Q}}$, and let $W_{\mathbb{Z}} = H_{\mathbb{Z}} \cap W_{\mathbb{Q}}$. If W has dimension e, the local system $(\bigwedge^e W)^{\otimes 2}$ is trivial. Indeed,

$$\bigwedge^{e} W \simeq \bigwedge^{e} W_{\mathbb{Z}} \otimes \mathbb{C},$$

and $\pi_1(S,s)$ can act on $(\bigwedge^e W_{\mathbb{Z}})_s$ only by ± 1.

Let V be a complex local subsystem of dimension d of $H_{\mathbb{C}}$, and let V' be a complement of V in W, by semisimplicity of W. One has

$$\bigwedge^{e} W \simeq \bigwedge^{d} V \otimes \bigwedge^{e-d} V'.$$

Apply (4.2.7) to the local subsystem $\bigwedge^d V \otimes \bigwedge^{e-d} V'$ of $\bigwedge^d H_{\mathbb{C}} \otimes \bigwedge^{e-d} H_{\mathbb{C}}$. We find that, for $t \in S$,

$$t(\bigwedge^{d} V^c \otimes \bigwedge^{e-d} V'^c) = \bigwedge^{d} tV^c \otimes \bigwedge^{e-d} tV'^c \subset \bigwedge^{d} H^c_{\mathbb{C}} \otimes \bigwedge^{e-d} H^c_{\mathbb{C}}$$

is locally constant. Hence $\bigwedge^d tV^c \subset \bigwedge^d H^c_{\mathbb{C}}$ and $tV^c \subset H^c_{\mathbb{C}}$ are locally constant.

By definition of W, one has $tV^c \subset W^c$: W^c is stable under S, and $W_{\mathbb{Q}}$ is a Hodge substructure of H. If ψ is a polarization of H, H is therefore the direct sum of W and of its orthogonal complement, both in $\mathcal{C}$, and the induction concludes.

Corollary (4.2.8). Under the hypotheses of (4.2.6):

(i) The algebra A of endomorphisms of the local system $H_{\mathbb{Q}}$ is semisimple. It admits a unique $\mathbb{Q}$-Hodge structure such that, at each point s, $A \otimes H_s \to H_s$ is a morphism of Hodge structures.

(ii) The center of A is of type $(0,0)$ and the composition law $A \otimes A \to A$ is a morphism of Hodge structures.

(iii) Let W be a complex local subsystem of dimension d of $H_{\mathbb{C}}$:

[2] Added in proof. Let S be a smooth connected scheme. A family of Hodge structures on S is a continuous family H of Hodge structures on S satisfying the following conditions: a) the Hodge filtration of $(H_{\mathbb{C}})_s$ varies holomorphically with s, i.e. corresponds to a filtration F of $H_{\mathcal{O}} = H_{\mathbb{Z}} \otimes \mathcal{O}$; b) the covariant derivative ∇ satisfies $\nabla F^p \subset \Omega^1_S \otimes F^{p-1}$. Let $\mathcal{C}$ be the category of those continuous families of $\mathbb{Q}$-Hodge structures on S which underlie a family of Hodge structures and whose homogeneous direct factors are polarizable. It is clear that $\mathcal{C}$ satisfies (4.2.2.1) to (4.2.2.3). From results of W. Schmid one can deduce that $\mathcal{C}$ satisfies (4.2.2.4). This theorem, of which (4.1.2) is a corollary, makes it possible to apply (4.2.6) and its corollaries to objects of $\mathcal{C}$.

a) For $t \in S$, $tW^c \subset H^c_{\mathbb{C}}$ is locally constant and defines a local subsystem tW isomorphic to W.

b) A power $(\bigwedge^d W)^{\otimes n}$ $(n \geq 1)$ of the local system $\bigwedge^d W$ is a trivial local system.

The algebra A is the commutant of $\pi_1(S,s)$ in $(H_{\mathbb{Q}})_s$. Its semisimplicity therefore follows from (4.2.6) and Bourbaki, Algebra, chap. 8, §5, no. 2, prop. 4. Moreover, A is the fiber of the largest constant local subsystem of $\mathrm{Hom}(H,H)$. Since $\mathrm{Hom}(H,H) \in \mathcal{C}$ ((4.2.3), (ii)), the existence of the Hodge structure in (i) follows from hypothesis (4.2.2.4). It is clear that the composition law is a morphism of Hodge structures. It follows that the center Z of A is a Hodge substructure: $Z_{\mathbb{C}}$ is bigraded, as the intersection of the kernels of the bihomogeneous morphisms $[x,\ ]$ for x bihomogeneous. Finally Z, and hence $Z_{\mathbb{C}}$, is semisimple, and for $(p,q) \neq (0,0)$, Z^{pq} is zero because it is nilpotent.

A complex local subsystem W of $H_{\mathbb{C}}$ is defined by a projector e of $A_{\mathbb{C}}$. For $t \in S$, tW^c is defined by the projector $t.e$, and is therefore locally constant. Since the group S is connected, to verify that tW is isomorphic to W, it suffices to verify that for t in a sufficiently small neighborhood of 1, the $\mathbb{C}[\pi_1(S,s)]$-module tW_s is isomorphic to W_s. Let H_i be the isotypic components of the $\mathbb{C}[\pi_1(S,s)]$-module $(H_{\mathbb{C}})_s$. One has

$$W_s = \bigoplus_i (W_s \cap H_i) \quad \text{and} \quad tW_s = \bigoplus_i (tW_s \cap H_i).$$

Moreover, for t sufficiently close to 1, tW_s is close to W_s in the Grassmannian, and hence

$$\dim(tW_s \cap H_i) \leq \dim(W_s \cap H_i). \tag{4.2.8.1}$$

In fact, since $\dim(tW_s) = \dim(W_s)$, equality holds in (4.2.8.1). The various isotypic components of W_s and tW_s therefore have the same respective lengths, and W_s is isomorphic to tW_s.

We prove (iii), b). It suffices to treat the case where H is homogeneous and W_s is a simple $\mathbb{C}[\pi_1(S,s)]$-module. Let H_1 be the isotypic component of $(H_{\mathbb{C}})_s$ which contains W_s. By a), H_1 is stable under S. If W_s has dimension d and H_1 has length k, one has

$$\bigwedge^{kd} H_1 \simeq (\bigwedge^{d} W_s)^{\otimes k}$$

as $\mathbb{C}[\pi_1(S,s)]$-modules. If χ is the character of $\pi_1(S,s)$ defined by $\bigwedge^d W$, the character χ_1 defined by $\bigwedge^{kd} H_1$ is therefore χ^k. Put $H_2 = H_1 + \overline{H_1}$. Since H_1 is stable under S, H_2 is defined by a real Hodge substructure of $(H_{\mathbb{R}})_s$. Every polarization form on H therefore induces on H_2 a nondegenerate bilinear form invariant under $\pi_1(S,s)$.

If $\dim(tH_2) = e$, the representation $(\bigwedge^e H_2)^{\otimes 2}$ of $\pi_1(S,s)$ is therefore trivial. Thus:

a) for H_1 real, χ^{2k} is trivial;

b) for H_1 nonreal, $(H_1 \neq \overline{H_1})$, $\chi^k \bar{\chi}^k$ is trivial.

In both cases one has $|\chi| = 1$.

The representation of $\pi_1(S,s)$ on $(H_{\mathbb{C}})_s$ is obtained by extension of scalars from a representation defined over $\mathbb{Q}$, namely $(H_{\mathbb{Q}})_s$. The conjugates of the representation W therefore occur in $(H_{\mathbb{C}})_s$; there are at most $N = \dim(H)$ of them, and for every automorphism σ of $\mathbb{C}$ one has

$$|\sigma(\chi)| = 1.$$

For $\gamma \in \pi_1(S,s)$, $\chi(\gamma)$ is an algebraic integer with at most N complex conjugates, all of absolute value 1; $\chi(\gamma)$ is therefore a k-th root of unity, with $k \leq N$. Hence $\chi^{N!} = 1$, which proves b).

Corollary (4.2.9). Let S be a smooth connected separated scheme, endowed with a base point $s \in S$. Let $f : X \to S$ be a morphism of schemes such that $R^i f_* \mathbb{Q}$ is a local system on S, let G be the Zariski closure of the image of $\pi_1(S,s)$ in $\mathrm{Aut}_{\mathbb{C}}((R^i f_* \mathbb{C})_s)$, and let G^0 be the identity component of G. Then:

a) If f is proper and smooth, G^0 is semisimple.

b) In general, G^0 has no quotient of multiplicative type, i.e. the radical of G^0 is unipotent.

We prove a). After replacing S by a finite étale covering, we reduce to the case $G = G^0$. By ((4.2.5), (iii)), the local system $R^i f_* \mathbb{Q}$ underlies an algebraic family H of Hodge structures, and hence is subject to (4.2.8) ((4.2.5), (i)). The group G is reductive, since it has a faithful semisimple representation, namely $(R^i f_* \mathbb{C})_s$. If G, assumed connected, were not semisimple, it would admit a nontrivial complex representation ρ of rank one. Since H_s is faithful, ρ would be a direct factor in $((H + H^*)^{\otimes m})_s$ for some m. This contradicts (4.2.8), (iii), b), because $(H + H^*)^{\otimes m}$ is algebraic and no tensor power of ρ is trivial.

Let π be a group, and consider the following property of a representation ρ of π on a finite-dimensional complex vector space V:

$(*)$ the identity component G^0 of the Zariski closure G of $\rho(\pi)$ in $\mathrm{Aut}(V)$ has no quotient of multiplicative type.

Lemma (4.2.10). Let $0 \to V' \to V \to V'' \to 0$ be an exact sequence of representations of the preceding type. Then V satisfies $(*)$ if and only if V' and V'' satisfy $(*)$.

Let G, G', and G'' be the groups defined by V, V', and V''. Like π, G leaves V' stable, whence a morphism

$$u = (u', u'') : G \longrightarrow G' \times G'',$$

with unipotent kernel, and such that u' and u'' are surjective. Let N be the identity component of the kernel of u''. Since u' is surjective, $u'(N)$ is distinguished in G', and therefore has no quotient of multiplicative type. If χ is a character of G^0, then χ factors through $u(G^0)$, is trivial on N, a power of χ factors through G''^0, and χ is trivial.

We prove (4.2.9), b), by dévissage and by induction on the relative dimension of f. We may suppose X reduced.

First suppose that X is quasi-projective. Then there exist a nonempty open subset U of S, a dense open subset X' of $X_U = f^{-1}(U)$ smooth over U, and a compactification $\overline{X}'$ of X' over U, proper and smooth over U. One can choose $\overline{X}'$ so that an open X'' of $\overline{X}'$ contains X' and is proper over X_U:

$$\begin{tikzcd}
X_U \arrow[ddrr, "f_U"'] & & X'' \arrow[ll, "p"'] \arrow[rr, "k"] \arrow[dd, "f''"] & & \overline{X}' \arrow[ddll, "\bar{f}'"] \\
 & X' \arrow[ul, "i"] \arrow[ur, "j"'] \arrow[urrr] \arrow[dr, "f'"] & & & \\
 & & U & &
\end{tikzcd}$$

We have long exact sequences of sheaves

$$\begin{gathered}
\to R^i f_{U*}(i_! \mathbb{Q}) \to R^i f_{U*}\mathbb{Q} \to R^i f_{U*}(\mathbb{Q}/i_!\mathbb{Q}) \to, \\
\to R^i f''_*(j_!\mathbb{Q}) \to R^i f''_*\mathbb{Q} \to R^i f''_*(\mathbb{Q}/j_!\mathbb{Q}) \to, \\
\to R^i f'_*\mathbb{Q} \to R^i \bar{f}'_*\mathbb{Q} \to R^i \bar{f}'_*(\mathbb{Q}/k_!\mathbb{Q}) \to .
\end{gathered}$$

For U sufficiently small, these sheaves are local systems and the induction hypothesis applies to $R^i f_{U*}(\mathbb{Q}/i_!\mathbb{Q})$, to $R^i f''_*(\mathbb{Q}/j_!\mathbb{Q})$, and to $R^i \bar{f}'_*(\mathbb{Q}/k_!\mathbb{Q})$. By a), condition $(*)$ is satisfied by $R^i \bar{f}'_*\mathbb{Q}$. By (4.2.10), it is therefore satisfied by $R^i f'_*\mathbb{Q}$, and by the dual local system $R^i f'_!\mathbb{Q}$ (Poincaré duality). By (4.2.10), condition $(*)$ is satisfied by $R^i f''_*(j_!\mathbb{Q})$, isomorphic to $R^i f'_!\mathbb{Q}$ since p is proper, and also by $R^i f''_*\mathbb{Q}$. Since the morphism $\pi_1(U) \to \pi_1(S)$ is surjective, $R^i f_*\mathbb{Q}$ satisfies (4.2.9), b).

To pass from the quasi-projective case to the general case, it suffices to use the Leray spectral sequence for a covering (U_i) of X by quasi-projective open subsets:

$$R^q f_*(\bigcap_i U_i, \mathbb{Q}) \Longrightarrow R^q f_*\mathbb{Q},$$

and (4.2.10).

4.3. Complement to [2]

In [2], (5.4) (footnote), I asserted without proof that:

Proposition (4.3.1). Let X be a proper smooth scheme over $\mathbb{C}$. If a C^∞ form on X satisfies $d'\alpha = d''\alpha = 0$ and is cohomologous to zero, then there exists β such that $\alpha = d'd''\beta$.

Here is the proof. As in loc. cit., we reduce to the case where α is bihomogeneous, of bidegree (p, q). Since the spectral sequence of the double complex $H^0(X, \mathcal{A}^{pq})$ degenerates, the exterior differential d is strictly compatible with the filtration by first degree p (1.3.2). If $\alpha \sim 0$, there therefore exists a form β_1 such that $d\beta_1 = \alpha$ and such that the components $\beta_1^{p'q'}$ of β_1 are zero for $p' < p$.

By complex conjugation, there similarly exists β_2 such that $d\beta_2 = \alpha$ and $\beta_2^{p'q'} = 0$ for $q' < q$. One has $d(\beta_1 - \beta_2) = 0$.

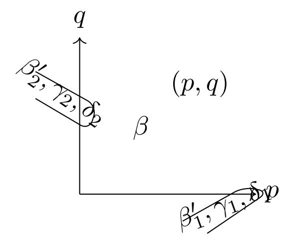
Let γ be a form satisfying $d'\gamma = d''\gamma = 0$ in the cohomology class of $\beta_1 - \beta_2$ (loc. cit.). Let γ_1, respectively γ_2, be the sum of the components of γ of type (p', q') with $p' \geq p$, respectively $q' \geq q$. Putting

$$\beta_1' = \beta_1 - \gamma_1, \qquad \beta_2' = \beta_2 + \gamma_2,$$

one still has $d\beta_1' = d\beta_2' = \alpha$; moreover, $\beta_1' - \beta_2' = \beta_1 - \beta_2 - \gamma$ is cohomologous to zero: there exists δ such that

$$d\delta = \beta_1' - \beta_2'.$$

Let δ_1 be the sum of the components of type (p', q') of δ with $p' < p - 1$, let δ_2 be the sum of the components for which $q' > q - 1$, and let β be the component of type $(p - 1, q - 1)$.



One has

$$\beta_2' = d\delta_2 + d''\beta \quad \text{and} \quad \alpha = d\beta_2' = dd\delta_2 + dd''\beta = d'd''\beta.$$

4.4. Homomorphisms of abelian schemes

We propose to show how, in some cases, one can remove from (4.1.3.2) the hypothesis of algebraicity at one point.

(4.4.1) For $f : X \to S$ a proper smooth morphism, denote by $R_i f_* \mathbb{Z}$ the local system on S^{an} of the homology of the fibers of f. At each point $s \in S$, $(R_i f_* \mathbb{Z})_s = H_i(X_s, \mathbb{Z})$ is endowed with a Hodge structure of weight $-i$, dual to the Hodge structure of $H^i(X_s, \mathbb{Z})$. This structure varies continuously with $s \in S$.

(4.4.2) If $f : X \to S$ is an abelian scheme, the exponential defines an exact sequence of sheaves on S^{an}

$$0 \to R_1 f_* \mathbb{Z} \to \mathrm{Lie}(X) \to X \to 0,$$

and the Hodge filtration is the short exact sequence

$$0 \to \mathrm{Ker}(\alpha) \to R_1 f_* \mathbb{Z} \otimes \mathcal{O} \xrightarrow{\alpha} \mathrm{Lie}(X) \to 0.$$

It is known that:

Reminder (4.4.3). Let S be a smooth scheme of finite type over $\mathbb{C}$. The construction (4.4.2) establishes an equivalence of categories between:

a) the category of abelian schemes over S;

b) the category of polarizable continuous families of Hodge structures H of type $(-1, 0) + (0, -1)$ on S, such that $H_{\mathbb{Z}}$ is torsion-free and the Hodge filtration varies holomorphically on S.

Essential surjectivity of the functor a)$\mapsto$b) follows from Borel [13]; full faithfulness follows from the fact that, for X_1 and X_2 two abelian schemes over S, the S-scheme $\mathrm{Hom}_S(X_1, X_2)$ is unramified over S, and therefore has the same algebraic and holomorphic sections.

(4.4.4) Let Z be a commutative field, H a central simple algebra over Z, and $*$ a Z-linear antiautomorphism of H. After extension of scalars, one has $H \simeq \mathrm{End}(V)$ for a vector space V over Z, and $*$ is then transposition with respect to a bilinear form Φ on V, unique up to scalar. Suppose that $*$ is involutive; the forms $\Phi(X,Y)$ and $\Phi(Y,X)$ are then proportional:

$$\Phi(X,Y) = \lambda\Phi(Y,X), \qquad \lambda^2 = 1.$$

Put $\varepsilon_* = \lambda$. If $[H:Z] = d^2$, then

$$\mathrm{Tr}(* : H \to H) = \varepsilon_* d.$$

Every Z-linear automorphism C of H is inner: $Cx = cxc^{-1}$. If C is involutive and commutes with $*$, the elements c^2 and cc^* are central, so that $c^* = \lambda c$ with λ central. Since $c^{**} = c$, one has $\lambda^2 = 1$. Put $\varepsilon_C = \lambda$.

One checks easily that the involution $C \circ *$ satisfies

$$\varepsilon_{C\circ *} = \varepsilon_C \varepsilon_*.$$

Reminder (4.4.5). Let X be a simple abelian variety over a field K and let H be the field $\mathrm{End}_K(X) \otimes \mathbb{Q}$. Every polarization of X defines a positive involution $*$ of H:

$$\mathrm{Tr}(hh^*) > 0 \qquad \text{if } h \neq 0.$$

If Z_0 is the subfield invariant under $*$ in the center Z of H, then Z_0 is totally real and one is in one of the following cases:

a) $Z = Z_0$; for every real place ρ of Z, $H \otimes_\rho \mathbb{R}$ is a matrix algebra over $\mathbb{R}$, and $\varepsilon_* = -1$.

b) $Z = Z_0$; for every real place ρ of Z, $H \otimes_\rho \mathbb{R}$ is a matrix algebra over $\mathbb{H}$, and $\varepsilon_* = -1$.

c) Z is a totally imaginary quadratic extension of Z_0.

(4.4.6) Let $f_i : X_i \to S$ $(i = 1, 2)$ be two abelian schemes over a smooth scheme S. The free abelian group

$$\mathrm{Hom}(R_1 f_{1*}\mathbb{Z}, R_1 f_{2*}\mathbb{Z}) = \Gamma\big(S, \mathrm{Hom}(R_1 f_{1*}\mathbb{Z}, R_1 f_{2*}\mathbb{Z})\big)$$

is then endowed with a Hodge structure of weight 0 (4.2.8). Moreover, by (2.1.11.1) and (4.4.3), the group $\mathrm{Hom}_S(X_1, X_2)$ is the component of type $(0,0)$ of this group.

(4.4.7) Let S be a nonempty smooth connected scheme with generic point η, and let $f : X \to S$ be an abelian scheme over S. The center Z of $H = \mathrm{End}(R_1 f_* \mathbb{Z})$ is then of type $(0,0)$ (4.2.8), and is therefore contained in $\mathrm{End}_S(X) \subset \mathrm{End}_{k(\eta)}(X_\eta)$.

A polarization of X defines an involution $*$ of H. Its restriction to $\mathrm{End}_S(X)$ is positive, so that Z is a product of totally real fields or quadratic imaginary extensions of totally real fields.

Let C denote the actions of $i \in S(\mathbb{R})$ on $H_\mathbb{R} = H \otimes \mathbb{R}$ and on the $(R_1 f_* \mathbb{R})_s$, for $s \in S$. On $H_\mathbb{R}$, $C^2 = 1$, and C commutes with $*$; on $(R_1 f_* \mathbb{R})_s$, $C^2 = -1$. If ψ is the alternating form on $R_1 f_* \mathbb{Z}$, with integral values, deduced from the polarization of X, then

$$\psi(hx, Cy) = \psi(x, h^* Cy) = \psi(x, C\,C(h^*)y).$$

It follows that

$$\mathrm{Tr}(h\,C(h^*)) > 0 \qquad \text{for } h \neq 0. \tag{4.4.7.1}$$

(4.4.8) Now suppose that X_η is simple, so that Z is a field. Let $\rho : Z \to \mathbb{C}$ be a complex embedding. The algebra

$$H_\rho = H \otimes_{Z,\rho} \mathbb{C}$$

is then a matrix algebra and a bigraded direct factor in $H_\mathbb{C} = H \otimes_\mathbb{Q} \mathbb{C}$. Choose an isomorphism

$$H_\rho \simeq \mathrm{End}(V_\rho). \tag{4.4.8.1}$$

Since the infinitesimal generators of S act on H_ρ by derivations, necessarily inner, it follows that V_ρ admits a bigrading, unique up to translation, such that (4.4.8.1) is a bigraded isomorphism. Let $(R_1f_*\mathbb{C})_\rho$ be the bigraded local system

$$(R_1f_*\mathbb{C}) \otimes_{Z\otimes\mathbb{C},\rho} \mathbb{C},$$

a direct factor of $R_1f_*\mathbb{C}$. Let

$$T_\rho = \mathrm{Hom}_{H_\rho}(V_\rho, (R_1f_*\mathbb{C})_\rho)$$

be the corresponding local system. One has

$$V_\rho \otimes_{\mathbb{C}} T_\rho \xrightarrow{\sim} (R_1f_*\mathbb{C})_\rho \tag{4.4.8.2}$$

as a bigraded isomorphism. Since $(R_1f_*\mathbb{C})_\rho$ has type $(-1,0)+(0,-1)$, one of the following conditions is satisfied:

a) V_ρ is bihomogeneous;

b) T_ρ is bihomogeneous.

It is clear that condition a), respectively b), is simultaneously satisfied for ρ and for $\bar{\rho}$. If condition a) is satisfied at every place, then H is of type $(0,0)$. If condition b) is satisfied at every place, then the Hodge structure on $R_1f_*\mathbb{Q}$ is locally constant.

(4.4.9) Now suppose that Z is totally real. For every real embedding φ of Z, put $\varepsilon_{H,\varphi} = +1$ if $H \otimes_{Z,\varphi} \mathbb{R}$ is a matrix algebra, and $\varepsilon_{H,\varphi} = -1$ otherwise. If a complex embedding ρ induces a real embedding φ, put $\varepsilon_{H,\rho} = \varepsilon_{H,\varphi}$.

Let $\rho : Z \to \mathbb{C}$ factor through $\varphi : Z \to \mathbb{R}$. Complex conjugation on H_ρ is induced by an antilinear automorphism σ_1 of V_ρ, unique up to a real scalar factor, with scalar square. Since σ_1 commutes with σ_1^2, σ_1^2 is real and, normalizing σ_1, one may suppose that $\sigma_1^2 = \pm 1$. Then

$$\sigma_1^2 = \varepsilon_{H,\rho}. \tag{4.4.9.1}$$

The automorphism of complex conjugation on

$$(R_1f_*\mathbb{C})_\rho = ((R_1f_*\mathbb{Q}) \otimes_{Z,\varphi} \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$$

is "compatible" with (4.4.8.2) and may be written

$$\sigma = \sigma_1 \otimes \sigma_2.$$

Since $\sigma^2 = 1$, by (4.4.9.1) one has

$$\sigma_2^2 = \sigma_1^2 = \varepsilon_{H,\rho}. \tag{4.4.9.2}$$

Let Φ be a polarization form on $R_1f_*\mathbb{Z}$. This form is necessarily of the type

$$\Phi = \mathrm{Tr}_{Z/\mathbb{Q}}(\psi),$$

where ψ is a Z-bilinear alternating form on $R_1f_*\mathbb{Q}$.

Let ψ_ρ be the induced form on $V_\rho \otimes_{\mathbb{C}} T_\rho$. The alternating form ψ_ρ is invariant under $\pi_1(S,s)$. Since T_ρ is irreducible, it can be written

$$\psi_\rho = A_\rho \otimes B_\rho, \tag{4.4.9.3}$$

with A_ρ on V_ρ, and B_ρ on T_ρ, invariant under $\pi_1(S,s)$.

Since T_ρ is irreducible, B_ρ is symmetric or alternating. Since ψ_ρ is alternating, A_ρ is then alternating or symmetric. Since the T_ρ are conjugate to one another, which case occurs is independent of ρ. Put

$$A_\rho(X,Y) = \varepsilon_V A_\rho(Y,X), \tag{4.4.9.4}$$

$$B_\rho(X,Y) = \varepsilon_T B_\rho(Y,X). \tag{4.4.9.5}$$

Thus

$$\varepsilon_V \varepsilon_T = -1 \tag{4.4.9.6}$$

and it is clear from (4.4.4) that, if $*$ is the involution of H deduced from ψ,

$$\varepsilon_* = \varepsilon_V. \tag{4.4.9.7}$$

In case (4.4.8), a), respectively b), normalize the bigradings so that V_ρ, respectively T_ρ, has bidegree $(0,0)$. Then for nonzero x, y in V_ρ and T_ρ:

$$\begin{aligned} &\text{case a)} \quad A_\rho(x, \sigma_1 x)\, B_\rho(y, C\sigma_2 y) > 0, \\ &\text{case b)} \quad A_\rho(x, C\sigma_1 x)\, B_\rho(y, \sigma_2 y) > 0. \end{aligned}$$

Normalizing A_ρ and B_ρ, of which only the tensor product is given, one may suppose that

$$\begin{aligned} &\text{case a)} \quad A_\rho(x, \sigma_1 x) > 0 \quad\ \ \text{and} \quad B_\rho(y, C\sigma_2 y) > 0, \\ &\text{case b)} \quad A_\rho(x, C\sigma_1 x) > 0 \quad \text{and} \quad B_\rho(y, \sigma_2 y) > 0. \end{aligned}$$

Moreover A_ρ and B_ρ, just like Φ, are invariant under the compact subtorus $\mathrm{Ker}(N)$ of S, and in particular under C. In case a) one has

$$0 < A_\rho(\sigma_1 x, \sigma_1 \sigma_1 x) = \varepsilon_V \varepsilon_{H,\rho} A_\rho(x, \sigma_1 x),$$

whence $\varepsilon_V \varepsilon_{H,\rho} = 1$. Similarly, in case b), one finds $\varepsilon_T \varepsilon_{H,\rho} = 1$.

Lemma (4.4.10). For $\varepsilon_T = \varepsilon_{H,\rho}$, one is in case b); for $\varepsilon_T = -\varepsilon_{H,\rho}$, one is in case a).

Proposition (4.4.11). Let S be a nonempty smooth connected scheme with generic point η, let $f : X \to S$ be an abelian scheme over S, let $H = \mathrm{End}(R_1 f_* \mathbb{Q})$, and let Z be the center of H. Suppose that:

(a) X_η is simple over $k(\eta)$.

(b) X does not become constant on any finite covering of S.

(c) One of the following conditions is satisfied (cf. (4.4.5)):

- (c1) Z is imaginary quadratic;
- (c2) Z is totally real, and for every real embedding $\varphi : Z \to \mathbb{R}$, $H \otimes_{Z,\varphi} \mathbb{R}$ is a matrix algebra over $\mathbb{R}$;
- (c3) Z is totally real, and for every real embedding $\varphi : Z \to \mathbb{R}$, $H \otimes_{Z,\varphi} \mathbb{R}$ is a matrix algebra over $\mathbb{H}$.

Then

$$\mathrm{End}(X) \xrightarrow{\sim} \mathrm{End}(R_1 f_* \mathbb{Z}).$$

We show that one is in case (4.4.8), a), or in case (4.4.8), b), at all places of Z simultaneously. This is clear from (4.4.8) under hypothesis (c1), and follows from (4.4.10) under hypotheses (c2) or (c3), because not only ε_T but also $\varepsilon_{H,\rho}$ is then independent of ρ.

Condition (4.4.8), b), cannot be satisfied at every place of Z, for the Hodge structure would then be locally constant, which by (4.1.3.3) and (4.4.3) violates b). Hence condition a) is satisfied at every place of Z, so that H is of type $(0,0)$ (4.4.8). In view of (4.4.3), this completes the proof.

Proposition (4.4.12). Let $f : X \to S$ be an abelian scheme over a smooth scheme S. The following conditions are equivalent:

(i) For every abelian scheme $g : Y \to S$, one has

$$\mathrm{Hom}_S(X, Y) \xrightarrow{\sim} \mathrm{Hom}_S(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z}).$$

(ii) Condition (i) is satisfied for $X = Y$, and the center Z of $\mathrm{End}_S(X) \otimes \mathbb{Q}$ has no complex place $\rho : Z \to \mathbb{C}$ such that the direct factor $R_1 f_* \mathbb{Q} \otimes_{Z,\rho} \mathbb{C}$ of $R_1 f_* \mathbb{C}$ is purely of Hodge type $(-1, 0)$.

One reduces to the case where S is nonempty and connected, with generic point η, and where X_η is a simple abelian variety over $k(\eta)$. If (i) or (ii) is satisfied, then

$$D = \mathrm{End}(X_\eta) \otimes \mathbb{Q}$$

is a division algebra, and, for $s \in S$, $(R_1 f_* \mathbb{Q})_s$ is an irreducible representation of $\pi_1(S, s)$.

To prove that (ii) implies (i), one may suppose that Y_η is simple and that $(R_1g_*\mathbb{Q})_s$ is isotypic of type $(R_1f_*\mathbb{Q})_s$. One then has an isomorphism of continuous families of $\mathbb{Q}$-Hodge structures, with D of type $(0,0)$,

$$R_1g_*\mathbb{Q} \simeq R_1f_*\mathbb{Q} \otimes_D \mathrm{Hom}(R_1f_*\mathbb{Q}, R_1g_*\mathbb{Q}).$$

One has $D \otimes \mathbb{C} \simeq M_n(Z \otimes \mathbb{C})$. If $e \in D \otimes \mathbb{C}$ is transformed, under such an isomorphism, into the idempotent e_{11}, one still has an isomorphism of bigraded vector spaces

$$(R_1g_*\mathbb{C})_s \simeq (R_1f_*\mathbb{C})_s \otimes_{Z\otimes\mathbb{C}} e\,\mathrm{Hom}(R_1f_*\mathbb{C}, R_1g_*\mathbb{C}).$$

If condition (ii) is satisfied, and if

$$e\big(\mathrm{Hom}(R_1f_*\mathbb{C}, R_1g_*\mathbb{C})\big)$$

is not purely of type $(0,0)$, then $(R_1g_*\mathbb{C})_s$ could not be purely of type $(-1,0)+(0,-1)$, which is absurd. It follows that $\mathrm{Hom}(R_1f_*\mathbb{Z}, R_1g_*\mathbb{Z})$ is purely of type $(0,0)$, and (i) follows from (4.4.3) and (2.1.11.1).

Let Z be a totally imaginary quadratic extension of a totally real number field Z^0. If

$$Z \otimes \mathbb{R} \simeq \mathbb{C} \oplus \cdots \oplus \mathbb{C},$$

define an action by homotheties of S on $Z \otimes \mathbb{R}$ by sending $s \in S(\mathbb{R}) \simeq \mathbb{C}^*$ to

$$(s^2N(s)^{-1}, 1, \ldots, 1).$$

The pair formed by Z and this action of S on $Z \otimes \mathbb{R}$ is a polarizable Hodge structure Z_ρ of type

$$(-1,1)+(0,0)+(1,-1),$$

depending on Z and on the chosen place $\rho : Z \to \mathbb{C}$.

Let X be an abelian scheme over S, with X_η simple. If the center Z of $\mathrm{End}(R_1f_*\mathbb{Q})$ is not totally real, it is a totally imaginary quadratic extension of a totally real number field. If the second hypothesis of (ii) is violated, the direct factor

$$R_1f_*\mathbb{Q} \otimes_Z Z_\rho$$

of the continuous family of polarizable Hodge structures $R_1f_*\mathbb{Q} \otimes_{\mathbb{Q}} Z_\rho$ is purely of type $(-1,0)+(0,-1)$.

By (4.4.3), the choice of an integral lattice in $R_1f_*\mathbb{Q} \otimes_Z Z_\rho$ defines an abelian scheme $g : Y \to S$, and $\mathrm{Hom}(R_1f_*\mathbb{Q}, R_1g_*\mathbb{Q})$ is not purely of type $(0,0)$, which violates (i).

Corollary (4.4.13). Let S be a smooth connected scheme with generic point η, and let $f : X \to S$ be an abelian scheme over S of relative dimension $g \leq 3$. Suppose that on no finite étale covering of S does X admit a constant abelian subscheme. Then, for every abelian scheme $g : Y \to S$, one has

$$\mathrm{Hom}(X,Y) \xrightarrow{\sim} \mathrm{Hom}(R_1f_*\mathbb{Z}, R_1g_*\mathbb{Z})$$

and

$$\mathrm{Hom}(Y,X) \xrightarrow{\sim} \mathrm{Hom}(R_1g_*\mathbb{Z}, R_1f_*\mathbb{Z}).$$

By duality, one is reduced to proving only the first assertion. One is also reduced to the case where X_η is simple. We verify that X satisfies one of the conditions (c) of (4.4.11). Put

$$H = M_c(D),$$

where D is a division algebra of rank b^2 over its center Z, which is known to be either totally real or totally imaginary. Putting $a = [Z : \mathbb{Q}]$, one has

$$ab^2c \mid 2g \leq 6. \tag{4.4.13.1}$$

1) If Z is totally imaginary and not imaginary quadratic, one must have $a = 2g = 4$. For $s \in S$, the commutant of Z acting on $(R_1f_*\mathbb{Q})_s$ is then commutative, so the action of $\pi_1(S,s)$ is abelian, hence factors through a finite group ((4.2.8), (iii), b), or (4.2.9)). This is absurd ((4.1.3.3) and (4.4.3)).

   With the notation of (4.4.8), one could also note that the T_ρ are then of rank one, so that condition (4.4.8), b), is satisfied at every place and the Hodge structure is locally constant.

2) If Z is totally real and if the conditions (c_1) and (c_2) are violated, then $a \geq 2$ and $b \geq 2$, which is absurd.

Let us verify that X satisfies the conditions of (4.4.12). If Z is imaginary quadratic, and if there exists $\rho : Z \to \mathbb{C}$ such that

$$R_1 f_* \mathbb{Q} \otimes_{Z,\rho} \mathbb{C}$$

is purely of Hodge type $(-1, 0)$, then the Hodge structure of $R_1 f_* \mathbb{Q}$ is locally constant, which is absurd. This proves (4.4.13).

(4.4.14) Let $f : X \to S$ be a polarized abelian scheme of dimension g over a smooth connected base S. If $s \in S$, the fundamental group $\pi_1(S, s)$ acts on the fiber $H^1(X_s, \mathbb{Z})$ of $R^1 f_* \mathbb{Z}$ at s. The polarization of X defines an alternating form on $H^1(X_s, \mathbb{Z})$, with values in $\mathbb{Z}(-1)$, nondegenerate over $\mathbb{Q}$, and the action of $\pi_1(S, s)$ respects this form. Consider the hypothesis:

> **(4.4.14.1)** The morphism defined by X from S into one of the irreducible components of the corresponding moduli scheme of polarized abelian varieties is dominant.

The following result was suggested to me by J.-P. Serre.

Corollary (4.4.15). Let S be a smooth connected scheme and let $f : X \to S$ be a polarized abelian scheme over S satisfying (4.4.14.1). Then, for every abelian scheme $g : Y \to S$, one has

$$\mathrm{Hom}(X, Y) \xrightarrow{\sim} \mathrm{Hom}(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z}).$$

Let $s \in S$. By (4.4.11) and (4.4.12), it suffices to show that the representation of $\pi_1(S, s)$ on $(R_1 f_* \mathbb{Q})_s$ is absolutely irreducible, so that $H = \mathbb{Q}$. This follows from the following lemma.

Lemma (4.4.16). If X satisfies (4.4.14.1), then the image of $\pi_1(S, s)$ in the group of symplectic automorphisms of $H^1(X_s, \mathbb{Z})$ is of finite index.

Let n be an integer. After replacing S by a surjective étale cover, defined by a subgroup of finite index in $\pi_1(S, s)$, one may suppose that the local system

$${}_nX = \mathrm{Ker}(n \cdot \mathrm{id} : X \to X)$$

is trivial on S. Replacing X by an isogenous abelian scheme, one is then further reduced to supposing that X is principally polarized. Suppose $n \geq 3$.

The abelian scheme X then defines a dominant morphism x from S to the "Jacobi" moduli scheme M_n of principally polarized abelian schemes endowed with a symplectic isomorphism between ${}_nX$ and $(\mathbb{Z}/n)^{2g}$; moreover, X is the inverse image under x of the universal abelian scheme over M_n. It is known that $\pi_1(M_n, x(s))$ maps isomorphically onto the subgroup of $\mathrm{Sp}(H^1(X_s, \mathbb{Z}))$ which is the kernel of reduction modulo n.

It remains only to verify:

Lemma (4.4.17). Let $p : S \to M$ be a dominant morphism of smooth connected schemes. The subgroup $p(\pi_1(S, s))$ of $\pi_1(M, p(s))$ is then of finite index.

Let t be a closed point of the generic fiber of p, and let T' be its closure in S. There exists a dense Zariski open subset U of M such that

$$T = p^{-1}(U) \cap T'$$

is an étale covering of U.

There is no loss of generality in taking $s \in T$. In the diagram

$$\begin{tikzcd} \pi_1(T, s) \arrow[r] \arrow[d, "a"'] & \pi_1(S, s) \arrow[d, "c"] \\ \pi_1(U, p(s)) \arrow[r, "b"'] & \pi_1(M, p(s)), \end{tikzcd}$$

the arrow a has image of finite index, because T is a finite étale cover, and the arrow b is surjective, because $M - U$ has real codimension ≥ 2. Thus the arrow c has image of finite index.

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A CONGRUENCE BETWEEN MULTINOMIAL COEFFICIENTS

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Let p be a prime number. For every integer $m \neq 0$, let $v(m)$ denote the exponent of the highest power of p which divides m. Put, moreover, $v(0) = \infty$.

For $\mathbf{n} = (n_1, \dots, n_k)$ a k-tuple of integers ≥ 0, put

$$|\mathbf{n}| = \sum_{i=1}^{k} n_i, \qquad a\mathbf{n} = (an_1, \dots, an_k), \qquad X^{\mathbf{n}} = \prod_{i=1}^{k} X_i^{n_i}, \qquad \mathbf{n}! = \prod_{i=1}^{k} n_i!,$$

and denote by $C(\mathbf{n})$ the coefficient of the monomial $X^{\mathbf{n}}$ in

$$\left(\sum_{i=1}^{k} X_i\right)^{|\mathbf{n}|}$$

(the multinomial coefficients).

Thus, for every integer $n \geq 0$,

$$\left(\sum_{i=1}^{k} X_i\right)^{n} = \sum_{|\mathbf{n}|=n} C(\mathbf{n}) X^{\mathbf{n}}, \tag{1}$$

$$C(\mathbf{n}) = \frac{|\mathbf{n}|!}{\mathbf{n}!}. \tag{2}$$

For $k = 2$ these are the binomial coefficients:

$$C(n_1, n_2) = \binom{n_1 + n_2}{n_1}. \tag{3}$$

Proposition. Let $\mathbf{n}$ be a k-tuple of integers ≥ 0, let $n = |\mathbf{n}|$, and let $l \geq 1$. If p is a prime number ≥ 5, then

$$C(p^l \mathbf{n}) \equiv C(\mathbf{n}) \pmod{p^{v(n)+3}}. \tag{4}$$

Example. For $p = 5$, $k = 2$, $l = 1$, and $\mathbf{n} = (1, 1)$, one has

$$\binom{10}{5} = 252 = \binom{2}{1} + 2 \cdot 5^3.$$

Proof. It is known, or follows from (2), that

$$\left(\sum_{i=1}^{k} X_i\right)^{p^l} = \left(\sum_{i=1}^{k} X_i^{p^l}\right) + pQ, \tag{5}$$

where Q is a polynomial with integral coefficients, of degree $< p^l$ in each variable. The binomial formula then gives

$$\left(\sum_{i=1}^{k} X_i\right)^{p^l n} = \left(\sum_{i=1}^{k} X_i^{p^l}\right)^{n} + \sum_{i=1}^{n} p^i \binom{n}{i} R_i, \tag{6}$$

with

$$R_i = \left(\sum_{j=1}^{k} X_j^{p^l}\right)^{n-i} Q^i. \tag{7}$$

For two polynomials A and B with integral coefficients, we shall write

$$A \doteq B \pmod{r}$$

if the coefficients of $X^{p^l \mathbf{n}}$ in A and B are congruent modulo r. With this notation, (4) is rewritten as

$$\left(\sum_{i=1}^{k} X_i\right)^{p^l n} \doteq \left(\sum_{i=1}^{k} X_i^{p^l}\right)^{n} \pmod{p^{v(n)+3}}. \tag{8}$$

Lemma 1.

$$v\binom{n}{i} \geq v(n) - v(i). \tag{9}$$

Let the additive group $\mathbb{Z}/(n)$ act on itself by translation.[1] If a subset P with i elements of $\mathbb{Z}/(n)$ has as stabilizer[2] a subgroup H of $\mathbb{Z}/(n)$, $H \cdot P = P$, then P is a union of cosets of H, and $\#H$ divides i. The cardinal

$$r_P = \frac{n}{\#H}$$

of the orbit of P under $\mathbb{Z}/(n)$, in $\mathcal{P}(\mathbb{Z}/(n))$, therefore satisfies

$$p^{v(n)-v(i)} \mid r_P.$$

Hence, since the set of subsets with i elements of $\mathbb{Z}/(n)$ is a disjoint union of orbits,

$$p^{v(n)-v(i)} \mid \binom{n}{i}.$$

Lemma 2. For $p \neq 2, 3$ and $i \geq 3$, one has

$$p^i \binom{n}{i} \equiv 0 \pmod{p^{v(n)+3}}. \tag{10}$$

Indeed,

$$v\left(p^i \binom{n}{i}\right) = i + v\binom{n}{i} \geq v(n) + (i - v(i))$$

by Lemma 1. For $3 \leq i < p$ one has $v(i) = 0$ and $i - v(i) \geq 3$. For $i \geq p$, one has

$$i - v(i) \geq i - \frac{\log i}{\log p}.$$

This last function of i is increasing for $i \geq p$, and is equal to $p - 1 \geq 3$ for $i = p$.

Lemma 3. The coefficient of $X^{p^l \mathbf{n}}$ in R_1 is zero.

None of the monomials $X^{\mathbf{r}}$ appearing in Q has an exponent $\mathbf{r}$ divisible by p^l. The polynomial

$$R_1 = \left(\sum_{j=1}^{k} X_j^{p^l}\right)^{n-1} Q$$

inherits this property.

It follows from Lemmas 2 and 3 that

$$\left(\sum_{i=1}^{k} X_i\right)^{p^l n} \doteq \left(\sum_{i=1}^{k} X_i^{p^l}\right)^{n} + \binom{n}{2} p^2 R_2 \pmod{p^{v(n)+3}}. \tag{11}$$

[1]Thus to every $x \in \mathbb{Z}/(n)$ is associated the permutation of $\mathbb{Z}/(n)$ which sends y to $x + y$.
[2]If a group G acts on a set E, the stabilizer of a subset P of E is the set of $g \in G$ for which P is stable.

This congruence already implies, by Lemma 1, that

$$C(p^l\mathbf{n}) \equiv C(\mathbf{n}) \pmod{p^{v(n)+2}}. \tag{12}$$

The only monomials $X^{\mathbf{r}}$ with exponent divisible by p^l which occur in $(pQ)^2$ are the monomials

$$X_i^{p^l} X_j^{p^l} \qquad (i < j).$$

The coefficient of each of them is

$$\sum_{a \neq 0, p^l} \binom{p^l}{a}\binom{p^l}{p^l - a} = \sum_a \binom{p^l}{a}^2 - 2 = \binom{2p^l}{p^l} - 2, \tag{13}$$

and

$$\binom{2p^l}{p^l} - 2 \equiv \binom{2p}{p} - 2 \pmod{p^3}$$

by (12).

Thus

$$\left(\sum_{i=1}^k X_i\right)^{p^l n} \doteq \left(\sum_{i=1}^k X_i^{p^l}\right)^n + \binom{n}{2}\left[\binom{2p}{p} - 2\right] R_2 \pmod{p^{v(n)+3}},$$

with R_2 a polynomial with integral coefficients. Since

$$v\binom{n}{2} \geq v(n)$$

by Lemma 1, the proposition follows from the next lemma.

Lemma 4. For p prime ≥ 5, one has

$$\binom{2p}{p} \equiv 2 \pmod{p^3}.$$

One has

$$p^{-1}\binom{p}{i} \equiv (-1)^{i-1} i^{-1} \pmod{p}$$

for $1 \leq i \leq p-1$, and hence

$$p^{-2}\left[\binom{2p}{p} - 2\right] = \sum_{i=1}^{p-1} p^{-2}\binom{p}{i}^2 \equiv \sum_{i=1}^{p-1} \frac{1}{i^2} \pmod{p}. \tag{14}$$

Define $z \in \mathbb{Z}/(p)$ by

$$z = \sum_{i \in (\mathbb{Z}/(p))^*} \frac{1}{i^2}.$$

For $u \in (\mathbb{Z}/(p))^*$ one has

$$z = \sum_{i \in (\mathbb{Z}/(p))^*} \frac{1}{(u^{-1}i)^2} = u^2 z$$

and

$$(u^2 - 1)z = 0.$$

Since $p \neq 2, 3$, there exists u such that $u^2 \neq 1$, and $z = 0$. Formula (14) therefore simplifies to

$$p^{-2}\left[\binom{2p}{p} - 2\right] \equiv 0 \pmod{p},$$

which completes the proof.

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The Weil conjecture for K3 surfaces

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1. Statement of the theorem

Let $\mathbb{F}_q$ be a field with q elements, let $\overline{\mathbb{F}}_q$ be an algebraic closure of $\mathbb{F}_q$, let $\varphi \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ be the Frobenius substitution $x \mapsto x^q$, and let $F = \varphi^{-1}$ be the "geometric Frobenius".

Let X be a scheme, separated and of finite type, over $\mathbb{F}_q$, and let $\overline{X}$ be the scheme over $\overline{\mathbb{F}}_q$ obtained from it by extension of scalars. For every closed point x of X, let $\deg(x) = [k(x) : \mathbb{F}_q]$ be the degree of the residue-field extension over $\mathbb{F}_q$. The zeta function $Z(X,t) \in \mathbb{Z}[[t]]$ is defined by

$$Z(X,t) = \prod_{x \in X} (1 - t^{\deg(x)})^{-1} \qquad (x \text{ a closed point of } X),$$

$$\log Z(X,t) = \sum_{n>0} \#X(\mathbb{F}_{q^n}) \frac{t^n}{n}.$$

For every prime number ℓ prime to q, the ℓ-adic cohomology with proper supports $H^i_c(\overline{X}, \mathbb{Q}_\ell)$ of $\overline{X}$ is a finite-dimensional vector space over $\mathbb{Q}_\ell$, on which $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ acts by transport of structure. By Grothendieck, one has

$$Z(X,t) = \prod_i \det(1 - Ft, H^i_c(\overline{X}, \mathbb{Q}_\ell))^{(-1)^{i+1}}.$$

Take X to be a nonsingular projective surface. In this case it is known that the factors

$$P_i(t) = \det(1 - Ft, H^i(\overline{X}, \mathbb{Q}_\ell))$$

of $Z(X,t)$ are polynomials with rational integral coefficients independent of ℓ. The Weil conjecture asserts here that the roots of $P_i(t)$ have complex absolute value $q^{-i/2}$. This conjecture is invariant under finite extension of the finite base field. For X as above, and $i \neq 2$, it follows from Weil [9]. If, for simplicity, X is assumed geometrically connected, one indeed has

$$P_0(t) = 1 - t, \qquad P_4(t) = 1 - q^2 t,$$

$$P_1(t) = \det(1 - \varphi t; T_\ell(\mathrm{Pic}^0(X)_{\mathrm{red}})), \qquad P_3(t) = P_1(qt).$$

Let k be a field of characteristic 0 and let X be a nonsingular projective geometrically connected surface over k. We say that X is a K3 surface if X is regular and has trivial canonical divisor, that is, if $H^1(X, \mathcal{O}) = 0$ and Ω^2_X is isomorphic to $\mathcal{O}_X$. For the miraculous properties of these surfaces, we refer to [8]. We shall mainly use the following facts.

1.1. Lemma. *(i) One has $h^{2,0} = h^{0,2} = 1$.*

(ii) One has $H^2(X, T^1_X) = 0$; infinitesimal deformations are unobstructed. The map induced by interior product

$$H^1(X, T^1_X) \otimes H^0(X, \Omega^2_X) \xrightarrow{\sim} H^1(X, \Omega^1_X) \tag{1.1.1}$$

is bijective, and $H^0(X, T^1_X) = 0$; X has no infinitesimal automorphisms.

One has

$$h^{2,0} =_{\mathrm{dfn}} \dim H^0(X, \Omega^2_X) = \dim H^0(X, \mathcal{O}_X) = 1,$$

*This article was written during a stay at the University of Warwick, which I thank for its hospitality.

and (i) follows by Hodge symmetry. Since $H^1(X, \mathcal{O}_X) = 0$, Hodge symmetry and Serre duality give

$$h^{0,1} = h^{1,0} = h^{2,1} = h^{1,2} = 0.$$

Since $\Omega^2_X = \mathcal{O}_X$, the interior-product isomorphism

$$T^1_X \otimes \Omega^2_X \xrightarrow{\sim} \Omega^1_X$$

induces isomorphisms $T^1_X \simeq \Omega^1_X$. The vanishing of $H^2(X, T^1_X)$ therefore follows from that of $h^{1,2}$, and it is clear that (1.1.1) is bijective. The same argument gives $H^0(X, T^1_X) = 0$.

1.2. Lemma. *Let Y_0 be a nonsingular projective surface over a finite field $\mathbb{F}_q$. The following conditions are equivalent.*

(i) *There exist a complete discrete valuation ring V of unequal characteristic, with finite residue field k, a proper smooth morphism $f_V : Y \to \mathrm{Spec}(V)$ whose generic fiber is a K3 surface, and $i : \mathbb{F}_q \to k$ such that $Y_0 \otimes_{\mathbb{F}_q} k \simeq Y \otimes_V k$.*

(ii) *There exist an integral scheme S whose function field has characteristic 0, a point $s \in S$, a proper smooth morphism $f : Y \to S$ whose generic fiber is a K3 surface, and $i : \mathbb{F}_q \to k(s)$ such that $Y_0 \otimes_{\mathbb{F}_q} k(s) \simeq Y_s$.*

The proof uses a standard passage to the limit. If the equivalent conditions of 1.2 are satisfied, we say that Y_0 lifts to a K3 surface.

1.3. Theorem. *A nonsingular projective surface over a finite field $\mathbb{F}_q$ which lifts to a K3 surface satisfies the Weil conjecture.*

This theorem was obtained independently by Pjateckii-Shapiro and Safarevitch. It applies, for example, to surfaces of degree 4 in $\mathbb{P}^3$. The Betti numbers of a K3 surface are $(1, 0, 22, 0, 1)$; the theorem therefore implies that

$$\left| \#X(\mathbb{F}_q) - \#\mathbb{P}^2(\mathbb{F}_q) \right| \leq 21q.$$

Grothendieck's conjectural theory of motives, in particular his theory of the "motivic Galois group", was very useful to me in constructing the proof. In that language, which will not be used below, the idea is roughly to construct, by reduction from $\mathbb{C}$ to $\mathbb{F}_q$, an abelian variety A and an injective "morphism" of motives

$$H^2(X) \hookrightarrow H^1(A) \otimes H^1(A),$$

in order to deduce the Weil conjecture for X from Weil's theorem for A. However, we do not define this "morphism" by an algebraic cycle; it is therefore not a morphism in the sense of [5]. In Hodge theory, its mode of definition has the following analogue, which will not be used below.

1.4. Lemma. *Let H be a variation of Hodge structures of weight n on a scheme S. If the local system $H_{\mathbb{Z}}$ has no global sections except the section s and its multiples, then at every point of S the section s is of type $(n/2, n/2)$.*

1.5. Notation. 1.5.1. Scalar extensions will be denoted by subscripts: if M_A is a module over a commutative ring with unit A, or a scheme over A, for example an algebraic group over A, then M_B will denote the module over the A-algebra B, or the scheme over B, obtained by extension of scalars.

1.5.2. If a group G acts on a set X, $g * x$ denotes the transform of x by g.

1.5.3. When an equality is the definition of one of its sides, it will be denoted $=_{\mathrm{dfn}}$.

2. Polarized Hodge structures

To handle Hodge structures, we shall use the formalism described in [4] and briefly recalled below.

2.1. Let $\underline{S}$ be the real algebraic group of invertible elements of the $\mathbb{R}$-algebra $\mathbb{C}$, let $w : \mathbb{G}_{m,\mathbb{R}} \to \underline{S}$ be the inclusion of $\mathbb{R}^*$ in $\mathbb{C}^*$, and let $t : \underline{S} \to \mathbb{G}_{m,\mathbb{R}}$ be the inverse of the norm; then $tw(x) = x^{-2}$. Let V be a finite-dimensional real vector space. It is equivalent to give:

a) $\rho : \underline{S} \to \mathrm{GL}(V)$ of weight n: $\rho w(x) * v = x^n v$;

b) a Hodge bigrading of weight n of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$:

$$V_{\mathbb{C}} = \bigoplus_{p+q=n} V^{p,q}, \qquad \overline{V^{p,q}} = V^{q,p};$$

c) a Hodge filtration of weight n of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$: this is a decreasing filtration such that

$$F^p(V_{\mathbb{C}}) \oplus \overline{F^{\,n-p+1}(V_{\mathbb{C}})} \xrightarrow{\sim} V_{\mathbb{C}}.$$

One has

$$F^p(V_{\mathbb{C}}) = \bigoplus_{p' \geq p} V^{p',q'};$$

for $z \in \underline{S}(\mathbb{R}) = \mathbb{C}^*$ and $v^{p,q} \in V^{p,q}$, one has

$$z * v^{p,q} = z^p \bar{z}^q v^{p,q}.$$

Put $C = \rho(i)$: this is Weil's operator C. One has $Cv^{p,q} = i^{p-q} v^{p,q}$. For $\mathscr{S} \subset \mathbb{Z} \times \mathbb{Z}$, a Hodge bigrading is said to be of type $\mathscr{S}$ if the Hodge numbers $h^{pq} = \dim V^{p,q}$ vanish for $(p,q) \notin \mathscr{S}$.

2.2. Here a Hodge structure H of weight n will mean a free finite-type $\mathbb{Z}$-module $H_{\mathbb{Z}}$ together with a Hodge bigrading of weight n of $H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$. The Tate Hodge structure $\mathbb{Z}(n)$ is the Hodge structure of type $\{(-n,-n)\}$ given by

$$\mathbb{Z}(n)_{\mathbb{C}} = \mathbb{C}, \qquad \mathbb{Z}(n)_{\mathbb{Z}} = (2\pi i)^n \mathbb{Z} \subset \mathbb{C}.$$

A polarization of a Hodge structure H of weight n is a morphism of Hodge structures

$$\psi : H \otimes H \longrightarrow \mathbb{Z}(-n)$$

such that, on $H_{\mathbb{R}} = H_{\mathbb{Z}} \otimes \mathbb{R}$, the real bilinear form

$$(2\pi i)^n \psi(x, Cy)$$

is symmetric and positive definite. The form ψ is then $(-1)^n$-symmetric.

A rational Hodge structure of weight n consists of a $\mathbb{Q}$-vector space $H_{\mathbb{Q}}$ and a Hodge bigrading of weight n of $H_{\mathbb{C}} = H_{\mathbb{Q}} \otimes \mathbb{C}$. Let $\mathbb{Q}(n)$ denote the rational Hodge structure underlying $\mathbb{Z}(n)$. The definition of polarizations is left to the reader.

The following statement reformulates a theorem of Riemann.

2.3. Scholium. *The functor $A \mapsto H^1(A, \mathbb{Z})$ identifies polarized abelian varieties with polarized Hodge structures of type $\{(0,1),(1,0)\}$.*

From the end of this section, we shall use only 2.11$(iii') \Rightarrow (ii)$. To prove this implication, one may restrict to connected groups, which makes 2.5 useless or evident.

2.4. Let G be a real linear algebraic group. We say that G is compact if $G(\mathbb{R})$ is compact and every connected component of $G(\mathbb{C})$ has a real point. The evident functor is then an equivalence from the category of algebraic linear representations of G to the category of linear representations of the compact group $G(\mathbb{R})$.

2.5. Lemma. *Every real algebraic subgroup H of a compact real algebraic group G is compact.*

It is known that the map
$$(g,\ell) \longmapsto g\cdot \exp(i\ell) : G(\mathbb{R}) \times \mathrm{Lie}(G) \longrightarrow G(\mathbb{C})$$
is bijective. It is enough to prove that, for every real subgroup H of G and every $h = g\cdot\exp(i\ell) \in H(\mathbb{C})$, one has $\ell \in \mathrm{Lie}(H)$. Since $\bar h \in H(\mathbb{C})$, one has
$$\exp(2i\ell) = \bar h^{-1} h \in H(\mathbb{C}).$$
Regard $G(\mathbb{C})$ as the set of real points of a real algebraic group. Let $\overline{J} \subset H(\mathbb{C})$ be the Zariski closure, in this group, of $J = \{\exp(ni\ell) \mid n \in 2\mathbb{Z}\}$. The elements $g \in J$, and hence the elements $g \in \overline{J}$, satisfy $g^{-1} = \bar g$, whence $\mathrm{Lie}(\overline{J}) \subset i\,\mathrm{Lie}(G)$. The Lie algebra $\mathrm{Lie}(\overline{J})$ is abelian, and $\exp(\mathrm{Lie}(\overline{J}))$ is a group, necessarily of finite index in $\overline{J}$. There are therefore n and $\ell' \in \mathrm{Lie}(\overline{J})$ such that
$$\exp(ni\ell) = \exp(\ell').$$
Thus $i\ell \in \mathrm{Lie}(\overline{J}) \subset \mathrm{Lie}(H)$ and $\ell \in \mathrm{Lie}(H)$, completing the proof.

2.6. Lemma. *Let G be a real algebraic group. The following conditions are equivalent:*

(i) *G is compact;*

(ii) *for every real, respectively complex, linear representation of G, there exists a positive-definite G-invariant symmetric bilinear form, respectively hermitian sesquilinear form;*

(iii) *for one faithful representation of G, there exists a form as in (ii).*

For G connected, these conditions are also equivalent to

(iii') *for one representation of G with finite kernel, there exists a form as in (ii).*

For G compact, G-invariance is equivalent to $G(\mathbb{R})$-invariance, and (ii) is classical. If (iii) holds, G is a subgroup of an orthogonal or unitary group, and one applies 2.5. Condition (iii') implies that $G(\mathbb{R})$ is compact, as a finite cover of a compact group; if G is connected, G is therefore compact.

2.7. Let G be a real linear algebraic group. For every involutive automorphism σ of G, let $G^{(\sigma)}$ be the real form of $G_{\mathbb{C}}$ defined by the antilinear involution $g \mapsto \sigma(\bar g)$. We say that σ is a Cartan involution if $G^{(\sigma)}$ is compact.

Let C be an element of $G(\mathbb{R})$ whose square is central. A real, respectively complex, representation of G will be called C-polarizable if there exists on V a G-invariant bilinear, respectively sesquilinear, form ψ such that $\psi(x, Cy)$ is symmetric, respectively hermitian, and positive definite. We then say that ψ is a C-polarization of V. A special case of the following lemma is stated without proof in [4].

2.8. Lemma. *a) Whether ψ is a C-polarization of V depends only on the $G(\mathbb{R})$-conjugacy class of C.*

b) The following conditions are equivalent:

(i) *$\mathrm{ad}\,C$ is a Cartan involution of G;*

(ii) *every real, respectively complex, representation of G is C-polarizable;*

(iii) *G has a faithful real, respectively complex, C-polarizable representation.*

If G is connected, these conditions are also equivalent to

(iii') *G has a real, respectively complex, C-polarizable representation with finite kernel.*

To prove a), note that $\psi(x, gCg^{-1}y) = \psi(g^{-1}x, Cg^{-1}y)$. A sesquilinear form ψ on a complex representation of G is G-invariant if and only if
$$\psi(gx, \bar g y) = \psi(x,y) \qquad \text{for } g \in G(\mathbb{C}).$$
This condition is equivalent to
$$\psi(gx, CC^{-1}\bar g Cy) = \psi(x, Cy) \qquad \text{for } g \in G(\mathbb{C})$$
(replace y by Cy), that is, to the $G^{(\mathrm{ad}\,C)}$-invariance of $\psi(x, Cy)$. The hermitian form in 2.8 therefore follows from 2.6. The non-hermitian form follows from it:

a) the complexification, respectively realification, of a real, respectively complex, C-polarizable representation is C-polarizable;

b) a polarization of V induces a polarization of every subrepresentation of V;

c) for V real, respectively complex, one has $V \hookrightarrow V \otimes_{\mathbb{R}} \mathbb{C}$.

2.9. Let G be a reductive group over $\mathbb{Q}$, and let t and w be morphisms defined over $\mathbb{Q}$

$$\mathbb{G}_m \xrightarrow{w} G \xrightarrow{t} \mathbb{G}_m,$$

with $tw(x) = x^{-2}$ and w central. Suppose also given a commutative diagram

$$\begin{tikzcd} \mathbb{G}_{m,\mathbb{R}} \arrow[r, "w"] \arrow[d, equal] & \underline{S} \arrow[r, "t"] \arrow[d, "h"] & \mathbb{G}_{m,\mathbb{R}} \arrow[d, equal] \\ \mathbb{G}_{m,\mathbb{R}} \arrow[r, "w"] & G_{\mathbb{R}} \arrow[r, "t"] & \mathbb{G}_{m,\mathbb{R}}. \end{tikzcd}$$

A representation $V_{\mathbb{Q}}$ of G, defined over $\mathbb{Q}$, is called homogeneous of weight n if $w(x) * v = x^n v$. Every representation is a direct sum of homogeneous representations. Let $V_{\mathbb{Q}}$ be a representation of weight n. Then h endows $V_{\mathbb{Q}}$ with a rational Hodge structure of weight n.

We shall regard $\mathbb{Q}(n)_{\mathbb{Q}}$ as a representation of G by the rule

$$g * x = t(g)^n x.$$

This convention is reasonable, since $\underline{S}$ acts on $\mathbb{Q}(n)_{\mathbb{R}}$ by the same formula. A polarization of a representation of weight n $V_{\mathbb{Q}}$ of G is a G-invariant form

$$\psi : V_{\mathbb{Q}} \otimes V_{\mathbb{Q}} \longrightarrow \mathbb{Q}(n)$$

such that $\psi(x, h(i)y)$ is a symmetric positive-definite form on $V_{\mathbb{R}}$. A polarization ψ, being G-invariant, is also $\underline{S}$-invariant, hence is a polarization of the rational Hodge structure $V_{\mathbb{Q}}$.

2.10. Lemma. *For the representation $V_{\mathbb{Q}}$ to be polarizable, it is necessary and sufficient that the representation $V_{\mathbb{R}}$ be $h(i)$-polarizable.*

Let $P_{\mathbb{Q}}$, respectively $P_{\mathbb{R}}$, be the space of G-invariant $(-1)^n$-symmetric bilinear forms on $V_{\mathbb{Q}}$, respectively on $V_{\mathbb{R}}$. One has $P_{\mathbb{R}} = \mathbb{R} \otimes P_{\mathbb{Q}}$. The $h(i)$-polarizations form an open subset of $P_{\mathbb{R}}$, and the polarizations of $V_{\mathbb{Q}}$ form the trace of this open subset on $P_{\mathbb{Q}}$. The lemma follows.

2.11. Proposition. *a) Whether ψ is a polarization of V depends only on the $G(\mathbb{R})$-conjugacy class of h.*

b) The following conditions are equivalent:

(i) $\operatorname{ad} h(i)$ *is a Cartan involution of* $\operatorname{Ker}(t)_{\mathbb{R}}$*;*

(ii) *every homogeneous representation of G is polarizable;*

(iii) *G has a faithful family of polarizable homogeneous representations.*

If G is connected, these conditions are equivalent to

(iii') *G has a polarizable representation ρ such that* $\operatorname{Ker}(\rho) \cap \operatorname{Ker}(t)$ *is finite.*

Assertion a) follows from 2.8.a). If G is connected, then $\operatorname{Ker}(t)$ is connected: otherwise t would be of the form t_0^n, with $n > 1$. Since $twx = x^{-2}$, one would have $n = 2$ and $w(-1) \notin \operatorname{Ker}(t)^0$. This is absurd, since

$$w(-1) \in h(\operatorname{Ker}(t : \underline{S} \to \mathbb{G}_{m,\mathbb{R}})),$$

which is connected. With this said, b) follows from 2.10 and 2.8.b).

3. Reminders on the spinor group

For the reminders contained in this section, one may consult [3].

3.1. Let V be a free module over a ring A, endowed with a quadratic form Q. The module V is identified with a submodule of the Clifford algebra $C(V)$; in $C(V)$ one has

$$v \cdot v = Q(v).$$

Let $C^+(V)$ denote the even part of $C(V)$.

If V is a free module over A, endowed with a symmetric bilinear form ψ, let $C(V)$, or $C(V, \psi)$, denote the Clifford algebra of V endowed with the quadratic form $\psi(x, x)$.

3.2. Suppose that A is a field of characteristic 0, and that Q is nondegenerate. Write $\mathrm{CSpin}(V)$ for the Clifford group. It is the algebraic group of invertible elements g of $C^+(V)$ such that $gVg^{-1} = V$. Define a morphism from $\mathrm{CSpin}(V)$ to $\mathrm{SO}(V)$ by

$$g \longmapsto (v \mapsto gvg^{-1}).$$

The kernel of this morphism is reduced to the scalars: one has an exact sequence of algebraic groups

$$0 \longrightarrow \mathbb{G}_m \xrightarrow{w} \mathrm{CSpin}(V) \longrightarrow \mathrm{SO}(V) \longrightarrow 0.$$

The spinor group $\mathrm{Spin}(V)$ is the algebraic subgroup of $\mathrm{CSpin}(V)$ which is the kernel of the spinor norm

$$N : \mathrm{CSpin}(V) \longrightarrow \mathbb{G}_m.$$

Let t denote the inverse of N; one has a commutative diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & & & \\
 & \mathrm{Spin}(V) \arrow[d] \arrow[dr] & & & \\
0 \arrow[r] & \mathbb{G}_m \arrow[r, "w"] \arrow[dr, "x \mapsto x^{-2}"'] & \mathrm{CSpin}(V) \arrow[r] \arrow[d, "t"] & \mathrm{SO}(V) \arrow[r] & 0 \\
 & & \mathbb{G}_m \arrow[d] & & \\
 & & 0 & &
\end{tikzcd} \tag{3.2.1}$$

By definition, one also has

$$\alpha : \mathrm{CSpin}(V) \longrightarrow C^+(V)^*. \tag{3.2.2}$$

3.3. Let $(C^+(V))_s$ denote the representation of $\mathrm{CSpin}(V)$ on $C^+(V)$ given by

$$g * x = \alpha(g) \cdot x. \tag{3.3.1}$$

Let $(C^+(V))_{\mathrm{ad}}$ denote the representation of $\mathrm{CSpin}(V)$ on $C^+(V)$ deduced from the action of $\mathrm{SO}(V)$ on $C^+(V)$ by transport of structure. One has

$$g *_{\mathrm{ad}} x = \alpha(g) \cdot x \cdot \alpha(g)^{-1}. \tag{3.3.2}$$

The action of $\mathrm{CSpin}(V)$ on $(C^+(V))_s$ is compatible with the right $C^+(V)$-module structure of $(C^+(V))_s$. One has an isomorphism of representations

$$\mathrm{End}_{C^+(V)}((C^+(V))_s) = (C^+(V))_{\mathrm{ad}}. \tag{3.3.3}$$

One has an isomorphism of representations

$$(C^+(V))_{\mathrm{ad}} = \bigoplus_i \bigwedge^{2i} V. \tag{3.3.4}$$

3.4. Suppose A algebraically closed. There are two cases.

1) $\dim(V)$ odd: $C^+(V)$ is here a matrix algebra. Let W be a simple $C^+(V)$-module; W is a spin representation of $\mathrm{CSpin}(V)$:

$$g * w = \alpha(g) \cdot w.$$

One has isomorphisms of representations

$$(C^+(V))_s = \text{a sum of copies of } W, \tag{3.4.1}$$

$$(C^+(V))_{\mathrm{ad}} = \mathrm{End}_A(W). \tag{3.4.2}$$

2) $\dim(V)$ even: $C^+(V)$ is here a product of two matrix algebras. Let W_1 and W_2 be two nonisomorphic simple $C^+(V)$-modules, and put $W = W_1 + W_2$; W_1 and W_2 are the half-spin representations. One has isomorphisms of representations

$$(C^+(V))_s = \text{a sum of copies of } W, \tag{3.4.3}$$

$$(C^+(V))_{\mathrm{ad}} = \mathrm{End}_A(W_1) \times \mathrm{End}_A(W_2). \tag{3.4.4}$$

In both cases, $C^+(V)$ is the bicommutant of the representation W.

3.5. Proposition. *Let Γ be a Zariski-dense subgroup of* $\mathrm{Spin}(V)$ *and let a be an automorphism of the A-algebra $C^+(V)$ which commutes with the action of Γ given by $\gamma \mapsto \alpha(\gamma) \cdot x \cdot \alpha(\gamma)^{-1}$. Then a is the identity.*

It is enough to treat the case where A is an algebraically closed field and Γ is the whole group Spin; indeed, commuting with a is a Zariski-closed condition. With the notation of 3.4, identify $C^+(V)$ with a subalgebra of $\mathrm{End}_A(W)$.

There is an automorphism $\bar{a}$ of W such that

$$a(x) = \bar{a}x\bar{a}^{-1} \qquad (x \in C^+(V)).$$

For $\gamma \in \mathrm{Spin}(V)$, by hypothesis one has

$$\alpha(\gamma)\bar{a}\alpha(\gamma)^{-1}\bar{a}^{-1} \cdot x \cdot \bar{a}\alpha(\gamma)\bar{a}^{-1}\alpha(\gamma)^{-1} = x \qquad (x \in C^+(V));$$

the commutator $(\alpha(\gamma), \bar{a}) = \alpha(\gamma)\bar{a}\alpha(\gamma)^{-1}\bar{a}^{-1}$ is therefore in the center of $C^+(V)$.

1) $\dim(V)$ odd: here $C^+(V) = \mathrm{End}_A(W)$, and $\det_W((\alpha(\gamma), \bar{a})) = 1$; hence $\alpha(\gamma)\bar{a}\alpha(\gamma)^{-1}\bar{a}^{-1}$ is a root of unity ν. Since $\mathrm{Spin}(V)$ is connected, and since for $\gamma = e$ one has $\nu = 1$, one even has

$$(\alpha(\gamma), \bar{a}) = 1, \qquad \text{i.e.} \qquad \alpha(\gamma)\bar{a} = \bar{a}\alpha(\gamma).$$

2) $\dim(V)$ even: here $C^+(V) = \mathrm{End}_A(W_1) \times \mathrm{End}_A(W_2)$, and $\bar{a}$ either permutes the W_i or preserves them. In both cases, from

$$\det_{W_i}(\alpha(\gamma)) = 1 \qquad (i = 1, 2)$$

one deduces that

$$\det_{W_i}((\alpha(\gamma), \bar{a})) = 1 \qquad (i = 1, 2),$$

and one concludes as above that

$$\alpha(\gamma)\bar{a} = \bar{a}\alpha(\gamma).$$

In both cases, $\bar{a}$ lies in the commutant of the representation, and therefore commutes with the bicommutant $C^+(V)$: a is the identity.

4. Construction of abelian varieties

In this paragraph and in part of the next, we present results of [7].

4.1. Let $V_{\mathbb{Z}}$ be a free $\mathbb{Z}$-module of rank $n+2$ endowed with a bilinear form of nonzero discriminant

$$B : V_{\mathbb{Z}} \otimes V_{\mathbb{Z}} \longrightarrow \mathbb{Z}.$$

Assume that B has index $(n+, 2-)$. We are interested in morphisms of weight 0, $h : \underline{S} \to \mathrm{SO}(V_{\mathbb{R}})$, such that

1) B is a polarization of the representation $V_{\mathbb{Q}}$ of $\mathrm{SO}(V_{\mathbb{Q}})$;
2) the Hodge numbers of $V_{\mathbb{C}}$ are
$$h^{-1,1} = h^{1,-1} = 1 \qquad \text{and} \qquad h^{0,0} = n.$$

4.2. A morphism h as above lifts uniquely to $\tilde{h} : \underline{S} \to \mathrm{CSpin}(V_{\mathbb{R}})$ making the diagram

$$\begin{tikzcd} \mathbb{G}_{m,\mathbb{R}} \arrow[r, "w"] \arrow[d, equal] & \underline{S} \arrow[r, "t"] \arrow[d, "\tilde{h}"] & \mathbb{G}_{m,\mathbb{R}} \arrow[d, equal] \\ \mathbb{G}_{m,\mathbb{R}} \arrow[r, "w"] & \mathrm{CSpin}(V_{\mathbb{R}}) \arrow[r, "t"] & \mathbb{G}_{m,\mathbb{R}} \end{tikzcd}$$

commutative.

4.3. Lemma. Relative to $\tilde{h}$, every representation of $\mathrm{CSpin}(V_{\mathbb{Q}})$ is polarizable (2.9).

Apply criterion 2.11 b)(iii') to the representation $V_{\mathbb{Q}}$ of $\mathrm{CSpin}(V_{\mathbb{Q}})$.

4.4. Lemma. The Hodge structure on $C^+(V_{\mathbb{Q}})_{\mathrm{ad}}$ defined by $\tilde{h}$ is of type $\{(-1,1),(0,0),(1,-1)\}$.

One has

$$C^+(V_{\mathbb{Q}})_{\mathrm{ad}} \simeq \bigwedge^2 V_{\mathbb{Q}},$$

and one concludes by using $h^{-1,1}(V_{\mathbb{Q}}) = 1$.

4.5. Proposition. The Hodge structure on $C^+(V_{\mathbb{Q}})_s$ defined by $\tilde{h}$ is of type $\{(0,1),(1,0)\}$. The Hodge structure $C^+(V_{\mathbb{Q}})_s$, endowed with the lattice $C^+(V_{\mathbb{Z}})$, therefore defines an abelian variety.

First suppose that n is odd. After extension of scalars to $\mathbb{C}$, $C^+(V_{\mathbb{Q}})$ becomes a matrix algebra

$$C^+(V_{\mathbb{C}}) \simeq \mathrm{End}(W).$$

On W, $\mathrm{CSpin}(V)_{\mathbb{C}}$ acts by $g \cdot w = \alpha(g)w$: this is the spin representation. The representation $C^+(V_{\mathbb{Q}})_s$ is a sum of $\dim(W)$ copies of W, so it is enough to prove that the weight-one representation W is purely of type $\{(0,1),(1,0)\}$. Otherwise, the representation $\mathrm{End}(W) \simeq C^+(V_{\mathbb{C}})_{\mathrm{ad}}$ could not be purely of type $\{(-1,1),(0,0),(1,-1)\}$.

For n even, similarly

$$C^+(V_{\mathbb{C}}) \simeq \mathrm{End}(W') \oplus \mathrm{End}(W''),$$

where W' and W'' are the semi-spin representations. As above, one proves that W' and W'' are of type $\{(0,1),(1,0)\}$, and hence so is $C^+(V_{\mathbb{Q}})_s$.

The second assertion follows from 2.3 and 3.3.

4.6. The construction given in this section hides the following fact, which we shall not use, but which is read from the Dynkin diagram and is the real reason everything works.

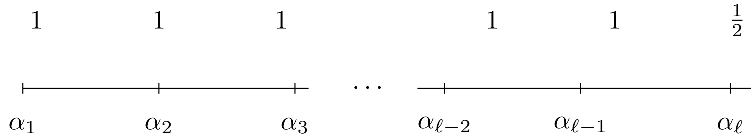
Let $V_{\mathbb{Z}}$ be a polarized Hodge structure of weight 0 and type $\{(1,-1),(0,0),(-1,1)\}$, with $h^{1,-1} = 0$. Let

$$r : \mathbb{G}_m \longrightarrow \mathrm{SO}(V_{\mathbb{C}})$$

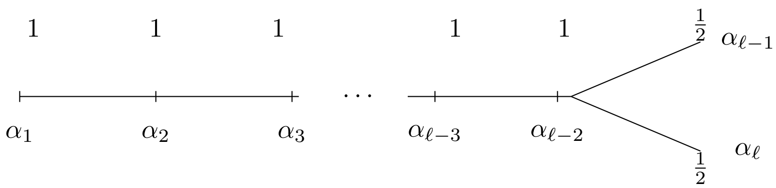
be the homomorphism such that $r(x)v^{pq} = x^p v^{pq}$ for v^{pq} of type (p,q). Let T be a maximal torus of $\mathrm{SO}(V_{\mathbb{C}})$ with a system of simple roots Φ, let $\widetilde{T}$ be the inverse image of T in $\mathrm{Spin}(V_{\mathbb{C}})$, and let $(w_\alpha)_{\alpha \in \Phi}$ be the fundamental weights of $\widetilde{T}$. The homomorphism r has a unique conjugate

$$r_0 : \mathbb{G}_m \longrightarrow T$$

such that the rational numbers $\langle w_\alpha, r_0\rangle \in \frac{1}{2}\mathbb{Z}$ are all positive. One checks that r_0 is the homomorphism $H_{\alpha_1} : \mathbb{G}_m \to T$ relative to the root denoted below by α_1. The numbers $\langle w_\alpha, r_0\rangle$ are as follows:



for $\mathrm{SO}(2, 2\ell - 1)$,



for $\mathrm{SO}(2, 2\ell - 2)$.

Let w be a dominant weight of a spin or semi-spin representation, and let σ be the opposition involution. The point is that

$$\langle w, r_0\rangle + \langle \sigma w, r_0\rangle = 1.$$

5. Construction of families of abelian varieties

We shall use the following theorem of A. Borel, to appear.

5.1. Theorem (A. Borel). Let X/Γ be the quotient of a Hermitian symmetric domain by a torsion-free arithmetic group. It is known [2] that X/Γ is naturally a quasi-projective algebraic variety. Let S be a reduced scheme over $\mathbb{C}$ and let $f : S^{an} \to X/\Gamma$ be a morphism of analytic spaces. Then f is algebraic.

Let S be smooth over $\mathbb{C}$. For the notion of a variation of Hodge structures $\underline{H}$ on S, and of a polarization of $\underline{H}$, we refer to Griffiths [6]. We shall denote by $H_\mathbb{Z}$ the local system of free $\mathbb{Z}$-modules on S defined by $\underline{H}$, and by H_s the Hodge structure which is the fiber of $\underline{H}$ at $s \in S$.

For X the Siegel half-space, Theorem 5.1 has the following corollary, which generalizes 2.3.

5.2. Scholium. The functor $(a : A \to S) \mapsto R^1 a_*\mathbb{Z}$ identifies polarized abelian schemes A over S with polarized variations of Hodge structures of type $\{(0, 1), (1, 0)\}$ on S.

5.3. In the category of variations of Hodge structures on S, all usual tensor operations make sense. Thus:

a) if $\underline{H}$ and $\underline{H}'$ are variations of Hodge structures, there are variations of Hodge structures

$$\underline{H} \otimes \underline{H}', \qquad \mathrm{Hom}(\underline{H}, \underline{H}'), \qquad \bigwedge\nolimits^n \underline{H}, \qquad \underline{H}^\vee, \qquad \mathrm{End}(\underline{H}), \dots;$$

b) if $\underline{H}$ is a variation of Hodge structures of weight 0, and ψ is a polarization of $\underline{H}$, or any symmetric morphism of Hodge structures $\underline{H} \otimes \underline{H} \to \mathbb{Z}$, then $C(\underline{H}, \psi)$ and $C^+(\underline{H}, \psi)$ are defined. At every point $t \in S$, with the notation of 3.4, one has

$$C^+(\underline{H}, \psi)_t = C^+(H_t, \psi_t).$$

5.4. Let $(V_\mathbb{Z}, B)$ be as in 4.1, and let X be the set of $h : \underline{S} \to \mathrm{SO}(V_\mathbb{R})$ of the type considered in 4.1. Giving such an h is equivalent to giving a two-dimensional subspace $V_\mathbb{R}^-$ of $V_\mathbb{R}$, on which B is negative definite, together with an orientation of $V_\mathbb{R}^-$: for $z = re^{i\theta} \in \mathbb{C}^*$, $h(z)$ induces the identity on the orthogonal complement of $V_\mathbb{R}^-$ and the rotation through angle 2θ on $V_\mathbb{R}^-$. The group $\mathrm{O}(V_\mathbb{R})$ acts on X by transport of structure, $(g * h)(z) = gh(z)g^{-1}$, and the description above makes it geometrically evident that X is a homogeneous space under $\mathrm{SO}(V_\mathbb{R})$. The stabilizer of $h \in X$ is a subgroup $\mathrm{SO}(2) \times \mathrm{SO}(n)$, the identity component of a maximal compact subgroup.

5.5. For $h \in X$, let F_h be the corresponding Hodge filtration of $V_\mathbb{C}$. The function $h \mapsto F_h^1$ identifies X with an open subset of the space of isotropic lines in $V_\mathbb{C}$. This construction endows X with a complex structure for which F_h is a holomorphic function of h. It is known that, for this structure, the two connected components of X are Hermitian symmetric domains.

5.6. Let $W_{\mathbb{Q}}$ be a weight-n representation of the algebraic group $\mathrm{CSpin}(V_{\mathbb{Q}})$: $w(x)y = x^n y$. Let $W_{\mathbb{Q}}$ also denote the constant local system on X with fiber $W_{\mathbb{Q}}$. The group $\mathrm{CSpin}(V_{\mathbb{Q}})$ acts on $(X, W_{\mathbb{Q}})$ by

$$\gamma * (w \text{ at } h \in X) = (\gamma w \text{ at } \gamma h \gamma^{-1} \in X).$$

Each $h \in X$ defines $h : \underline{S} \to \mathrm{CSpin}(V_{\mathbb{R}})$, hence a $\mathbb{Q}$-Hodge structure on the fiber of $W_{\mathbb{Q}}$ at $h \in X$. Thus one obtains a variation of $\mathbb{Q}$-Hodge structures of weight n on X, again denoted $W_{\mathbb{Q}}$:

a) the Hodge filtration is a holomorphic function of h;

b) the transversality axiom is satisfied; we shall use it only in a trivial case.

This construction is compatible with tensor products. In particular, a polarization $\psi : W_{\mathbb{Q}} \otimes W_{\mathbb{Q}} \to \mathbb{Q}(-n)$ of the representation $W_{\mathbb{Q}}$ defines a polarization of $W_{\mathbb{Q}}$.

Let $W_{\mathbb{Z}}$ be an integral lattice in $W_{\mathbb{Q}}$ and let Γ be a subgroup of $\mathrm{CSpin}(V_{\mathbb{Q}})$, discrete in $\mathrm{CSpin}(V_{\mathbb{R}})$, acting freely on X and satisfying $\Gamma W_{\mathbb{Z}} = W_{\mathbb{Z}}$. Then the constant local system $W_{\mathbb{Z}}$ on X, with fiber $W_{\mathbb{Z}}$, underlies a Γ-equivariant variation of Hodge structure W on X as above. Passing to the quotient, it defines a variation of Hodge structures W on X/Γ.

If $W_{\mathbb{Q}}$ is a representation of $\mathrm{SO}(V_{\mathbb{Q}})$, the same construction applies to discrete subgroups of $\mathrm{SO}(V_{\mathbb{Q}})$.

5.7. Proposition. Let $(\underline{H}, \psi)$ be a polarized variation of Hodge structures of type $\{(-1,1),(0,0),(1,-1)\}$, with $h^{1,-1} = 1$, on a smooth connected scheme S. There exist:

a) a finite surjective étale cover $u : S_1 \to S$;

b) an abelian scheme A over S_1;

c) a $\mathbb{Z}$-algebra C, and $\mu : C \to \mathrm{End}_{S_1}(A)$;

d) an isomorphism of variations of Hodge structures

$$u : C^+(\underline{H}, \psi) \xrightarrow{\sim} \mathrm{End}_C(R^1 a_* \mathbb{Z})$$

inducing an isomorphism of local systems of algebras

$$u_{\mathbb{Z}} : C^+(H_{\mathbb{Z}}, \psi) \xrightarrow{\sim} \mathrm{End}_C(R^1 a_* \mathbb{Z}).$$

Let $(V_{\mathbb{Z}}, B)$ be a $\mathbb{Z}$-module endowed with a symmetric bilinear form, isomorphic to the $(H_{s\mathbb{Z}}, \psi)$, and resume the notation of 5.4. Let n be an integer,

$$\Gamma = \{\gamma \in \mathrm{SO}(V_{\mathbb{Z}}) \mid \gamma \equiv 1 \pmod{n}\}$$

and

$$\widetilde{\Gamma} = \{\gamma \in \mathrm{CSpin}(V_{\mathbb{Q}}) \mid \alpha(\gamma) \equiv 1 \pmod{n} \text{ in } C^+(V_{\mathbb{Z}})\}.$$

Assume n large enough that Γ and $\widetilde{\Gamma}$ are torsion-free.

There exists a finite étale cover $v : S_1' \to S$ such that the local system $v^* H_{\mathbb{Z}/n\mathbb{Z}}$ is trivial. Choose an isomorphism σ_n between this local system and the constant local system $V_{\mathbb{Z}/n\mathbb{Z}}$. Let $s \in S_1'$ and let $\sigma : V_{\mathbb{Z}} \xrightarrow{\sim} H_{s\mathbb{Z}}$ be an isometry such that $\sigma = \sigma_n \pmod{n}$. The Hodge structure h_s of H_s defines $\sigma^{-1}(h_s) \in X$. The image of $\sigma^{-1}(h_s)$ in X/Γ depends only on s. Thus one defines

$$m : S_1' \longrightarrow X/\Gamma$$

and an isomorphism compatible with polarizations

$$\underline{H} \simeq m^* \underline{V}.$$

Put

$$S_1 = S_1' \times_{X/\Gamma} X/\widetilde{\Gamma} :$$

$$
\begin{tikzcd}
 & S_1 \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\
S \arrow[r, "v"'] & S'_1 \arrow[d, "m"] & X/\widetilde{\Gamma} \arrow[dl] \\
 & X/\Gamma &
\end{tikzcd}
$$

On S_1 one has:

a) an isomorphism $(vp_1)^*\underline{H} \simeq (mp_1)^*\underline{V}$;

b) a variation of Hodge structure, the inverse image under p_2 of the one on $X/\widetilde{\Gamma}$ defined by the representation $(C^+(V_{\mathbb{Q}}))_s$ and the lattice $C^+(V_{\mathbb{Z}})$.

By 5.2, 5.6, and 4.5, this variation of Hodge structures corresponds to an abelian scheme $a : A \to S_1$.

Let C be the algebra $C^+(V_{\mathbb{Z}})$. The abelian scheme constructed in b) has complex multiplication by C, and the variation of Hodge structures $\mathrm{End}_C(R^1a_*\mathbb{Z})$ is

$$p_2^*((C^+(V_{\mathbb{Z}}))_{\mathrm{ad}}).$$

Combining this with a), one obtains an isomorphism of local systems of algebras and of variations of Hodge structures

$$C^+((vp_1)^*\underline{H}) \simeq \mathrm{End}_C(R^1a_*\mathbb{Z}).$$

This proves 5.7.

6. K3 surfaces

6.1. A polarized $K3$ surface over a field K of characteristic 0 is a $K3$ surface over K, say Y, endowed with an element $\eta \in \mathrm{NS}(Y) = \mathrm{Pic}(Y)$ which is the class of an ample invertible sheaf. For $K = \mathbb{C}$, we shall identify $\mathrm{NS}(Y)$ with a subset of $H^2(Y, \mathbb{Z})$.

6.2. A family of $K3$ surfaces parametrized by a characteristic-zero scheme S is a proper flat morphism $f : Y \to S$ whose fibers are $K3$ surfaces. A polarization of $f : Y \to S$ is a section η of $\mathrm{Pic}_S(Y)$ which, for every $s \in S$, is a polarization of $Y_s = f^{-1}(s)$.

For every prime number ℓ, identify η with a section of $R^2f_*\mathbb{Z}_\ell$; let $P^2f_*\mathbb{Z}_\ell$ denote the orthogonal complement of η for the cup product; it is a sub-$\mathbb{Z}_\ell$-sheaf of $R^2f_*\mathbb{Z}_\ell$. The cup product

$$R^2f_*\mathbb{Z}_\ell \otimes R^2f_*\mathbb{Z}_\ell \longrightarrow R^4f_*\mathbb{Z}_\ell$$

and the trace morphism

$$R^4f_*\mathbb{Z}_\ell \xrightarrow{\sim} \mathbb{Z}_\ell(-2)$$

induce

$$\psi_\ell : P^2f_*\mathbb{Z}_\ell(1) \otimes P^2f_*\mathbb{Z}_\ell(1) \longrightarrow \mathbb{Z}_\ell. \tag{6.2.1}$$

6.3. Take S to be a scheme of finite type over $\mathbb{C}$. Then $R^2f_*\mathbb{Z}$ is a variation of Hodge structures on S; η is everywhere of type $(1,1)$ and its orthogonal complement $P^2f_*\mathbb{Z}$ is again a variation of Hodge structures. As above, the cup product defines

$$\psi : P^2f_*\mathbb{Z}(1) \otimes P^2f_*\mathbb{Z}(1) \longrightarrow \mathbb{Z}. \tag{6.3.1}$$

This time, the opposite of ψ is a polarization of the variation of Hodge structures $P^2f_*\mathbb{Z}(1)$. Moreover, ψ_ℓ is deduced from ψ via the isomorphism

$$(P^2f_*\mathbb{Z}(1)_{\mathbb{Z}}) \otimes \mathbb{Z}_\ell \simeq P^2f_*\mathbb{Z}_\ell(1).$$

For every $s \in S$, the fiber $(P^2f_*\mathbb{Z}(1))_s$ is the primitive part of the cohomology $H^2(Y_s, \mathbb{Z})$. The Hodge structure on $(P^2f_*\mathbb{Z}(1))_s$ is of type $\{(-1,1),(0,0),(1,-1)\}$, with $h^{-1,1} = 1$ ((1.1)(i)). The action of $\pi_1(S,s)$ on $H^2(Y_s,\mathbb{Z})$ induces

$$\pi : \pi_1(S,s) \longrightarrow \mathrm{O}(P^2(Y_s,\mathbb{Z})(1)_{\mathbb{Z}}, \psi). \tag{6.3.2}$$

6.4. Proposition. For every complex polarized $K3$ surface Y_0, there exists a family of polarized $K3$ surfaces $f : Y \to S$ as above, with $Y_s \simeq Y_0$ for some $s \in S$, and such that the image of π (6.3.2) is of finite index.

Let $f^\wedge : Y^\wedge \to T^\wedge$ be the versal formal deformation of Y_0. Since $H^0(Y_0, T^1) = 0$, $T^\wedge$ is even a formal moduli scheme for Y_0, but this is unimportant. Since $H^2(Y_0, T^1) = 0$ ((1.1)(i)), $T^\wedge$ is the spectrum of a power-series ring over $\mathbb{C}$. From $f^\wedge$ one obtains a non-polarized variation of Hodge structures on $T^\wedge$. By (1.1.i), it is the universal deformation of the Hodge structure $H^2(Y_0, \mathbb{Z})$ endowed with the cup product.

Let $\tilde{\eta}$ be the image of η under the composite map

$$H^2(Y_0, \mathbb{Z}) \longrightarrow H^2_{DR}(Y^\wedge/T^\wedge) \longrightarrow R^2 f^\wedge_* \mathcal{O}.$$

Let $S^\wedge$ be the formal subscheme of $T^\wedge$ with equation $\tilde{\eta} = 0$, and let $f^\wedge : Y^\wedge \to S^\wedge$ be induced by $f^\wedge$. Since $h^{0,2} = 1$, $S^\wedge \subset T^\wedge$ is defined by one equation; this equation belongs to a regular system of parameters, so $S^\wedge$ is a power-series ring. On $S^\wedge$, η comes from an ample invertible sheaf $\mathcal{O}(1)$ on $Y^\wedge$; hence $f^\wedge$ is algebraizable. Moreover, on $S^\wedge$, the polarized variation of Hodge structure

$$(P^2(Y^\wedge/S^\wedge, \mathbb{Z})(1), \psi)$$

is a universal deformation of $(P^2(Y_0, \mathbb{Z})(1), \psi)$.

Put $S^\wedge = \mathrm{Spec}(A^\wedge)$, and express $A^\wedge$ as an inductive limit of finite-type algebras over $\mathbb{C}$. Since $Y^\wedge$ is algebraizable, there exists a Cartesian diagram

$$\begin{tikzcd} Y^\wedge \arrow[r] \arrow[d, "f^\wedge"'] & Y \arrow[d, "f"] \\ S^\wedge \arrow[r] & S \end{tikzcd}$$

with S of finite type over $\mathbb{C}$ and Y a polarized family of $K3$ surfaces over S; one does not suppose that $S^\wedge$ is a completion of S. One may take S normal and connected; let $s \in S$ be the image of the closed point of $S^\wedge$.

In fact, using Artin's approximation theorem [1], one can find S such that $S^\wedge$ is a completion of S. Then S is smooth in a neighborhood of s. The simplifications resulting from such a choice are mostly psychological.

We show that $f : Y \to S$ satisfies 6.2. After replacing S by a finite étale cover, one defines, as in 5.7, a holomorphic map

$$m : S \longrightarrow X/\Gamma$$

from S to a moduli space of polarized Hodge structures such that the inverse image of the universal family of Hodge structures on X/Γ is $(P^2(Y/S, \mathbb{Z})(1), \psi)$.

By Borel (5.1), m is a morphism of schemes. The morphism m is dominant, because the composite $m^\wedge : S^\wedge \to X/\Gamma$ is dominant. It follows, for example from P. Deligne, Théorie de Hodge II, Publ. Math. IHES 40, Lemma 4.4.17, that $m(\pi_1(S, s))$ is of finite index in

$$\pi_1(X/\Gamma, m(s)) = \Gamma,$$

itself of finite index in the orthogonal group.

6.5. Proposition. Let Y_0 be a polarized $K3$ surface over a field K of characteristic 0. There exist:

a) a scheme S of finite type over $\mathbb{Q}$, and a family $f : Y \to S$ of polarized $K3$ surfaces, parametrized by S;

b) a finite extension K' of K and $v : \mathrm{Spec}(K') \to S$ such that v^*Y is isomorphic to $Y_0 \otimes_K K'$;

c) an abelian scheme $a : A \to S$ over S;

d) a $\mathbb{Z}$-algebra C and $\mu : C \to \mathrm{End}_S(A)$;

e) an isomorphism of sheaves of $\mathbb{Z}_\ell$-algebras on S

$$u : C^+(P^2 f_* \mathbb{Z}_\ell(1), \psi_\ell) \xrightarrow{\sim} \mathrm{End}_C(R^1 a_* \mathbb{Z}_\ell).$$

One is reduced to the case where K is of finite type over $\mathbb{Q}$. Choose a complex embedding $\sigma : K \to \mathbb{C}$ and apply 6.4 to $Y_0 \otimes_{K,\sigma} \mathbb{C}$ to obtain a family of polarized $K3$ surfaces $f_1 : Y_1 \to S_1$. Apply 5.7 to f_1; one obtains a family $f_2 : Y_2 \to S_2$ still satisfying the conclusion of 6.4, and, over S_2, an abelian scheme $a_2 : A_2 \to S_2$ with complex multiplication $\mu_2 : C \to \mathrm{End}_{S_2}(A_2)$ and an isomorphism

$$u_2 : C^+(P^2 f_{2*}\mathbb{Z}(1), \psi) \xrightarrow{\sim} \mathrm{End}_C(R^1 a_{2*}\mathbb{Z})$$

of local systems of algebras. The system of complex schemes and morphisms

$$\Xi_2 = (S_2, f_2, Y_2, a_2, A_2, \mu_2)$$

is defined by finitely many equations. Hence there exists a finite-type extension K_3 of K, with $K \subset K_3 \subset \mathbb{C}$, and, over K_3, a system

$$\Xi_3 = (S_3; f_3, Y_3, a_3, A_3, \mu_3)$$

which, after extension of scalars from K_3 to $\mathbb{C}$, gives Ξ_2.

6.5.1. Lemma. There exists a unique isomorphism of sheaves of $\mathbb{Z}_\ell$-algebras

$$u_3 : C^+(P^2 f_{3*}\mathbb{Z}_\ell(1), \psi_\ell) \xrightarrow{\sim} \mathrm{End}_C(R^1 a_{3*}\mathbb{Z}_\ell).$$

It is enough to verify that such an isomorphism exists and is unique after extension of scalars to an algebraic closure of K_3, for instance $\mathbb{C}$. Existence holds by construction, and uniqueness follows from 3.5. Indeed, for $s \in S_2$, the image of $\pi_1(S_2, s)$ in $\mathrm{O}(P^2(Y_{2s}, \mathbb{Z}), \psi)$ has finite index and hence contains a Zariski-dense subgroup of the group SO.

Let T be an integral scheme of finite type over $\mathbb{Q}$, with function field K_3. For suitable T, S_3 is the generic fiber of a system

$$\Xi = (S, f, Y, a, A, \mu)$$

over T, and u_3 extends to

$$u : C^+(P^2 f_*\mathbb{Z}_\ell(1), \psi_\ell) \xrightarrow{\sim} \mathrm{End}_C(R^1 a_*\mathbb{Z}_\ell).$$

One easily verifies the existence of K' as in b), and this completes the proof.

6.6. We prove Theorem 1.3. Let $f_V : Y_V \to \mathrm{Spec}(V)$ be as in 1.2(i). Let K be the fraction field of V, and apply 6.5 to the surface

$$Y_K = Y \otimes_V K$$

over K, endowed with any polarization. After replacing V by its normalization in a finite extension of K, one may suppose that there exists

$$\Xi = (S, f, Y, a, A, \mu, u)$$

as in 6.5, such that Y_K is the inverse image of Y under $v : \mathrm{Spec}(K) \to S$. By inverse image, one deduces an abelian variety with complex multiplication (A_K, μ) over K.

Let $\overline{K}$ be an algebraic closure of K, and let $\overline{k}$ be the corresponding algebraic closure of k, the residue field of the normalization of V in $\overline{K}$. From u, one obtains an isomorphism of $\mathrm{Gal}(\overline{K}/K)$-modules

$$C^+(P^2(Y_{\overline{K}}, \mathbb{Z}_\ell)(1), \psi_\ell) \xrightarrow{\sim} \mathrm{End}_C(H^1(A_{\overline{K}}, \mathbb{Z}_\ell)). \tag{6.6.1}$$

Since Y_V is smooth over V, the inertia group I acts trivially on

$$H^2(Y_{\overline{K}}, \mathbb{Z}_\ell) \simeq H^2(Y_{\overline{k}}, \mathbb{Z}_\ell).$$

The polarization defines an invariant η in $H^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$, with orthogonal complement $P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$, and I acts trivially on

$$C^+(P^2(Y_{\overline{K}}, \mathbb{Z}_\ell)(1), \psi_\ell) \simeq C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell).$$

Since I acts trivially on $\mathrm{End}_C(H^1(A_{\overline{K}}, \mathbb{Z}_\ell))$, and commutes with the complex multiplication, it acts through the center of $C \otimes \mathbb{Q}_\ell$. On the other hand, it is known that a subgroup of finite index in I acts on $H^1(A_{\overline{K}}, \mathbb{Z}_\ell)$ unipotently; it follows that I acts through one of its finite quotients. After again replacing K by a finite extension, one may therefore suppose that A_K has good reduction over V.

For $A_{\overline{k}}$ the special fiber of its reduction, (6.6.1) reduces to an isomorphism of $\mathrm{Gal}(\overline{k}/k)$-modules

$$C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell) \xrightarrow{\sim} \mathrm{End}_C(H^1(A_{\overline{k}}, \mathbb{Z}_\ell)). \tag{6.6.2}$$

By Weil's theory for A_k, the eigenvalues of Frobenius acting on

$$C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell)$$

are algebraic numbers all of whose complex conjugates have absolute value 1. One has

$$\bigwedge^2(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)) \subset C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell),$$

and, since $\dim P^2(Y_{\overline{k}}, \mathbb{Z}_\ell) > 2$ in the present case, it follows that the eigenvalues of Frobenius acting on $P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$ also have complex absolute value 1. The same assertion holds for $H^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$, and the theorem follows.

7. Remarks

7.1. Let H be a rational Hodge structure. The Mumford–Tate group of H is the smallest algebraic subgroup G of $\mathrm{GL}(H_{\mathbb{Q}})$, defined over $\mathbb{Q}$, such that $G(\mathbb{R})$ contains the image of $h : \underline{S} \to \mathrm{GL}(H_{\mathbb{R}})$. If H is polarizable, G is reductive. This definition differs from Mumford's: if H has weight $n \neq 0$, our G contains the homotheties. By 2.9, every homogeneous representation of G is endowed with a natural Hodge structure.

One of the key points in the proof of 1.3 was that the Hodge structure $H^2(X, \mathbb{Q})$, for X a $K3$, can be expressed in terms of abelian varieties, in the following sense.

7.2. Definition. A rational Hodge structure H is expressible in terms of abelian varieties if it belongs to the smallest category of rational Hodge structures which is stable under direct factors, direct sums, and tensor products, and which contains the $H^1(A, \mathbb{Q})$ for A an abelian variety, as well as the $\mathbb{Q}(n)$.

In this subsection, we show that this property of $K3$ surfaces is exceptional.

Consider the following property of a rational Hodge structure, with Mumford–Tate group G:

($*$) the Hodge structure of the adjoint representation $\mathrm{Lie}(G)$ of G is purely of type $\{(-1, 1), (0, 0), (1, -1)\}$.

7.3. Proposition. If a rational Hodge structure H is expressible in terms of abelian varieties, then condition ($*$) is satisfied.

Indeed:

(a) the Hodge structures $H^1(A, \mathbb{Q})$ and $\mathbb{Q}(n)$ satisfy ($*$);

(b) condition ($*$) is stable under direct factors, direct sums, and tensor products.

7.4. Analogous arguments show that if H is expressible in terms of abelian varieties, then the reductive group $G_{\mathbb{C}}$ has no factors of exceptional type.

7.5. Proposition. Let $\underline{H}$ be a polarized variation of Hodge structures on an irreducible scheme S. For $s \in S$ outside a meagre subset of S, there exists a finite-index subgroup of $\pi_1(S, s)$ whose image in $\mathrm{GL}(H_{s\mathbb{Z}})$ is contained in the Mumford–Tate group of H_s.

At every point of S, the Mumford–Tate group G_s of H_s may be described as follows. It is the algebraic subgroup of $\mathrm{GL}(H_{s\mathbb{Q}})$ which fixes all tensors $t \in H_{s\mathbb{Q}}^{\otimes n} \otimes H_{s\mathbb{Q}}^{*\otimes n}$, $n > 0$, that are rational of type $(0, 0)$. From this description one easily deduces that, outside a meagre subset M of S, G_s is locally constant: for $s \notin M$, the largest subspace of type $(0, 0)$

$$V(n)_s \subset H_{s\mathbb{Q}}^{\otimes n} \otimes H_{s\mathbb{Q}}^{*\otimes n}$$

is locally constant and defines a local system $V(n)$ on S. This local system underlies a polarized variation of Hodge structures of type $(0, 0)$; hence on $V(n)_s$ there exists a positive-definite quadratic form invariant under $\pi_1(S, s)$, and $\pi_1(S, s)$ acts on $V(n)_s$ through a finite group. One concludes by observing that there exists a finite sequence of integers n_i such that G_s is the algebraic subgroup of $\mathrm{GL}(H_{s\mathbb{Q}})$ acting trivially on the $V(n_i)_s$.

7.6. Consider a Lefschetz pencil of hypersurfaces of degree d and odd dimension $2r-1$:

$$\begin{tikzcd} X \arrow[r] \arrow[d, "f"'] & \mathbb{P}^{2r}(\mathbb{C}) \times \mathbb{P}^1(\mathbb{C}) \arrow[dl, "\mathrm{pr}_2"] \\ \mathbb{P}^1(\mathbb{C}). & \end{tikzcd}$$

Let $U = \mathbb{P}^1(\mathbb{C})$ be the complement of the finite set of $t \in \mathbb{P}^1(\mathbb{C})$ such that the hypersurface $X_t = f^{-1}(t)$ is singular, and let $s \in U$. The cup product is an alternating form ψ on $H^{2r-1}(X_s, \mathbb{Z})$, respected by the monodromy, whence

$$m : \pi_1(U, s) \longrightarrow \mathrm{Sp}(H^{2r-1}(X_s, \mathbb{Z}), \psi).$$

By a theorem of Kazhdan and Margulis, the image of m is Zariski-dense in the symplectic group. By 7.5, for s outside a meagre, in fact countable, subset of $\mathbb{P}^1(\mathbb{C})$, the Mumford–Tate group of the rational Hodge structure $H^{2r-1}(X_s, \mathbb{Q})$ is therefore the whole group of symplectic similitudes. If this Hodge structure is not of Hodge level at most one, i.e. if some h^{pq} with $|q-p| \neq 1$ is nonzero, it follows from 7.3 that it is not expressible in terms of abelian varieties.

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COMPLETE INTERSECTIONS OF HODGE LEVEL ONE

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Introduction

Throughout this article, let $n = 2m + 1$ be a fixed odd integer > 3, and let $\underline{a} = (a_1, \ldots, a_v)$ be a fixed sequence of integers > 1, such that the smooth complete intersections

$$V_n(\underline{a}) \subset \mathbb{P}^{n+v}(\mathbb{C})$$

of dimension n and multidegree $\underline{a}$ have Hodge level 1. This means that, for such a complete intersection X, namely the intersection of v hypersurfaces of degrees $a_1, \ldots, a_v$ which are smooth and meet transversely along X,

a) for $p + q = n$ and $|q - p| > 1$, one has $H^q(X, \Omega^p_X) = 0$;

b) $H^m(X, \Omega^{m+1}_X) \neq 0$.

Recall that, by Lefschetz,

c) $H^*(X, \mathbb{Z})$ is torsion-free;

d) for k odd, $k \neq n$, one has $H^k(X, \mathbb{Z}) = 0$;

e) for $p \leq n$, $H^{2p}(X, \mathbb{Z})$ has rank 1 and is purely of type (p, p); if $\eta \in H^2(X, \mathbb{Z})$ is the cohomology class of a hyperplane section, then $H^{2p}(X, \mathbb{Q}) = \mathbb{Q} \cdot \eta^p$.

The list of possible values of $(n, \underline{a})$ appears in [10].

The intermediate Jacobian $J(X)$ of X is, by definition, the complex torus

$$J(X) = H_n(X, \mathbb{Z}) \backslash H^n(X, \mathbb{C}) / H^{m+1,m}(X) = H^{m,m+1} / H_n(X, \mathbb{Z}).$$

The cup product on $H^n(X, \mathbb{Z})$ defines a Riemann form on $J(X)$, which is therefore a principally polarized abelian variety. We prove that the construction $X \mapsto J(X)$ is purely algebraic, i.e. independent of the topology of $\mathbb{C}$. In particular, if X is defined over a field $k \subset \mathbb{C}$, then $J(X)$ is defined over the same field k.

When $n = 1$, a case excluded below only for convenience, $J(X)$ is the Jacobian of X. For $V_3(3)$, $J(X)$ is the Jacobian of the Fano surface of X (see [1]). For $n = 3$ and X unirational, Manin's methods [8] make it possible to express $J(X)$ in terms of Jacobians. For $V_n(2, 2)$, if $(L_t)_{t \in \mathbb{P}^1}$ is a Lefschetz pencil of hyperplane sections and if C is the covering of $\mathbb{P}^1$ whose points are the linear subspaces of dimension $(n-1)/2$ in the L_t, then $J(X)$ is, up to isogeny, a direct factor of the Jacobian of C. In the other cases I know no direct interpretation of $J(X)$. According to Lieberman [7], the Hodge conjecture implies that $J(X)$ is the quotient of the set of algebraic cycles on X, of codimension $(n+1)/2$ and homologous to zero, by a suitable equivalence relation.

As a corollary of the theory one obtains, for n and $\underline{a}$ fixed as above, that smooth complete intersections of dimension n and multidegree $\underline{a}$ over a finite field $\mathbb{F}_q$ satisfy the Weil conjecture. For $V_5(3)$ and $V_3(4)$, this result is, to my knowledge, new.

1. Over $\mathbb{C}$

1.1. Let X be a nonsingular connected projective complex algebraic variety of dimension k. Duality for coherent algebraic sheaves gives a purely algebraic isomorphism

$$H^k(X, \Omega^k_X) \xrightarrow{\sim} \mathbb{C}. \tag{1.1.1}$$

On the other hand, Hodge theory, here GAGA plus the Poincaré lemma, gives an isomorphism

$$H^k(X, \Omega^k_X) \xrightarrow{\sim} H^{2k}(X, \mathbb{C}). \tag{1.1.2}$$

The image, under (1.1.1), of integral cohomology $H^{2k}(X, \mathbb{Z}) \subset H^{2k}(X, \mathbb{C})$ is the lattice $(2\pi i)^{-k}\mathbb{Z} \subset \mathbb{C}$.

This leads to the definition of the Tate Hodge structure $\mathbb{Z}(s)$ as the rank-one Hodge structure, pure of type $(-s,-s)$, with integral lattice

$$\mathbb{Z}(s)_{\mathbb{Z}} = (2\pi i)^s\mathbb{Z} \subset \mathbb{Z}(s)_{\mathbb{C}} = \mathbb{C}.$$

The cup product defines a morphism of Hodge structures

$$H^k(X,\mathbb{Z}) \otimes H^k(X,\mathbb{Z}) \longrightarrow \mathbb{Z}(-k). \tag{1.1.3}$$

1.2. Recall that $\mathbb{Z}/r(1)$ denotes the group of r-th roots of unity. We identify

$$\mathbb{Z}(1)_{\mathbb{Z}} \otimes \mathbb{Z}/r \simeq \mathbb{Z}/r(1)$$

by the map $x \mapsto \exp(x/r)$. For every prime number ℓ, put

$$\mathbb{Z}_\ell(1) = \varprojlim \mathbb{Z}/\ell^a(1),$$

with transition maps $x \mapsto x^\ell$. The exponential map identifies $\mathbb{Z}(1)_{\mathbb{Z}} \otimes \mathbb{Z}_\ell$ with $\mathbb{Z}_\ell(1)$, and

$$\mathbb{Z}(s)_{\mathbb{Z}} \otimes \mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(s) := \mathbb{Z}_\ell(1)^{\otimes s}.$$

If (1.1.3) is tensored with $\mathbb{Z}_\ell$, the resulting morphism

$$H^k(X,\mathbb{Z}_\ell) \otimes H^k(X,\mathbb{Z}_\ell) \longrightarrow \mathbb{Z}_\ell(-k) \tag{1.2.1}$$

is just the cup product in ℓ-adic cohomology. It has a purely algebraic meaning.

1.3. One has

$$H^k(X,\mathbb{Z}) \otimes \mathbb{C} = H^k(X,\mathbb{C}) =_{\text{dfn}} H^k_{\text{DR}}(X). \tag{1.3.1}$$

Here and below we shall always identify $\mathbb{Z}(k)_{\mathbb{C}} = \mathbb{Z}(k)_{\mathbb{Z}} \otimes \mathbb{C}$ with $\mathbb{C}$ by the defining isomorphism which sends the integral lattice to $(2\pi i)^k\mathbb{Z}$. Tensor (1.1.3) with $\mathbb{C}$. The resulting morphism

$$H^k_{\text{DR}}(X) \otimes H^k_{\text{DR}}(X) \longrightarrow \mathbb{C} \tag{1.3.2}$$

is just the cup product in De Rham cohomology. It has a purely algebraic meaning.

1.4. Now take X to be a complete intersection $V_n(\underline{a})$. Tensor (1.1.3) by $\mathbb{Z}(2m)$, and multiply the resulting morphism by $(-1)^m$. One obtains

$$\psi : H^n(X)(m) \otimes H^n(X)(m) \longrightarrow \mathbb{Z}(-1). \tag{1.4.1}$$

By hypothesis, the Hodge structure $H^n(X)(m)$ is of type $(0,1)+(1,0)$. The form ψ is a polarization of $H^n(X)(m)$, and, by Poincaré duality for X, the alternating form $\psi_{\mathbb{Z}}$ on the integral lattice $H^n(X)(m)_{\mathbb{Z}}$ has discriminant one.

Therefore there exists a unique principally polarized abelian variety $J(X)$, endowed with an isomorphism of polarized Hodge structures

$$\alpha_{\mathbb{Z}} : H^n(X)(m) \xrightarrow{\sim} H^1(J(X)).$$

We call $J(X)$ the intermediate Jacobian of X.

1.5. Let S be a smooth scheme of finite type over $\mathbb{C}$, and let $f : X \to S$ be a proper smooth morphism whose fibers are complete intersections $V_n(\underline{a})$. Then $R^nf_*\mathbb{Z}(m)$ is a variation of polarized Hodge structures on S, of type $(0,1)+(1,0)$ (see [5], §2). By a theorem of Borel (see [4], 5.1 and 5.2, for the statement), it defines a principally polarized abelian scheme

$$u : J(X/S) \longrightarrow S,$$

endowed with an isomorphism of variations of polarized Hodge structures

$$\alpha : R^nf_*\mathbb{Z}(m) \xrightarrow{\sim} R^1u_*\mathbb{Z}.$$

This construction expresses that $J(X)$ varies algebraically with X. It would not be difficult to extend it to the case where S is an arbitrary scheme over $\mathbb{C}$.

The analogues in ℓ-adic and De Rham cohomology of the variation of Hodge structures $R^nf_*\mathbb{Z}(m)$ and of α are the following.

1.6. Tensoring with $\mathbb{Z}_\ell$, α defines an isomorphism of $\mathbb{Z}_\ell$-sheaves

$$\alpha_\ell : R^n f_* \mathbb{Z}_\ell(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}_\ell. \tag{1.6.1}$$

The ℓ-adic cup product, multiplied by $(-1)^m$, is a morphism of $\mathbb{Z}_\ell$-sheaves

$$\psi_\ell : R^n f_* \mathbb{Z}_\ell(m) \otimes R^n f_* \mathbb{Z}_\ell(m) \longrightarrow \mathbb{Z}_\ell(-1), \tag{1.6.2}$$

and α_ℓ transforms ψ_ℓ into the polarization form

$$s\psi_\ell : R^1 u_* \mathbb{Z}_\ell \otimes R^1 u_* \mathbb{Z}_\ell \longrightarrow \mathbb{Z}_\ell(-1). \tag{1.6.3}$$

The forms (1.6.2) and (1.6.3) have a purely algebraic meaning.

1.7. The locally free coherent analytic sheaf $R^n f_* \mathbb{Z} \otimes \mathcal{O}$ underlies the coherent algebraic sheaf

$$R^n f_* \Omega^\bullet_{X/S} =_{\mathrm{dfn}} H^n_{\mathrm{DR}}(X/S).$$

The evident integrable connection on $R^n f_* \mathbb{Z} \otimes \mathcal{O}$ is identified with the Gauss–Manin connection on $H^n_{\mathrm{DR}}(X/S)$. Tensoring α with $\mathcal{O}$ gives an isomorphism of analytic sheaves

$$\alpha_{\mathrm{DR}} : H^n_{\mathrm{DR}}(X/S) \xrightarrow{\sim} H^1_{\mathrm{DR}}(J(X/S)/S). \tag{1.7.1}$$

The Gauss–Manin connections of $H^n_{\mathrm{DR}}(X/S)$ and $H^1_{\mathrm{DR}}(J(X/S)/S)$ are regular ([5], 14, or [2], II, §7), and α_{DR} is horizontal. Hence α_{DR} is algebraic ([2], II, 5.9).

The De Rham cup product, multiplied by $(-1)^m$, is a horizontal morphism

$$\psi_{\mathrm{DR}} : H^n_{\mathrm{DR}}(X/S) \otimes H^n_{\mathrm{DR}}(X/S) \longrightarrow \mathcal{O}_S, \tag{1.7.2}$$

and α_{DR} transforms ψ_{DR} into the polarization form

$$s\psi_{\mathrm{DR}} : H^1_{\mathrm{DR}}(J(X/S)/S) \otimes H^1_{\mathrm{DR}}(J(X/S)/S) \longrightarrow \mathcal{O}_S. \tag{1.7.3}$$

1.8. Let $S_\mathbb{Z}$ be the subscheme of the Hilbert scheme over $\mathbb{Z}$ parametrizing smooth complete intersections of dimension n and multidegree $\underline{a}$ in $\mathbb{P}^{n+v}$. Let

$$f : X_\mathbb{Z} \longrightarrow S_\mathbb{Z}$$

be the universal family of $V_n(\underline{a})$ parametrized by $S_\mathbb{Z}$. For every ring A, let $f : X_A \to S_A$ denote the morphism obtained from f by extension of scalars from $\mathbb{Z}$ to A.

Put $G = \mathrm{PGL}_{n+v+1}$. The group scheme G over $\mathbb{Z}$ acts on $X_\mathbb{Z}/S_\mathbb{Z}$ through its action on $\mathbb{P}^{n+v}$.

1.9. Proposition. Let $g : Y \to T$ be a proper smooth morphism whose geometric fibers are complete intersections $V_n(\underline{a})$.

(i) Étale-locally on T, Y is isomorphic to the intersection of v hypersurfaces $H_1, \ldots, H_v$, of degrees $a_1, \ldots, a_v$, in projective space $\mathbb{P}^{n+v}$ over T.

(ii) In particular, étale-locally on T, there exist $u : T \to S_\mathbb{Z}$ and a T-isomorphism $t : Y \xrightarrow{\sim} u^*(X_\mathbb{Z}/S_\mathbb{Z})$.

(iii) If (u', t') is another system as in (ii), then, étale-locally on T, there exists $a \in G(T)$ such that $(u', t') = a(u, t)$.

(iv) $S_\mathbb{Z}$ is smooth over $\mathrm{Spec}(\mathbb{Z})$.

In this section, the only corollary of 1.9 that we shall not use is the smoothness of $S_\mathbb{C}$. One could even avoid it by subsequently replacing $S_\mathbb{Z}$ by the scheme $S'_\mathbb{Z}$ introduced below.

For every geometric fiber $Y_{\bar{t}}$ of g, one has $H^1(Y_{\bar{t}}, \mathcal{O}) = H^2(Y_{\bar{t}}, \mathcal{O}) = 0$, so that $\mathrm{Pic}_T(Y)$ is étale over T. Moreover, $\mathrm{Pic}(Y_{\bar{t}}) \simeq \mathbb{Z}$, generated by the class of the unique very ample invertible sheaf $\mathcal{O}(1)$ of degree $\prod a_i$. Étale-locally on T, there is thus a relatively ample invertible sheaf of degree $\prod a_i$ on Y, denoted $\mathcal{O}(1)$. Locally on T, it is unique. Since $H^1(Y_{\bar{t}}, \mathcal{O}(k)) = 0$ (FAC), $g_*\mathcal{O}(k)$ is locally free and commutes with base change. Hence $\mathcal{O}(1)$ is very ample, and any choice of a basis of $g_*\mathcal{O}(1)$ gives a diagram

$$Y \hookrightarrow \mathbb{P}^{n+v}_T \longrightarrow T$$

with $j^*\mathcal{O}(1) = \mathcal{O}(1)$.

It remains to prove that $j(Y)$ is an intersection of type (i); then (iii) follows from the uniqueness of $\mathcal{O}(1)$. The multiplication maps on homogeneous sections are surjective, hence so are the maps $\pi_*\mathcal{O}(k) \to g_*\mathcal{O}(k)$, and the kernels of these epimorphisms of locally free sheaves are locally free. The assertion follows readily.

Let $S'_{\mathbb{Z}}$ be the scheme over $\mathbb{Z}$ parametrizing v-tuples of hypersurfaces $H_i \subset \mathbb{P}^{n+v}$, with $\deg(H_i) = a_i$. Let $S'^0_{\mathbb{Z}}$ be the open subscheme of $S'_{\mathbb{Z}}$ parametrizing tuples (H_i) for which $\bigcap H_i$ is smooth of dimension n. The scheme $S'^0_{\mathbb{Z}}$ is smooth over $\mathrm{Spec}(\mathbb{Z})$, being open in a product of projective spaces. The evident map from $S'^0_{\mathbb{Z}}$ to $S_{\mathbb{Z}}$ is surjective by (i). It is smooth: using (i), one readily verifies an infinitesimal criterion of smoothness. Hence $S_{\mathbb{Z}}$ is smooth over $\mathrm{Spec}(\mathbb{Z})$.

Put $S = S_{\mathbb{C}}$, $X = X_{\mathbb{C}}$, and let $f : X \to S$ be the universal family of 1.8.

1.10. Fundamental lemma. Let $a : A \to S$ be an abelian scheme over S.

(i) If there exists a prime number ℓ and an isomorphism of $\mathbb{Q}_\ell$-sheaves

$$v_\ell : R^1u_*\mathbb{Q}_\ell \xrightarrow{\sim} R^1a_*\mathbb{Q}_\ell,$$

or if there exists a horizontal isomorphism

$$v_{\mathrm{DR}} : H^1_{\mathrm{DR}}(J(X/S)/S) \xrightarrow{\sim} H^1_{\mathrm{DR}}(A/S),$$

then A is isomorphic to $J(X/S)$. The isomorphism

$$v' : J(X/S) \xrightarrow{\sim} A$$

is unique up to sign.

(ii) If, in addition, A is endowed with a principal polarization, then v' is compatible with the polarizations. If v_ℓ, respectively v_{DR}, as in (i), is compatible with the polarization forms, then v_ℓ, respectively v_{DR}, is induced by v' or by $-v'$.

We shall use the following theorem of Lefschetz.

1.11. Theorem (Lefschetz). Let $s \in S$, and let $X_s = f^{-1}(s)$.

(i) The monodromy representation of $\pi_1(S, s)$ on $H^n(X_s, \mathbb{Q})$ is absolutely irreducible.

(ii) For every prime ℓ, the monodromy representation of $\pi_1(S, s)$ on $H^n(X_s, \mathbb{Z}/\ell)$ is absolutely irreducible.

For an exposition in ℓ-adic cohomology of this theorem, following very closely Lefschetz's proof (cf. [6], chap. V, III, p. 106), see SGA 7 bis.

We prove 1.10. Consider the monodromy representations

$$\rho_J : \pi_1(S, s) \to \mathrm{GL}(H^1(J(X_s), \mathbb{Z})) \simeq \mathrm{GL}(H^n(X_s, \mathbb{Z})),$$

$$\rho_A : \pi_1(S, s) \to \mathrm{GL}(H^1(A_s, \mathbb{Z})).$$

Hypothesis (i) implies that, after extension of scalars to $\mathbb{Q}_\ell$ or to $\mathbb{C}$, these representations become isomorphic. Isomorphism of two representations is invariant under extension of the base field; hence there is a $\pi_1(S, s)$-equivariant isomorphism

$$v'' : H^1(J(X_s), \mathbb{Q}) \xrightarrow{\sim} H^1(A_s, \mathbb{Q}).$$

By 1.11(i), the image of $v''^{-1}(H^1(A_s, \mathbb{Z}))$ in $H^1(J(X_s), \mathbb{Z}/\ell)$ is all or nothing. It follows that a rational multiple of v'' is an isomorphism

$$v'_{\mathbb{Z}} : H^1(J(X_s), \mathbb{Z}) \xrightarrow{\sim} H^1(A_s, \mathbb{Z}).$$

By 1.11(i), $v'_{\mathbb{Z}}$ is uniquely determined up to sign. By [3], 4.4.11 and 4.4.12, $v'_{\mathbb{Z}}$ comes from an isomorphism of abelian schemes

$$v' : J(X/S) \xrightarrow{\sim} A,$$

which, like $v'_{\mathbb{Z}}$, is unique up to sign. We recall the principle of the proof. Interpret $v'_{\mathbb{Z}}$ as a section of the local system $\mathrm{Hom}(R^1u_*\mathbb{Z}, R^1a_*\mathbb{Z})$; by 1.11(i), it is the unique section of this local system up to a scalar. It is also known ([3], 4.12) that the module of all sections defines, at each point of S, a Hodge substructure of

$\mathrm{Hom}(H^1(J(X_s),\mathbb{Z}),H^1(A_s,\mathbb{Z}))$. It follows that $v'_{\mathbb{Z}}$ is everywhere of Hodge type $(0,0)$, i.e. a morphism of Hodge structures, and the assertion follows.

We prove (ii). By 1.11(i), every polarization ψ' of $J(X/S)$ is a rational multiple $b\psi$ of the given polarization. If ψ' is principal, then b is a positive invertible integer, i.e. $\psi'=\psi$, proving the first assertion. For the second, a priori $v_\ell = bv'_\ell$, respectively $v_{\mathrm{DR}} = bv'_{\mathrm{DR}}$, and compatibility with the forms ψ_ℓ or ψ_{DR} gives $b^2=1$, i.e. $b=\pm1$.

1.12. By 1.11(i), the isomorphism α_ℓ is characterized up to sign by the fact that it carries ψ_ℓ to $s\psi_\ell$ (1.6). It would be easy to adapt, and simplify, the arguments of [4], §6, to deduce from 1.11 that complete intersections $V_n(\underline{a})$ over finite fields satisfy the Weil conjectures. The same method would prove the following, perhaps more general, result.

1.13. Theorem. Let

$$X \hookrightarrow \mathbb{P}^N_{\mathrm{Spec}(V)}$$

be a smooth projective variety of dimension $2m+2$ over a discrete valuation ring V of unequal characteristic and finite residue field $\mathbb{F}_q$. Let X_0 and X_K be the special and general fibers. Suppose that the new part of the cohomology of the smooth hyperplane sections H of X_K has level ≤ 1 in the following sense:

$$(*)\quad \text{for } p+q=2m+1 \text{ and } |q-p|>1,\quad \mathrm{Coker}\big(H^q(X_K,\Omega^p_{X_K})\to H^q(H,\Omega^p_H)\big)=0.$$

Then, for every smooth hyperplane section H_0 of X_0, the eigenvalues of Frobenius on

$$\mathrm{Coker}\big(H^{2m+1}(X_0,\mathbb{Z}_\ell)\to H^{2m+1}(H_0,\mathbb{Z}_\ell)\big)$$

are algebraic integers all of whose complex conjugates have absolute value $q^{(2m+1)/2}$. These algebraic integers are divisible by q^m.

2. Over $\mathbb{Q}$

2.1. Let ℓ_0 be a prime number. For every extension K of $\mathbb{Q}$, consider the following problem:

(J,K) Construct a principally polarized abelian scheme $u: J_K\to S_K$ over S_K, endowed with an isomorphism

$$\alpha_{\ell_0}: R^n f_*\mathbb{Z}_{\ell_0}(m) \xrightarrow{\sim} R^1u_*\mathbb{Z}_{\ell_0}$$

which transforms ψ_{ℓ_0}, defined as in (1.6.2), into the polarization form.

For $K=\mathbb{C}$, problem (J,K) has, by 1.10, a solution unique up to unique isomorphism.

2.2. A general principle says that a problem P, defined over a subfield k of $\mathbb{C}$, and having over $\mathbb{C}$ a solution unique up to unique isomorphism, has a solution, equally unique, over k. Here this gives an abelian scheme

$$u: J_{\mathbb{Q}}\to S_{\mathbb{Q}}$$

over $S_{\mathbb{Q}}$, endowed with a principal polarization $s\psi$ and with an isomorphism α_{ℓ_0} as in (J,K). Moreover, the triple $(J_{\mathbb{Q}},s\psi,\alpha_{\ell_0})$ is unique up to unique isomorphism.

2.3. The preceding principle is justified as follows.

a) If the problem P is to find a number a having some geometric property, the principle is evident: k is the field of invariants of $\mathrm{Aut}(\mathbb{C}/k)$, and a solution a of P over $\mathbb{C}$, being unique, is invariant under $\mathrm{Aut}(\mathbb{C}/k)$. “Geometric” means invariant under extension of scalars.

b) If the problem P is to find a vector subspace of a vector space defined over k, having some geometric property, apply the same argument, in the form Bourbaki, Algebra, chap. 2, §8, th. 1, to $\mathrm{Aut}(\mathbb{C}/k)$. It is not required that V be finite-dimensional.

c) If the problem P is to find a subscheme Y of an affine scheme W defined over k, having some geometric properties, apply b): the ideal of Y is a vector subspace of the coordinate ring.

d) If, in c), W were not affine, one reduces to c) by gluing.

e) For every integer $b \geq 3$, let M_b be the scheme over $S_{\mathbb{Z}}[1/b]$ whose sections over any $S_{\mathbb{Z}}[1/b]$-scheme $t : T \to S_{\mathbb{Z}}[1/b]$ classify principally polarized abelian schemes $a : A \to T$ endowed with an isomorphism

$$t^* R^n f_* \mathbb{Z}/b(m) \xrightarrow{\sim} R^1 a_* \mathbb{Z}/b,$$

compatible with the intersection forms. A solution of problem (J, K) is interpreted as a section over S_K of

$$M_\infty = \varprojlim_b M_b.$$

Apply d) to this section, viewed as a subscheme of $(M_\infty)_K$.

2.4. Remark. When problem P has a solution over a finite extension of k, the principle stated above is proved by Galois descent. One can often reduce to this case using the finite-extension principle EGA IV, 9.1.1.

Let ℓ be a prime number. Let K be the function field of $S_{\mathbb{Q}}$, and let $\overline{K}$ be an algebraic closure of K. Every $\mathbb{Z}_\ell$-sheaf F on $S_{\mathbb{Q}}$ defines an ℓ-adic representation of $\mathrm{Gal}(\overline{K}/K)$ on the geometric generic fiber $F_{\bar{\eta}}$ of F.

2.5. Lemma. The ℓ-adic representations of $\mathrm{Gal}(\overline{K}/K)$ defined by $R^n f_* \mathbb{Z}_\ell(m)$ and by $R^1 u_* \mathbb{Z}_\ell$ have the same character.

Let U be a Zariski open subset of $S_{\mathbb{Z}}$, containing $S_{\mathbb{Q}}$, on which ℓ and ℓ_0 are invertible and such that $J_{\mathbb{Q}}$ is the restriction to $S_{\mathbb{Q}}$ of an abelian scheme $u' : J' \to U$. The ℓ-adic representations are then unramified outside U: they factor through a representation of $\pi_1(U, \bar{\eta})$. The same applies to their ℓ_0-adic analogues. Each closed point s of U defines a conjugacy class of Frobenius substitutions $\varphi_s \in \pi_1(U, \bar{\eta})$. One has

$$\mathrm{Tr}(\varphi_s^{-1}, (R^1 u_* \mathbb{Z}_\ell)_{\bar{\eta}}) = \mathrm{Tr}(\varphi_s^{-1}, (R^1 u_* \mathbb{Z}_{\ell_0})_{\bar{\eta}}) \in \mathbb{Z}, \tag{2.5.1}$$

the trace of the Frobenius endomorphism of J_s. Similarly,

$$\mathrm{Tr}(\varphi_s^{-1}, (R^n f_* \mathbb{Z}_\ell(m))_{\bar{\eta}}) = \mathrm{Tr}(\varphi_s^{-1}, (R^n f_* \mathbb{Z}_{\ell_0}(m))_{\bar{\eta}}) \in \mathbb{Q}, \tag{2.5.2}$$

and both sides are, by the Lefschetz trace formula in ℓ-adic cohomology,

$$q_s^{-m} \left(\# \mathbb{P}^{n+v}(k(s)) - \# X_s(k(s)) \right).$$

By hypothesis the equality of traces holds for ℓ_0, and by (2.5.1) and (2.5.2) it holds for ℓ. One concludes by observing that the trace is a continuous function on $\pi_1(U, \bar{\eta})$, and, by Chebotarev's density theorem, the Frobenius elements φ_s are dense in $\pi_1(U, \bar{\eta})$.

2.6. Lemma. The representations of $\mathrm{Gal}(\overline{K}/K)$ on $R^n f_* \mathbb{Q}_\ell(m)$ and on $R^n f_* \mathbb{Z}/\ell(m)$ are irreducible.

Indeed, the image of $\mathrm{Gal}(\overline{K}/K)$ in $\mathrm{GL}(R^n f_* \mathbb{Q}_\ell(m))$, or in $\mathrm{GL}(R^n f_* \mathbb{Z}/\ell(m))$, contains the image of the geometric fundamental group $\pi_1(S_{\overline{\mathbb{Q}}}, \bar{\eta})$. By 1.11, the action of the latter is already irreducible.

From 2.5 and 2.6 it follows that the ℓ-adic representations of $\mathrm{Gal}(\overline{K}/K)$ on $(R^n f_* \mathbb{Z}_\ell(m))_{\bar{\eta}}$ and on $(R^1 u_* \mathbb{Z}_\ell)_{\bar{\eta}}$ are isomorphic. Every isomorphism between these representations comes from an isomorphism

$$\alpha'_\ell : R^n f_* \mathbb{Z}_\ell(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}_\ell.$$

By 1.11, after extending scalars from $\mathbb{Q}$ to $\mathbb{C}$, α'_ℓ becomes an ℓ-adic multiple of the isomorphism (1.6.1): $\alpha'_\ell = b_\ell \alpha_\ell$, with $b_\ell \in \mathbb{Z}_\ell^*$. Over $\mathbb{Q}$, put $\alpha_\ell = b_\ell^{-1} \alpha'_\ell$.

2.7. We have thus obtained:

a) a principally polarized abelian scheme $J_{\mathbb{Q}}$ over $S_{\mathbb{Q}}$;

b) for every prime ℓ, an isomorphism compatible with the intersection forms

$$\alpha_\ell : R^n f_* \mathbb{Z}_\ell(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}_\ell;$$

c) an isomorphism

$$J_{\mathbb{Q}} \times_{S_{\mathbb{Q}}} S_{\mathbb{C}} \xrightarrow{\sim} J(X_{\mathbb{C}}/S_{\mathbb{C}})$$

which identifies the preceding isomorphisms with the isomorphisms (1.6.1).

2.8. For every extension K of $\mathbb{Q}$, let W_K be the K-vector space of horizontal morphisms

$$\alpha' : H^n_{\mathrm{DR}}(X_K/S_K) \longrightarrow H^1_{\mathrm{DR}}(J_K/S_K).$$

One has $W_K = W_{\mathbb{Q}} \otimes K$. For $K = \mathbb{C}$, $W_{\mathbb{C}}$ is one-dimensional, and its nonzero elements are isomorphisms. Hence there exists an isomorphism

$$\alpha' : H^n_{\mathrm{DR}}(X_{\mathbb{Q}}/S_{\mathbb{Q}}) \xrightarrow{\sim} H^1_{\mathrm{DR}}(J_{\mathbb{Q}}/S_{\mathbb{Q}}).$$

Moreover, α' is unique up to multiplication by $\lambda \in \mathbb{Q}^*$. Over $\mathbb{C}$, one verifies:

a) α' respects the Hodge filtrations on the two sides: $\alpha'(F^{m+1}) = F^1$;

b) α' transforms ψ_{DR} into a multiple of $s\psi_{\mathrm{DR}}$:

$$\alpha'(\psi_{\mathrm{DR}}) = \mu(\alpha')\, s\psi_{\mathrm{DR}}, \qquad \mu(\alpha') \in \mathbb{Q}^*.$$

One has $\mu(\lambda\alpha') = \lambda^{-2}\mu(\alpha')$.

2.9. Proposition. The multiplier $\mu(\alpha')$ is positive.

This is checked after extension of scalars from $\mathbb{Q}$ to $\mathbb{R}$.

2.10. Conjecture. The multiplier $\mu(\alpha')$ is a square.

This conjecture is equivalent to the following statements:

$$\text{for a suitable choice of } \alpha',\ \text{one has } \alpha'(\psi_{\mathrm{DR}}) = s\psi_{\mathrm{DR}}; \tag{2.10.1}$$

$$\text{over } \mathbb{C},\ \text{the isomorphism deduced from } \alpha' \text{ transforms rational cohomology into rational cohomology;} \tag{2.10.2}$$

$$\text{via 2.7 c), the isomorphism } \alpha_{\mathrm{DR}} \text{ of (1.7.1) is defined over } \mathbb{Q}. \tag{2.10.3}$$

That this conjecture follows from both the Hodge conjecture and the Tate conjecture is meagre consolation.

Recall that

$$G = \mathrm{PGL}_{n+v+1}$$

acts on $X_{\mathbb{Q}}/S_{\mathbb{Q}}$ (1.8).

2.11. Lemma. There exists a unique action of G on $J_{\mathbb{Q}}/S_{\mathbb{Q}}$ extending the action of G on $S_{\mathbb{Q}}$. For this action, the isomorphisms α_ℓ of 2.7 b) are equivariant.

For every extension K of $\mathbb{Q}$, consider the following problem. If

$$\beta : G_K \times S_K \longrightarrow S_K$$

is the action of G_K on S_K, find an isomorphism

$$a : \beta^* J_K \xrightarrow{\sim} \mathrm{pr}_2^*\, J_K$$

which defines an action of G_K on J_K/S_K. Since G is connected, the α_ℓ are automatically equivariant. From this, or from Mumford [9], 6.1, p. 115, it follows that a, if it exists, is unique. By the principle stated in 2.2, it is enough to show that a exists over $\mathbb{C}$. This follows from the fact that, over $\mathbb{C}$, the intermediate Jacobian is defined up to unique isomorphism, and from the fact that $\beta^*(X_K/S_K)$ is given as isomorphic to $\mathrm{pr}_2^*(X_K/S_K)$.

From 1.9 and 2.11 one obtains the following theorem.

2.12. Theorem. Up to unique isomorphism, there exists a unique functor J of the following type, endowed with the following additional data:

a) for T a scheme of characteristic 0 and $g : Y \to T$ a proper smooth morphism with fibers $V_n(\underline{a})$, $u : J(Y/T) \to T$ is a principally polarized abelian scheme;

b) J is compatible with base change $c : T' \to T$: isomorphisms $J(c^*Y/T') \xrightarrow{\sim} c^*J(Y/T)$ are given;

c) compatible isomorphisms

$$\alpha_\ell : R^n g_* \mathbb{Z}_\ell(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}_\ell$$

are given;

d) when restricted to smooth schemes over $\mathbb{C}$, the functor (J, α_ℓ) becomes isomorphic to the intermediate-Jacobian functor constructed in §1.

3. Over $\mathbb{Z}$

The theory over $\mathbb{Z}$ is as unsatisfactory as it is in De Rham cohomology (2.8 to 2.10).

3.1. Proposition. There exists a proper birational morphism

$$b : S \longrightarrow S_{\mathbb{Z}}$$

such that:

a) b induces an isomorphism over $S_{\mathbb{Q}}$;

b) there is a principally polarized abelian scheme J over S, whose restriction to $S_{\mathbb{Q}}$ is $J_{\mathbb{Q}}$.

We use the following lemma.

3.2. Lemma. Let V be a discrete valuation ring, with fraction field K of characteristic 0, and let $g : \mathrm{Spec}(V) \to S_{\mathbb{Z}}$ be a morphism. Then the abelian variety over K obtained by pullback from $J_{\mathbb{Q}}$ has good reduction.

Let $\overline{K}$ be an algebraic closure of K, let X_V be the inverse image of X over V, let J_K be the inverse image of J over $\mathrm{Spec}(K)$, and let $X_{\overline{K}}$, $J_{\overline{K}}$ be the inverse images over $\overline{K}$. Let $k \geq 3$ be an integer invertible on V. Since X_V is smooth over V, the specialization theorem in ℓ-adic cohomology implies that the $\mathrm{Gal}(\overline{K}/K)$-module $H^n(X_{\overline{K}}, \mathbb{Z}/k)(m)$ is unramified. By 2.7, this module is isomorphic to $H^1(J_{\overline{K}}, \mathbb{Z}/k)$, and it remains to apply the Néron–Ogg–Shafarevich criterion for good reduction.

A short, but too complicated, method for deducing 3.1 from 3.2 is the following. Let $k \geq 3$, and let M_k be as in 2.3 e). Then $J_{\mathbb{Q}}$ defines a section of M_k over $S_{\mathbb{Q}}$. Let S_k be the closure of this section in M_k. To prove that $b : S_k \to S_{\mathbb{Z}}[1/k]$ is proper, it is enough to test the valuative criterion of properness on discrete valuation rings whose fraction fields have characteristic 0; indeed, $S_{k,\mathbb{Q}}$ is dense in S_k. For these rings, it suffices to apply 3.2.

It is easy to check that over $\mathrm{Spec}(\mathbb{Z}[1/k_1k_2])$, S_{k_1} coincides with S_{k_2}. The S_k glue to form the desired S.

3.3. I conjecture that in 3.1 one may take $S = S_{\mathbb{Z}}$. If this were so, 2.12 would remain true over $\mathbb{Z}$.

One verifies easily:

3.4. Proposition. Let $a : J \to S$ be as in 3.1. For each prime number ℓ, the isomorphism α_ℓ of 2.7 extends over $S[1/\ell]$ to

$$\alpha_\ell : R^n f_* \mathbb{Z}_\ell(m) \xrightarrow{\sim} R^1 a_* \mathbb{Z}_\ell.$$

3.5. The Weil conjecture for $V_n(\underline{a})$. Using 3.4 one can give a variant of the proof, indicated in 1.12, of the Weil conjecture for the $V_n(\underline{a})$.

Let X be a smooth complete intersection of dimension n and multidegree $\underline{a}$ in $\mathbb{P}^{n+v}(\mathbb{F}_q)$. It corresponds to a point $x \in S_{\mathbb{Z}}(\mathbb{F}_q)$.

a) There exists a point $\tilde{x} \in S(\mathbb{F}_q)$ above x. Indeed, one may lift X to a complete intersection over the Witt vectors $W(\mathbb{F}_q)$, which lifts x to $x' \in S_{\mathbb{Z}}(W(\mathbb{F}_q))$. The point x' defines a point of $S(W(\mathbb{F}_q) \otimes \mathbb{Q})$, and, since b is proper, this point extends to a point of $S(W(\mathbb{F}_q))$. Reducing gives $\tilde{x}$.

b) Let J be the abelian variety over $\mathbb{F}_q$, the fiber of J at $\tilde{x}$. If X and J are obtained from X and J by extension of scalars from $\mathbb{F}_q$ to an algebraic closure $\overline{\mathbb{F}}_q$, then 3.4 gives an isomorphism

$$H^n(X, \mathbb{Z}_\ell)(m) \xrightarrow{\sim} H^1(J, \mathbb{Z}_\ell).$$

This isomorphism commutes with the geometric Frobenius endomorphisms, so

$$\det(1 - Ft, H^n(X, \mathbb{Z}_\ell)) = \det(1 - q^m Ft, H^1(J, \mathbb{Z}_\ell)). \tag{3.5.1}$$

The Weil conjecture for X therefore follows from Weil's theorem [11] for J.

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BUILDINGS OF GENERALIZED BRAID GROUPS

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Introduction

The main result of this work is the following.

Theorem. Let V be a finite-dimensional real vector space, let $\mathcal{A}$ be a finite set of homogeneous hyperplanes of V, let $V_{\mathbb{C}}$ be the complexification of V, and put

$$Y = V_{\mathbb{C}} - \bigcup_{M \in \mathcal{A}} M_{\mathbb{C}}.$$

Assume that the connected components of

$$V - \bigcup_{M \in \mathcal{A}} M$$

are open simplicial cones. Then Y is a $K(\pi, 1)$.

Let V be as above, and let $W \subset \mathrm{GL}(V)$ be a finite group generated by reflections. Suppose that no nonzero vector of V is fixed by W:

$$V^W = 0.$$

Let φ be any Euclidean structure on V invariant under W, and let $\mathcal{A}$ be the set of hyperplanes M such that the orthogonal reflection with respect to M belongs to W. It is then known that $(V, \mathcal{A})$ satisfies the hypothesis of the theorem, and that W acts freely on the corresponding space Y_W. The quotient

$$X_W = Y_W / W$$

is therefore also a $K(\pi, 1)$.

This result had been conjectured by Brieskorn. It is new only for W of type H_3, H_4, E_6, E_7, or E_8 (Brieskorn [2]). We also give a new proof that the fundamental group of X_W is the generalized braid group corresponding to W (Brieskorn [3]).

Reduced to the special case considered above, the plan of the proof is as follows.

a) We construct a building $I(W)$ on which W acts, and a set $\mathcal{S}$ of spheres in $I(W)$, isomorphic to the unit sphere of V. The group W acts strictly simply transitively on $\mathcal{S}$.

b) We show that, up to homotopy, $I(W)$ is the bouquet of the spheres $S \in \mathcal{S}$.

c) We relate X_W and $I(W)$.

The description of $\mathcal{S}$ and the possibility of c) appeared during a conversation with Brieskorn and Tits in the spring of 1970. The ideas required to establish b) were supplied to me by Garside [4], whom I often follow very closely.

In Section 4 we determine the center of $\widetilde{W}$ and solve the word problem and the conjugacy problem in $\widetilde{W}$. These results were obtained independently by E. Brieskorn and K. Saito, who, like us, use Garside's methods [4].

The geometric language used and the techniques of proof are essentially due to Tits. I was able to become familiar with his ideas by attending one of his seminars and through conversations. I am pleased to express my gratitude to him here.

0. Notation

(0.1) $L^+(D)$: the free monoid with unit generated by a set D.

(0.2) $L(D)$: the free group generated by a set D.

(0.3) $\overline{A}$: the closure of a subset A of a topological space.

(0.4) For x, y in a monoid with unit and $n \geq 0$, put

$$\mathrm{prod}(n; x, y) = xyxy\cdots \quad (n \text{ factors});$$

thus

$$\mathrm{prod}(2n; x, y) = (xy)^n, \qquad \mathrm{prod}(2n+1; x, y) = (xy)^n x.$$

1. Galleries

(1.1) Let $\mathcal{A}$ be a finite set of hyperplanes, called walls, in a finite-dimensional real vector space V. We shall use the terminology of [1], V, §1: walls, chambers, faces, facets; the facets form a partition of V. The support P of a facet F is the intersection of the walls containing F; F is open in P. For every chamber A and every wall M, let $D_M(A)$ denote the set of chambers on the same side of M as A. If A and B are chambers, denote by $D(A, B)$ the intersection of the $D_M(A)$ for those M for which A and B are on the same side of M. Departing from the terminology of [1], IV, 1, ex. 15, we shall say that two chambers A and B are adjacent if $A \neq B$ and A and B have a face in common. We shall constantly use the following elementary facts.

(1.2) Lemma.

(i) Let M be a wall of a chamber B. There is a unique chamber B', adjacent to B, having M as a wall. The wall M is the only wall separating B from B'.

(ii) Let B_1, B_2, B_3 be three chambers, and let $\mathcal{M}(B_i, B_j)$ be the set of walls separating B_i from B_j. Then

$$\mathcal{M}(B_1, B_3) = \big(\mathcal{M}(B_1, B_2) - \mathcal{M}(B_2, B_3)\big) \cup \big(\mathcal{M}(B_2, B_3) - \mathcal{M}(B_1, B_2)\big).$$

A gallery of length n ($n \geq 0$) with source A and target B is a sequence of chambers $(C_0, \ldots, C_n)$ with $A = C_0$, C_{i+1} adjacent to C_i for $0 \leq i < n$, and $C_n = B$. The composite of two galleries $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_m)$ is defined if $C_n = C'_0$ and is then

$$GG' = (C_0, \ldots, C_n, C'_1, \ldots, C'_m).$$

If G is a gallery with source A, then $A.G$ denotes its target.

The opposite gallery G^* of $G = (C_0, \ldots, C_n)$ is $(C_n, \ldots, C_0)$. One has $(GG')^* = G'^* G^*$. The antipodal gallery $-G$ is the sequence of antipodal chambers

$$-G = (-C_0, \ldots, -C_n).$$

One has $-(GG') = (-G)(-G')$ and $-(G^*) = (-G)^*$.

The distance $d(A, B)$ from a chamber A to a chamber B is the least length of a gallery from A to B. A gallery from A to B is minimal if its length is $d(A, B)$.

(1.3) Proposition. The distance $d(A, B)$ is the number of walls separating A from B. A gallery G from A to B is minimal if and only if it crosses once each wall which separates A from B, and crosses no other wall.

It is clear that every gallery from A to B crosses every wall separating A from B. It remains only to prove the existence of a gallery G from A to B of length equal to the number k of walls separating them. We argue by induction on k. The intersection of the $D_M(A)$ for M a wall of A is reduced to A. Hence either $A = B$, in which case one takes $G = (A)$, or there is a wall M of A separating A from B. Let A' be the chamber adjacent to A with wall M. The wall M is the only wall separating A from A', so a wall N separates A' from B if and only if $M \neq N$ and N separates A from B. By induction there is a gallery G' of length $k-1$ from A' to B, and we take $G = (A, A')G'$.

(1.4) Corollary. Let A, B, C be three chambers. The following conditions are equivalent:

(i) $d(A,C) = d(A,B) + d(B,C)$; equivalently, there exists a minimal gallery from A to C passing through B.

(ii) A wall separates A from C if and only if it separates A from B or B from C. In other words,

$$\mathcal{M}(A,C) = \mathcal{M}(A,B) \cup \mathcal{M}(B,C).$$

(iii) $B \in D(A,C)$.

(1.5) Proposition. Let P be an intersection of walls, let $V_P = V/P$, let $\mathrm{pr}_P : V \to V_P$ be the projection, and let $\mathcal{A}_P$ be the set of hyperplanes N of V_P such that $\mathrm{pr}_P^{-1}(N)$ is a wall. Denote by π'_P the unique map from facets of $(V, \mathcal{A})$ to facets of $(V_P, \mathcal{A}_P)$ such that $\mathrm{pr}_P(F) \subset \pi'_P(F)$.

(i) The map π'_P respects incidence $F_1 \subset \overline{F_2}$: if $F_1 \subset \overline{F_2}$, then

$$\mathrm{codim}(F_1 \text{ in } \overline{F_2}) \geq \mathrm{codim}(\pi'_P(F_1) \text{ in } \overline{\pi'_P(F_2)}).$$

It carries chambers to chambers: if A and B are adjacent, then either $\pi'_P(A) = \pi'_P(B)$ or $\pi'_P(A)$ and $\pi'_P(B)$ are adjacent.

(ii) Let F be a facet with support P. The restriction of π'_P to the set of facets E of $(V, \mathcal{A})$ such that $F \subset \overline{E}$ is bijective. Both this restriction and its inverse π_F preserve codimension, incidence $E_1 \subset \overline{E_2}$, and hence adjacency of chambers. We shall again write π_F for the inverse bijection between intersections of walls in V_P and intersections of walls in V which contain P.

(iii) If $C = \pi_F(C')$, then for every chamber X of $(V, \mathcal{A})$,

$$\pi'_P D(C,X) = D(C', \pi'_P(X)).$$

For $X = \pi_F(X')$, one has

$$\pi_F(D(C',X')) = D(C,X), \qquad \pi_F(\mathcal{M}(C',X')) = \mathcal{M}(C,X),$$

and π_F induces a bijection between minimal galleries from C' to X' and from C to X.

The verification is left to the reader; (iii) follows from 1.4(ii).

(1.6) Notation.

(i) If F is a facet with support P, then $V_P, \mathcal{A}_P, \mathrm{pr}_P, \pi_P$ will also be written $V_F, \mathcal{A}_F, \mathrm{pr}_F, \pi_F$.

(ii) Let P be an intersection of walls and C a chamber. Suppose that there is a facet of C, in $\overline{C}$, with support P. It is then unique; call it $F(P)$, and put

$$C.A(P) = \pi_{F(P)}(-\pi'_P(C)).$$

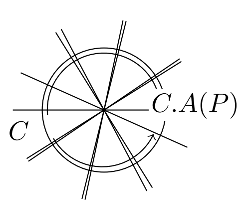
(1.7) Lemma.

(i) A wall separates C from $C.A(P)$ if and only if it contains P.

(ii) If M is a wall of C, then $C.A(M)$ is the unique chamber adjacent to C with wall M.

(iii) Let $M \neq N$ be two walls of C, and suppose that their intersection P contains a facet of C open in P. There are exactly two minimal galleries from C to $C.A(P)$. One begins with $(C, C.A(M))$, the other with $(C, C.A(N))$.

Assertion (iii) is checked by reduction to rank two, cf. 1.5(iii), where the picture is



(1.7.1)

From now on we assume the following condition.

(1.8) Hypothesis. The chambers are open simplicial cones. Equivalently, every chamber is the set of points with coordinates > 0 in a suitable basis of V.

(1.9.1) Remark. Let P be an intersection of walls.

(i) $(V_P, \mathcal{A}_P)$ still satisfies (1.8);

(ii) the set $\mathcal{A}^P$ of traces on P of the walls not containing P still satisfies (1.8).

If $\mathcal{A}$ is defined by a Weyl group W, see the introduction, then $\mathcal{A}^P$ is in general no longer of that type.

(1.9.2) Remark. Hypothesis (1.8) ensures that, if P is an intersection of walls of a chamber C, the condition of (1.6) is satisfied. The chamber $C.A(P)$ is therefore defined.

(1.10) Definition. Two galleries G and G' with the same endpoints are equivalent, written $G \sim G'$, if there exists a sequence of galleries $G = G_0, \ldots, G_n = G'$ $(n \geq 0)$ such that G_{j+1} is obtained from G_j as follows:

a) there are decompositions $G_j = E_1 F E_2$ and $G_{j+1} = E_1 F' E_2$;

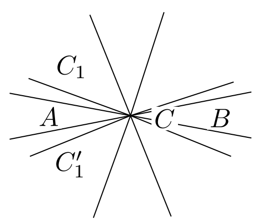
b) there is a chamber C and two walls M, N of C such that F and F' are the two minimal galleries from C to $C.A(M \cap N)$.

Composition of galleries, the operations $G \mapsto G^*$ and $G \mapsto -G$, the source and target maps, and the length function are compatible with this equivalence relation and therefore pass to the quotient. We shall generally denote equivalence classes of galleries by lowercase letters. We say that a class e begins, respectively ends, with a class f if there exists g with $e = fg$, respectively $e = gf$.

(1.11) Proposition. Equivalent galleries cross each wall the same number of times.

(1.12) Proposition. Two minimal galleries with the same endpoints are equivalent.

Let $G = (C_0, \ldots, C_n)$ and $G' = (C_0, C_1', \ldots, C_n')$ have endpoints A and B. We argue by induction on the length of the galleries. The case $n = 0$ is trivial, and the induction hypothesis permits us to suppose $C_1 \neq C_1'$. Let M and M' be the walls separating $A = C_0 = C_0'$ from C_1 and C_1', and put $C = A.A(M \cap M')$. The walls M and M' separate A from B. By reduction to rank two (1.5)(iii), every wall separating A from C also separates A from B.



(drawing in $V_{M \cap M'}$; we omit writing $\pi'_{M \cap M'}(\ )$).

Let F (respectively F') be the minimal gallery from A to C passing through C_1 (respectively through C_1'), and let E be a minimal gallery from C to B. The galleries FE and $F'E$ are minimal and equivalent. The minimal galleries G and FE, respectively G' and $F'E$, begin with (A, C_1), respectively with (A, C_1'). The induction hypothesis implies that they are equivalent, and (1.12) follows.

(1.13) Notation.

(i) Let $u(A, B)$ denote the equivalence class of minimal galleries from A to B.

(ii) If I is a set of walls of a chamber C, with intersection P, put

$$A(P) = u(C, C.A(P))$$

(cf. (1.9.2)). If I is the set of all walls, put $A = u(C, -C)$.

The interest of the notation lies in its ambiguity with respect to C.

(1.14) Proposition. Let A be a chamber and let $\mathfrak{a}$ be a set of gallery classes of bounded length and source A. Suppose that $\mathfrak{a}$ satisfies condition (i):

(ia) $(A) \in \mathfrak{a}$;

(ib) if $gh \in \mathfrak{a}$, then $g \in \mathfrak{a}$;

(ic) let g have target B, and let M, N be two walls of B. If $gA(M)$ and $gA(N)$ are in $\mathfrak{a}$, then $gA(M \cap N)$ is in $\mathfrak{a}$.

Then:

(ii) there is a unique gallery class x with source A such that

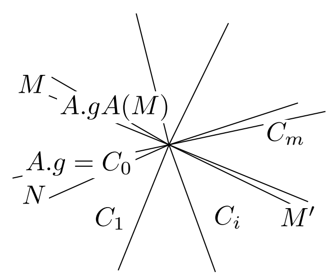
$$\mathfrak{a} = \{g \mid x \text{ begins with } g\}.$$

Uniqueness is clear: if x begins with y and $x \neq y$, then y has strictly smaller length than x. Let x be of maximal length in $\mathfrak{a}$. To see that it works, it is enough to prove the following assertion:

(∗) Let g and M be such that (a) x begins with g, (b) x does not begin with $gA(M)$, and $gA(M) \in \mathfrak{a}$. Then there exist g' and M' satisfying again (a) and (b), with g' strictly longer than g.

Since $gA(M) \in \mathfrak{a}$, maximality of x implies $g \neq x$, hence x begins with $gA(N)$ for a suitable N, necessarily distinct from M. By (ic), $gA(M \cap N) \in \mathfrak{a}$.

Since $gA(M \cap N)$ begins with $gA(M)$, x does not begin with $gA(M \cap N)$. Let $(C_0, \ldots, C_m)$ be the minimal gallery from $A.g$ to $A.gA(M \cap N)$ which begins with $(A.g, A.gA(N))$. Let i $(0 < i < m)$ be the largest index such that x begins with $g' = g(C_0, \ldots, C_i)$. Then g' and the wall M' between C_i and C_{i+1} satisfy (a)(b).



(projection into $V_{M \cap N}$).

(1.15) Proposition. Let A be a chamber and let $\mathfrak{B}$ be a set of chambers. For $\mathfrak{B}$ to satisfy condition (i):

(ia) $A \in \mathfrak{B}$;

(ib) if $B \in \mathfrak{B}$, then $D(A, B) \subset \mathfrak{B}$;

(ic) let M, N be two walls of a chamber B. If A and B are on the same side of M and of N, and $B.A(M)$ and $B.A(N)$ belong to $\mathfrak{B}$, then $B.A(M \cap N) \in \mathfrak{B}$,

it is necessary and sufficient that:

(ii) there exists a unique chamber C such that $\mathfrak{B} = D(A, C)$.

It is easy to check that (ii) implies (i). To prove the converse, apply (1.14) to the set $\mathfrak{a}$ of the $u(A, B)$ for $B \in \mathfrak{B}$, and then apply (1.4)(i) to the gallery class $x = u(A, X)$ obtained.

(1.16) Corollary. Let A and B be adjacent chambers separated by the wall M, and let C be on the same side of M as B. There exists C' such that

$$D(A, C) \cap D_M(B) = D(B, C').$$

(1.17) Corollary.

(i) Let A, C_1, C_2 be three chambers. There exists C such that

$$D(A, C) = D(A, C_1) \cap D(A, C_2). \tag{1.17.1}$$

(ii) Let A and C_1 be two chambers, and let M be a wall of A. There exists C' such that

$$D(A, C') = D(A, C_1) \cap D_M(A). \tag{1.17.2}$$

Indeed, (ii) is the special case of (i) with $C_2 = -(A.A(M))$.

(1.18) Lemma. Let M, M', M'' be three distinct walls of a chamber A, and put

$$A_1 = A.A(M), \qquad B = A.A(M \cap M' \cap M''),$$

$$B' = A.A(M \cap M'), \qquad B'' = A.A(M \cap M'').$$

For any chamber C, if $B' \in D(A_1, C)$ and $B'' \in D(A_1, C)$, then $B \in D(A_1, C)$.

The facet F of A open in $M \cap M' \cap M''$ is a facet of all the chambers A, A_1, B, B' and B''. By (1.5)(iii), we may reduce to the case $\dim V = 3$. By (1.17)(ii), we may also suppose that A_1 and C are on the same side of M. Under these hypotheses, if $B', B'' \in D(A_1, C)$, then M' and M'' separate A_1 from C, while A_1 and C remain on the same side of M. Thus A and C are separated by M, M' and M'', and $C = -A = B$.

The following key result is inspired by [4].

(1.19) Proposition.

(i) Let A, B, C be three chambers, F a gallery from A to B, and G_1, G_2 two galleries from B to C. If $FG_1 \sim FG_2$, then $G_1 \sim G_2$.

(ii) Similarly, if $G_1E \sim G_2E$, then $G_1 \sim G_2$.

(iii) Let g be an equivalence class of galleries with source A. There exists a chamber C such that g begins with $u(A, B)$ if and only if $B \in D(A, C)$.

Statement (ii) follows from (i) by passing to opposite galleries.

Let $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_n)$ be two galleries of length n with the same endpoints. If $n \geq 1$, set $G_1 = (C_1, \ldots, C_n)$ and $G'_1 = (C'_1, \ldots, C'_n)$. If in addition $C_1 \neq C'_1$, let M and M' be the walls separating $C_0 = C'_0$ from C_1 and C'_1, and put $A = C_0.A(M \cap M')$.

Consider the following relation $R_n(G, G')$. It means that one of the following holds:

a) $n = 0$;

b) $n \neq 0$, $C_1 = C'_1$, and $G_1 \sim G'_1$;

c) $n \neq 0$, $C_1 \neq C'_1$, and there exists a gallery F from A to $C_n = C'_n$ such that

$$G_1 \sim u(C_1, A)F, \qquad G'_1 \sim u(C'_1, A)F.$$

We prove by induction on n that (A_n): R_n is an equivalence relation.

Assume (A_i) for $i < n$, and prove the three intermediate assertions below.

(1.19.1) For galleries of length $i < n$, $R_i(G, G')$ is equivalent to $G \sim G'$.

The implication $R_i(G, G') \Rightarrow G \sim G'$ is trivial, and if $G \sim G'$ then, using the notation of (1.10), one has $R_i(G_j, G_{j+1})$.

(1.19.2) Assertion (i) is true for composites FG_i of length $< n$. This follows from (1.19.1) and the second clause in the definition of the relations R_i.

(1.19.3) Assertion (iii) is true for g of length $< n$.

This follows from (1.15) applied to the set $\mathfrak{B}$ of chambers B such that g begins with $u(A, B)$. The hypothesis (ic) of (1.15) is checked using (1.19.2), (1.18), and the third clause in the definition of the R_i.

It remains to prove that if G, G' and G'' are three galleries of length n from A to C, with $R_n(G, G')$ and $R_n(G, G'')$, then $R_n(G', G'')$. If $n = 0$, or if $C_1 = C'_1$, or if $C_1 = C''_1$, this is trivial.

For $n > 0$, let M, M', M'' be the walls separating $A = C_0 = C'_0 = C''_0$ from C_1, C'_1, C''_1. There are two cases.

Case 1. $C_1 \neq C_1' = C_1''$. Let $B = A.A(M \cap M')$. By hypothesis,

$$(C_1, \dots, C_n) \sim u(C_1, B)F' \sim u(C_1, B)F'',$$

$$(C_1', \dots, C_n') \sim u(C_1', B)F', \qquad (C_1'', \dots, C_n'') \sim u(C_1'', B)F''.$$

By (1.19.2), $F' \sim F''$, hence $R_n(G', G'')$.

Case 2. C_1, C_1', C_1'' are pairwise distinct. Put

$$B_1 = A.A(M' \cap M''), \quad B' = A.A(M \cap M'), \quad B'' = A.A(M \cap M''),$$

and $B = A.A(M \cap M' \cap M'')$. Then

$$G_1 \sim u(C_1, B')F' \sim u(C_1, B'')F'',$$

$$G_1' \sim u(C_1', B')F', \qquad G_1'' \sim u(C_1'', B'')F''.$$

Thus the class g_1 of G_1 begins with $u(C_1, B')$ and $u(C_1, B'')$. By (1.19.3) and Lemma (1.18), g_1 begins with $u(C_1, B)$:

$$G_1 \sim u(C_1, B)F.$$

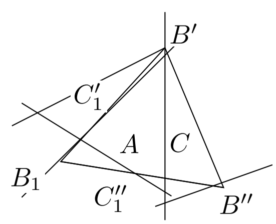
By (1.19.2), $F' \sim u(B', B)F$ and $F'' \sim u(B'', B)F$. Hence

$$G_1' \sim u(C_1', B)F, \qquad G_1'' \sim u(C_1'', B)F.$$

Therefore

$$G_1' \sim u(C_1', B_1)u(B_1, B)F, \qquad G_1'' \sim u(C_1'', B_1)u(B_1, B)F,$$

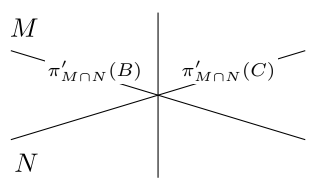
and so $R_n(G', G'')$.



(a drawing on the unit sphere of $V_{M \cap M' \cap M''}$).

(1.20) Corollary. The conditions (i) and (ii) of (1.14) are equivalent.

Assume (ii) and prove (ic). Under the hypotheses of (ic), if $x = gh$, with h of source B, then (1.19)(i) implies that h begins with $A(M)$ and $A(N)$. Let C be the chamber guaranteed by (1.19)(iii) for h. Since M and N separate B from C, $\pi'_{M \cap N}(C)$ can only be $\pi'_{M \cap N}(B).A$.



And (ic) follows from (1.5)(iii).

(1.21) Corollary. For every chamber A, let $n(A, i)$ be the number of gallery classes with source A and length i. Put

$$f_A = \sum_0^\infty n(A, i)t^i \in \mathbb{Z}[[t]].$$

For every chamber A, and for P running through the intersections of walls of A, including $\{0\}$ and V, one has

$$\sum_P (-1)^{\operatorname{codim} P} t^{d(A, A.A(P))} f_{A.A(P)} = 1.$$

This expresses the facts that:

a) the number of gallery classes of length i which begin with $A(P)$ is

$$n(A.A(P), i - d(A, A.A(P)));$$

b) gallery classes which begin with both $A(P)$ and $A(Q)$ begin with $A(P \cap Q)$.

(1.22) Algorithm. Let $G = (A_0, \ldots, A_n)$ be a gallery of length $n > 1$, let $G_1 = (A_1, \ldots, A_n)$, let M be the wall separating A_0 from A_1, let C be the chamber whose existence is guaranteed by (1.19)(iii) for G, and let C_1 be the analogous chamber for G_1. One can compute C recursively using the formula

$$D(A_1, C) = D_M(A_1) \cap D(A_1, C_1). \tag{1.22.1}$$

Since $A_1 \in D(A, C)$, the wall M separates A from C, and $C \in D_M(A_1)$. By (1.19)(i), one also has $D(A_1, C) \subset D(A_1, C_1)$, giving one inclusion. Finally, if $B \in D_M(A_1) \cap D(A_1, C_1)$, then G_1 begins with $u(A_1, B)$ and G begins with

$$u(A, A_1)u(A_1, B) = u(A, B),$$

which proves (1.22.1).

(1.23) Corollary. Let G and H be two composable galleries. Let A be the source of G, B the source of H, and let C be as in (1.19)(iii), so that

$$H \sim u(B, C)H'.$$

Then, for every chamber D, the class of GH begins with $u(A, D)$ if and only if the class of $Gu(B, C)$ begins with $u(A, D)$.

(1.24) Proposition. Let A be a chamber, let P be an intersection of walls of A, let $F = F(P)$ (1.6), and let g be an equivalence class of galleries with source A. There exists a gallery class g' of $(V_P, \mathcal{M}_P)$, with source $\pi_F^{-1}(A)$, such that, for every gallery H of $(V_P, \mathcal{M}_P)$, g begins with $\pi_F(H)$ if and only if g' begins with H.

Let $\mathcal{S}$ be the set of gallery classes h with source $\pi_F^{-1}(A)$ in $(V_P, \mathcal{M}_P)$ such that g begins with $\pi_F(h)$. Apply (1.14) to $(V_P, \mathcal{M}_P)$ and to $\mathcal{S}$. By (1.19), condition (i) is satisfied, cf. the proof of (1.20), and (1.24) follows from (1.14)(ii).

(1.25) The set of galleries may be regarded as the set of arrows of a category $\mathrm{Gal}_0(V, \mathcal{M})$ whose objects are the chambers. Similarly, equivalence classes of galleries are the arrows of a quotient category $\mathrm{Gal}^+(V, \mathcal{M})$. Let A, B, C be three chambers, let E be a gallery from A to B, and let F be a gallery from B to C. Although the general categorical convention is different, we shall continue to write EF for the composite of E and F. The law $G \mapsto G^*$, respectively $G \mapsto -G$, is an anti-equivalence, respectively an equivalence, of $\mathrm{Gal}_0(V, \mathcal{M})$, or of $\mathrm{Gal}^+(V, \mathcal{M})$, with itself, inducing the identity, respectively $C \mapsto -C$, on objects.

When no confusion is possible, we shall write simply Gal_0 and Gal^+ for $\mathrm{Gal}_0(V, \mathcal{M})$ and $\mathrm{Gal}^+(V, \mathcal{M})$, or for a category $\mathrm{Gal}_0(V_P, \mathcal{M}_P)$ or $\mathrm{Gal}^+(V_P, \mathcal{M}_P)$. By abuse of language, we shall often call the arrows of Gal^+ galleries.

(1.26) Lemma.

(i) In Gal^+, for every gallery g, one has

$$g\Delta = \Delta(-g).$$

(ii) For any g and h with source A in Gal^+, there exists n such that $g\Delta^n$ begins with h. If h is composed of k galleries $u(A_i, A_{i+1})$, one may take $n = k$.

By induction, it is enough to prove (i) when g has length one. For $g = (B, C)$, one has

$$g\Delta = u(B, C)u(C, -C) = u(B, C)u(C, -B)u(-B, -C) = u(B, -B)u(-B, -C) = \Delta(-g).$$

For every chamber B, $\Delta = u(A, -A)$ begins with $u(A, B)$. Hence (ii) follows from (i) by induction on k.

The following proposition follows at once from (1.26), from its transform by $*$, and from (1.19)(i),(ii).

(1.27) Proposition.

(i) In Gal^+, the set of all arrows admits a left and right calculus of fractions.

(ii) Let $\mathrm{Gal}(V, \mathcal{M})$, or simply Gal, be the category, a groupoid, obtained from Gal^+ by making all arrows invertible. Call positive those arrows of Gal lying in the image of Gal^+. The canonical functor from Gal^+ to Gal is faithful, and every arrow g of Gal can be written in the forms

$$g = g_1 \Delta^{-n} = \Delta^{-n} g_2$$

with g_1 and g_2 positive.

If a category C has only one object A, and if the monoid $\mathrm{Hom}(A, A)$ is cancellative, to say that all arrows of C admit a left and right calculus of fractions means that $\mathrm{Hom}(A, A)$ satisfies Ore's condition on the left and on the right. The calculus of fractions used in (1.27) and below is an immediate generalization of Ore's theory for embedding such a monoid in a group.

(1.28) For every wall M, the number of times that $g \in \mathrm{Gal}^+$ crosses M is well-defined (1.11). This function of g extends additively to $g \in \mathrm{Gal}$. More generally, for every intersection of walls P, one could use the universal property of Gal to define functors

$$\pi_P : \mathrm{Gal}(V, \mathcal{M}) \longrightarrow \mathrm{Gal}(V_P, \mathcal{M}_P).$$

(1.29) Let A be a chamber, let P be an intersection of walls of A, and let $F = F(P)$ (1.6). The function π_F induces a functor

$$\pi_F : \mathrm{Gal}_0(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}_0(V, \mathcal{M}). \tag{1.29.1}$$

The image of this functor is stable under equivalence, and it induces a faithful functor

$$\pi_F : \mathrm{Gal}^+(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}^+(V, \mathcal{M}). \tag{1.29.2}$$

From the calculus of fractions and (1.19)(i) follows the following proposition.

(1.30) Proposition. Under the preceding hypotheses, the functor deduced from (1.29.2)

$$\pi_F : \mathrm{Gal}(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}(V, \mathcal{M})$$

is faithful.

(1.31) Proposition. Let C be a chamber; let I and J be two sets of walls of C; put $K = I \cup J$; let P, Q, R be the intersections of the walls in I, J, K, and let $F(P), F(Q), F(R)$ be the corresponding facets in C (1.6). The facet $F(R)$ is the smallest facet containing $F(P)$ and $F(Q)$. In the groupoid $\mathrm{Gal}(V, \mathcal{M})$, one has

$$\pi_{F(P)}(\mathrm{Gal}(V_P, \mathcal{M}_P)) \cap \pi_{F(Q)}(\mathrm{Gal}(V_Q, \mathcal{M}_Q)) = \pi_{F(R)}(\mathrm{Gal}(V_R, \mathcal{M}_R)).$$

A chamber having $F(P)$ and $F(Q)$ as facets also has $F(R)$ as a facet. This proves (1.31) for objects.

Let $g \in \mathrm{Hom}_{\mathrm{Gal}}(A, B)$, and suppose that

$$g = \pi_{F(P)}(g_1) = \pi_{F(Q)}(g_2).$$

By (1.27)(ii), one has

$$g_1 = \Delta(P)^{-n} g_1', \qquad g_2 = \Delta(Q)^{-n} g_2',$$

with g_i' positive, for $n > 0$ sufficiently large. Let $g_1 = \pi_{F(P)}(g_1')$ and $g_2 = \pi_{F(Q)}(g_2')$. By (1.26)(ii), transformed by $*$, $\Delta^n \Delta(P)^{-n}$ is positive, and in $\mathrm{Gal}^+(V, \mathcal{M})$ one has

$$(\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2. \tag{1.31.1}$$

(1.31.2) Lemma. If $h \in \mathrm{Hom}_{\mathrm{Gal}^+}(A,B)$ begins both with $\Delta^n\Delta(P)^{-n}$ and with $\Delta^n\Delta(Q)^{-n}$, then h begins with $\Delta^n\Delta(R)^{-n}$.

We deduce (1.31) from (1.31.2). By (1.31.2), one has

$$(\Delta^n\Delta(P)^{-n})g_1 = (\Delta^n\Delta(Q)^{-n})g_2 = (\Delta^n\Delta(R)^{-n})g_3$$

with g_3 positive. Hence $g = \Delta(R)^{-n}g_3$. The positive gallery $g_3 = \Delta(R)^n g$ lies in the image of both $\pi_{F(P)}$ and $\pi_{F(Q)}$. By (1.28), it therefore crosses zero times every wall not containing P or Q, i.e. every wall not containing R. It thus lies in the image of $\pi_{F(R)}$.

We prove (1.31.2). Let $A(P) = A.\Delta.A(P)$, and similarly for Q and R. The gallery $\Delta\Delta(P)^{-1}$ with source A is $u(A, A(P))$; the one with source $A(P)$ is $u(A(P), A)$. By (1.26)(i), $\Delta\Delta(P) = \Delta(P)\Delta$. Hence

$$\Delta^n\Delta(P)^{-n} = u(A, A(P))\, u(A(P), A)\, u(A, A(P)) \cdots \tag{1.31.3}$$

with n factors.

(1.31.4) Lemma. $A(R) \in D(A(P), A(Q))$.

With the notation of (1.2), $\mathcal{M}(A, A(P))$ is the set of walls containing P, and $\mathcal{M}(A, A(Q))$ is the set of those containing Q. By (1.2), $\mathcal{M}(A(P), A(Q))$ is the set of walls containing P or Q, but not R, namely

$$\mathcal{M}(A(P), A(R)) \cup \mathcal{M}(A(Q), A(R)),$$

and one applies (1.4).

By (1.19)(iii), a gallery class which begins with $\Delta\Delta(P)^{-1}$ and $\Delta\Delta(Q)^{-1}$ also begins with $\Delta\Delta(R)^{-1}$, and (1.31.2) follows by induction from the following lemma.

(1.31.5) Lemma. Let P and R be intersections of walls of a chamber A, with $P \subset R$. Let h be in Gal^+ with source A. If h begins both with $\Delta^n\Delta(P)^{-n}$, $n > 0$, and with $\Delta\Delta(R)^{-1}$, then h begins with

$$(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-1}.$$

One may assume $n > 2$. Put $h = (\Delta\Delta(P)^{-1})h'$. With the preceding notation, h' then begins both with $\Delta\Delta(P)^{-1} = u(A(P), A)$ and with $u(A(P), A(R))$. Apply (1.19)(iii). The chamber C of loc. cit. is then separated from $A(P)$ by every wall M separating $A(P)$ from A, i.e. $M \not\supset P$, or $A(P)$ from $A(R)$, i.e. $M \supset P$ and $M \not\supset R$. Thus C is separated from $A(P)A$ only by walls $M \supset R$, and

$$C \in D(A(P)A, A(P)A(R)).$$

It follows that h' begins both with $\Delta\Delta(R)^{-1}$ and with $(\Delta\Delta(P)^{-1})^{n-1}$. Induction on n shows that h' begins with

$$(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-2},$$

so that h begins with

$$(\Delta\Delta(P)^{-1})(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-2}.$$

One concludes by noting that

$$\Delta\Delta(P)^{-1}\Delta\Delta(R)^{-1} = \Delta\Delta(R)^{-1}\Delta\Delta(P)^{-1}.$$

(1.32) Proposition. For F a facet of a chamber C, generating an intersection of walls P, one has

$$\pi_F^{-1}(\mathrm{Gal}^+(V, \mathcal{M})) = \mathrm{Gal}^+(V_P, \mathcal{M}_P) \subset \mathrm{Gal}(V_P, \mathcal{M}_P).$$

Suppose that $\pi_F(\Delta^{-n}g) = h$ lies in $\mathrm{Gal}^+(V, \mathcal{M})$. Then $\pi_F(g) = \Delta(P)^n h$, and by (1.28) every gallery H representing h crosses only walls passing through F. Thus $h = \pi_F(h_1)$, with h_1 in $\mathrm{Gal}^+(V_P, \mathcal{M}_P)$, and the conclusion follows from (1.30).

2. Buildings

(2.1) Let $(V, \mathcal{M})$ be as in §1, satisfying (1.8), and let $r = \dim V$. Let S be the sphere of radius one in V, for some Euclidean structure on V; more intrinsically, one could take $S = (V - \{0\})/\mathbb{R}_+^*$. The hyperplanes $M \in \mathcal{M}$ cut out a triangulation of S, and we shall again denote by S both the corresponding simplicial space and its geometric realization. We transfer to S the terminology used for V: chambers, facets, adjacency, galleries.

(2.2) Choose a fundamental chamber A_0 in S. We denote by I^+ the space constructed below; it depends on $V, \mathcal{M}, A_0$, and is equipped with $q : I^+ \to S$.

a) Let $\mathcal{G}^+$ be the set of equivalence classes of galleries g with source A_0,

$$\mathcal{G}^+ = \coprod_B \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, B).$$

b) Let Z^+ be the disjoint sum, indexed by $g \in \mathcal{G}^+$, of the closed simplex of S which is the closure of the open simplex of dimension $r-1$ equal to the target of g:

$$Z^+ = \coprod_{g \in \mathcal{G}^+} (\mathrm{target}\, g)^-.$$

Let q' be the evident map from Z^+ to S.

c) I^+, with q, is a quotient of Z^+, with q'. If G is a gallery from A_0 to B, and if C is a chamber adjacent to B, one glues $(\mathrm{target}\, g)^-$ and $(\mathrm{target}\, g(BC))^-$ along the common closed face of their images in S.

(2.3) The space I^+ is decomposed into facets, images of facets of Z^+, each mapping bijectively to a facet of S. Its chambers, respectively faces and vertices, are its facets of dimension $r-1$, respectively $r-2$ and 0.

The vertices of a facet F are the vertices contained in the closure $\overline{F}$ of F. The chambers of I^+ are indexed by $\mathcal{G}^+$; $A_0.g$ denotes the one indexed by g, and A_0 the one indexed by (A_0).

Let $g \in \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, A)$, let $A = A_0.g$, and let $H = (C_0, \dots, C_n)$ be a gallery from A to B, with class h. Put

$$A.h = A_0.gh.$$

The chambers $A.(C_0, \dots, C_i)$, $0 \le i \le n$, form a gallery in I^+. Such galleries will be called positive.

(2.4) Lemma. Let F be a facet of S, let A and B be two chambers of S having F as a facet, and let $g \in \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, A)$, $h \in \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, B)$. The following conditions are equivalent:

(a) in I^+, the facets $q^{-1}(F)$ of $A_0.g$ and $A_0.h$ coincide;

(b) $g^{-1}h \in \mathrm{Hom}_{\mathrm{Gal}}(A, B)$ lies in $\pi_F(\mathrm{Gal})$;

(c) there are g_1 and h_1 in $\pi_F(\mathrm{Gal}^+)$ such that $gg_1 = hh_1$.

Condition (b) is an equivalence relation between galleries from A_0 to a chamber having F as a facet. This gives (a) $\Rightarrow$ (b). That (b) implies (c) follows from (1.26)(i). Finally, (c) $\Rightarrow$ (a) is immediate from the definitions.

From (2.4), (a) $\Leftrightarrow$ (b), and from (1.31), one immediately obtains:

(2.5) Proposition. Each facet of I^+ is uniquely determined by the set of its vertices. Thus I^+ may be described as the geometric realization of the following simplicial scheme.

a) For x a vertex of S, let $\mathcal{G}(x)$ be the set of g in Gal^+ with source A_0 and with target a chamber having x as a vertex. Let $\mathcal{G}(x)/\pi_x$ be the quotient of $\mathcal{G}(x)$ by the equivalence relation (2.4)(b), for $F = x$. Then the set of vertices is

$$\coprod_x \mathcal{G}(x)/\pi_x.$$

b) A set E of vertices spans a simplex if and only if there exists a gallery g with source A_0 such that, for $y \in E$, g lies in $\mathcal{G}(qy)$ and y is the image of g in $\mathcal{G}(qy)/\pi_{qy}$.

(2.6) Proposition. Let F be a facet of I^+. There exists a gallery class g such that, for F to be a facet of a chamber $B = A_0.h$, it is necessary and sufficient that

$$h = g\,\pi_F(h')$$

for a suitable h' in Gal^+.

Take g to be a gallery of minimum length such that F is a facet of $A = A_0.g$. Let $B = A_0.h$ be a chamber having F as a facet. By (2.4)(a) $\Leftrightarrow$ (c), there are galleries h_1 and h_2 in $(V_{qF}, \mathcal{M}_{qF})$ with

$$g\pi_F(h_1) = h\pi_F(h_2).$$

Apply the transform by $*$ of (1.24). By the minimality of g, one finds that h_1 ends with h_2, say $h_1 = h'h_2$. By (1.19)(ii), $g\pi_F(h') = h$, which proves (2.6).

(2.7) Notation. If A is a chamber of I^+, denote by $S(A)$ the union of the closed chambers $(A.(qA, B))^-$, where B runs through the chambers of S. One checks that $q|S(A)$ is an isomorphism from $S(A)$ to S.

(2.8) Definition. $\overline{I}^+$ is the space obtained from I^+ by filling in the spheres $S(A)$.

More precisely, let B^r be a ball whose boundary is S. If one wants to work simplicially, one takes B^r to be the cone on the simplicial space S. Define $\overline{I}^+$ from I^+ by attaching a family of copies $b(A)$ of B^r, indexed by the chambers of I^+.

(2.9.2) Lemma. Suppose that there exists a chamber $B = A_0.h$ in $(I^+)_{n+1}$, distinct from $A = A_0.g$, having F as a facet. Then there exists a chamber $B' = A_0.h'$ in $(I^+)_n$, and a wall M of qB', containing qF, such that

$$A = B'.\Delta(M).$$

Let g_0 be the gallery considered in (2.6). One has

$$g = g_0\pi_F(g_1), \qquad h = g_0\pi_F(h_1).$$

The hypotheses imply that $\lg(g_1) \neq 0$. For a suitable wall M of qA, one then has $g_1 = h'\Delta(M)$, and one puts $B' = \pi_F h'$.

For the closed chamber $\overline{A}$, $\overline{A} \cap (I^+)_n$ is the union of a nonempty set of closed faces, and in $(I^+)_{n+1}$ the points of

$$\overline{A} - (\overline{A} \cap (I^+)_n)$$

belong only to the single closed chamber $\overline{A}$. There are two cases.

Case 1. $\overline{A} \cap (I^+)_n \neq \partial\overline{A}$. In this case, there is no sphere $S(B)$ with

$$A \subset S(B) \subset (I^+)_{n+1}.$$

We take $\varphi_t|\overline{A}$ to be a collapse of $\overline{A}$ onto $\overline{A} \cap (I^+)_n$.

Case 2. $\overline{A} \cap (I^+)_n = \partial\overline{A}$. Put $A = A_0.g$. For every wall M of qA, it follows from (2.7.2) that g ends with $\Delta(M)$. By (1.19)(iii), g therefore ends with Δ: $A = B.\Delta$, with B uniquely determined by A, by (1.19)(ii). In passing from $(\overline{I}^+)_{n+1}$ to $(\overline{I}^+)_n$, A and the interior of the ball $b(B)$ disappear. We take $\varphi_t|b(B)$ to be a collapse of $b(B)$ onto $S(B) - A$. As seen in case 1, every ball which disappears is of the preceding type. This completes the construction of φ_t, and proves (2.7).

(2.10) The space I, with $q : I \to S$, is defined like I^+, replacing Gal^+ by Gal:

a) put $\mathcal{G} = \coprod_B \mathrm{Hom}_{\mathrm{Gal}}(A_0, B)$;

b) put $Z = \coprod_{g\in\mathcal{G}} (\mathrm{target}\, g)^-$;

c) let q' be the evident map from Z to S. The space I, with q, is a quotient of Z, with q'. If $g \in \mathcal{G}$ has target B and C is adjacent to B, one glues $(\mathrm{target}\, g)^-$ and $(\mathrm{target}\, g(BC))^-$ along the common closed face of their images in S.

Equivalently, if two chambers B and C have a common facet F, if $g \in \mathrm{Hom}_{\mathrm{Gal}}(A_0, B)$, and if $h \in \mathrm{Hom}_{\mathrm{Gal}}(A_0, C)$, then, in I, the closed facets $q'^{-1}(F)^-$ of $(\mathrm{target}\, g)^-$ and of $(\mathrm{target}\, h)^-$ are identified if and only if

$$hg^{-1} \in \pi_F\, \mathrm{Hom}_{\mathrm{Gal}}(\pi_F^{-1}B, \pi_F^{-1}C).$$

(2.11) As for I^+, one defines chambers, faces, and open or closed facets of I. One defines A_0, and $A.h$, for A a chamber and h in Gal with source qA, as in (2.3). The analogue of (2.4)(a) $\Leftrightarrow$ (b) is here trivially true. As in (2.5), one deduces from (1.31) that each facet of I is uniquely determined by its set of vertices, and I appears as the geometric realization of a simplicial scheme with a description of type (2.5). For every chamber A of I, define a sphere $S(A)$ as in (2.7); q induces an isomorphism from $S(A)$ to S. As in (2.8), define a building $\overline{I}$, equipped with $q : \overline{I} \to B^r$, by filling in all the spheres $S(A)$.

(2.12) Besides q and its chamber decomposition, I has the following additional structure. A composition $B.g$ is defined, which associates to a chamber B of I, whose image in S is B, and to $g \in \mathrm{Hom}_{\mathrm{Gal}}(B, C)$, a chamber of I whose image in S is C. The construction $g \mapsto B.g$ identifies the analogue of I obtained by choosing B as fundamental chamber with the space I. In particular,

$$B.(gh) = (B.g).h.$$

(2.13) For every chamber A of I above A_0, define a map

$$i_A : I^+ \longrightarrow I$$

and similarly $i_A : \overline{I}^+ \to \overline{I}$, compatible with the projection to S, respectively B^r, by sending the closed chamber $(A.g)^-$ of I^+, respectively the ball $b(A.g)$ of $\overline{I}^+$, to the chamber, respectively ball, of the same name in I, respectively $\overline{I}$.

(2.14) Proposition.

(i) i_A identifies I^+ with a closed subspace of I, and $\overline{I}^+$ with a closed subspace of $\overline{I}$.

(ii) One has

$$I = \varinjlim i_{A_0.\Delta^{-2n}}(I^+), \qquad \overline{I} = \varinjlim i_{A_0.\Delta^{-2n}}(\overline{I}^+).$$

The assertions (i) or (ii) for $\overline{I}^+$ and $\overline{I}$ follow from the corresponding assertions for I^+ and I.

We prove (i) for I^+ and I. Let B and C be two chambers of S with a common facet F. Let $g \in \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, B)$ and $h \in \mathrm{Hom}_{\mathrm{Gal}^+}(A, C)$. Suppose that the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are equal in I. Then

$$g^{-1}h = \pi_F(e) \quad \text{for} \quad e \in \mathrm{Hom}_{\mathrm{Gal}}(\pi_F^{-1}B, \pi_F^{-1}C).$$

By (1.26), write $e = e_1 e_2^{-1}$, with e_i in Gal^+. By (1.29), $g\pi_F(e_1) = h\pi_F(e_2)$ in Gal^+, and the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are therefore already equal in I^+.

For (ii), it is enough to note that every gallery g with source A_0 in Gal can be written

$$g = \Delta^{-2n} g'$$

with g' in Gal^+.

From (2.14) one obtains $\pi_i(I) = \varinjlim \pi_i(I^+)$. Since I is a CW-complex, (2.9) gives:

(2.15) Theorem. The space $\overline{I}$ is contractible.

The space I is therefore the bouquet of the spheres $S(A)$, as A ranges through the chambers of I.

3. Coverings

(3.1) Let $V_{\mathbb{C}}$ be the complexification of V, and let $M_{\mathbb{C}}$ be the complexification of M, for $M \in \mathcal{M}$. Put

$$Y = V_{\mathbb{C}} - \bigcup_{M \in \mathcal{M}} M_{\mathbb{C}}.$$

In this section, we construct by gluing a space above Y. The gluing data will be described using I and $\overline{I}$. The facets of S and the facets of V other than $\{0\}$ are in canonical bijection, and we shall constantly pass from one to the other. In general we shall use the same notation for the corresponding facets; when necessary the correspondence will be denoted $F \mapsto [F]$.

(3.2) Lemma. Let A, B, C be three chambers of I, with $C \subset S(A) \cap S(B)$. Then q induces an isomorphism from $S(A) \cap S(B)$ to the intersection of S with the closed half-spaces $D'_M(qC)$, containing qC, bounded by a wall M which separates qA from qB.

The intersection of the $D'_M(qC)$ is a union of closed chambers. If C' is one of them, then qA and qB lie on the same side of every wall separating qC from C'. It follows that the lifts, to $S(A)$ or $S(B)$, of a minimal gallery from qC to C' coincide, and $(C')^- \subset q(S(A) \cap S(B))$.

Conversely, if a facet F is not in the intersection of the $D'_M(qC)$, then there is a wall M such that $F \subset M$, separating qA from qB and F from qC. Suppose, for contradiction, that $F \subset q(S(A) \cap S(B))$. Let D be a chamber of S having F as a facet, and let $(C_0, C_1, \ldots, C_{n+1})$ be a gallery from qC to D. Let M_i be the wall separating C_i from C_{i+1}, and let α_i, respectively β_i, be $+1$ or -1 according as C_{i+1} and A, respectively B, are or are not on the same side of M_i. By hypothesis, F is a facet of

$$C.\Delta(M_0)^{\alpha_0} \cdots \Delta(M_n)^{\alpha_n}$$

and of

$$C.\Delta(M_0)^{\beta_0} \cdots \Delta(M_n)^{\beta_n}.$$

By (2.4) and (1.24), since $M \not\supset F$, this gives contradictory equalities: one side equals 1, the other -1.

(3.3) Lemma. Let $\Re$ and $\Im$ be the real and imaginary part maps from $V_{\mathbb{C}}$ to V. Let $v \in V_{\mathbb{C}}$, and let F be the facet of V such that $\Re v \in F$. For v to belong to Y, it is necessary and sufficient that $\mathrm{pr}_F(\Im v)$ lie in a chamber of $(V_F, \mathcal{M}_F)$.

This is clear.

What follows is based on the following intuition: to every walk in I, performed by following a gallery of the form

$$\Delta(M_1)^{\varepsilon_1} \cdots \Delta(M_k)^{\varepsilon_k}, \qquad \varepsilon_i = \pm 1,$$

there corresponds a path, or path class, in Y. Its image under $\Re$ in V passes from chamber to chamber, crossing successively the walls $M_1, \ldots, M_k$ at points m_i. For $\varepsilon_i = 1$, respectively -1, the path passes through one of the two components R of $\Re^{-1}(m_i) \cap Y$, in bijection by $\Im$ with $V - M_i$: the one such that $\Im R$ contains, respectively does not contain, the preceding chamber.

(3.4) Notation.

(i) If C is a chamber of $(V, \mathcal{M})$, let $Y(C)$ be the open subset of Y consisting of the $x \in V_{\mathbb{C}}$ satisfying the following condition: if F is the facet of V such that $\Re x \in F$, then $\mathrm{pr}_F(\Im x) \in \pi_F^{-1}(C)$.

(ii) Let A and B be two chambers of I. If $S(A) \cap S(B)$ contains a chamber C, temporarily denote by $Y'(A, B)$ the intersection of the open half-spaces $D'_M(qC)^\circ$ for M as in 3.2. Put

$$Y(A, B) = Y(qA) \cap Y(qB) \cap \Re^{-1}(Y'(A, B)) \subset V_{\mathbb{C}}.$$

If $S(A) \cap S(B)$ contains no chamber, put $Y(A, B) = \varnothing$.

(iii) For a facet F of V, the star $\mathrm{Et}(F)$ is the union of the open facets G of $(V, \mathcal{M})$ such that $F \subset \overline{G}$.

(iv) For a facet F of V and a chamber C of $(V_F, \mathcal{M}_F)$, put

$$V(F, C) = \{\, v \in V_{\mathbb{C}} \mid \Re v \in \mathrm{Et}(F),\ \mathrm{pr}_F(\Im v) \in C \,\}.$$

(3.5) Lemma. Let G be a facet of V. The $V(F, C)$, for F a facet such that $G \subset \overline{F}$ and C a chamber of V_F, form an open covering of

$$Y \cap \Re^{-1}(\mathrm{Et}\, G).$$

In particular, taking $G = \{0\}$, the $V(F, C)$ cover Y.

If $x \in Y \cap \Re^{-1}(\mathrm{Et}\, G)$ and $\Re x \in F$, then x lies in one of the $V(F, C)$, by (3.3).

(3.6) Let Y' be the disjoint sum, indexed by the balls $b(A)$ of $\overline{I}$,

$$Y' = \coprod_{b(A)} Y(qA).$$